

**Description of GATE Valve (GA12-DI-F600):**

Type: Gate, Resilient Wedge, Flanged
Sub-Type: with Indicator Post Style Stuffing Box
Valve Size: NPS 2 -to- NPS 24
Class: ASME 125
Pressure: to 12" NPS = 350 psig (24 bar) max; 14" -24" NPS = 250 psig (17.6 bar)
Ends: Flanged FF (Flat Faced)
Body: Ductile Iron (ASTM A536)
Bonnet: Ductile Iron (ASTM A536)
Wedge: Ductile Iron/Rubber Encapsulated
Bore-Port: Full
Bonnet Type: Bolted
Stem Design: Non Rising Stem (N.R.S.)
Stem: 316 SS Stem
Top Nut: 2" Square Operating Nut
Operation: Left-to-Open (CCW), Right-to-Close (CW)
Coating: Fusion Bonded Epoxy - interior and exterior surfaces
Body-Bonnet Bolting: Bolts/Nuts: MFG STD.
Note: Any working pressure above 250 psig requires special gaskets per AWWA C111.

The standards are:

DESIGN: AWWA C515
ENDS: ASME B16.1, CL125
RATING: MFG. STD.
TESTING: API 598
APPROVALS: FM 1120/1130, UL 262 Listed
CERTIFIED: NSF 61 and NSF 372
DIMENSIONAL: ASME B16.10

Approved Vendors: Mueller, Model P-2361-6, CL125, N.R.S., Flanged Ends

Description of GATE Valve (GA12-DI-F602):

Type: Gate, Resilient Wedge, MJ Joints
Sub Type: with Indicator Post Style Stuffing Box
Option: with M.J. Accessories
Valve Size: NPS 2 -to- NPS 24
Class: ASME 125
Pressure: to 12" NPS = 350 psig (24 bar) max; 14" -24" NPS = 250 psig (17.6 bar)
Ends: Mechanical Joint (MJ)
Body: Ductile Iron (ASTM A536)
Bonnet: Ductile Iron (ASTM A536)
Wedge: Ductile Iron/Rubber Encapsulated
Bore-Port: Full
Bonnet Type: Bolted
Stem Design: Non Rising Stem (N.R.S.)
Stem: 316 SS Stem
Top Nut: 2" Square Operating Nut
Operation: Left-to-Open (CCW), Right-to-Close (CW)
Coating: Fusion Bonded Epoxy - interior and exterior surfaces
Body-Bonnet Bolting: Bolts/Nuts: MFG STD (316 SS)
Note: Any working pressure above 250 psig requires special gaskets per AWWA C111.

The standards are:

DESIGN: AWWA C515
ENDS: ASME B16.1, CL125
TESTING: MFG STD
APPROVALS: FM 1120/1130, UL 262 Listed
CERTIFIED: NSF 61 and NSF 372
DIMENSIONAL: MFG STD

Approved Vendors: Mueller, Model P-2361-20, CL125, N.R.S., with MJ Accesories

Description of GATE Valve (GA25-DI-F620):

Type: Gate, Resilient Wedge, Flanged
 Sub-Type: with Indicator Post Style Stuffing Box
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 250**
 Pressure: to 12" NPS = 350 psig (24 bar) max; 14" -24" NPS = 250 psig (17.6 bar)
 Ends: Flanged FF (Flat Faced)
 Body: Ductile Iron (ASTM A536)
 Bonnet: Ductile Iron (ASTM A536)
 Wedge: Ductile Iron/Rubber Encapsulated
 Bonnet Type: Bolted
 Bore-Port: Full
 Stem Design: Non Rising Stem (N.R.S.)
 Stem: 316 SS Stem
 Top Nut: 2" Square Operating Nut
 Operation: Left-to-Open (CCW), Right-to-Close (CW)
 Coating: Fusion Bonded Epoxy - interior and exterior surfaces
 Body-Bonnet Bolting: Bolts: MFG STD.
 Nuts: MFG STD.

The standards are:

DESIGN: AWWA C515
 ENDS: ASME B16.1, CL250
 RATING: MFG. STD.
 TESTING: MFG STD
 APPROVALS: FM 1120/1130, UL 262 Listed
 CERTIFIED: NSF 61 and NSF 372
 DIMENSIONAL: MFG STD

Approved Vendors: Mueller, Model P-2361-12, **CL250**, N.R.S., Flanged Ends

Description of GATE Valve (GA25-DI-F622):

Type: Gate, Resilient Wedge, MJ
 Sub-Type: with Indicator Post Style Stuffing Box
 Option: with M.J. Accessories
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 250**
 Pressure: to 12" NPS = 350 psig (24 bar) max; 14" -24" NPS = 250 psig (17.6 bar)
 Ends: Mechanical Joint (MJ)
 Body: Ductile Iron (ASTM A536)
 Bonnet: Ductile Iron (ASTM A536)
 Wedge: Ductile Iron/Rubber Encapsulated
 Bonnet Type: Bolted
 Bore-Port: Full
 Stem Design: Non Rising Stem (N.R.S.)
 Stem: 316 SS Stem
 Top Nut: 2" Square Operating Nut
 Operation: Left-to-Open (CCW), Right-to-Close (CW)
 Coating: Fusion Bonded Epoxy - interior and exterior surfaces
 Body-Bonnet Bolting: Bolts: MFG STD (316 SS)
 Nuts: MFG STD (316 SS)

The standards are:

DESIGN: AWWA C515
 ENDS: ASME B16.1, CL250
 RATING: ASME MFG STD
 TESTING: MFG STD
 APPROVALS: FM 1120/1130, UL 262 Listed
 CERTIFIED: NSF 61 and NSF 372
 DIMENSIONAL: MFG STD

Approved Vendors: Mueller, Model P-2361-20, **CL250**, N.R.S., with MJ Accessories

Description of FIRE HYDRANT (FH12-DI-F600):

Type: Fire Hydrant
 Hydrant Size: 5-1/4" Main Valve Opening
 Class: ASME 125
 Pressure: 250 psig (16.7 bar)
 Ends: Flanged, FF
 Post Type: Dry Barrel Design
 Operating Nut: Bronze (ASTM B62)
 Bonnet: Cast Iron (ASTM A126, CL B)
 Upper Barrel: Cast Iron (ASTM A126, CL. B)
 Lower Barrel: Cast Iron (ASTM A126, CL. B)
 Lower Shoe (Entry): Cast Iron (ASTM A126, CL B)
 Drain Valves Dual bronze drain valves
 Connection #1: (1) 4" Storz Pumper Connection w/non-locking caps
 Connection #2: (2) 2-1/2" Hose connections
 Operation: Left-to-Open (CCW); Right-to-Close (CW)

The standards are:

DESIGN: AWWA C502
 ENDS: ASME B16.1, CL125
 RATING: MFG STD.
 TESTING: MFG STD.
 DIMENSIONAL: MFG STD.
 LISTINGS: UL 246 and FM 1510

Approved Vendors: Mueller, Super Centurion, Model, A-423, FF, 5-1/4" Main Valve
 Specify Burial Depth

Description of FIRE HYDRANT (FH12-DI-M602):

Type: Fire Hydrant
 Hydrant Size: 5-1/4" Main Valve Opening
 Class: ASME 125
 Pressure: 250 psig (16.7 bar)
 Ends: Mechanical Joint (MJ), Horizontal
 Post Type: Dry Barrel Design
 Operating Nut: Bronze (ASTM B62)
 Bonnet: Cast Iron (ASTM A126, CL B)
 Upper Barrel: Cast Iron (ASTM A126, CL. B)
 Lower Barrel: Cast Iron (ASTM A126, CL. B)
 Lower Shoe (Entry): Cast Iron (ASTM A126, CL B)
 Drain Valves Dual bronze drain valves
 Connection #1: (1) 4" Storz Pumper Connection w/non-locking caps
 Connection #2: (2) 2-1/2" Hose connections
 Operation: Left-to-Open (CCW); Right-to-Close (CW)

The standards are:

DESIGN: AWWA C502
 ENDS: ASME B16.1, CL125
 RATING: MFG STD.
 TESTING: MFG STD.
 DIMENSIONAL: MFG STD.
 LISTINGS: UL 246 and FM 1510

Approved Vendors: Mueller, Super Centurion, Model A-423, MJ Joint, 5-1/4" Main Valve
 Specify Burial Depth

**Description of CHECK Valve (CHS12-DI-F600):**

Type: Check, Swing
Design Type: Swing
Valve Size: NPS 3 -to- NPS 16
Service: -29 to 60 °C (-20 to 140 °F)
Rating: 250 psig (17 bar)
Class: ASME 125
Ends: Flanged FF
Body: Ductile Iron (ASTM A536, Gr. 65-45-12)
Cover: Ductile Iron
Disc: Steel w/EPDM coating
Bore: Full Port (FP)
Cover/Bonnet Type: Bolted
Body-Bonnet Gaskets: Rubber (NBR)
Body-Bonnet Bolting: Bolts/Nuts: MFG STD
Coating: Fusion Bonded Epoxy - Internal & External
Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN:
ENDS: ASME B16.1, CL 125
RATING: MFG. STD.
TESTING: MFG STD
CERTIFICATIONS: NSF 61
DIMENSIONAL: MFG. STD

Approved Vendors: Mueller/Pratt, RD-Series Check Valve, CL 125, FF

Description of INDICATOR POST (IP-"Valve Size"-F600): Example: IP-6"-F600

Type: Indicator Post, Vertical
Use on Valve Size: 2" to 24" NPS Valves
Class: N/A
Ends: Bell Mounts to Stuffing Box Flange
Post Head: Gray Iron (ASTM A126, Class B)
Bell: Gray Iron
Upper Barrel: Carbon Steel
Lower Barrel: PVC, DR 14
Upper/Lower Stem: Carbon Steel
Adjustment: Post Adjusting Screws
Operation: Standard = Position for Left-hand (CCW) Opening Valves
..... Optional = Position for Right-hand (CW) Opening Valves

The standards are:

DESIGN: MFG STD.
ENDS: MFG STD.
RATING: N/A (Atmospheric)
TESTING: N/A
DIMENSIONAL: MFG STD.

Approved Vendors: 4" to 14" NPS = Mueller, Model A-20806-{Order Length}
..... 16" to 24" NPS = Mueller, Model A-20807-{Order Length}
..... Order Length (Min. & Max.) = per Mueller Catalog Calculation
..... based on Trench Depth & Mueller Mfg. Tables (Table #1 and #2)

Rev.	RevDate	RevDescription	By:	Check:
0	20 SEPT 2023	Issued for Release	MFS	SBS

***** Start of Normal Carbon Steel Ball Valves *****

Description of BALL Valve (BA01CB500): (BA15-CS-F500)

Type: Ball, Floating, RP
 Valve Size: NPS 1.0 -to- NPS 8
 Service: -29 to 204 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flangedp RF
 Body: CS (ASTM A216-WCB)
 Trim: 316 SS
 Seat: RPTFE or TFM 1600
 Stem: 316 SS
 Body Type: Split, 2-Piece
 Design: Floating Ball, Fire Tested API 607
 Bore-Port: **Reduced Port (RP)**
 Operation: Lever to 6" NPS; **Gear Op. at 8" NPS**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern = Short Pattern (1" to 4" NPS)
 ASME B16.10 – Short Pattern (6" or 8" NPS) *** MUST be Short Pattern ***

Approved Vendors:

(1-1/2" to 6" NPS Only) Apollo (Cast, CL150, RP): Series 88A-140-24, Lever Op.
 (1" to 8" NPS) Bray, Flow-Tek, Series RF15 (Cast, CL150, RP): 15-{Size}-11001-CSJG

Description of BALL Valve (BA01CB501): (BA15-CS-F501)

Type: Ball, Floating, FP
 Valve Size: NPS 0.5 -to- NPS 8
 Service: -29 to 204 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: CS (ASTM A216-WCB)
 Trim: 316 SS
 Seat: RPTFE or TFM 1600
 Stem: 316 SS
 Body Type: Split, 2-Piece
 Design: Floating Ball, Fire Tested API 607
 Bore-Port: **Full Port (FP)**
 Operation: Lever to 6" NPS; **Gear Op. at 8" NPS**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern (All Sizes)

Approved Vendors:

(1/2" to 6" NPS) APOLLO (Cast, CL150, FP): Series 88A-240-24, Level to 6" NPS
 (1/2" to 8" NPS) Bray, Flow-Tek, Series F15 (Cast, CL150, FP): 15-{Size}-11000-CSJG
 Lever Op. to 6" NPS; Gear Op at 8" NPS

**Description of BALL Valve (BA01CB502):**

(BAT15-CS-F502)

Type: Ball, Trunnion, RP
Valve Size:..... NPS 6 -to- NPS 12
Temperature: -26 to 204 °C (-15 to 400 °F)
Class: **ASME 150**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Trim: 316 SS
Seat: RPTFE
Stem/Trunnion 316 SS
Springs: Alloy X-750
Body Type: Side-Entry, 2-Piece
Design: **Trunnion Ball**, Fire Tested API 607
Bore-Port: **Reduced Port (RP)**
Operation:..... **Gear**
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite
O-Rings: Viton

The standards are:

DESIGN:..... ASME B16.34 or API 6D
ENDS: ASME B16.5
RATING:..... ASME B16.34
TESTING:..... API 6D
DIMENSIONAL:..... **ASME B16.10 - Long Pattern (6" to 12" NPS)**
..... ***** No short-pattern valves allowed *****

Approved Vendors:

(6" to 12" NPS) WKM (CL150, Cast, RP): {Size x Size}-370D6-150-RFxRF-24-TVF-23-WG

Description of BALL Valve (BA01CB503):

(BAT15-CS-F503)

Type: Ball, Trunnion, FP
Valve Size:..... NPS 6 -to- NPS 16
Service: -26 to 204 °C (-15 to 400 °F)
Class: **ASME 150**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Trim: 316 SS
Seat: RPTFE
Stem/Trunnion:..... 316 SS
Springs: Alloy X-750
Body Type: Side-Entry, 2-Piece
Design: **Trunnion Ball**, Fire Tested API 607
Bore-Port: **Full Port (FP)**
Operation:..... **Gear**
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite
O-Rings: Viton

The standards are:

DESIGN:..... ASME B16.34 or API 6D
ENDS: ASME B16.5
RATING:..... ASME B16.34
TESTING:..... API 6D
DIMENSIONAL:..... ASME B16.10 - Long Pattern

Approved Vendors:

(6" to 16" NPS) WKM (CL150, Cast, FP): {Size}-370D6-150-RFxRF-24-TVF-23-WG

Description of BALL Valve (BA01CB504):

(BA15-CS-F504)

Type: Ball, Floating, FP
 Valve Size:..... NPS 4 -to- NPS 10
 Service: -29 to 204 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: CS (ASTM A216-WCB)
 Trim: 316 SS
 Seat: RPTFE or TFM 1600
 Stem: 316 SS
 Body Type: Split, 2-Piece
 Design: Floating Ball, Fire Tested API 607
 Bore-Port: **Full Port (FP)**
 Operation:..... **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN:..... API 608
 ENDS: ASME B16.5
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... ASME B16.10 - Long Pattern (4" to 10" NPS)

Approved Vendors:

(4" to 10" NPS) Apollo (Cast, CL150, FP): Series 88A-240-24-MG, Gear Oper
 (4" to 10" NPS) Bray, Flow-Tek, Series F15 (Cast, CL150, FP): 15-{Size}-11000-CSJG

Description of BALL Valve (BA01CB505):

(BA15-CS-F505)

Type: Ball, Floating, RP
 Valve Size:..... NPS 4 -to- NPS 10
 Service: -29 to 204 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: CS (ASTM A216-WCB)
 Trim: 316 SS
 Seat: RPTFE or TFM 1600
 Stem: 316 SS
 Body Type: Split, 2-Piece
 Design: Floating Ball, Fire Tested API 607
 Bore-Port: **Reduced Port (RP)**
 Operation:..... **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN:..... API 608
 ENDS: ASME B16.5
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... ASME B16.10 - Long Pattern = Short Pattern (4" NPS)
 ASME B16.10 – Short Pattern (6" to 10" NPS) ==> MUST be Short Pattern

Approved Vendors:

(4" to 6" NPS) Apollo (Cast, CL150, RP): Series 88A-140-24-MG, Gear Oper
 (4" to 10" NPS) Bray, Flow-Tek, Series RF15 (Cast, CL150, RP): 15-{Size}-11001-CSJG

**Description of BALL Valve (BA03CB500):**

(BA30-CS-F500)

Type: Ball, Floating, RP
Valve Size: NPS **1.0** -to- NPS **8**
Service: -29 to 204 °C (-20 to 400 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Trim: 316 SS
Seat: RPTFE or TFM 1600
Stem: 316 SS
Body Type: Split, 2-Piece
Design: Floating Ball, Fire Tested API 607
Bore-Port: **Reduced (Standard) Port (RP)**
Operation: [Lever to 6" NPS](#); [Gear Op. at 8" NPS](#)
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite

The standards are:

DESIGN: API 608
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: [ASME B16.10 - Long Pattern \(1" to 6"\)](#)
..... [ASME B16.10 – Short Pattern \(8"\) ==> MUST be Short Pattern ***](#)

Approved Vendors:

[\(1" to 6" NPS Only\)](#) Apollo (Cast, CL300, RP): Series 88A-740-24, Lever Op.
[\(1" to 8" NPS\)](#) [Bray, Flow-Tek, Series RF30](#) (Cast, CL300, RP): 15-{Size}-11021-CSJG
..... [** Gear Op. at 8" NPS **](#)

Description of BALL Valve (BA03CB501):

(BA30-CS-F501)

Type: Ball, Floating, FP
Valve Size: NPS 0.5 -to- NPS **8**
Service: -29 to 204 °C (-20 to 400 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Trim: 316 SS
Seat: RPTFE or TFM 1600
Stem: 316 SS
Body Type: Split, 2-Piece
Design: Floating Ball, Fire Tested API 607
Bore-Port: **Full Port (FP)**
Operation: [Lever to 6" NPS](#); [Gear Oper. At 8" NPS](#)
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite

The standards are:

DESIGN: API 608
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 - Long Pattern (All Sizes)

Approved Vendors:

[\(1/2" to 6" NPS\)](#) Apollo (Cast, CL300, FP): Series 88A-940-24, Lever Op.
[\(1/2" to 8" NPS\)](#) [Bray, Flow-Tek, Series F30](#) (Cast, CL300, FP): 15-{Size}-11002-CSJG

**Description of BALL Valve (BA03CB502):**

(BAT30-CS-F502)

Type: Ball, Trunnion, RP
Valve Size: NPS 6 -to- NPS 12
Service: -26 to 204 °C (-15 to 350 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Trim: 316 SS
Seat: RPTFE
Stem/Trunnion 316 SS
Springs: Alloy X-750
Body Type: Side-Entry, 2-Piece
Design: **Trunnion Ball**, Fire Tested API 607
Bore-Port: Reduced Port (RP)
Operation: **Gear**
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite
O-Rings: Viton

The standards are:

DESIGN: ASME B16.34 or API 6D
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 6D
DIMENSIONAL: ASME B16.10 - **Long Pattern** (NPS 6 -to- NPS 12)

Approved Vendors: WKM (CL300, Cast, RP): {Size x Size}-370D6-300-RFxRF-24-TVF-23-WG

Description of BALL Valve (BA03CB503):

(BAT30-CS-F503)

Type: Ball, Trunnion, FP
Valve Size: NPS 6 -to- NPS 16
Service: -26 to 204 °C (-15 to 350 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Trim: 316 SS
Seat: RPTFE
Stem/Trunnion 316 SS
Springs: Alloy X-750
Body Type: Side-Entry, 2-piece
Design: **Trunnion Ball**, Fire Tested API 607
Bore-Port: **Full Port (FP)**
Operation: **Gear**
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite
O-Rings: Viton

The standards are:

DESIGN: ASME B16.34 or API 6D
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 6D
DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendors: WKM (CL300, Cast, FP): {Size}-370D6-300-RFxRF-24-TVF-23-WG

**Description of BALL Valve (BA03CB504):**

(BA30-CS-F504)

Type: Ball, Floating, FP
Valve Size: NPS 4 -to- NPS 10
Service: -29 to 204 °C (-20 to 400 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Trim: 316 SS
Seat: RPTFE or TFM 1600
Stem: 316 SS
Body Type: Split, 2-piece
Design: Floating Ball, Fire Tested API 607
Bore-Port: **Full Port (FP)**
Operation: **Gear**
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite

The standards are:

DESIGN: API 608
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 - Long Pattern (All Sizes)

Approved Vendors:

(4" to 10" NPS) Apollo (Cast, CL300, FP): Series 88A-940-24-MG, Gear Oper
(4" to 10" NPS) Bray, Flow-Tek, [Series F30](#) (Cast, CL300, FP): 15-{Size}-11002-CSJG

Description of BALL Valve (BA03CB505):

(BA30-CS-F505)

Type: Ball, Floating, RP
Valve Size: NPS 4 -to- NPS 10
Service: -29 to 204 °C (-20 to 400 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Trim: 316 SS
Seat: RPTFE or TFM 1600
Stem: 316 SS
Body Type: Split, 2-Piece
Design: Floating Ball, Fire Tested API 607
Bore-Port: **Reduced Port (RP)**
Operation: **Gear**
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite

The standards are:

DESIGN: API 608
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 - Long Pattern = Short Pattern (up to 6" NPS)
..... ASME B16.10 – Short Pattern (8" to 10" NPS) ==> **MUST** be Short Pattern

Approved Vendors:

(4" to 6" NPS) Apollo (Cast, CL300, RP): Series 88A-740-24-MG, Gear Oper
(4" to 10" NPS) Bray, Flow-Tek, [Series RF30](#) (Cast, CL300, RP): 15-{Size}-11021-CSJG

Description of BALL Valve (BA06CB500):

(BA60-CS-F500)

Type: Ball, Floating, RP
 Valve Size:..... NPS 1.0 -to- NPS 8
 Service: -29 to 204 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: CS (ASTM A216-WCB)
 Trim: 316 SS
 Seat: RPTFE or TFM 1600
 Stem: 316 SS
 Body Type: Split, 2-Piece
 Design: Floating Ball, Fire Tested API 607, AED Seals
 Bore-Port: Reduced Port (RP)
 Operation:..... Lever to 4" NPS; Gear Op. 6" & 8" NPS
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN:..... API 608
 ENDS: ASME B16.5
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... ASME B16.10 - Long Pattern (All Sizes)

Approved Vendors:

(1" to 8" NPS) WKM (Cast, CL600, RP): {Size x Size}-320F-600-RFxRF-24-CGF-23-WH, Lever

Description of BALL Valve (BA06CB501):

(BA60-CS-F501)

Type: Ball, Floating, FP
 Valve Size:..... NPS 0.5 -to- NPS 6
 Service: -29 to 204 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: CS (ASTM A216-WCB)
 Trim: 316 SS
 Seat: RPTFE or TFM 1600
 Stem: 316 SS
 Body Type: Split, 2-Piece
 Design: Floating Ball, Fire Tested API 607, AED Seals
 Bore-Port: **Full Port (FP)**
 Operation:..... Lever to 4" NPS; Gear Op. at 6" NPS
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN:..... API 608
 ENDS: ASME B16.5
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... ASME B16.10 - Long Pattern (All Sizes)

Approved Vendors:

(1/2" to 6" NPS) WKM (Cast, CL600, FP): {Size}-320F-600-RFxRF-24-CGF-23-WH, Lever Op.

Description of BALL Valve (BA06CB502):

(BAT60-CS-F502)

Type: Ball, Trunnion, RP
 Valve Size: NPS 3 -to- NPS 12
 Service: -26 to 204 °C (-15 to 400 °F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: CS (ASTM A216-WCB or ASTM A105N)
 Trim: 316 SS
 Seat: RPTFE
 Stem/Trunnion 316 SS
 Springs: Alloy X-750
 Body Type: Side-Entry, 2-Piece
 Design: **Trunnion Ball**, Fire Tested API 607, AED Seals
 Bore-Port: Reduced Port (RP)
 Operation: **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite
 O-Rings: Viton AED

The standards are:

DESIGN: ASME B16.34 or API 6D
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 6D
 DIMENSIONAL: ASME B16.10 - Long Pattern (= Short Pattern), all sizes

Approved Vendors:

(3" to 12" NPS) WKM (Cast, CL600, RP): {Size x Size}-370D6-600-RFxRF-24-TVF-23-WG

Description of BALL Valve (BA06CB503):

(BAT60-CS-F503)

Type: Ball, Trunnion, FP
 Valve Size: NPS 2 -to- NPS 16
 Service: -26 to 204 °C (-15 to 400 °F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: CS (ASTM A216-WCB or ASTM A105N)
 Trim: 316 SS
 Seat: RPTFE
 Stem/Trunnion 316 SS
 Springs: Alloy X-750
 Body Type: Side-Entry, 2-Piece
 Design: **Trunnion Ball**, Fire Tested API 607, AED Seals
 Bore-Port: **Full Port (FP)**
 Operation: **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite
 O-Rings: Viton AED

The standards are:

DESIGN: ASME B16.34 or API 6D
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 6D
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendors:

(2" to 16" NPS) WKM (Cast, CL600, FP): {Size}-370D6-600-RFxRF-24-TVF-23-WG

Description of BALL Valve (BA06CB504):

(BAT60-CS-F504)

Type: Ball, Trunnion, FP
 Valve Size:..... NPS 6 -to- NPS 24
 Service: -26 to 121 °C (-15 to 250 °F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: CS (ASTM A216-WCB or ASTM A105N)
 Trim: 316 SS
 Seat: Specify Devlon or Nylon (see Vendor info below)
 Stem/Trunnion 316 SS
 Springs: Alloy X-750
 Body Type: Side-Entry, 3-Piece
 Design: **Trunnion Ball**, Fire Tested API 607, AED Seals
 Bore-Port: **Full Port (FP)**
 Operation:..... **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite
 O-Rings: HNBR
 Note: Check to see that Seat material is compatible with intended service media used with this valve.

The standards are:

DESIGN:..... ASME B16.34 or API 6D
 ENDS: ASME B16.5
 RATING:..... ASME B16.34
 TESTING:..... API 6D
 DIMENSIONAL:..... ASME B16.10 - Long Pattern

Approved Vendors:..... WKM (CL600, Cast, FP) {Size}-D5 600 RFxRF 24 DRF 23 WGA (Devlon)
 Warren (CL600, Cast, 3-Piece, FP): {Size}-6605F-T35DHRG (Devlon Seats)
 Warren (CL600, Cast, 3-Piece, FP): {Size}-6605F-T35NHRG (Nylon Seats)

Description of BALL Valve (BA06CB300 or BA03CB300 ==> BA60-CS-S300)

Note: BA03CB300 should be BA06CB300 (since there are no CL300 Socket Weld Valves)

Type: Ball, Floating, FP
 Valve Size:..... NPS 0.5 -to- NPS 2
 Service: -29 to 204 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: **SW**
 Body: CS (ASTM A216-WCB or Forged CS)
 Trim: 316 SS
 Seat: RPTFE or TFM 1600
 Stem: 316 SS
 Body Type: 3-Piece Bolted
 Design: Floating Ball, Fire Tested API 607
 Bore-Port: **Full Port (FP)**
 Operation:..... Lever
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN:..... ASME B16.34
 ENDS: ASME B16.11
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... MFG STD

Approved Vendors:..... Apollo (CL600, Forged, FP): {Size}-83B-240-24

Description of BALL Valve (BA60-CS-T210)

Type: Ball, Floating, FP
 Valve Size:..... NPS 0.5 -to- NPS 2
 Service: -29 to 204 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: FNPT
 Body: CS (ASTM A216-WCB)
 Trim: 316 SS
 Seat: RPTFE or TFM 1600
 Stem: 316 SS
 Body Type: 3-Piece
 Design: Floating Ball, Fire Tested, API 607
 Bore-Port: **Full Port (FP)**
 Operation:..... Lever
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN:..... ASME B16.34
 ENDS: ASME B1.20.1
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... MFG STD.

Approved Vendors:..... Apollo (CL600, Cast, FP): {Size}-83B-140-24, Lever Op,

Description of BALL Valve (BA60-CS-X210)

Type: Ball, Floating, FP
 Valve Size:..... NPS 0.5 -to- NPS 2
 Service: -29 to 204 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: SW x FNPT
 Body: CS (ASTM A216-WCB)
 Trim: 316 SS
 Seat: RPTFE or TFM 1600
 Stem: 316 SS
 Body Type: 3-Piece
 Design: Floating Ball, Fire Tested, API 607
 Bore-Port: **Full Port (FP)**
 Operation:..... Lever
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN:..... ASME B16.34
 ENDS: SW; ASME B16.11, THRD; ASME B1.20.1
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... MFG STD.

Approved Vendors:..... Apollo (CL600, Cast, FP): {Size}-83B-140-24-63, Lever Op,

***** Start of Low Temperature Carbon Steel Ball Valves *****

Description of BALL Valve (BA01CA500): (BA15-CL-F500)

Type: Ball, Floating Ball, Low-Temperature, RP

Valve Size: NPS 1.5 -to- NPS 6

Service: Low Temperature -46 to 204 °C (-50 to 400 °F)

Class: **ASME 150**

Ends: Flanged RF

Body: ITCS (ASTM A352-LCB)

Trim: 316 SS (ASTM A276-316)

Seat: RPTFE

Stem: 316 SS

Body Type: Split, 2-Piece

Design: Floating Ball, Fire Tested API 607,
..... Elastomers Not Permitted except for Weather Seals

Bore-Port: Reduced Port (RP)

Operation: Lever

Stem-Packing/Seals: Flexible Graphite

Body Gaskets: Flexible Graphite

The standards are:

DESIGN: API 608

ENDS: ASME B16.5

RATING: ASME B16.34

TESTING: API 598

DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendors: Apollo (CL150, Cast, RP): {Size}-88L-140-24 (1.5" to 4"), Lever Op.

Description of BALL Valve (BA01CA501): (BA15-CL-F501)

Type: Ball, Floating Ball, Low-Temperature, FP

Valve Size: NPS 0.5 -to- NPS 6

Service: Low Temperature -46 to 204 °C (-50 to 400 °F)

Class: **ASME 150**

Ends: Flanged RF

Body: ITCS (ASTM A352-LCB)

Trim: 316 SS (ASTM A276-316)

Seat: RPTFE

Stem: 316 SS

Body Type: Split, 2-Piece

Design: Floating Ball, Fire Tested API 607,
..... Elastomers Not Permitted except for Weather Seals

Bore-Port: **Full Port (FP)**

Operation: Lever

Stem-Packing/Seals: Flexible Graphite

Body Gaskets: Flexible Graphite

The standards are:

DESIGN: API 608

ENDS: ASME B16.5

RATING: ASME B16.34

TESTING: API 598

DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendors: APOLLO (CL150, Cast, FP): {Size}-Series 88L-240-24, Lever Oper

Description of BALL Valve (BA01CA502):

(BA15-CL-F502)

Type: Ball, Floating Ball, Low-Temperature, RP
 Valve Size:..... NPS 4 -to- NPS 8
 Service: Low Temperature -46 to 204 °C (-50 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: ITCS (ASTM A352-LCB)
 Trim: 316 SS
 Seat: RPTFE
 Stem: 316 SS
 Body Type: Split, 2Piece
 Design: Floating Ball, Fire Tested API 607,
 Elastomers Not Permitted except for Weather Seals
 Bore-Port: Reduced Port (RP)
 Operation:..... **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN:..... API 608
 ENDS: ASME B16.5
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... ASME B16.10 - Long Pattern

Approved Vendors:..... APOLLO (CL150, Cast, RP): Series 88L-140-24-MG, Gear Oper.

Description of BALL Valve (BA01CA503):

(BA15-CL-F503)

Type: Ball, Floating Ball, Low-Temperature, FP
 Valve Size:..... NPS 4 -to- NPS 10
 Service: Low Temperature -46 to 204 °C (-50 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: ITCS (ASTM A352-LCB)
 Trim: 316 SS
 Seat: RPTFE
 Stem: 316 SS
 Body Type: Split, 2-Piece
 Design: Floating Ball, Fire Tested API 607,
 Elastomers Not Permitted except for Weather Seals
 Bore-Port: **Full Port (FP)**
 Operation:..... **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN:..... API 608
 ENDS: ASME B16.5
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... ASME B16.10 - Long Pattern

Approved Vendors:..... APOLLO (CL150, Cast, FP): Series 88L-240-24-MG, Gear Oper.

**Description of BALL Valve (BA01CA504):** (BAT15-CL-F504)

Type: Ball, Trunnion, Low-Temperature, FP
Valve Size: NPS 6 -to- NPS 16
Service: Low Temperature -46 to 204 °C (-50 to 400 °F)
Class: **ASME 150**
Ends: Flanged RF
Body: ITCS (ASTM A350-LF2-CL-1 or ASTM A352-LCB)
Trim: 316 SS
Seat: RPTFE;
Stem/Trunnion: 316 SS
Springs: Alloy X-750
Body Type: Side-Entry, 2-Piece
Design: **Trunnion Ball**, Fire Tested API 607,
Bore-Port: **Full Port (FP)**
Operation: **Gear**
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite
O-Rings: MFG STD

The standards are:

DESIGN: ASME B16.34 or API 6D
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 6D
DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendors: WKM (CL150, Cast, FP): {Size}-370D6-150-RFxRF-37-TWF-23-WG

Description of BALL Valve (BA01CA505): (BAT15-CL-F505)

Type: Ball, Trunnion, Low-Temperature (RP)
Valve Size: NPS 6 -to- NPS 12
Service: Low Temperature -46 to 204 °C (-50 to 400 °F)
Class: **ASME 150**
Ends: Flanged RF
Body: ITCS (ASTM A350-LF2-CL-1 or ASTM A352-LCB)
Trim: 316 SS
Seat: RPTFE
Stem/Trunnion: 316 SS
Springs: Alloy X-750
Body Type: Side Entry, 2-Piece
Design: **Trunnion Ball**, Fire Tested API 607,
..... Elastomers Not Permitted except for Weather Seals
Bore-Port: Reduced Port (RP)
Operation: **Gear**
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 6D
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 6D
DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendors: WKM (CL150, Cast, RP): {Size x Size}-370D6-150-RFxRF-37-TWF-23-WG

Description of BALL Valve (BA03CA500):

(BA30-CL-F500)

Type: Ball, Floating, Low Temperature, RP
 Valve Size: NPS 0.5 -to- NPS 6
 Service: Low Temperature -46 to 204 °C (-50 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: ITCS (ASTM A352-LCB)
 Trim: 316 SS
 Seat: RPTFE
 Stem: 316 SS
 Body Type: Split, 2-Piece
 Design: Floating Ball, Fire Tested API 607,
 Elastomers Not Permitted except for Weather Seals
 Bore-Port: Reduced Port (RP)
 Operation: Lever
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 – Long Pattern

Approved Vendors: APOLLO (CL300, Cast, RP): Series 88L-740-24, Lever Oper.

Description of BALL Valve (BA03CA501):

(BA30-CL-F501)

Type: Ball, Floating, Low Temperature, FP
 Valve Size: NPS 0.5 -to- NPS 4
 Service: Low Temperature -46 to 204 °C (-50 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: ITCS (ASTM A352-LCB)
 Trim: 316 SS
 Seat: RPTFE
 Stem: 316 SS
 Body Type: Split, 2-Piece
 Design: Floating Ball, Fire Tested API 607,
 Elastomers Not Permitted except for Weather Seals
 Bore-Port: **Full Port (FP)**
 Operation: Lever
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendors: APOLLO (CL300, Cast, FP): Series 88L-940-24, Lever Op.

Description of BALL Valve (BA03CA502): (BAT30-CL-F502)

Type: Ball, Trunnion, Low Temperature, RP
 Valve Size:..... NPS 8 -to- NPS 12
 Service: Low Temperature -46 to 204 °C (-50 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: ITCS (ASTM A350-LF2-CL-1 or ASTM A352-LCB)
 Trim: 316 SS
 Seat: RPTFE
 Stem/Trunnion:..... 316 SS
 Springs: Alloy X-750
 Body Type: Side-Entry, 2-Piece
 Design: **Trunnion Ball**, Fire Tested API 607,
 Elastomers Not Permitted except for Weather Seals
 Bore-Port:..... Reduced Port (RP)
 Operation:..... **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite
 O-Rings: Viton

The standards are:

DESIGN:..... ASME B16.34 or API 6D
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING:..... API 6D
 DIMENSIONAL:..... ASME B16.10

Approved Vendors:..... WKM (CL300, Cast, RP): {Size x Size}-370D6-300-RFxRF-37-TWF-23-WG

Description of BALL Valve (BA03CA503): (BAT30-CL-F503)

Type: Ball, Trunnion, Low Temperature, FP
 Valve Size:..... NPS 6 -to- NPS 16
 Service: Low Temperature -46 to 204 °C (-50 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: ITCS (ASTM A350-LF2-CL-1 or ASTM A352-LCB)
 Trim: 16 SS
 Seat: RPTFE
 Stem/Trunnion:..... 316 SS
 Springs: Alloy X-750
 Body Type: Side-Entry, 2-Piece
 Design: **Trunnion Ball**, Fire Tested API 607,
 Elastomers Not Permitted except for Weather Seals
 Bore-Port:..... **Full Port (FP)**
 Operation:..... **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite
 O-Rings: MFG STD

The standards are:

DESIGN:..... ASME B16.34 or API 6D
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING:..... API 6D
 DIMENSIONAL:..... ASME B16.10 - Long Pattern

Approved Vendors:..... WKM (CL300, Cast, FP): {Size}-370D6-300-RFxRF-37-TWF-23-WG

Description of BALL Valve (BA03CA504): (BA30-CL-F504)
 Type: Ball, Floating, Low Temperature, FP
 Valve Size: NPS 3 -to- NPS 8
 Service: Low Temperature -46 to 204 °C (-50 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: ITCS (ASTM A352-LCB)
 Trim: 316 SS
 Seat: RPTFE
 Stem: 316 SS
 Body Type: Split, 2-Piece
 Design: Floating Ball, Fire Tested API 607,
 Elastomers Not Permitted except for Weather Seals
 Bore-Port: **Full Port (FP)**
 Operation: **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendors: APOLLO (CL300, Cast, FP): Series 88L-940-24-MG, Gear Op.

Description of BALL Valve (BA03CA505): (BA30-CL-F505)
 Type: Ball, Floating, Low Temperature, RP
 Valve Size: NPS 4 -to- NPS 10
 Service: Low Temperature -46 to 204 °C (-50 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: ITCS (ASTM A352-LCB)
 Trim: 316 SS
 Seat: RPTFE
 Stem: 316 SS
 Body Type: Split, 2-Piece
 Design: Floating Ball, Fire Tested API 607,
 Elastomers Not Permitted except for Weather Seals
 Bore-Port: **Reduced Port (RP)**
 Operation: **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendors: APOLLO (CL300, Cast, RP): Series 88L-740-24-MG, Gear Op

Description of BALL Valve (BA06CA500):

(BA60-CL-F500)

Type: Ball, Floating, Low Temperature, RP
 Valve Size: NPS 0.75 -to- NPS 4
 Service: Low Temperature -46 to 204 °C (-50 to 400 °F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: ITCS (ASTM A352-LCB)
 Trim: 316 SS
 Seat: RPTFE
 Stem: 316 SS
 Body Type: Split, 2-Piece
 Design: Floating Ball, Fire Tested API 607,
 Bore-Port: Reduced Port (RP)
 Operation: Lever
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendors: WKM (CL600, Cast, RP): {Size x Size}-320F-600-RFxRF-37-CGF-23-WH

Description of BALL Valve (BA06CA501):

(BA60-CL-F501)

Type: Ball, Floating, Low Temperature, FP
 Valve Size: NPS 0.5 -to- NPS 3
 Service: Low Temperature -46 to 204 °C (-50 to 400 °F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: ITCS (ASTM A352-LCB)
 Trim: 316 SS
 Seat: RPTFE
 Stem: 316 SS
 Body Type: Split, 2-Piece
 Design: Floating Ball, Fire Tested API 607,
 Elastomers Not Permitted except for Weather Seals
 Bore-Port: **Full Port (FP)**
 Operation: Lever
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendors: WKM (CL600, Cast, FP): {Size}-320F-600-RFxRF-37-CGF-23-WH

**Description of BALL Valve (BA06CA502):** (BAT60-CL-F502)

Type: Ball, Trunnion, Low-Temperature, RP
Valve Size: NPS 3 -to- NPS 12
Service: Low Temperature -46 to 204 °C (-50 to 400 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: ITCS (ASTM A350-LF2-CL-1 or ASTM A352-LCB)
Trim: 316 SS
Seat: RPTFE
Stem/Trunnion: 316 SS
Springs: Alloy X-750
Body Type: Side-Entry, 2-Piece
Design: **Trunnion Ball**, Fire Tested API 607,
..... Elastomers Not Permitted except for Weather Seals
Bore-Port: Reduced Port (RP)
Operation: **Gear**
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite
O-Rings: MFG STD

The standards are:

DESIGN: ASME B16.34 or API 6D
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 6D
DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendors: WKM (CL600, Cast, RP): {Size x Size}-370D6-600-RFxRF-37-TWF-23-WG

Description of BALL Valve (BA06CA503): (BAT60-CL-F503)

Type: Ball, Trunnion, Low-Temperature, FP
Valve Size: NPS 2 -to- NPS 16
Service: Low Temperature -46 to 204 °C (-50 to 400 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: ITCS (ASTM A350-LF2-CL-1 or ASTM A352-LCB)
Trim: 316 SS
Seat: RPTFE
Stem/Trunnion: 316 SS
Springs: Alloy X-750
Body Type: Side-Entry, 2-Piece
Design: **Trunnion Ball**, Fire Tested API 607,
..... Elastomers Not Permitted except for Weather Seals
Bore-Port: **Full Port (FP)**
Operation: **Gear**
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite
O-Rings: MFG STD

The standards are:

DESIGN: ASME B16.34 or API 6D
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 6D
DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendors: WKM (CL600, Cast, FP): {Size}-370D6-600-RFxRF-37-TWF-23-WG



Rev.	RevDate	RevDescription	By	Checked
0	11 SEPT 2023	Issued for release	MFS	SBS
1	08 NOV 2023	Update Valve Tags, Minor edits	MFS	SBS
2	26 MAR 2024	Add port size designations, Add Valve Tags	MFS	WGS
3	22 MAY 2024	Update RP Valve Information	MFS	MFS

End Document #: GCTLR00-TSLA-ME-SPC-00002

Valve Specification for Carbon Steel Ball Valves

Description of BUTTERFLY Valve (BF01CB700)

BUD15-CS-L700

Type:..... Butterfly, Double Offset
 Valve Size:..... NPS 3 -to- NPS 4
 Temperature:..... -29 to +205 °C (-20 to 400 °F)
 Class:..... ASME 150
 Ends:..... Lugged
 Body:..... CS (ASTM A216-WCB)
 Disc/Seat:..... 316 SS / RPTFE
 Stem:..... 17-4 PH SS
 Body Type:..... Lugged, Bolt Holes Tapped
 Design (Other):..... Fire Tested API 607
 Operation:..... Lever
 Stem-Packing:..... Flexible Graphite
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN:..... API 609 Category B
 ENDS:..... ASME B16.5
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... API 609 Category B

Approved Vendors:..... Bray Butterfly Valve, Double Offset: 41-{Size}-11001-468
 CS, Lugged (tapped), Trim Code = 468, Lever Oper

Description of BUTTERFLY Valve (BF01CB701)

BUD15-CS-L701

Type:..... Butterfly, Double Offset
 Valve Size:..... NPS 3 -to- NPS 24
 Temp Limits:..... -29 to +205 °C (-20 to 400 °F)
 Class:..... ASME 150
 Ends:..... Lugged
 Body:..... CS (ASTM A216-WCB)
 Disc/Seat:..... CS - 316 SS Overlay / RPTFE
 Stem:..... 17-4 SS
 Body Type:..... Lugged, Bolt Holes Tapped
 Design (Other):..... Fire Tested API 607
 Operation:..... Gear
 Stem-Packing:..... Flexible Graphite
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN:..... API 609 Category B
 ENDS:..... ASME B16.5
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... API 609 Category B

Approved Vendors:..... Bray (Double-offset): 41-{Size}-11001-468
 CS, Lugged (tapped), Trim Code = 468, Gear Oper
 Flow Seal(Lugged): {Size}-1LA-121-BFF (to 12")
 {Size}-1LC-121-BFF on (14" to 24")

Description of BUTTERFLY Valve (BF01CB702)

BUD15-CS-L702

Type: Butterfly, Double Offset
 Valve Size: NPS 26 -to- NPS 48
 Temp Limits: -29 to +205 °C (-20 to 400 °F)
 Class: ASME 150
 Ends: Lugged
 Body: CS (ASTM A216-WCB)
 Disc/Seat: CS - 316 SS Overlay / RPTFE
 Stem: 17-4 PH SS
 Body Type: Lugged, Bolt Holes Tapped
 Design (Other): Fire Tested API 607
 Operation: Gear
 Stem-Packing: Flexible Graphite
 Notes: 1. Gaskets are required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609 Category B
 ENDS: ASME B16.47 Series B
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: Bray (Double Offset): 41-{Size}-11001-468
 CS, Lugged (tapped), B16.47 Series B
 Crane (Flow Seal): {Size}-1LC-121-BFF

Description of BUTTERFLY Valve (BF01CB703)

BUD15-CS-L703

Type: Butterfly, Double Offset
 Valve Size: NPS 26 -to- NPS 48
 Temp Limits: -29 to +205 °C (-20 to 400 °F)
 Class: ASME 150
 Ends: Lugged
 Body: CS (ASTM A216-WCB)
 Disc/Seat: CS - 316 SS Overlay / RPTFE
 Stem: 17-4 PH SS
 Body Type: Lugged, Bolt Holes Tapped
 Design (Other): Fire Tested API 607
 Operation: Gear
 Stem-Packing: Flexible Graphite
 Notes: 1. Gaskets are required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609 Category B
 ENDS: ASME B16.47 Series A
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: Bray (Double Offset): 41-{Size}-11001-468
 CS, Lugged (tapped), B16.47 Series A

Description of BUTTERFLY Valve (BF03CB700)

BUD30-CS-L700

Type:..... Butterfly, Double Offset
 Valve Size:..... NPS 3 -to- NPS 24
 Service: -29 to +205 °C (-20 to 400 °F)
 Class: ASME 300
 Ends: Lugged
 Body: CS (ASTM A216-WCB)
 Disc/Seat: CS - 316 SS Overlay / RPTFE
 Stem: 17-4 PH SS
 Body Type: Lugged, Bolt Holes Tapped
 Design (Other):..... Fire Tested API 607
 Operation:..... Gear
 Stem-Packing: Flexible Graphite
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN:..... API 609 Category B
 ENDS: ASME B16.5
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... API 609 Category B

Approved Vendors:..... Bray (Double-offset): 43-{Size}-11001-468
 CS, Lugged (tapped), Trim Code = 468, Gear Op.
 Flow Seal (Double-offset): {Size}-3LA-121-BFF, Gear Op.

Description of BUTTERFLY Valve (BF03CB701)

BUD30-CS-L701

Type:..... Butterfly, Double Offset
 Valve Size:..... NPS 26 -to- NPS 48
 Service: -29 to +205 °C (-20 to 400 °F)
 Class: ASME 300
 Ends: Lugged
 Body: CS (ASTM A216-WCB)
 Disc/Seat: CS - 316 SS Overlay / RPTFE
 Stem: 17-4 PH SS
 Body Type: Lugged, Bolt Holes Tapped
 Design (Other):..... Fire Tested API 607
 Operation:..... Gear
 Stem-Packing: Flexible Graphite
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN:..... API 609 Category B
 ENDS: ASME B16.47 Series B
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... MFG STD

Approved Vendors:..... Bray (Series 43): 43-{Size}-11001-468
 CS, Lugged (tapped), B16.47 Series B, Gear Oper, Trim Code = 468
 Crane (Flow Seal): {Size}-3LC-121-BFF

Description of BUTTERFLY Valve (BF03CB702)

BUD30-CS-L702

Type:..... Butterfly, Double Offset
 Valve Size:..... NPS 26 -to- NPS 48
 Service: -29 to +205 °C (-20 to 400 °F)
 Class: ASME 300
 Ends: Lugged
 Body: CS (ASTM A216-WCB)
 Disc/Seat: CS - 316 SS Overlay / RPTFE
 Stem: 17-4 PH SS
 Body Type: Lugged, Bolt Holes Tapped
 Design (Other):..... Fire Tested API 607
 Operation:..... Gear
 Stem-Packing: Flexible Graphite
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN:..... API 609 Category B
 ENDS: ASME B16.47 Series A
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... MFG STD

Approved Vendors:.....

Description of BUTTERFLY Valve (BF06CB700)

BUD60-CS-L700

Type:..... Butterfly, Double Offset
 Valve Size:..... NPS 3 -to- NPS 24
 Service: -29 to +205 °C (-20 to 400 °F)
 Class: ASME 600
 Ends: Lugged
 Body: CS (ASTM A216-WCB)
 Disc/Seat: CS - 316 SS Overlay / RPTFE
 Stem: 17-4 PH SS
 Body Type: Lugged, Bolt Holes Tapped
 Design (Other):..... Fire Tested API 607
 Operation:..... Gear
 Stem-Packing: Flexible Graphite
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN:..... API 609 Category B
 ENDS: ASME B16.5
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... API 609 Category B

Approved Vendors:..... Bray (Double Offset): 45-{Size}-11001-468
 Lugged (tapped), Trim = 468, Gear Oper.
 Crane (Flow Seal): {Size}-6LC-121-BFF,Lugged, Gear Oper.

Description of BUTTERFLY Valve (BF06CB701)

BUD60-CS-L701

Type:..... Butterfly, Double Offset
 Valve Size:..... NPS 26 -to- NPS 36
 Service: -29 to +205 °C (-20 to 400 °F)
 Class: ASME 600
 Ends: Lugged
 Body: CS (ASTM A216-WCB)
 Disc/Seat: CS - 316 SS Overlay / RPTFE
 Stem: 17-4 PH SS
 Body Type: Lugged, Bolt Holes Tapped
 Design (Other):..... Fire Tested API 607
 Operation:..... Gear
 Stem-Packing: Flexible Graphite
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN:..... API 609 Category B
 ENDS: ASME B16.47 Series B
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... MFG STD

Approved Vendors:.....

Description of BUTTERFLY Valve (BF06CB702)

BUD60-CS-L702

Type:..... Butterfly, Double Offset
 Valve Size:..... NPS 26 -to- NPS 48
 Service: -29 to +205 °C (-20 to 400 °F)
 Class: ASME 600
 Ends: Lugged
 Body: CS (ASTM A216-WCB)
 Disc/Seat: CS - 316 SS Overlay / RPTFE
 Stem: 17-4 PH SS
 Body Type: Lugged, Bolt Holes Tapped
 Design (Other):..... Fire Tested API 607
 Operation:..... Gear
 Stem-Packing: Flexible Graphite
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN:..... API 609 Category B
 ENDS: ASME B16.47 Series A
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... MFG STD

Approved Vendors:.....

***** End of Normal Temp Double-Offset Butterfly Valves *****

***** Start of Low Temperature Butterfly Valves *****

Description of BUTTERFLY Valve (BF03CA700) BUD30-CL-L700

Type:..... Butterfly, Double Offset, Low Temperature
Valve Size:..... NPS 3 -to- NPS 24
Temperature:..... Low Temperature -46 to 205 °C (-50 to 400 °F)
Class:..... ASME 300
Ends:..... Lugged
Body:..... ITCS (ASTM A352-LCB)
Disc/Seat:..... ITCS - 316 SS Overlay / RPTFE
Stem:..... 17-4 PH SS
Stem-Packing:..... Flexible Graphite
Body Type:..... Lugged, Bolt Holes Tapped
Design (Other):..... Fire Tested API 607
Operation:..... 3" to 4" NPS = Lever
..... 6" to 24" NPS = Gear

Notes: 1. Gaskets are required on each side of Butterfly Valve
..... 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN:..... API 609 Category B
ENDS:..... ASME B16.5
RATING:..... ASME B16.34
TESTING:..... API 598
DIMENSIONAL:..... API 609 Category B

Approved Vendors:..... Bray (Series 43): 43-{Size}-11001-468
..... ITCS, Lugged (tapped), Trim Code = 468

Description of BUTTERFLY Valve (BF03CA701) BUD30-CL-L701

Type:..... Butterfly, Double Offset, Low Temperature
Valve Size:..... NPS 26 -to- NPS 48
Temperature:..... Low Temperature -46 to 205 °C (-50 to 400 °F)
Class:..... ASME 300
Ends:..... Lugged
Body:..... ITCS (ASTM A352-LCB)
Disc/Seat:..... ITCS - 316 SS Overlay / RPTFE
Stem:..... 17-4 PH SS
Body Type:..... Lugged, Bolt Holes Tapped
Design (Other):..... Fire Tested API 607
Operation:..... Gear
Stem-Packing:..... Flexible Graphite

Notes: 1. Gaskets are required on each side of Butterfly Valve
..... 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN:..... API 609 Category B
ENDS:..... ASME B16.47 Series B
RATING:..... ASME B16.34
TESTING:..... API 598
DIMENSIONAL:..... MFG STD

Approved Vendors:..... Approved Vendors: Bray (Series 43): 43-{Size}-11001-468, Gear Oper
..... ITCS, Lugged (tapped), Trim Code = 468

Description of BUTTERFLY Valve (BF03CA702)

BUD30-CL-L702

Type: Butterfly, Double Offset, Low Temperature
 Valve Size: NPS 26 -to- NPS 48
 Service: Low Temperature -46 to 205 °C (-50 to 400 °F)
 Class: ASME 300
 Ends: Lugged
 Body: ITCS (ASTM A352-LCB)
 Disc/Seat: ITCS - 316 SS Overlay / RPTFE
 Stem: 17-4 PH SS
 Body Type: Lugged, Bolt Holes Tapped
 Design (Other): Fire Tested API 607
 Operation: Gear
 Stem-Packing: Flexible Graphite
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609 Category B
 ENDS: ASME B16.47 Series A
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors:

Description of BUTTERFLY Valve (BF06CA700)

BUD60-CL-L700

Type: Butterfly, Double Offset, Low Temperature
 Valve Size: NPS 3 -to- NPS 24
 Service: Low Temperature -46 to 205 °C (-50 to 400 °F)
 Class: ASME 600
 Ends: Lugged
 Body: ITCS (ASTM A352-LCB)
 Disc/Seat: ITCS - 316 SS Overlay / RPTFE
 Stem: 17-4 PH SS
 Body Type: Lugged, Bolt Holes Tapped
 Design (Other): Fire Tested API 607
 Operation: Gear
 Stem-Packing: Flexible Graphite
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609 Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609 Category B

Approved Vendors: Bray (Series 45): 45-{Size}-11001-468
 ITCS, Lugged (tapped), Gear Oper.

Description of BUTTERFLY Valve (BF06CA701)

BUD60-CL-L701

Type:..... Butterfly, Double Offset, Low Temperature
 Valve Size:..... NPS 26 -to- NPS 36
 Service: Low Temperature -46 to 205 °C (-50 to 400 °F)
 Class: ASME 600
 Ends: Lugged
 Body: ITCS (ASTM A352-LCB)
 Disc/Seat: ITCS - 316 SS Overlay / RPTFE
 Stem: 17-4 PH SS
 Body Type: Lugged, Bolt Holes Tapped
 Design (Other):..... Fire Tested API 607
 Operation:..... Gear
 Stem-Packing: Flexible Graphite
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN:..... API 609 Category B
 ENDS: ASME B16.47 Series B
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... MFG STD

Approved Vendors:.....

Description of BUTTERFLY Valve (BF06CA702)

BUD60-CL-L702

Type:..... Butterfly, Double Offset, Low Temperature
 Valve Size:..... NPS 26 -to- NPS 48
 Service: Low Temperature -46 to 205 °C (-50 to 400 °F)
 Class: ASME 600
 Ends: Lugged
 Body: ITCS (ASTM A352-LCB)
 Disc/Seat: ITCS - 316 SS Overlay / RPTFE
 Stem: 17-4 PH SS
 Body Type: Lugged, Bolt Holes Tapped
 Design (Other):..... Fire Tested API 607
 Operation:..... Gear
 Stem-Packing: Flexible Graphite
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN:..... API 609 Category B
 ENDS: ASME B16.47 Series A
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... MFG STD

Approved Vendors:.....



Document #: GCTLR00-TSLA-ME-SPC-00003

Valve Specification for CS Butterfly Valves

Rev.	RevDate	RevDescription	By:	Check:
0	15 SEPT 2023	Issued for Release	MFS	SBS
1	08 NOV 2023	Valve Trim Updates. Add Gasket Requirement Notes	MFS	SBS

End Document #: GCTLR00-TSLA-ME-SPC-00003

Valve Specification for CS Butterfly Valves

***** Dual-Plate Check Valves *****

Description of CHECK Valve (CD01CB500): (CHD15-CS-B500)
 Type: Check, Dual-Plate, Retainerless (Double-Flanged)
 Valve Size: NPS 6 -to- NPS 24
 Class: **ASME 150**
 Ends: Double Flanged, RF
 Body: CS (ASTM A216-WCB)
 Trim: API 594 **TRIM #1** (410 SS Trim)
 Plate & Hinge Pin 410 SS
 Springs: Alloy X-750
 Stop Pin: 410 SS
 Body Type: Double Flanged
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check): {Size}-H15-SPF3-169 (Trim #1)
 Velan (Proquip): PF-{Size}-002E18-3EB
 Keckley: {Size}-DF2R-LC-6IX

Description of CHECK Valve (CD01CB512): CHD15-CS-B512
 Type: Check, Dual Plate, Retainerless (Double-Flanged)
 Valve Size: NPS 8 -to- NPS 24
 Class: ASME 150
 Ends: Double-Flanged, RF
 Body: CS (ASTM A216-WCB)
 Trim: API 594 **TRIM 10** (316 SS Trim)
 Plate & Hinge Pin: 316 SS
 Springs: Inconel X-750
 Stop Pin: SS
 Body Type: **Double-Flanged**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek, Dbl-flanged): {Size}-H15-SPF3-233 (Trim #10)

Description of CHECK Valve (CD01CB700): (CHD15-CS-W700)

Type: Check, [Dual-Plate, Retainerless](#)
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 150**
 Ends: **Wafer**
 Body: CS (ASTM A216-WCB)
 Trim: API 594 **TRIM #1** (410 SS Trim)
[Plate & Hinge Pin](#) [410 SS](#)
 Springs: Alloy X-750
 Stop Pin: 410 SS
 Body Type: **Wafer**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check): {Size}-H15-SPF-169 (Trim #1)
 Velan (Proquip): PW-{Size}-002E18-3EB

Description of CHECK Valve (CD01CB702): (CHD15-CS-L702)

[Note: For Sizes 8" to 24" NPS, use Valve: CD01CB512](#)

Type: Check, [Dual-Plate, Retainerless](#)
 Valve Size: NPS 2 -to- NPS 6
 Class: **ASME 150**
 Ends: **Lugged**
 Body: CS (ASTM A216-WCB)
 Trim: API 594 **TRIM #1**
[Plate & Hinge Pin](#) [410 SS](#)
 Springs: Alloy X-750
 Stop Pin: 410 SS
 Body Type: **Lugged**, Bolt Holes Through-Drilled
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check): {Size}-H15-SPF2-169 (Trim #1)
 Velan (Proquip): PM-{Size}-002E18-3EB

Description of CHECK Valve (CD01CB704): (CHD15-CS-W704)

Type: Check, [Dual-Plate, Retainerless](#)
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 150**
 Ends: **Wafer**
 Body: CS (ASTM A216-WCB)
 Trim: API 594 **TRIM #1**
[Plate & Hinge Pin](#) [410 SS](#)
 Springs: Alloy X-750
 Stop Pin: 410 SS
 Body Type: **Wafer**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check): {Size}-H15-SPF-169 (Trim #1)
 Velan (Proquip): PW-{Size}-002E18-3EB

Description of CHECK Valve (CD03CB500):

(CHD30-CS-B500)

Type: Check, **Dual-Plate, Retainerless (Double-Flanged)**
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 300**
 Ends: **Double Flanged (RF)**
 Body: CS (ASTM A216-WCB)
 Trim: API 594 **TRIM #1** (410 SS Trim)
Plate & Hinge Pin **410 SS**
 Springs: Alloy X-750
 Stop Pin: 410 SS
 Body Type: Double Flanged
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check): {Size}-H30-SPF3-169 (Trim #1)
 Velan (Proquip): PF-{Size}-031D19-3DB (Trim #1)

Description of CHECK Valve (CD03CB512):

CHD30-CS-B512

Type: Check, Dual Plate, Retainerless (Double-Flanged)
 Valve Size: NPS 12 -to- NPS 24
 Class: ASME 300
 Ends: Double-Flanged, RF
 Body: CS (ASTM A216-WCB)
 Trim: API 594 **TRIM 10** (316 SS Trim)
Plate & Hinge Pin **316 SS**
 Springs: Inconel X-750
 Stop Pin: 316 SS
 Body Type: **Double-Flanged**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek, Dbl-flanged): {Size}-H30-SPF3-233 (Trim #10)

Description of CHECK Valve (CD03CB700): (CHD30-CS-W700)

Type: Check, [Dual-Plate, Retainerless](#)
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 300**
 Ends: **Wafer**
 Body: CS (ASTM A216-WCB)
 Trim: API 594 **TRIM #1**
[Plate & Hinge Pin](#) **410 SS**
 Springs: Alloy X-750
 Stop Pin: 410 SS
 Body Type: **Wafer**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check): {Size}-H30-SPF-169 (Trim #1)
 Velan (Proquip): PW-{Size}-102E18-3EB

Description of CHECK Valve (CD03CB702): (CHD30-CS-L702)

[Note: For Sizes 12" to 24" NPS, use Valve: CD03CB512](#)

Type: Check, [Dual-Plate, Retainerless](#)
 Valve Size: NPS 2 -to- NPS 10
 Class: **ASME 300**
 Ends: **Lugged**
 Body: CS (ASTM A216-WCB)
 Trim: API 594 **TRIM #1**
[Plate & Hinge Pin](#) **410 SS**
 Springs: Alloy X-750
 Stop Pin: 410 SS
 Body Type: **Lugged**, Bolt Holes Through-Drilled
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check): {Size}-H30-SPF2-169 (Trim #1)
 Velan (Proquip): PM-{Size}-102E18-3EB

Description of CHECK Valve (CD03CB703): (CHD30-CS-W703)

Type: Check, [Dual-Plate](#), [Retainerless](#)
 Valve Size:..... NPS 2 -to- NPS 24
 Class: **ASME 300**
 Ends: **Wafer**
 Body: CS (ASTM A216-WCB)
 Trim: API 594 **TRIM #8** (410 SS & Hard Faced)
[Plate & Hinge Pin:](#) [410 SS](#)
 Springs: Alloy X-750
 Stop Pin: 410 SS
 Body Type: **Wafer**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward

The standards are:

DESIGN:..... API 594 Type A
 ENDS: ASME B16.5
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... API 594

Approved Vendors:..... Crane (Cast, Duo-Check): {Size}-H30-SPF-131E (Trim #8)
 Velan (Proquip): PW-{Size}-102E18-3EB (Trim #12)

Description of CHECK Valve (CD06CB500):

(CHD60-CS-B500)

Type: Check, **Dual-Plate, Retainerless (Double-Flanged)**
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 600**
 Ends: Double Flanged (RF)
 Body: CS (ASTM A216-WCB)
 Trim: API 594 **TRIM #1** (410 SS)
Plate & Hinge Pin **410 SS**
 Springs: Alloy X-750
 Stop Pin: 410 SS
 Body Type: **Double Flanged**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check): {Size}-H60-SPF3-169 (Trim #1)
 Velan (Proquip): PF-{Size}-202E18-EB

Description of CHECK Valve (CD06CB512):

CHD60-CS-B512

Type: Check, Dual Plate, Retainerless (Double-Flanged)
 Valve Size: NPS 12 -to- NPS 24
 Class: ASME 600
 Ends: Double-Flanged, RF
 Body: CS (ASTM A216-WCB)
 Trim: API 594 **TRIM 10** (316 SS Trim)
Plate & Hinge Pin **316 SS**
 Springs: Inconel X-750
 Stop Pin: 316 SS
 Body Type: **Double-Flanged**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek, Dbl-flanged): {Size}-H60-SPF3-233 (Trim #10)

Description of CHECK Valve (CD06CB700): (CHD60-CS-W700)

Type: Check, [Dual-Plate, Retainerless](#)
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 600**
 Ends: **Wafer**
 Body: CS (ASTM A216-WCB)
 Trim: API 594 **TRIM #1**
[Plate & Hinge Pin](#) **410 SS**
 Springs: Alloy X-750
 Stop Pin: 410 SS
 Body Type: **Wafer**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward
 The standards are:
 DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check): {Size}-H60-SPF-169 (Trim #1)
 Velan (Proquip): PW-{Size}-202E18-3EB

Description of CHECK Valve (CD06CB702): (CHD60-CS-L702)

[Note: For Sizes 12" to 24" NPS, use Valve: CD06CB512](#)
 Type: Check, [Dual-Plate, Retainerless](#)
 Valve Size: NPS 2 -to- NPS 10
 Class: **ASME 600**
 Ends: **Lugged**
 Body: CS (ASTM A216-WCB)
 Trim: API 594 **TRIM #1** (410 SS Trim)
[Plate & Hinge Pin](#) **410 SS**
 Springs: Alloy X-750
 Stop Pin: 410 SS
 Body Type: **Lugged**, Bolt Holes Through-Drilled
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward
 The standards are:
 DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check, Lugged): {Size}-H60-SPF2-169 (Trim #1)
 Velan (ProQuip): PM-{Size}-202E18-3EB

Description of CHECK Valve (CD06CB704): (CHD60-CS-W704)

Type: Check, [Dual-Plate](#), [Retainerless](#)
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 600**
 Ends: **Wafer**
 Body: CS (ASTM A216-WCB)
 Trim: API 594 **TRIM #5**
[Plate](#): [Hard Faced Plate](#)
[Seat](#): [Hard Faced Seat](#)
 Springs: Alloy X-750
 Stop Pin: 316 SS
 Body Type: **Wafer**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward
 The standards are:
 DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check): {Size}-H60-SPF-491 (Trim #5)
 Velan (Proquip): PW-{Size}-202E18-3EB

Description of CHECK Valve (CD09CB500): (CHD90-CS-B500)

Type: Check, [Dual-Plate](#), [Retainerless](#)
 Valve Size: NPS 6 -to- NPS 24
 Class: **ASME 900**
 Ends: Double Flanged (RF)
 Body: CS (ASTM A216-WCB)
 Trim: API 594 **TRIM #1**
[Plate & Hinge Pin](#) [410 SS](#)
 Springs: Alloy X-750
 Stop Pin: 410 SS
 Body Type: Double Flanged
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward

The standards are:
 DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check): {Size}-H90-SPF3-169 (Trim #1)
 Velan (Proquip): PF-{Size}-702E18-3EB

Description of CHECK Valve (CD09CB700): (CHD90-CS-W700)

Type: Check, [Dual-Plate](#), [Retainerless](#)
 Valve Size: NPS 3 -to- NPS 24
 Class: **ASME 900**
 Ends: **Wafer**
 Body: CS (ASTM A216-WCB)
 Trim: API 594 **TRIM #1**
[Plate & Hinge Pin](#) [410 SS](#)
 Springs: Alloy X-750
 Stop Pin: 410 SS
 Body Type: **Wafer**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward
 The standards are:
 DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check): {Size}-H90-SPF-169 (Trim #1)
 Velan (Proquip): PW-{Size}-702E18-3EB

Description of CHECK Valve (CD09CB701): (CHD90-CS-L701)

Type: Check, [Dual-Plate](#), [Retainerless](#)
 Valve Size: NPS 3 -to- NPS 24
 Class: **ASME 900**
 Ends: **Lugged**
 Body: CS (ASTM A216-WCB)
 Trim: API 594 **TRIM #1**
[Plate & Hinge Pin](#) [410 SS](#)
 Springs: Alloy X-750
 Stop Pin: 410 SS
 Body Type: **Lugged**, Bolt Holes Through-Drilled
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward

The standards are:
 DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check, Lugged): {Size}-H90-SPF2-169 (Trim #1)
 Velan (Proquip): PM-{Size}-702E18-3B

Description of CHECK Valve (CD15CB700): (CHD1.5K-CS-L700)

Type: Check, [Dual-Plate](#), [Retainerless](#)
Valve Size: NPS 2 -to- NPS 24
Class: **ASME 1500**
Ends: **Wafer**
Body: CS (ASTM A216-WCB)
Trim: API 594 **TRIM #1**
[Plate & Hinge Pin](#) [410 SS](#)
Springs: Alloy X-750
Stop Pin: 410 SS
Body Type: **Wafer**
Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward
The standards are:
DESIGN: API 594 Type A
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check, Wafer): {Size}-H150-SPF-169 (Trim #1)
..... Velan (Proquip): PW-{Size}-302E18-3EB

Description of CHECK Valve (CD15CB701): (CHD1.5K-CS-L701)

Type: Check, [Dual-Plate](#), [Retainerless](#)
Valve Size: NPS 2 -to- NPS 24
Class: **ASME 1500**
Ends: **Lugged**
Body: CS (ASTM A216-WCB)
Trim: API 594 **TRIM #1**
[Plate & Hinge Pin](#) [410 SS](#)
Springs: Alloy X-750
Stop Pin: 410 SS
Body Type: **Lugged**, Bolt Holes Through-Drilled
Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward

The standards are:
DESIGN: API 594 Type A
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check): {Size}-H15-SPF2-169 (Trim #1)
..... Velan (Proquip): PM-{Size}-302E18-3B

***** **Piston Lift Check Valves** *****

Description of CHECK Valve (CL01CB500): (CHP15-CS-F500)

Type: Check, **Piston Lift, SP**
 Design Type: Piston Lift
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 150**
 Ends: Flanged RF
 Body: CS (ASTM A105N)
 Cover/Bonnet: CS (ASTMA105N)
 Trim: API 602 **TRIM #8** (410 SS & Hard Faced)
 Springs: None
 Port: **Standard Port (SP)**
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: (DSI (Forged, SP): 1331-A8 (no spring)
 Bonney Forge (Forged, SP): L1-40-1-RF (no spring)

Description of CHECK Valve (CL01CB501): (CHP15-CS-F501)

Type: Check, **Piston Lift, SP**
 Design Type: Piston Lift
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 150**
 Ends: Flanged RF
 Body: CS (ASTM A105N)
 Cover/Bonnet: CS (ASTMA105N)
 Trim: API 602 **TRIM #8** (410 SS & Hard Faced)
 Springs: Alloy X-750
 Port: **Standard Port (SP)**
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: (DSI, Forged, SP): 1331-A8 (w/spring)
 Bonney Forge (Forged, SP): L1-40-1-RF (w/spring)

Description of CHECK Valve (CL03CB500):

(CHP30-CS-F500)

Type: Check, [Piston Lift, SP](#)
 Design Type: Piston Lift
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 300**
 Ends: Flanged RF
 Body: CS (ASTM A105N)
 Cover/Bonnet: CS (ASTMA105N)
 Trim: API 602 **TRIM #8**
 Springs: None
[Port:](#) [Standard Port \(SP\)](#)
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: (DSI, Forged): 2331-A8 (no springs)
 Bonney Forge: L3-40-1-RF (no springs)

Description of CHECK Valve (CL03CB501):

(CHP30-CS-F501)

Type: Check, [Piston Lift, SP](#)
 Design Type: Piston Lift
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 300**
 Ends: Flanged RF
 Body: CS (ASTM A105N)
 Cover/Bonnet: CS (ASTMA105N)
 Trim: API 602 **TRIM #8**
 Springs: Alloy X-750
[Port:](#) [Standard Port \(SP\)](#)
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: (DSI, Forged): 2331-A8 (w/spring)
 Bonney Forge: L3-40-1-RF (w/spring)

Description of CHECK Valve (CL06CB500):

(CHP60-CS-F500)

Type: Check, [Piston Lift, SP](#)
 Design Type: Piston Lift
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 600**
 Ends: Flanged RF
 Body: CS (ASTM A105N)
 Cover/Bonnet: CS (ASTMA105N)
 Trim: API 602 **TRIM #8**
 Springs: None
[Port: Standard Port \(SP\)](#)
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up
 The standards are:
 DESIGN: API 602
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10
 Approved Vendors: (DSI, Forged): 3331-A8 (no spring)
 Bonney Forge: L6-40-1-RF (no spring)

Description of CHECK Valve (CL06CB501):

(CHP60-CS-F501)

Type: Check, [Piston Lift, SP](#)
 Design Type: Piston Lift
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 600**
 Ends: Flanged RF
 Body: CS (ASTM A105N)
 Cover/Bonnet: CS (ASTMA105N)
 Trim: API 602 **TRIM #8**
 Springs: Alloy X-750
[Port: Standard Port \(SP\)](#)
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up
 The standards are:
 DESIGN: API 602
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10
 Approved Vendors: (DSI, Forged): 3331-A8 (w/spring)
 Bonney Forge: L6-40-1-RF (w/spring)

Description of CHECK Valve (CL08CB000 ==> CHP80-CS-T210)

Type: Check, [Piston Lift](#), [SP](#)
 Design Type: Piston Lift
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **API 800**
 Ends: Threaded
 Body: CS (ASTM A105N)
 Cover/Bonnet: CS (ASTMA105N)
 Trim: API 602 **TRIM #8**
 Bore: Standard Port (SP)
 Springs: None
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up
 The standards are:
 DESIGN: API 602
 ENDS: ASME B1.20.1
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: (DSI, Forged, FNPT): 4321-A8 (no spring)
 Bonney Forge: HL-40-1-T (no spring)

Description of CHECK Valve (CL08CB001):

(CHP80-CS-T211)

Type: Check, [Piston Lift](#), [SP](#)
 Design Type: Piston Lift
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **API 800**
 Ends: Threaded
 Body: CS (ASTM A105N)
 Cover/Bonnet: CS (ASTMA105N)
 Bore: Standard Port (SP)
 Trim: API 602 **TRIM #8**
 Springs: Alloy X-750
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:
 DESIGN: API 602
 ENDS: ASME B1.20.1
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: (DSI, Forged, FNPT): 4321-A8 (w/spring)
 Bonney Forge: HL-40-1-T (w/spring)

Description of CHECK Valve (CL08CB300):

(CHP80-CS-S300)

Type: Check, **Piston Lift, SP**
 Design Type: Piston Lift
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **API 800**
 Ends: SW
 Body: CS (ASTM A105N)
 Cover/Bonnet: CS (ASTMA105N)
 Trim: API 602 **TRIM #8**
 Bore: Standard Port (SP)
 Springs: None
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: (DSI, Forged, Trim #8): 4311-A8 (no spring)
 Bonney Forge: HL-40-1-SW (no spring)

Description of CHECK Valve (CL08CB301):

(CHP80-CS-S301)

Type Check, **Piston Lift, SP**
 Design Type: Piston Lift
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **API 800**
 Ends: SW
 Body: CS (ASTM A105N)
 Cover/Bonnet: CS (ASTMA105N)
 Trim: API 602 **TRIM #5** (410 SS & Hardfaced Seat/Disc)
 Bore: Standard Port (SP)
 Springs: None
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: (DSI, Forged, Trim #5): 4311-A5 (no spring)
 Bonney Forge: HL-40-2-SW (no spring)

Description of CHECK Valve (CL08CB302):

(CHP80-CS-S302)

Type: Check, **Piston Lift, SP**
 Design Type: Piston Lift
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **API 800**
 Ends: SW
 Body: CS (ASTM A105N)
 Cover/Bonnet: CS (ASTMA105N)
 Trim: API 602 **TRIM #8** (410 SS & Hard Faced)
 Bore: Standard Port (SP)
 Springs: Alloy X-750
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: (DSI, Forged): 4311-A8 (w/spring)
 Bonney Forge: HL-40-1-SW (w/spring)

Description of CHECK Valve (CL08CB303):

(CHP80-CS-S303)

Type: Check, **Piston Lift, SP**
 Design Type: Piston Lift
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **API 800**
 Ends: SW
 Body: CS (ASTM A105N)
 Cover/Bonnet: CS (ASTMA105N)
 Trim: API 602 **TRIM #5**
 Bore: Standard Port (SP)
 Springs: Alloy X-750
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: (DSI, Forged): 4311-A5 (w/spring)
 Bonney Forge: L1-40-2-SW (w/spring)

Description of CHECK Valve (CHP80-CS-T101):

(CHP80-CS-T101)

Type: Check, **Piston Lift, SP**
 Design Type: Piston Lift
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 800**
 Ends: FNPT
 Body: CS (ASTM A105N)
 Cover/Bonnet: CS (ASTM A105N)
 Port: Standard Port (SP)
 Trim: API 602 **TRIM #8**
 Springs: None
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: Mfg.Std.

Approved Vendors: (DSI, Forged): 4321-A8 (no spring)
 Bonney Forge: HL-40-1-T (no spring)

Description of CHECK Valve (CL15CB300):

(CHP1.5K-CS-S300)

Type: Check, **Piston Lift, SP**
 Design Type: Piston Lift
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 1500**
 Ends: SW
 Body: CS (ASTM A105N)
 Cover/Bonnet: CS (ASTMA105N)
 Trim: API 602 **TRIM #8**
 Bore: Standard Port (SP)
 Springs: None
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: (DSI, Forged): 6311-A8 (no spring)
 Bonney Forge: 9HL-40-1-SW (no spring)

Description of CHECK Valve (CL15CB301):

(CHP1.5K-CS-S301)

Type: Check, [Piston Lift](#), [SP](#)
 Design Type: Piston Lift
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 1500**
 Ends: SW
 Body: CS (ASTM A105N)
 Cover/Bonnet: CS (ASTMA105N)
 Trim: API 602 **TRIM #5**
 Bore: Standard Port (SP)
 Springs: None
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: (DSI, Forged): 6311-A5 (no spring)
 Bonney Forge: 9HL-40-2-SW (no spring)

Description of CHECK Valve (CL15CB302):

(CHP1.5K-CS-S302)

Type: Check, [Piston Lift](#), [SP](#)
 Design Type: Piston Lift
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 1500**
 Ends: SW
 Body: CS (ASTM A105N)
 Cover/Bonnet: CS (ASTMA105N)
 Trim: API 602 **TRIM #8**
 Bore: Standard Port (SP)
 Springs: Alloy X-750
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: ASME B16.34 and API 594 Type B, as applicable
 ENDS: ASME B16.11
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: (DSI, Forged): 6311-A8 (w/spring)
 Bonney Forge: 9HL-40-1-SW (w/spring)

Description of CHECK Valve (CL15CB303):

(CHP1.5K-CS-S303)

Type: Check, **Piston Lift, SP**
 Design Type: Piston Lift
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 1500**
 Ends: SW
 Body: CS (ASTM A105N)
 Cover/Bonnet: CS (ASTMA105N)
 Trim: API 602 **TRIM #5**
 Bore: Standard Port (SP)
 Springs: Alloy X-750
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: (DSI, Forged): 6311-A5 (w/spring)
 Bonney Forge: 9HL-40-2-SW (w/spring)

***** Swing Check Valves *****

Description of CHECK Valve (CS01CB500): (CHS15-CS-F500)

Type: Check, [Swing, FP](#)
 Design Type: Swing
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 150**
 Ends: Flanged RF
 Body: CS (ASTM A216-WCB)
 Cover/Bonnet: CS (ASTMA216-WCB)
 Trim: API 594 **TRIM #8**
 Bore: Full Port (FP)
 Disc Washer, Disc Nut, Cotter Pin: 410 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: ASTM A193-B7
 Nuts: ASTM A194-2H
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL150, FP): 137-XU
 Crane (Cast, CL150, FP): 147-XU
 Velan (Cast, CL150, FP): F-{Size}-0114C-02TY

Description of CHECK Valve (CS03CB500): (CHS30-CS-F500)

Type: Check, [Swing, FP](#)
 Design Type: Swing
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 300**
 Ends: Flanged RF
 Body: CS (ASTM A216-WCB)
 Cover/Bonnet: CS (ASTMA216-WCB)
 Trim: API 594 **TRIM #8**
 Bore: Full Port (FP)
 Disc Washer, Disc Nut, Cotter Pin: 410 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: ASTM A193-B7
 Nuts: ASTM A194-2H
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL300, FP): 149-XU (2" to 24" NPS)
 Crane (Cast, CL300, FP): 159-XU
 Velan (Cast, CL300, FP): F-{Size}-1114C-02TY

Description of CHECK Valve (CS06CB500):

(CHS60-CS-F500)

Type: Check, **Swing, FP**
 Design Type: Swing
 Valve Size: NPS 2 -to- NPS 20
 Class: **ASME 600**
 Ends: Flanged RF
 Body: CS (ASTM A216-WCB)
 Cover/Bonnet: CS (ASTMA216-WCB)
 Trim: API 594 **TRIM #8** (was #1)
 Bore: Full Port (FP)
 Disc Washer, Disc Nut, Cotter Pin: 410 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: ASTM A193-B7
 Nuts: ASTM A194-2H
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL600, FP): 165-XU (2" to 16" NPS)
 Crane (Cast, CL600, FP): 172-XU (2" to 20" NPS)
 Velan (Cast, CL600, FP): F-{Size}-2114C-02TY

Description of CHECK Valve (CHS80-CS-S300):

Type: Check, **Swing, FP**
 Design Type: **Swing**
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **API 800**
 Ends: **SW**
 Body: CS (ASTM A105N)
 Cover/Bonnet: CS (ASTMA105N)
 Trim: API 602 **TRIM #8**
 Bore: Full Port (FP)
 Springs: **None**
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: (DSI, Forged, Trim #8): 4411-FB-A8
 Bonney Forge: H-60-1-SW (no spring)

Description of CHECK Valve (CHS80-CS-T107):

(CHS80-CS-T107)

Type: Check, [Swing, SP](#)
 Design Type: Swing
 Valve Size: NPS 0.5 -to-NPS 2
 Class: **ASME 800**
 Ends: FNPT
 Body: CS (ASTM A105N)
 Cover/Bonnet: CS (ASTM A105N)
 Port: **Standard Port (SP)**
 Trim: API 602 **TRIM #8**
 Springs: None
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: [ASME B16.11](#)
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: Mfg.Std.

Approved Vendors: DSI (Forged, RP): 4421-A8
 Bonney Forge: HL-60-1-T

Description of CHECK Valve (CHS80-CS-T108 ==> CHS80-CS-T210)

Type: Check, [Swing, FP](#)
 Design Type: Swing
 Valve Size: NPS 0.5 -to-NPS 2
 Class: **ASME 800**
 Ends: FNPT
 Body: CS (ASTM A105N)
 Cover/Bonnet: CS (ASTM A105N)
 Port: **Full Port (FP)**
 Trim: API 602 TRIM 8
 Springs: None
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: [ASME B16.11](#)
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: Mfg.Std.

Approved Vendors: DSI (Forged, FP): 4421-FB-A8
 Bonney Forge: H-60-1-T

Description of CHECK Valve (CS09CB500):

(CHS90-CS-F500)

Type: Check, [Swing, FP](#)
 Design Type: Swing
 Valve Size: NPS 2 -to- NPS 12
 Class: **ASME 900**
 Ends: Flanged RF
 Body: CS (ASTM A216-WCB)
 Cover/Bonnet: CS (ASTMA216-WCB)
 Trim: API 594 **TRIM #8** (Was #1)
 Bore: Full Port (FP)
 Disc Washer, Disc Nut, Cotter Pin: 410 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: ASTM A193-B7
 Nuts: ASTM A194-2H
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL900, FP): 177-XU (2" to 10" NPS)
 Crane (Cast, CL900, FP): 187-XU (2" to 12" NPS)
 Velan (Cast, CL900, FP): F-{Size}-7114C-02TY

Description of CHECK Valve (CS15CB500):

(CHS1.5K-CS-F500)

Type: Check, [Swing, FP](#)
 Design Type: Swing
 Valve Size: NPS 2 -to- NPS 10
 Class: **ASME 1500**
 Ends: Flanged RF
 Body: CS (ASTM A216-WCB)
 Cover/Bonnet: CS (ASTMA216-WCB)
 Trim: API 594 **TRIM #8** (was #1)
 Bore: Full Port (FP)
 Disc Washer, Disc Nut, Cotter Pin: 410 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: ASTM A193-B7
 Nuts: ASTM A194-2H
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL1500, FP): 189-XU (2" to 8" NPS)
 Crane (Cast, CL1500, FP): 199-XU (2" to 10" NPS)
 Velan (Cast, CL1500, FP): F-{Size}-3114C-02TY

***** Low Temperature Check Valves *****

Description of CHECK Valve (CD01CA500): (CHD15-CL-B500)

Type: Check, [Dual-Plate](#), [Low-Temperature](#)
Design Type: Dual Plate, Retainerless
Valve Size: NPS 2 -to- NPS 24
Service: Low Temperature -46 to 343°C (-50 to 650°F)
Class: **ASME 150**
Ends: Double Flanged (RF)
Body: ITCS (ASTM A352-LCB or ASTM A352-LCC)
Trim: API 594 **TRIM #10** (316 SS)
Springs: Alloy X-750
Stop Pin: 316 SS
Body Type: Double Flanged
Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward

The standards are:

DESIGN: API 594 Type A
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check): {Size}-H15-GCPF3-233 (Trim #10)
..... Velan (Proquip): PF-{Size}-002E17-3DB

Description of CHECK Valve (CD01CA700): (CHD15-CL-W700)

Type: Check, [Dual-Plate](#), [Low Temperature](#)
Design Type: Dual Plate, Retainerless
Valve Size: NPS 2 -to- NPS 24
Service: Low Temperature -46 to 343°C (-50 to 650°F)
Class: **ASME 150**
Ends: **Wafer**
Body: ITCS (ASTM A352-LCB or ASTM A352-LCC)
Trim: API 594 **TRIM #10** (316 SS)
Springs: Alloy X-750
Stop Pin: 316 SS
Body Type: **Wafer**
Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward

The standards are:

DESIGN: API 594 Type A
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check): {Size}-H15-GCPF3-233 (Trim #10)
..... Velan (Proquip): PW-{Size}-026E17-3DB
..... Keckley: {Size}-031D17-3DB

Description of CHECK Valve (CD03CA500): (CHD30-CL-B500)

Type: Check, [Dual-Plate](#), [Low-Temperature](#)
 Design Type: Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Service: Low Temperature -46 to 343°C (-50 to 650°F)
 Class: **ASME 300**
 Ends: Double Flanged (RF)
 Body: ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Trim: API 594 **TRIM #10** (316 SS)
 Springs: Alloy X-750
 Stop Pin: 316 SS
 Body Type: Double Flanged
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check): {Size}-H30-GPF3-233 (Trim #10)
 Velan: PF-{Size}-131D19-3DB

Description of CHECK Valve (CD03CA700): (CHD30-CL-W700)

Type: Check, [Dual-Plate](#), [Low-Temperature](#)
 Design Type: Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Service: Low Temperature -46 to 343°C (-50 to 650°F)
 Class: **ASME 300**
 Ends: **Wafer**
 Body: ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Trim: API 594 **TRIM #10** (316 SS)
 Springs: Alloy X-750
 Stop Pin: 316 SS
 Body Type: **Wafer**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check): {Size}-H30-GPF3-233 (Trim #10)
 Velan (Proquip): PW-{Size}-131D19-3DB

Description of CHECK Valve (CD03CA702): (CHD30-CL-L702)

Type: Check, **Dual-Plate, Low Temperature**
 Design Type: Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Service: Low Temperature -46 to 343°C (-50 to 650°F)
 Class: **ASME 300**
 Ends: **Lugged**
 Body: ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Trim: API 594 **TRIM #10** (316 SS)
 Springs: Alloy X-750
 Stop Pin: 316 SS
 Body Type: **Lugged**, Bolt Holes Through-Drilled
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check): {Size}-H30-GPF2-233 (Trim #10)
 Velan (Proquip): PM-{Size}-131D19-3DB

Description of CHECK Valve (CD06CA500): (CHD60-CL-B500)

Type: Check, [Dual-Plate](#), [Low-Temperature](#)
 Design Type: Dual Plate, Retainerless
 Valve Size: NPS 8 -to- NPS 24
 Service: Low Temperature -46 to 343°C (-50 to 650°F)
 Class: **ASME 600**
 Ends: Double Flanged (RF)
 Body: ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Trim: API 594 **TRIM #10**
 Springs: Alloy X-750
 Stop Pin: 316 SS
 Body Type: Double Flanged
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward
 The standards are:
 DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check): {Size}-H60-GCPF3-233 (Trim #10)
 Velan (Proquip): PF-{Size}-031D19-3DB (Trim #12)

Description of CHECK Valve (CD06CA700): (CHD60-CL-W700)

Type: Check, [Dual-Plate](#), [Low-Temperature](#)
 Design Type: Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Service: Low Temperature -46 to 343°C (-50 to 650°F)
 Class: **ASME 600**
 Ends: **Wafer**
 Body: ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Trim: API 594 **TRIM #10**
 Springs: Alloy X-750
 Stop Pin: 316 SS
 Body Type: **Wafer**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward

The standards are:
 DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check): {Size}-H60-GCPF3-233 (Trim #10)
 Velan (Proquip): PW-{Size}-231D17-3DB (Trim #12)

Description of CHECK Valve (CD06CA702): (CHD60-CL-L702)

Type: Check, **Dual-Plate, Low-Temperature**
 Design Type: Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Service: Low Temperature -46 to 343°C (-50 to 650°F)
 Class: **ASME 600**
 Ends: **Lugged**
 Body: ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Trim: API 594 **TRIM #10**
 Springs: Alloy X-750
 Stop Pin: 316 SS
 Body Type: **Lugged**, Bolt Holes Through-Drilled
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendors: Crane (Cast, Duo-Check, Lugged): {Size}-H60-GPF2-233 (Trim #10)
 Velan (Proquip): PM-{Size}-231D19-3DB (Trim #12 ??)

Description of CHECK Valve (CL08CA300): (CHP80-CL-S300)

Type: Check, **Piston Lift, Low-Temperature, SP**
 Design Type: Piston Lift
 Valve Size: NPS 0.5 -to- NPS 2
 Service: Low Temperature -46 to 343°C (-50 to 650°F)
 Class: **API 800**
 Ends: SW
 Body: ITCS (ASTM A350-LF2-CL-1)
 Cover/Bonnet: ITCS (ASTMA350-LF2-CL-1)
 Trim: API 602 **TRIM #12**
 Bore: Standard Port (SP)
 Springs: None
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: (DSI, Forged): 4311-F12 (no spring)
 Bonney Forge: L1-40-L3S-SW (no spring)

Description of CHECK Valve (CL08CA301): (CHP80-CL-S301)

Type: Check, **Piston Lift, Low-Temperature, SP**
 Design Type: Piston Lift
 Valve Size: NPS 0.5 -to- NPS 2
 Service: Low Temperature -46 to 343°C (-50 to 650°F)
 Class: **API 800**
 Ends: SW
 Body: ITCS (ASTM A350-LF2-CL-1)
 Cover/Bonnet: ITCS (ASTMA350-LF2-CL-1)
 Trim: API 602 **TRIM #12**
 Bore: Standard Port (SP)
 Springs: Alloy X-750
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: (DSI, Forged): 4311-F12 (w/spring)
 Bonney Forge: L1-40-L3S-SW (w/spring)

Description of CHECK Valve (CS01CA500): (CHS15-CL-F500)

Type: Check, **Swing, Low-Temperature, FP**
 Design Type: Swing
 Valve Size: NPS 2 -to- NPS 24
 Service: Low Temperature -46 to 343°C (-50 to 650°F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Cover/Bonnet: ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Trim: API 594 **TRIM #10**
 Bore: Full Port (FP)
 Disc Washer, Disc Nut, Cotter Pin: 316 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, FP): 137-L-LCC
 Crane (Cast, FP): 147-L-2
 Velan (Cast, FP): F-{Size}-0114C-25MX

Description of CHECK Valve (CS03CA500): (CHS30-CL-F500)

Type: Check, [Swing, Low-Temperature, FP](#)
 Design Type: Swing
 Valve Size: NPS 2 -to- NPS 24
 Service: Low Temperature -46 to 343°C (-50 to 650°F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Cover/Bonnet: ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Trim: API 594 TRIM 10
 Bore: Full Port (FP)
 Disc Washer, Disc Nut, Cotter Pin: 316 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, FP): 149-L-LCC
 Crane (Cast, FP): 159-L-2
 Velan (Cast, FP): F-{Size}-1114C-25MX

Description of CHECK Valve (CS06CA500): (CHS60-CL-F500)

Type: Check, [Swing, Low-Temperature, FP](#)
 Design Type: Swing
 Valve Size: NPS 2 -to- NPS 20
 Service: Low Temperature -46 to 343°C (-50 to 650°F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Cover/Bonnet: ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Trim: API 594 TRIM 10
 Bore: Full Port (FP)
 Disc Washer, Disc Nut, Cotter Pin: 316 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, FP): 165-L-LCC (2" to 16" NPS)
 Crane (Cast, FP): 175-L-2 (2" to 20" NPS)
 Velan (Cast, FP): F-{Size}-2114C-25MX



Document #: GCTLR00-TSLA-ME-SPC-00004

Valve Specification for CS Check Valves

Rev.	RevDate	Description	By:	Check:
0	15 SEPT 2023	Issued for Release	MFS	SBS
1	26 MAR 2024	Categorize by Valve Type, Add Double-Flanged Valves	MFS	WGS

End Document #: GCTLR00-TSLA-ME-SPC-00004

Valve Specification for CS Check Valves

**Description of GATE Valve (GA01CB500):****GA15-CS-F500**

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 150**
Ends: Flanged RF
Body: CS (ASTM A105N)
Bonnet: CS (ASTM A105N)
Trim: API 602 Trim 8
Wedge Type: Solid
Bonnet Type: Bolted
Bore-Port: **Standard Port (RP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 602
..... Nuts: API 602

The standards are:

DESIGN: API 602
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Forged, CL150, RP): 1131-A8
..... Bonney Forge (Forged, CL150, RP): L1-10-1-RF

Description of GATE Valve (GA01CB501):**GA15-CS-F501**

Type: Gate
Valve Size: NPS 2 -to- NPS 24
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 150**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Bonnet: CS (ASTM A216-WCB)
Trim: API 600 Trim 8
Wedge Type: Solid or Flexible (NPS 2 -to- NPS 4), Flexible (NPS 6 and Greater)
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 600
..... Nuts: API 600

The standards are:

DESIGN: API 600
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL150, FP): 37-XUF, Handwheel Op.
..... Crane (Cast, CL150, FP): 47-XUF, Handwheel Op.
..... Velan (Cast, CL150, FP): F-{Size}-0064C-02TY, Handwheel Op.
..... SCV (Cast, CL150, FP): GAT-{Size}-01CH4-H1, Handwheel Op.

**Description of GATE Valve (GA01CB503):****GA15-CS-F503**

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 150**
Ends: Flanged RF
Body: CS (ASTM A105N)
Bonnet: CS (ASTM A105N)
Trim: API 602 Trim 8
Wedge Type: Solid
Bonnet Type: Bolted
Bore-Port: **Standard Port (RP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
The standards are:
DESIGN: API 602
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: Bonney Forge (Forged, CL150, RP): L1-10-1-RF

Description of GATE Valve (GA01CB504):**GA15-CS-F504**

Type: Gate
Valve Size: NPS 8 -to- NPS 24
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 150**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Bonnet: CS (ASTM A216-WCB)
Trim: API 600 Trim 8
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Gear
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 600
..... Nuts: API 600
The standards are:
DESIGN: API 600
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL150, FP): G-37-XUF (8" to 24"), Gear Op.
..... Crane (Cast, CL150, FP): 47-XUF (8" to 10"), Gear Op.
..... SCV (Cast, CL150, FP): GAT-{Size}-01CH4-G1, Gear Op.

Description of GATE Valve (GA01CB505):

GA15-CS-F505

Type: Gate
 Valve Size: NPS 26 -to- NPS 48
 Temperature: -29 to 343 °C (-20 to 649 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: CS (ASTM A216-WCB)
 Bonnet: CS (ASTM A216-WCB)
 Trim: API 600 Trim 8
 Wedge Type: Flexible
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Gear
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 600
 Nuts: API 600
 The standards are:
 DESIGN: API 600, as applicable
 ENDS: ASME B16.47 Series B
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 (NPS 26 -to- NPS 36)
 MFG STD (NPS 38 -to- NPS 48)

Approved Vendors:

Description of GATE Valve (GA01CB506):

GA15-CS-F506

Type: Gate
 Valve Size: NPS 26 -to- NPS 48
 Temperature: -29 to 343 °C (-20 to 649 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: CS (ASTM A216-WCB)
 Bonnet: CS (ASTM A216-WCB)
 Trim: API 600 Trim 8
 Wedge Type: Flexible
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Gear
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 600
 Nuts: API 600
 The standards are:
 DESIGN: API 600, as applicable
 ENDS: ASME B16.47 Series A
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 (NPS 26 -to- NPS 36)
 MFG STD (NPS 38 -to- NPS 48)

Approved Vendors:

**Description of GATE Valve (GA03CB500):****GA30-CS-F500**

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: CS (ASTM A105N)
Bonnet: CS (ASTM A105N)
Trim: API 602 Trim 8
Wedge Type: Solid
Bonnet Type: Bolted
Bore-Port: Standard
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 602
..... Nuts: API 602

The standards are:

DESIGN: API 602
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Forged, CL300, RP): 2131-A8
..... Bonney Forge (Forged, CL300, RP): L3-10-1-RF

Description of GATE Valve (GA03CB501):**GA30-CS-F501**

Type: Gate
Valve Size: NPS 2 -to- NPS 24
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Bonnet: CS (ASTM A216-WCB)
Trim: API 600 Trim 8
Wedge Type: Solid or Flexible (NPS 2 -to- NPS 4), Flexible (NPS 6 and Greater)
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 600
..... Nuts: API 600

The standards are:

DESIGN: API 600
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL300, FP): 23-XUF (2" to 6"), Handwheel Op.
..... Crane (Cast, CL300, FP): 33-XUF (2" to 10"), Handwheel Op.
..... SCV (Cast, CL300, FP): GAT-{Size}-03CH4-H1, Handwheel Op.

Description of GATE Valve (GA03CB503):

GA30-CS-F503

Type: Gate
 Valve Size:..... NPS 0.5 -to- NPS 2
 Temperature:..... -29 to 343 °C (-20 to 649 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: CS (ASTM A105N)
 Bonnet: CS (ASTM A105N)
 Trim: API 602 Trim 8
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (RP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 The standards are:
 DESIGN:..... API 602
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL:..... ASME B16.10
 Approved Vendors:..... DSI (Forged, CL300, RP): 2133-A8
 Bonney Forge (Forged, CL300, RP): L3-10-1-RF

Description of GATE Valve (GA03CB504):

GA30-CS-F504

Type: Gate
 Valve Size:..... NPS 4 -to- NPS 24
 Temperature:..... -29 to 343 °C (-20 to 649 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: CS (ASTM A216-WCB)
 Bonnet: CS (ASTM A216-WCB)
 Trim: API 600 Trim 8
 Wedge Type: Flexible
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Gear
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:..... SS/Graphite
 Body-Bonnet Bolting:..... Bolts: API 600
 Nuts: API 600
 The standards are:
 DESIGN:..... API 600
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL:..... ASME B16.10
 Approved Vendors:..... DSI (Cast, CL300, FP): G-23-XUF (4" to 24"), Gear Op.
 Crane (Cast, CL300, FP): 33-XUF-2 (4" to 10"), Gear Op.
 SCV (Cast, CL300, FP): GAT-{Size}-03CH4-G1 (8" to 24"), Gear Op.

Description of GATE Valve (GA03CB505):

GA30-CS-F505

Type: Gate
 Valve Size:..... NPS 26 -to- NPS 48
 Temperature:..... -29 to 343 °C (-20 to 649 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: CS (ASTM A216-WCB)
 Bonnet: CS (ASTM A216-WCB)
 Trim: API 600 Trim 8
 Wedge Type: Flexible
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation:..... Gear
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:..... SS/Graphite
 Body-Bonnet Bolting:..... Bolts: API 600
 Nuts: API 600
 The standards are:
 DESIGN:..... API 600, as applicable
 ENDS: ASME B16.47 Series B
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... ASME B16.10 (NPS 26 -to- NPS 36)
 MFG STD (NPS 38 -to- NPS 48)

Approved Vendors:.....

Description of GATE Valve (GA03CB506):

GA30-CS-F506

Type: Gate
 Valve Size:..... NPS 26 -to- NPS 48
 Temperature:..... -29 to 343 °C (-20 to 649 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: CS (ASTM A216-WCB)
 Bonnet: CS (ASTM A216-WCB)
 Trim: API 600 Trim 8
 Wedge Type: Flexible
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation:..... Gear
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:..... SS/Graphite
 Body-Bonnet Bolting:..... Bolts: API 600
 Nuts: API 600
 The standards are:
 DESIGN:..... API 600, as applicable
 ENDS: ASME B16.47 Series A
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... ASME B16.10 (NPS 26 -to- NPS 36)
 MFG STD (NPS 38 -to- NPS 48)

Approved Vendors:.....

**Description of GATE Valve (GA06CB500):****GA60-CS-F500**

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 600**
Ends: Flanged, RF
Body: CS (ASTM A105N)
Bonnet: CS (ASTM A105N)
Trim: API 602 Trim 8
Wedge Type: Solid
Bonnet Type: Bolted
Bore-Port: Standard
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 602
..... Nuts: API 602

The standards are:

DESIGN: API 602
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Forged, CL600, RP): 3131-A8
..... Bonney Forge (Forged, CL600, RP): L6-10-1-RF

Description of GATE Valve (GA06CB501):**GA60-CS-F501**

Type: Gate
Valve Size: NPS 2 -to- NPS 24
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 600**
Ends: **Flanged, RF**
Body: CS (ASTM A216-WCB)
Bonnet: CS (ASTM A216-WCB)
Trim: API 600 Trim 8
Wedge Type: Solid or Flexible (NPS 2 -to- NPS 4), Flexible (NPS 6 and Greater)
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 600
..... Nuts: API 600

The standards are:

DESIGN: API 600
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL600, FP): 66-XUF (3" to 24"), Handwheel Op.
..... Crane (Cast, CL600, FP): 76-XUF (3" to 24"), Handwheel Op
..... SCV (Cast, CL600, FP): GAT-{Size}-06L3S-H1 (3" to 24"), Handwheel Op

**Description of GATE Valve (GA06CB502):****GA60-CS-F502**

Type: Gate
Valve Size: NPS 2 -to- NPS 24
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Bonnet: CS (ASTM A216-WCB)
Trim: API 600 **Trim 5**
Wedge Type: Solid or Flexible (NPS 2 -to- NPS 4), Flexible (NPS 6 and Greater)
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 600
..... Nuts: API 600

The standards are:

DESIGN: API 600
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL600, FP): G-66-XUF (3" to 24"), Handwheel Op.
..... Crane (Cast, CL600, FP): 76-XUF (3" to 24"), Handwheel Op

Description of GATE Valve (GA06CB504):**GA60-CS-F504**

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: CS (ASTM A105N)
Bonnet: CS (ASTM A105N)
Trim: API 602 Trim 5
Wedge Type: Solid
Bonnet Type: Welded
Bore-Port: **Standard Port (RP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 602
..... Nuts: API 602

The standards are:

DESIGN: API 602
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: (Forged, CL600, RP)

Description of GATE Valve (GA06CB506):

GA60-CS-F506

Type: Gate
Valve Size: NPS 3 -to- NPS 24
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Bonnet: CS (ASTM A216-WCB)
Trim: API 600 Trim 8
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Gear
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 600
..... Nuts: API 600
The standards are:
DESIGN: API 600
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10
Approved Vendors: DSI (Cast, CL600, FP): G-66-XUF (3" to 24"), Gear Op.
..... Crane (Cast, CL600, FP): 76-XUF (3" to 24"), Gear Op
..... SCV (Cast, CL600, FP): GAT-{Size}-06CH4-G1, Gear Op

Description of GATE Valve (GA06CB507):

GA60-CS-F507

Type: Gate
Valve Size: NPS 26 -to- NPS 48
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Bonnet: CS (ASTM A216-WCB)
Trim: API 600 Trim 8
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Gear
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 600
..... Nuts: API 600
The standards are:
DESIGN: API 600, as applicable
ENDS: ASME B16.47 Series B
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 (NPS 26 -to- NPS 30)
..... MFG STD (NPS 32 -to- NPS 48)
Approved Vendors:

**Description of GATE Valve (GA06CB508):****GA60-CS-F508**

Type: Gate
Valve Size: NPS 3 -to- NPS 24
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Bonnet: CS (ASTM A216-WCB)
Trim: API 600 **Trim 5**
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Gear
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 600
..... Nuts: API 600

The standards are:

DESIGN: API 600
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL600, FP): G-66-XUF (3" to 24"), Gear Op.
..... Crane (Cast, CL600, FP): 76-XUF (3" to 24"), Gear Op

Description of GATE Valve (GA06CB509):**GA60-CS-F509**

Type: Gate
Valve Size: NPS 26 -to- NPS 48
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Bonnet: CS (ASTM A216-WCB)
Trim: API 600 Trim 5
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Gear
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 600
..... Nuts: API 600

The standards are:

DESIGN: API 600, as applicable
ENDS: ASME B16.47 Series B
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 (NPS 26 -to- NPS 30)
..... MFG STD (NPS 32 -to- NPS 48)

Approved Vendors:

**Description of GATE Valve (GA06CB510):****GA60-CS-F510**

Type: Gate
Valve Size: NPS 26 -to- NPS 48
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Bonnet: CS (ASTM A216-WCB)
Trim: API 600 Trim 5
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Gear
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 600
..... Nuts: API 600

The standards are:

DESIGN: API 600, as applicable
ENDS: ASME B16.47 Series A
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 (NPS 26 -to- NPS 30)
..... MFG STD (NPS 32 -to- NPS 48)

Approved Vendors:

Description of GATE Valve (GA06CB511):**GA60-CS-F511**

Type: Gate
Valve Size: NPS 26 -to- NPS 48
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Bonnet: CS (ASTM A216-WCB)
Trim: API 600 Trim 8
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Gear
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 600
..... Nuts: API 600

The standards are:

DESIGN: API 600, as applicable
ENDS: ASME B16.47 Series A
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 (NPS 26 -to- NPS 30)
..... MFG STD (NPS 32 -to- NPS 48)

Approved Vendors:

Description of Gate Valve (GA80-CS-T101)

Type: Gate
 Valve Size: NPS 0.5 -to-NPS 2
 Temperature: -29 to 343 °C (-20 to 649 °F)
 Class: **API 800**
 Ends: FNPT
 Body: CS (ASTM A105N)
 Bonnet: CS (ASTM A105N)
 Trim: API 602 Trim 8
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (RP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602

The standards are:

DESIGN: API 602
 ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

VENDORS: DSI (Forged, CL800, RP): 4121-A8
 Bonney Forge (Forged, CL800, RP): HL-10-1-T

Description of Gate Valve (GA80-CS-T102)

Type: Gate
 Valve Size: NPS 0.5 -to-NPS 2
 Temperature: -29 to 343 °C (-20 to 649 °F)
 Class: **API 800**
 Ends: FNPT
 Body: CS (ASTM A105N)
 Bonnet: CS (ASTM A105N)
 Trim: API 602 Trim 8
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Full Port**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602

The standards are:

DESIGN: API 602
 ENDS: THRD: ASME B1.20.1
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

VENDORS: DSI (Forged, CL800, FP): 4121-FB-A8
 Bonney Forge (Forged, CL800, FP): H-10-1-T

Description of GATE Valve (GA80-CS-X210)

Type: Gate
 Valve Size: NPS 0.5 -to- NPS 2
 Temperature: -29 to 343 °C (-20 to 649 °F)
 Class: **API 800**
 Ends: SW x FNPT
 Body: CS (ASTM A105N)
 Bonnet: CS (ASTM A105N)
 Trim: API 602 Trim 8
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (RP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 The standards are:
 DESIGN: API 602
 ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: DSI (Forged, CL800, RP): 4101-A8
 Bonney Forge (Forged, CL800, RP): HL-10-1-SW/T

Description of GATE Valve (GA80-CS-T210)

Type: Gate
 Valve Size: NPS 0.5 -to- NPS 2
 Temperature: -29 to 343 °C (-20 to 649 °F)
 Class: **API 800**
 Ends: FNPT
 Body: CS (ASTM A105N)
 Bonnet: CS (ASTM A105N)
 Trim: API 602 Trim 8
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (RP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 The standards are:
 DESIGN: API 602
 ENDS: THRD: ASME B1.20.1
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: DSI (Forged, CL800, RP): 4121-A8
 Bonney Forge (Forged, CL800, RP): HL-10-1-T

**Description of GATE Valve (GA08CB200):****GA80-CS-X200**

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **API 800**
Ends: SW x FNPT
Body: CS (ASTM A105N)
Bonnet: CS (ASTM A105N)
Trim: API 602 Trim 8
Wedge Type: Solid
Bonnet Type: Bolted
Bore-Port: **Standard Port (RP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 602
..... Nuts: API 602
The standards are:
DESIGN: API 602
ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
RATING: API 602
TESTING: API 598
DIMENSIONAL: MFG STD

Approved Vendors: DSI (Forged, CL800, RP): 4101-A8
..... Bonney Forge (Forged, CL800, RP): HL-10-1-SW/T

Description of GATE Valve (GA08CB201):**GA80-CS-X201**

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **API 800**
Ends: SW x FNPT
Body: CS (ASTM A105N)
Bonnet: CS (ASTM A105N)
Trim: API 602 Trim 5
Wedge Type: Solid
Bonnet Type: Welded (Not bolted bonnet)
Bore-Port: **Standard Port (RP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
The standards are:
DESIGN: API 602
ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
RATING: API 602
TESTING: API 598
DIMENSIONAL: MFG STD

Approved Vendors: DSI (Forged, CL800, RP): 4103-A5
..... Bonney Forge (Forged, CL800, RP): WL-10-2-SW/T

**Description of GATE Valve (GA08CB202):****GA80-CS-X202**

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **API 800**
Ends: SW x FNPT
Body: CS (ASTM A105N)
Bonnet: CS (ASTM A105N)
Trim: API 602 Trim 8
Wedge Type: Solid
Bonnet Type: Welded
Bore-Port: **Standard Port (RP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
The standards are:
DESIGN: API 602
ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
RATING: API 602
TESTING: API 598
DIMENSIONAL: MFG STD

Approved Vendors: DSI (Forged, CL800, RP): 4103-A8
..... Bonney Forge (Forged, CL800, RP): WL-10-1-SW/T

Description of GATE Valve (GA08CB300):**GA80-CS-S300**

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **API 800**
Ends: SW
Body: CS (ASTM A105N)
Bonnet: CS (ASTM A105N)
Trim: API 602 Trim 8
Wedge Type: Solid
Bonnet Type: Bolted
Bore-Port: Standard
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 602
..... Nuts: API 602
The standards are:
DESIGN: API 602
ENDS: ASME B16.11
RATING: API 602
TESTING: API 598
DIMENSIONAL: MFG STD

Approved Vendors: DSI (Forged, CL800, RP): 4111-A8
..... Bonney Forge (Forged, CL800, RP): HL-10-1-SW

**Description of GATE Valve (GA08CB301):****GA80-CS-S301**

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **API 800**
Ends: SW
Body: CS (ASTM A105N)
Bonnet: CS (ASTM A105N)
Trim: API 602 **Trim 5**
Wedge Type: Solid
Bonnet Type: Welded
Bore-Port: **Standard Port (RP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 602
..... Nuts: API 602
The standards are:
DESIGN: API 602
ENDS: ASME B16.11
RATING: API 602
TESTING: API 598
DIMENSIONAL: MFG STD
Approved Vendors: DSI (Forged, CL800, RP): 4113-A5
..... Bonney Forge (Forged, CL800, RP): WL-10-2-SW

Description of GATE Valve (GA08CB302):**GA80-CS-S302**

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **API 800**
Ends: SW
Body: CS (ASTM A105N)
Bonnet: CS (ASTM A105N)
Trim: API 602 Trim 8
Wedge Type: Solid
Bonnet Type: Welded
Bore-Port: Standard
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
The standards are:
DESIGN: API 602
ENDS: ASME B16.11
RATING: API 602
TESTING: API 598
DIMENSIONAL: MFG STD
Approved Vendors: DSI (Forged, CL800, RP): 4113-A8
..... Bonney Forge (Forged, CL800, RP): WL-10-1-SW

**Description of GATE Valve (GA08CB303):****GA80-CS-S303**

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **API 800**
Ends: SW
Body: CS (ASTM A105N)
Bonnet: CS (ASTM A105N)
Trim: API 602 Trim 8
Wedge Type: Solid
Bonnet Type: Bolted
Bore-Port: **FULL Port (FP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 602
..... Nuts: API 602

The standards are:

DESIGN: API 602
ENDS: ASME B16.11
RATING: API 602
TESTING: API 598
DIMENSIONAL: MFG STD

Approved Vendors: DSI (Forged, CL800, FP): 4111-FB-A8
..... Bonney Forge (Forged, CL800, FP): H-10-1-SW

Description of GATE Valve (GA09CB500):

GA90-CS-F500

Type: Gate
 Valve Size: NPS 3 -to- NPS 16
 Temperature: -29 to 343 °C (-20 to 649 °F)
 Class: **ASME 900**
 Ends: Flanged RF
 Body: CS (ASTM A216-WCB)
 Bonnet: CS (ASTM A216-WCB)
 Trim: API 600 Trim 8
 Wedge Type: Solid or Flexible (NPS 2 -to- NPS 4), Flexible (NPS 6 and Greater)
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 600
 Nuts: API 600

The standards are:

DESIGN: API 600
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL900, FP): 73-XUF (3" to 12"), Handwheel Op.
 Crane (Cast, CL900, FP): 83-XUF (3" to 16"), Handwheel Op.
 SCV (Cast, CL900, FP): GAT-{Size}-09CH4-H1 (3" to 24"), Handwheel Op.

Description of GATE Valve (GA09CB501):

GA90-CS-F501

Type: Gate
 Valve Size: NPS 3 -to- NPS 24
 Temperature: -29 to 343 °C (-20 to 649 °F)
 Class: **ASME 900**
 Ends: Flanged RF
 Body: CS (ASTM A216-WCB)
 Bonnet: CS (ASTM A216-WCB)
 Trim: API 600 Trim 5
 Wedge Type: Solid or Flexible (NPS 2 -to- NPS 4), Flexible (NPS 6 and Greater)
 Bonnet Type: Pressure Seal
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: MFG STD
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD

The standards are:

DESIGN: ASME B16.34 and MSS SP-144
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors:

.

**Description of GATE Valve (GA09CB502):****GA90-CS-F502**

Type: Gate
Valve Size: NPS 4 -to- NPS 24
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 900**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Bonnet: CS (ASTM A216-WCB)
Trim: API 600 Trim 8
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Gear
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 600
..... Nuts: API 600

The standards are:

DESIGN: API 600
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL900, FP): G-73-XUF (4" to 12"), Gear Op.
..... Crane (Cast, CL900, FP): 83-XUF (4" to 10"), Gear Op.
..... SCV (Cast, CL900, FP): GAT-{Size}-09CH4-H1, Gear Op.

Description of GATE Valve (GA09CB503):**GA90-CS-F503**

Type: Gate
Valve Size: NPS 3 -to- NPS 24
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 900**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Bonnet: CS (ASTM A216-WCB)
Trim: API 600 Trim 5
Wedge Type: Flexible with Cavity Vent to High-Pressure Side (Indicated) Bonnet Type:
..... Pressure Seal
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Gear
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: MFG STD
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD

The standards are:

DESIGN: ASME B16.34 and MSS SP-144
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors:

**Description of GATE Valve (GA15CB200):****GA1.5K-CS-X200**

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 1500**
Ends: SW x FNPT
Body: CS (ASTM A105N)
Bonnet: CS (ASTM A105N)
Trim: API 602 Trim 8
Wedge Type: Solid
Bonnet Type: Bolted
Bore-Port: **Standard Port (RP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 602
..... Nuts: API 602
The standards are:
DESIGN: API 602
ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: MFG STD

Approved Vendors: DSI (Forged, CL1500, RP): 6101-A8
..... Bonney Forge (Forged, CL1500, RP): 9HL-10-1-SW/T

Description of GATE Valve (GA15CB201):**GA1.5K-CS-X201**

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 1500**
Ends: SW x FNPT
Body: CS (ASTM A105N)
Bonnet: CS (ASTM A105N)
Trim: API 602 Trim 5
Wedge Type: Solid
Bonnet Type: Welded
Bore-Port: **Standard Port (RP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
The standards are:
DESIGN: API 602
ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: MFG STD

Approved Vendors: DSI (Forged, CL1500, RP): 6103-A5
..... Bonney Forge (Forged, CL1500, RP): 9WL-10-2-SW/T



Description of GATE Valve (GA15CB300):

GA1.5K-CS-S300

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 1500**
Ends: SW
Body: CS (ASTM A105N)
Bonnet: CS (ASTM A105N)
Trim: API 602 Trim 8
Wedge Type: Solid
Bonnet Type: Bolted
Bore-Port: **Standard Port (RP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 602
..... Nuts: API 602

The standards are:

DESIGN: API 602
ENDS: ASME B16.11
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: MFG STD

Approved Vendors: DSI (Forged, CL1500, RP): 6111-A8
..... Bonney Forge (Forged, CL1500, RP): 9HL-10-1-SW

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 1500**
Ends: SW
Body: CS (ASTM A105N)
Bonnet: CS (ASTM A105N)
Trim: API 602 Trim 5
Wedge Type: Solid
Bonnet Type: Welded
Bore-Port: **Standard Port (RP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited

The standards are:

DESIGN: API 602
ENDS: ASME B16.11
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: MFG STD

Approved Vendors: DSI (Forged, CL1500, RP): 6113-A5
..... Bonney Forge (Forged, CL1500, RP): 9WL-10-2-SW



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Valve Specification for CS Gate Valves

Description of GATE Valve (GA15CB500):

GA1.5K-CS-F500

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 1500**
Ends: Flanged RF
Body: CS (ASTM A105N)
Bonnet: CS (ASTM A105N)
Trim: API 602 Trim 8
Wedge Type: Solid
Bonnet Type: Bolted
Bore-Port: **Standard Port (RP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 602
..... Nuts: API 602
The standards are:
DESIGN: API 602
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 - Globe Pattern (NPS 0.5 -to- NPS 0.75)
..... ASME B16.10 (NPS 1 -to- NPS 2)

Approved Vendors: DSI (Forged, CL1500, RP): 6131-A8

**Description of GATE Valve (GA15CB501):****GA1.5K-CS-F501**

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 1500**
Ends: Flanged RF
Body: CS (ASTM A105N)
Bonnet: CS (ASTM A105N)
Trim: API 602 Trim 5
Wedge Type: Solid
Bonnet Type: Welded
Bore-Port: **Standard Port (RP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
The standards are:
DESIGN: API 602
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 - Globe Pattern (NPS 0.5 -to- NPS 0.75)
..... ASME B16.10 (NPS 1 -to- NPS 2)

Approved Vendors: DSI (Forged, CL1500, RP): 6133-A5

Description of GATE Valve (GA15CB502):**GA1.5K-CS-F502**

Type: Gate
Valve Size: NPS 2 -to- NPS 24
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 1500**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Bonnet: CS (ASTM A216-WCB)
Trim: API 600 Trim 8
Wedge Type: Solid or Flexible (NPS 2 -to- NPS 4), Flexible (NPS 6 and Greater)
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 600
..... Nuts: API 600

The standards are:
DESIGN: API 600
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors:

**Description of GATE Valve (GA15CB503):****GA1.5K-CS-F503**

Type: Gate
Valve Size: NPS 2 -to- NPS 12
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 1500**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Bonnet: CS (ASTM A216-WCB)
Trim: API 600 Trim 5
Wedge Type: Solid or Flexible (NPS 2 -to- NPS 4), Flexible (NPS 6 and Greater)
Bonnet Type: Pressure Seal
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: MFG STD
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD

The standards are:

DESIGN: ASME B16.34 and MSS SP-144
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors:

Description of GATE Valve (GA15CB504):**GA1.5K-CS-F504**

Type: Gate
Valve Size: NPS 3 -to- NPS 24
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 1500**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Bonnet: CS (ASTM A216-WCB)
Trim: API 600 Trim 8
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Gear
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: API 600
..... Nuts: API 600

The standards are:

DESIGN: API 600
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors:



Description of GATE Valve (GA15CB505):

GA1.5K-CS-F505

Type: Gate
Valve Size: NPS 3 -to- NPS 24
Temperature: -29 to 343 °C (-20 to 649 °F)
Class: **ASME 1500**
Ends: Flanged RF
Body: CS (ASTM A216-WCB)
Bonnet: CS (ASTM A216-WCB)
Trim: API 600 Trim 5
Wedge Type: Flexible with Cavity Vent to High-Pressure Side (Indicated) Bonnet Type:
..... Pressure Seal
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Gear
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: MFG STD
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD

The standards are:

DESIGN: ASME B16.34 and MSS SP-144
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors:

***** Low Temp Gate Valves *****

Description of GATE Valve (GA01CA500): GA15-CL-F500

Type: Gate
 Valve Size:..... NPS 0.5 -to- NPS 2
 Service: Low Temperature -46 to 343 °C (-50 to 649 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: ITCS (ASTM A350-LF2-CL-1)
 Bonnet: ITCS (ASTM A350-LF2-CL-1)
 Trim: API 602 Trim 12
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (RP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: API 602
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10
 Approved Vendors: DSI (Forged, CL150, RP): 1131-F12
 Bonney Forge (Forged, CL150, RP): L1-10-LS-RF

Description of GATE Valve (GA01CA501): GA15-CL-F501

Type: Gate
 Valve Size:..... NPS 2 -to- NPS 24
 Service: Low Temperature -46 to 343 °C (-50 to 649 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Bonnet: ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Trim: API 600 Trim 12
 Wedge Type: Solid or Flexible (NPS 2 -to- NPS 4), Flexible (NPS 6 and Greater)
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: API 600
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10
 Approved Vendors: DSI (Cast, CL150, FP): 37-LUF-LCC (2" to 12"), Handwheel
 Crane (Cast, CL150, FP): 47-LUF-2 (2" to 10"), Handwheel
 SCV (Cast, CL150, FP): GAT-{Size}-01L3S-H1 (2" to 24"), Handwheel

**Description of GATE Valve (GA01CA503):****GA15-CL-F503**

Type: Gate
Valve Size: NPS 8 -to- NPS 24
Service: Low Temperature -46 to 343 °C (-50 to 649 °F)
Class: **ASME 150**
Ends: Flanged RF
Body: ITCS (ASTM A352-LCB or ASTM A352-LCC)
Bonnet: ITCS (ASTM A352-LCB or ASTM A352-LCC)
Trim: API 600 Trim 12
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Gear
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD

The standards are:

DESIGN: API 600
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL150, FP): 37-LUF-LCC (8" to 12"), Gear Op.
..... Crane (Cast, CL150, FP): 47-LUF-2 (8" to 10"), Gear Op.
..... SCV (Cast, CL150, FP): GAT-{Size}-01L3S-H1 (8" to 24"), Gear Op.

Description of GATE Valve (GA03CA500):**GA30-CL-F500**

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Service: Low Temperature -46 to 343 °C (-50 to 649 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: ITCS (ASTM A350-LF2-CL-1)
Bonnet: ITCS (ASTM A350-LF2-CL-1)
Trim: API 602 Trim 12
Wedge Type: Solid
Bonnet Type: Bolted
Bore-Port: **Standard Port (RP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD

The standards are:

DESIGN: API 602
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Forged, CL300, RP): 2131-F12
..... Bonney Forge (Forged, CL300, RP): L3-10-L3S-RF

Description of GATE Valve (GA03CA501):

GA30-CL-F501

Type: Gate
Valve Size: NPS 2 -to- NPS 24
Service: Low Temperature -46 to 343 °C (-50 to 649 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: ITCS (ASTM A352-LCB or ASTM A352-LCC)
Bonnet: ITCS (ASTM A352-LCB or ASTM A352-LCC)
Trim: API 600 Trim 12
Wedge Type: Solid or Flexible (NPS 2 -to- NPS 4), Flexible (NPS 6 and Greater)
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD

The standards are:

DESIGN: API 600
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL300, FP): 23-LUF, Handwheel Op
..... Crane (Cast, CL300, FP): 33-LUF, Handwheel Op.
..... SCV (Cast, CL300, FP): GAT-{Size}-03L3S-H1, Handwheel Op

Description of GATE Valve (GA03CA503):

GA30-CL-F503

Type: Gate
Valve Size: NPS 4 -to- NPS 24
Service: Low Temperature -46 to 343 °C (-50 to 649 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: ITCS (ASTM A352-LCB or ASTM A352-LCC)
Bonnet: ITCS (ASTM A352-LCB or ASTM A352-LCC)
Trim: API 600 Trim 12
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD

The standards are:

DESIGN: API 600
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL300, FP): 23-LUF (4" to 24"), Handwheel Op.
..... Crane (Cast, CL300, FP): 33-LUF (4" to 10"), Handwheel Op.
..... SCV (Cast, CL300, FP): GAT-{Size}-01L3S-H1, Handwheel Op.

**Description of GATE Valve (GA06CA500):****GA60-CL-F500**

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Service: Low Temperature -46 to 343 °C (-50 to 649 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: ITCS (ASTM A350-LF2-CL-1)
Bonnet: ITCS (ASTM A350-LF2-CL-1)
Trim: API 602 Trim 12
Wedge Type: Solid
Bonnet Type: Bolted
Bore-Port: **Standard Port (RP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD

The standards are:

DESIGN: API 602
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Forged, CL600, RP): 3131-F12
..... Bonney Forge (Forged, CL600, RP): L6-10-L3S-RF

Description of GATE Valve (GA06CA501):**GA60-CL-F501**

Type: Gate
Valve Size: NPS 2 -to- NPS 24
Service: Low Temperature -46 to 343 °C (-50 to 649 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: ITCS (ASTM A352-LCB or ASTM A352-LCC)
Bonnet: ITCS (ASTM A352-LCB or ASTM A352-LCC)
Trim: API 600 Trim 12
Wedge Type: Solid or Flexible (NPS 2 -to- NPS 4), Flexible (NPS 6 and Greater)
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD

The standards are:

DESIGN: API 600
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL600, FP): 66-LUF (8" to 12"), Handwheel Op.
..... Crane (Cast, CL600, FP): 76-LUF (2" to 10"), Handwheel Op.
..... SCV (Cast, CL600, FP): GAT-{Size}-06L3S-H1 (8" to 24"), Handwheel Op.

**Description of GATE Valve (GA06CA503):****GA60-CL-F503**

Type: Gate
Valve Size: NPS 3 -to- NPS 24
Service: Low Temperature -46 to 343 °C (-50 to 649 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: ITCS (ASTM A352-LCB or ASTM A352-LCC)
Bonnet: ITCS (ASTM A352-LCB or ASTM A352-LCC)
Trim: API 600 Trim 12
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Gear
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD

The standards are:

DESIGN: API 600
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL600, FP): G-66-XUF (3" to 24"), Gear Op.
..... Crane (Cast, CL600, FP): 76-XUF (3" to 24"), Gear Op
..... SCV (Cast, CL600, FP): GAT-{Size}-06L3S-H1 (3" to 24"), Gear Op

Description of GATE Valve (GA06CA504):**GA60-CL-F504**

Type: Gate
Valve Size: NPS 26 -to- NPS 48
Service: Low Temperature -46 to 343 °C (-50 to 649 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: ITCS (ASTM A352-LCB or ASTM A352-LCC)
Bonnet: ITCS (ASTM A352-LCB or ASTM A352-LCC)
Trim: API 600 Trim 12
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Gear
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD

The standards are:

DESIGN: API 600, as applicable
ENDS: ASME B16.47 Series B
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 (NPS 26 -to- NPS 30)
..... MFG STD (NPS 32 -to- NPS 48)

Approved Vendors:

**Description of GATE Valve (GA06CA505):****GA60-CL-F505**

Type: Gate
Valve Size: NPS 26 -to- NPS 48
Service: Low Temperature -46 to 343 °C (-50 to 649 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: ITCS (ASTM A352-LCB or ASTM A352-LCC)
Bonnet: ITCS (ASTM A352-LCB or ASTM A352-LCC)
Trim: API 600 Trim 12
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Gear
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN: API 600, as applicable
ENDS: ASME B16.47 Series A
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 (NPS 26 -to- NPS 30)
..... MFG STD (NPS 32 -to- NPS 48)

Approved Vendors:

Description of GATE Valve (GA08CA200):**GA80-CL-X200**

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Service: Low Temperature -46 to 343 °C (-50 to 649 °F)
Class: **API 800**
Ends: SW x FNPT
Body: ITCS (ASTM A350-LF2-CL-1)
Bonnet: ITCS (ASTM A350-LF2-CL-1)
Trim: API 602 Trim 12
Wedge Type: Solid
Bonnet Type: Bolted
Bore-Port: **Standard Port (RP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN: API 602
ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
RATING: API 602
TESTING: API 598
DIMENSIONAL: MFG STD

Approved Vendors: DSI (Forged, CL800, RP): 4101-F12
..... Bonney Forge (Forged, CL800, RP): HL-10-L3S-SW/T



Document #: GCTLR00-TSLA-ME-SPC-00006

Valve Specification for CS Gate Valves

Description of GATE Valve (GA08CA300):

GA80-CL-S300

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Service: Low Temperature -46 to 343 °C (-50 to 649 °F)
Class: **API 800**
Ends: SW
Body: ITCS (ASTM A350-LF2-CL-1)
Bonnet: ITCS (ASTM A350-LF2-CL-1)
Trim: API 602 Trim 12
Wedge Type: Solid
Bonnet Type: Bolted
Bore-Port: **Standard Port (RP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD

The standards are:

DESIGN: API 602
ENDS: ASME B16.11
RATING: API 602
TESTING: API 598
DIMENSIONAL: MFG STD

Approved Vendors: DSI (Forged, CL800, RP): 4111-F12
..... Bonney Forge (Forged, CL800, RP): HL-10-L3S-SW

Rev.	RevDate	RevDescription	By:	Check:
0	15 SEPT 2023	Issued for Release	MFS	SBS
1	18 SEPT 2023	Add Misc. Valves	MFS	SBS
2	08 NOV 2023	Add small-bore (RP) forged gate valves	MFS	SBS
3	22 MAY 2024	Add small-bore (FP) forged gate valve	MFS	MFS

End Document #: GCTLR00-TSLA-ME-SPC-00006

Valve Specification for CS Gate Valves

Description of GLOBE Valve (GL01CB500):

GL15-CS-F500

Type:Globe
 Valve Size:.....NPS 2 -to- NPS 16
 Class:ASME 150
 Ends:Flanged RF
 Body:CS (ASTM A216-WCB)
 Bonnet:CS (ASTM A216-WCB)
 Trim:API 623 TRIM 8
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts API 623
Nuts API 623

The standards are:

DESIGN:.....API 623
 ENDS:ASME B16.5
 RATING:ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10

Approved Vendors:.....DSI (Cast, FP): 133-XU (2" to 10"), Handwheel Op.
SCV (Cast, FP): GLB-{Size}-01C4S-H1, Handwheel Op.

Description of GLOBE Valve (GL01CB501):

GL15-CS-F501

Type:Globe
 Valve Size:.....NPS 0.5 -to- NPS 2
 Class:ASME 150
 Ends:Flanged RF
 Body:CS (ASTM A105N)
 Bonnet:CS (ASTM A105N)
 Trim:API 602 TRIM 8
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts API 602
Nuts API 602

The standards are:

DESIGN:.....API 602
 ENDS:ASME B16.11
 RATING:API 602
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10

Approved Vendors:.....DSI (Forged, RP): 1231-A8
Bonney Forge: L1-30-1-RF

Description of GLOBE Valve (GL01CB502):

GL15-CS-F502

Type:Globe
 Valve Size:.....NPS 6 -to- NPS 16
 Class:ASME 150
 Ends:Flanged RF
 Body:CS (ASTM A216-WCB)
 Bonnet:CS (ASTM A216-WCB)
 Trim:API 623 TRIM 8
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Gear
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts API 623
Nuts API 623

The standards are:

DESIGN:.....API 623
 ENDS:ASME B16.5
 RATING:ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10

Approved Vendors:.....DSI (Cast, FP): G-133-XU (6" to 12"), Gear Op.
Crane (Cast, FP): 143-XU (6" to 16"), Gear Op.

Description of GLOBE Valve (GL03CB500):

GL30-CS-F500

Type:Globe
 Valve Size:.....NPS 0.5 -to- NPS 2
 Class:ASME 300
 Ends:Flanged RF
 Body:CS (ASTM A105N)
 Bonnet:CS (ASTM A105N)
 Trim:API 602 TRIM 8
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts API 602
Nuts API 602

The standards are:

DESIGN:.....API 602
 ENDS:ASME B16.11
 RATING:API 602
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10

Approved Vendors:.....DSI (Forged, RP): 3231-A8
Bonney Forge: L3-30-1-RF

Description of GLOBE Valve (GL03CB501):

GL30-CS-F501

Type:Globe
 Valve Size:.....NPS 2 -to- NPS 8
 Class:ASME 300
 Ends:Flanged RF
 Body:CS (ASTM A216-WCB)
 Bonnet:CS (ASTM A216-WCB)
 Trim:API 623 TRIM 8
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts API 623
Nuts API 623

The standards are:

DESIGN:.....API 623
 ENDS:ASME B16.5
 RATING:ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10 (NPS 2 -to- NPS 8)

Approved Vendors:.....DSI (Cast, FP): 141-XU (2" to 8"), Handwheel Op.
Crane (Cast, FP): 151-XU, Handwheel Op.

Description of GLOBE Valve (GL03CB502):

GL30-CS-F502

Type:Globe
 Valve Size:.....NPS 4 -to- NPS 16
 Class:ASME 300
 Ends:Flanged RF
 Body:CS (ASTM A216-WCB)
 Bonnet:CS (ASTM A216-WCB)
 Trim:API 623 TRIM 8
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Gear
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts API 623
Nuts API 623

The standards are:

DESIGN:.....API 623
 ENDS:ASME B16.5
 RATING:ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10 (NPS 2 -to- NPS 12)
MFG STD (NPS 14 -to- NPS 16)

Approved Vendors:.....DSI (Cast, FP): G-141-XU (4" to 12"), Gear Op.
Crane (Cast, FP): 151-XU (4" to 16"), Gear Op.
SCV (Cast): GLB-{Size}-03C4S-G1, Gear Op.

Description of GLOBE Valve (GL06CB500):

GL60-CS-F500

Type:Globe
 Valve Size:.....NPS 2 -to- NPS 8
 Class:ASME 600
 Ends:Flanged RF
 Body:CS (ASTM A216-WCB)
 Bonnet:CS (ASTM A216-WCB)
 Trim:API 623 TRIM 8
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts API 623
Nuts API 623

The standards are:

DESIGN:.....API 623
 ENDS:ASME B16.5
 RATING:ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10 - Long Pattern (NPS 2 -to- NPS 10)
MFG STD (NPS 12 -to- NPS 14)

Approved Vendors:.....DSI (Cast, FP): 161-XU (2" to 8"), Handwheel Op.
Crane (Cast, FP): 171-XU, Handwheel Op.

Description of GLOBE Valve (GL06CB501):

GL60-CS-F501

Type:Globe
 Valve Size:.....NPS 0.5 -to- NPS 2
 Class:ASME 600
 Ends:Flanged RF
 Body:CS (ASTM A105N)
 Bonnet:CS (ASTM A105N)
 Trim:API 602 TRIM 8
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts API 602
Nuts API 602

The standards are:

DESIGN:.....API 602
 ENDS:ASME B16.11
 RATING:API 602
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10

Approved Vendors:.....DSI (Forged, RP): 3231-A8
Bonney Forge (Forged, RP): L6-30-1-RF
 .

Description of GLOBE Valve (GL06CB502):

GL60-CS-F502

Type:Globe
 Valve Size:.....NPS 3 -to- NPS 14
 Class:ASME 600
 Ends:Flanged RF
 Body:CS (ASTM A216-WCB)
 Bonnet:CS (ASTM A216-WCB)
 Trim:API 623 TRIM 8
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Gear
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts API 623
Nuts API 623

The standards are:

DESIGN:.....API 623
 ENDS:ASME B16.5
 RATING:ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10 - Long Pattern (NPS 2 -to- NPS 10)
MFG STD (NPS 12 -to- NPS 14)

Approved Vendors:.....DSI (Cast, FP): G-161-XU (3" to 8"), Gear Op.
Crane (Cast, FP): 171-XU (3" to 14"), Gear Op.
SCV (Cast, FP): GLB-{Size}-06C4S-G1, Gear Op.

Description of GLOBE Valve (GL08CB300):

GL80-CS-S300

Type:Globe
 Valve Size:.....NPS 0.5 -to- NPS 2
 Class:API 800
 Ends:SW
 Body:CS (ASTM A105N)
 Bonnet:CS (ASTM A105N)
 Trim:API 602 TRIM 8
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts API 602
Nuts API 602

The standards are:

DESIGN:.....API 602
 ENDS:ASME B16.11
 RATING:API 602
 TESTING:.....API 598
 DIMENSIONAL:.....MFG STD

Approved Vendors:.....DSI (Forged, RP): 4211-A8
Bonney Forge: HL-30-1-SW

Description of GLOBE Valve (GL08CB301):

GL80-CS-S301

Type:Globe
 Valve Size:.....NPS 0.5 -to- NPS 2
 Class:API 800
 Ends:SW
 Body:CS (ASTM A105N)
 Bonnet:CS (ASTM A105N)
 Trim:API 602 TRIM 5
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts API 602
Nuts API602

The standards are:

DESIGN:.....API 602
 ENDS:ASME B16.11
 RATING:.....API 602
 TESTING:.....API 598
 DIMENSIONAL:.....MFG STD

Approved Vendors:.....DSI (Forged, RP): 4211-A5
Bonney Forge: HL-30-2-SW

Description of GLOBE Valve (GL08CB302):

GL80-CS-S302

Type:Globe
 Valve Size:.....NPS 0.5 -to- NPS 2
 Class:API 800
 Ends:SW
 Body:CS (ASTM A105N)
 Bonnet:CS (ASTM A105N)
 Trim:API 602 TRIM 8
 Body Type:T-body
 Bonnet Type:Welded
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited

The standards are:

DESIGN:.....API 602
 ENDS:ASME B16.11
 RATING:.....API 602
 TESTING:.....API 598
 DIMENSIONAL:.....MFG STD

Approved Vendors:.....DSI (Forged, RP): 4213-A8
Bonney Forge: WL-30-1-SW



The standards are:

Description of GLOBE Valve (GL80-CS-X210):

The standards are:

Valve Specification

Description of GLOBE Valve (GL80-CS-T101):

Type:Globe
 Valve Size:.....NPS 0.5 -to-NPS 2
 Class:API 800
 Ends:FNPT
 Body:CS (ASTM A105N)
 Bonnet:CS (ASTM A105N)
 Port:.....**Standard Port**
 Trim:API 602 TRIM 8
 Body Type:T-body
 Bonnet Type: Bolted
 Disc Type:..... Conical
 Stem Design:.....OS&Y
 Operation:.....Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts API 623
Nuts API 623

The standards are:

DESIGN:.....API 623
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....Mfg.Std.
 VENDORS:.....Bonney Forge: HL-30-1-T
DSI: 4221-A8

Description of GLOBE Valve (GL80-CS-T102):

Type:Globe
 Valve Size:.....NPS 0.5 -to-NPS 2
 Class:API 800
 Ends:FNPT
 Body:CS (ASTM A105N)
 Bonnet:CS (ASTM A105N)
 Port:.....**Full Port**
 Trim:API 602 TRIM 8
 Body Type:T-body
 Bonnet Type: Bolted
 Disc Type:..... Conical
 Stem Design:.....OS&Y
 Operation:.....Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts API 623
Nuts API 623

The standards are:

DESIGN:.....API 623
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....Mfg.Std.
 VENDORS:.....Bonney Forge: H-30-1-T
DSI: 4221-FB-A8

Description of GLOBE Valve (GL09CB500):

GL90-CS-F500

Type:Globe
Valve Size:.....NPS 2 -to- NPS 6
Class:ASME 900
Ends:Flanged RF
Body:CS (ASTM A216-WCB)
Bonnet:CS (ASTM A216-WCB)
Trim:API 623 TRIM 8
Body Type:T-body
Bonnet Type:Bolted
Disc Type:.....Conical
Stem Design:.....OS&Y
Operation:.....Handwheel
Stem-Packing:.....Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:.....SS/Graphite
Body-Bonnet Bolting:.....Bolts API 623
.....Nuts API 623

The standards are:

DESIGN:.....API 623
ENDS:ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598.
DIMENSIONAL:.....ASME B16.10 - Long Pattern

Approved Vendors:.....DSI (Cast, FP): 173-XU (2" to 6"), Handwheel Op.
.....Crane (Cast, FP): 183-XU (2" to 6"), Handwheel Op.
.....SCV (Cast): GLB-{Size}-09C4S-H1, Handwheel Op.

Description of GLOBE Valve (GL09CB501):

GL90-CS-F501

Type:Globe
Valve Size:.....NPS 3 -to- NPS 12
Class:ASME 900
Ends:Flanged RF
Body:CS (ASTM A216-WCB)
Bonnet:CS (ASTM A216-WCB)
Trim:API 623 TRIM 8
Body Type:T-body
Bonnet Type:Bolted
Disc Type:.....Conical
Stem Design:.....OS&Y
Operation:.....Gear
Stem-Packing:.....Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:.....SS/Graphite
Body-Bonnet Bolting:.....Bolts API 623
.....Nuts API 623

The standards are:

DESIGN:.....API 623
ENDS:ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10 - Long Pattern

Approved Vendors:.....DSI (Cast, FP): G-173-XU (3" to 8"), Gear Op.
.....Crane (Cast, FP): 183-XU (3" to 12"), Gear Op.

Description of GLOBE Valve (GL15CB300):

GL1.5K-CS-S300

Type:Globe
 Valve Size:.....NPS 0.5 -to- NPS 2
 Class:ASME 1500
 Ends:SW
 Body:CS (ASTM A105N)
 Bonnet:CS (ASTM A105N)
 Trim:API 602 TRIM 8
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts API 602
Nuts API602
 The standards are:
 DESIGN:.....API 602
 ENDS:ASME B16.11
 RATING:ASME B16.34
 TESTING:API 598
 DIMENSIONAL:.....MFG STD

Approved Vendors:DSI (Forged, RP): 6211-A8
Bonney Forge: 9WL-30-1-SW

Description of GLOBE Valve (GL15CB301):

GL1.5K-CS-S301

Type:Globe
 Valve Size:.....NPS 0.5 -to- NPS 2
 Class:ASME 1500
 Ends:SW
 Body:CS (ASTM A105N)
 Bonnet:CS (ASTM A105N)
 Trim:API 602 TRIM 5
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts API 602
Nuts API602
 The standards are:
 DESIGN:.....API 602
 ENDS:ASME B16.11
 RATING:ASME B16.34
 TESTING:API 598
 DIMENSIONAL:.....MFG STD

Approved Vendors:DSI (Forged, RP): 6211-A5
Bonney Forge: 9HL-30-2-SW

Description of GLOBE Valve (GL15CB500):

GL1.5K-CS-F500

Type:Globe
 Valve Size:.....NPS 2 -to- NPS 6
 Class:ASME 1500
 Ends:Flanged RF
 Body:CS (ASTM A216-WCB)
 Bonnet:CS (ASTM A216-WCB)
 Trim:API 623 TRIM 8
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts API 623
Nuts API 623

The standards are:

DESIGN:.....API 623
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10 - Long Pattern

Approved Vendors:.....DSI (Cast, FP): 179-XU, Handwheel Op.
Crane (Cast, FP): 189-XU, Handwheel Op.

Description of GLOBE Valve (GL15CB501):

GL1.5K-CS-F501

Type:Globe
 Valve Size:.....NPS 3 -to- NPS 12
 Class:ASME 1500
 Ends:Flanged RF
 Body:CS (ASTM A216-WCB)
 Bonnet:CS (ASTM A216-WCB)
 Trim:API 623 TRIM 8
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Gear
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts API 623
Nuts API 623

The standards are:

DESIGN:.....API 623
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10 - Long Pattern

Approved Vendors:.....DSI (Cast, FP): G-179-XU, Gear Op.
Crane (Cast, FP): 189-XU, Gear Op.

***** Low Temperature CS Globe Valves *****

Description of GLOBE Valve (GL01CA500):

GL15-CL-F500

Type:Globe
 Valve Size:.....NPS 2 -to- NPS 16
 Service:Low Temperature -46 to 343 °C (-50 to 649 °F)
 Class:ASME 150
 Ends:Flanged RF
 Body:ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Bonnet:ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Trim:API 623 TRIM 12
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts MFG STD
Nuts MFGSTD

The standards are:

DESIGN:.....API 623
 ENDS:ASME B16.5
 RATING:ASME B16.34
 TESTING:API 598
 DIMENSIONAL:.....ASME B16.10

Approved Vendors:.....DSI (Cast, FP): 133-LU (2" to 12"), Handwheel
SCV (Cast, FP): GLB-{Size}-01L3S-H1, Handwheel

Description of GLOBE Valve (GL01CA501):

GL15-CL-F501

Type:Globe
 Valve Size:.....NPS 6 -to- NPS 16
 Service:Low Temperature -46 to 343 °C (-50 to 649 °F)
 Class:ASME 150
 Ends:Flanged RF
 Body:ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Bonnet:ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Trim:API 623 TRIM 12
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Gear
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts MFG STD
Nuts MFGSTD

The standards are:

DESIGN:.....API 623
 ENDS:ASME B16.5
 RATING:ASME B16.34
 TESTING:API 598
 DIMENSIONAL:.....ASME B16.10

Approved Vendors:.....DSI (Cast, FP): G-133-LU (6" to 12"), Gear Op.
SCV (Cast, FP): GLB-{Size}-01L3S-G1, Gear Op.

Description of GLOBE Valve (GL03CA500):

GL30-CL-F500

Type:Globe
 Valve Size:.....NPS 2 -to- NPS 10
 Service:Low Temperature -46 to 343 °C (-50 to 649 °F)
 Class:ASME 300
 Ends:Flanged RF
 Body:ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Bonnet:ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Trim:API 623 TRIM 12
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts MFG STD
Nuts MFGSTD

The standards are:

DESIGN:.....API 623
 ENDS:ASME B16.5
 RATING:ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10 (NPS 2 -to- NPS 12)
MFG STD (NPS 14 -to- NPS 16)

Approved Vendors:.....DSI (Cast, FP): 141-LU-LCC (2" to 10"), Handwheel Op.
SCV (Cast, FP): GLB-{Size}-03L3S-H1, Handwheel Op.

Description of GLOBE Valve (GL03CA501):

GL30-CL-F501

Type:Globe
 Valve Size:.....NPS 4 -to- NPS 16
 Service:Low Temperature -46 to 343 °C (-50 to 649 °F)
 Class:ASME 300
 Ends:Flanged RF
 Body:ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Bonnet:ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Trim:API 623 TRIM 12
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Gear
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts MFG STD
Nuts MFGSTD

The standards are:

DESIGN:.....API 623
 ENDS:ASME B16.5
 RATING:ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10 (NPS 2 -to- NPS 12)
MFG STD (NPS 14 -to- NPS 16)

Approved Vendors:.....DSI (Cast, FP): G-141-LU-LCC (4" to 12"), Gear Op.
SCV (Cast, FP): GLB-{Size}-03L3S-G1, Gear Op.

Description of GLOBE Valve (GL06CA500):

GL60-CL-F500

Type:Globe
 Valve Size:.....NPS 2 -to- NPS 8
 Service:Low Temperature -46 to 343 °C (-50 to 649 °F)
 Class:ASME 600
 Ends:Flanged RF
 Body:ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Bonnet:ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Trim:API 623 TRIM 12
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts MFG STD
Nuts MFGSTD

The standards are:

DESIGN:.....API 623
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10 - Long Pattern (NPS 2 -to- NPS 10)
MFG STD (NPS 12 -to- NPS 14)

Approved Vendors:.....DSI (Cast, FP): 161-LU-LCC (2" to 8"), Handwheel Op.
SCV (Cast): GLB-{Size}-06L3S-H1, Handwheel Op.

Description of GLOBE Valve (GL06CA501):

GL60-CL-F501

Type:Globe
 Valve Size:.....NPS 3 -to- NPS 14
 Service:Low Temperature -46 to 343 °C (-50 to 649 °F)
 Class:ASME 600
 Ends:Flanged RF
 Body:ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Bonnet:ITCS (ASTM A352-LCB or ASTM A352-LCC)
 Trim:API 623 TRIM 12
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:.....Conical
 Stem Design:.....OS&Y
 Operation:.....Gear
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts MFG STD
Nuts MFGSTD

The standards are:

DESIGN:.....API 623
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10 - Long Pattern (NPS 2 -to- NPS 10)
MFG STD (NPS 12 -to- NPS 14)

Approved Vendors:.....DSI (Cast, FP): G-161-LU-LCC (3" to 8"), Gear Op.
SCV (Cast, FP):GLB-{Size}-06L3S-G1, Gear Op.

Description of GLOBE Valve (GL08CA300):

GL80-CL-S300

Type:Globe
 Valve Size:.....NPS 0.5 -to- NPS 2
 Service:Low Temperature -46 to 343 °C (-50 to 649 °F)
 Class:API 800
 Ends:SW
 Body:ITCS (ASTM A350-LF2-CL-1)
 Bonnet:ITCS (ASTM A350-LF2-CL-1)
 Trim:API 602 TRIM 12
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:.....Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:.....SS/Graphite
 Body-Bonnet Bolting:.....Bolts MFG STD
Nuts MFG STD
 The standards are:
 DESIGN:.....API 602
 ENDS:ASME B16.11
 RATING:API 602
 TESTING:.....API 598
 DIMENSIONAL:.....MFG STD
 Approved Vendors:.....DSI (Forged, RP): 4211-F12
Bonney Forge: HL-30-3-SW



Rev.	RevDate	RevDescription	By:	Check:
0	15 SEPT 2023	Issued for Release	MFS	SBS
1	08 NOV 2023	Add Small-bore Valve Tags	MFS	SBS

Description of KNIFE Valve (KF15-CS-W700)

Type: Knife Gate, Two-Piece, Bolted
 Valve size: NPS 2 -to- NPS 16
 Service: Slurry
 Class: ASME 150
 Range: -20 °F to +248 °F (-29 °C to +120 °C)
 Rating: 150 psig (10 bar)
 Ends: Wafer
 Body: CS (ASTM A216-WCB)
 Stem: Stainless steel
 Gate: 316 SS
 Perimeter Seat: EPDM, Steel Reinforced
 Packing Seal: PTFE impregnated fiber
 Flow: Bi-directional
 Design (Other): **Closed Body**, zero leakage
 Operation: 2" NPS to 8" NPS = Handwheel, Direct Mounted
 > 8" NPS = Bevel Gear
 Body Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN MSS SP-81
 Face-to-Face MSS SP-81
 ENDS ASME B16.5, CL 150
 RATING ASME B16.34
 TESTING: MSS SP-81

Approved Vendor: Bray, Knife Gate, **Model 752**, 10 bar, CL150, 2-Piece CS Body,
 316 SS Gate, EPDM Seat, 17-4 SS Stem, MSS SP-81

Description of KNIFE Valve (KF15-CS-W702)

Type: Knife Gate, Two-Piece, Bolted
 Valve size: NPS 2 -to- NPS 16
 Service: Slurry
 Class: ASME 150
 Range: -20 °F to +248 °F (-29 °C to +120 °C)
 Rating: **240 psig** (16 bar)
 Ends: Wafer
 Body: CS (ASTM A216-WCB)
 Stem: Stainless steel
 Gate: 316 SS
 Perimeter Seat: EPDM, Steel Reinforced
 Packing Seal: PTFE impregnated fiber
 Flow: Bi-directional
 Design (Other): **Closed Body**, zero leakage
 Operation: 2" NPS to 8" NPS = Handwheel, Direct Mounted
 > 8" NPS = Bevel Gear
 Body Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN MSS SP-81
 Face-to-Face MSS SP-81
 ENDS ASME B16.5, CL 150
 RATING ASME B16.34
 TESTING: MSS SP-81

Approved Vendor: Bray, Knife Gate, **Model 752**, 16 bar, CL150, 2-Piece CS Body,
 316 SS Gate, EPDM Seat, 17-4 SS Stem, SP-81

Description of KNIFE Valve (KF15-CS-F510)

Type:	Knife Gate, Two-Piece, Bolted
Valve size:	NPS 3 -to- NPS 24
Service:	Slurry, Heavy Duty
Class:	ASME 150
Range:	-20 °F to +302 °F (-29 °C to +150 °C)
Rating:	100 psig (10 bar)
Ends:	Flanged w/Elastomer (uses Cap Screws not Stud Bolts)
Body:	CS (ASTM A216-WCB)
Stem:	Stainless steel
Gate:	316 Stainless Steel
Sleeve:	EPDM (Elastomer)
Flow:	Bi-directional
Design (Other):	Push-through (with bottom flush plate & tapped ports)
Operation:	2" NPS to 8" NPS = Handwheel, Direct Mounted
	> 8" NPS = Bevel Gear
Body Bolting:	Bolts: MFG STD / Nuts: MFG STD
Notes:	Install valve between either RF or FF companion flanges.

The standards are:

DESIGN.....	MFG STD
Face-to-Face.....	MFG STD
ENDS	ASME B16.5, CL 150
RATING	ASME B16.34
TESTING.....	MFG STD

Approved Vendor:.....Bray, Heavy Duty Knife Gate, Model 762, 10 bar, CL150,
.....2-Piece bolted CS Body, 316 SS Gate, EPDM Seat, 17-4 SS Stem

Description of KNIFE Valve (KF15-CS-L721)

Type:	Knife Gate, Bolted
Valve size:	NPS 2 -to- NPS 24
Service:	Slurry
Class:	ASME 150
Range:	-20 °F to +250 °F (-29 °C to +121 °C)
Ends:	Wafer
Body:	CS (ASTM A216-WCB)
Gate:	316 Stainless Steel
Seals:	EPDM
Scrapers:	Phenolic
Flow:	Bi-directional
Design (Other):	Closed Body, Guided Shear Gate
Operation:	2" NPS to 8" NPS = Handwheel
	> 8" NPS = Bevel Gear
Body Bolting:	Bolts: MFG STD
	Nuts: MFG STD

The standards are:

DESIGN.....MFG STD
Face-to-FaceMSS SP-135, Long Pattern
ENDSASME B16.5
RATINGASME B16.34

Approved Vendor:.....DSS Severe Service Knife Gate, Model SSKGV
SV1L-{Size}-CS-SS-EP-1-{HW or BG}

**Description of KNIFE Valve (KF15-DI-L708)**

Type: Knife Gate
Valve size: NPS 2 -to-NPS 24
Service: Slurry
Class: ASME 150
Ratings: 100 psig CWP
Ends: Lugged
Body: Ductile iron A536, Epoxy coated, EPDM rubber sleeve
Flow: Bi-directional
Trim: Stem stainless steel
Disc type: Gate, material Super Duplex S2507
Design (Other): Open bottom end
Operation: Handwheel, DN200 and bigger with gear
Body-Bolting: Bolts: MFG STD
..... Nuts: MFG STD
Notes: Housing is drilled & tapped to match ASME B16.5 Flanges

The standards are:

DESIGN MFG STD
ENDS ASME B16.5
RATING ASME B16.34
DIMENSIONAL MFG STD

Approved Vendor: Clarkson, Model KGD

Rev.	Rev. Date	Rev. Description	By	Checked
0	18 SEPT 2023	Issued for release	MFS	SBS

Description of NEEDLE Valve (NE6K-CS-T100)

Type: Needle, General Service, Soft Seat, H1 Style
 Valve size: [NPS 0.5 -to- NPS 1.0](#)
 Service: Nitrogen, water, etc.
 Class: N/A (6000 psig @ 200 F)
 Ends: MNPT x FNPT (Male x Female)
 Body: Carbon Steel (ASTM A108)
 Trim: 303 SS (ASTM A581-303)
 Body type:..... T-body
 Disc type:..... Needle
 Seat: Delrin (standard)
 Flow Direction:..... Bi-directional
 Orifice Size: 3/8" Dia. Orifice (see P/N's below)
 Operation:..... T-Handle
 Stem-Packing: Teflon Packing

The standards are:

DESIGN..... ASME B16.34
 ENDS MFG STD
 RATING ASME B16.34
 DIMENSIONAL..... MFG STD

Vendor: Anderson Greenwood, H1 Style, MNPT x FNPT
 1/2" Size (3/8" Port) = [H1-VDC-44](#) (Delrin Seat)
 3/4" Size (3/8" Port) = H1-VDC-66Q (Delrin Seat)
 1" Size (3/8" Port) = H1-VDC-88Q (Delrin Seat)

Description of NEEDLE Valve (NE6K-CS-T101)

Type: Needle, General Service, Soft Seat, H1 Style
 Valve size: [NPS 0.5 -to- NPS 1.0](#)
 Service: Nitrogen, water, etc.
 Class: N/A (6000 psig @ 200 F)
 Ends: FNPT x FNPT (Female x Female)
 Body: Carbon Steel (ASTM A108)
 Trim: 303 SS (ASTM A581-303)
 Body type:..... T-body
 Disc type:..... Needle
 Seat: Delrin (standard)
 Flow Direction:..... Bi-directional
 Orifice Size: 3/8" Dia. Orifice (see P/N's below)
 Operation:..... T-Handle
 Stem-Packing: Teflon Packing

The standards are:

DESIGN..... ASME B16.34
 ENDS MFG STD
 RATING ASME B16.34
 DIMENSIONAL..... MFG STD

Vendor: Anderson Greenwood, H1 Style, FNPT x FNPT
 1/2" Size (3/8" Port) = H1-VDC-4 (Delrin Seat)
 3/4" Size (3/8" Port) = H1-VDC-6Q (Delrin Seat)
 1" Size (3/8" Port) = H1-VDC-8Q (Delrin Seat)

Description of NEEDLE Valve (NE6K-CS-T102)

Type: Needle, Soft Seat, H1 Style
 Valve size: NPS 0.25 -to- NPS 0.5
 Service: Nitrogen, water, etc.
 Class: N/A (6000 psig @ 200 F)
 Ends: MNPT x FNPT
 Body: Carbon Steel (ASTM A108)
 Trim: 303 SS (ASTM A581-303)
 Body type:..... T-body
 Disc type:..... Needle
 Seat: Delrin (standard)
 Flow Direction:..... Bi-directional
 Orifice Size: 3/16 or 1/4" Dia. Orifice (see P/N's below)
 Operation:..... T-Handle
 Stem-Packing: Teflon Packing

The standards are:

DESIGN ASME B16.34
 ENDS MFG STD
 RATING ASME B16.34
 DIMENSIONAL MFG STD

Vendor: Anderson Greenwood, H1 Style, MNPT x FNPT
 1/4" Size (3/16" Port) = H1-VDC-22 (Delrin Seat)
 1/2" Size (3/16" Port) = H1-VDC-44Q (Delrin Seat)
 *****
 1/2" Size (1/4" Port) = H1-VDC-44QR (Delrin Seat)

Description of NEEDLE Valve (NE6K-CS-T103)

Type: Needle, Soft Seat, H1 Style
 Valve size: NPS 0.25 -to- NPS 0.5
 Service: Nitrogen, water, etc.
 Class: N/A (6000 psig @ 200 F)
 Ends: FNPT x FNPT
 Body: Carbon Steel (ASTM A108)
 Trim: 303 SS (ASTM A581-303)
 Body type:..... T-body
 Disc type:..... Needle
 Seat: Delrin (standard)
 Flow Direction:..... Bi-Directional
 Orifice Size: 3/16" (std.) or 1/4" Dia. (opt.) Orifice
 Operation:..... T-Handle
 Stem-Packing: Teflon Packing

The standards are:

DESIGN ASME B16.34
 ENDS MFG STD
 RATING ASME B16.34
 DIMENSIONAL MFG STD

Vendor: Anderson Greenwood, H1 Style, FNPT x FNPT
 1/4" Size (3/16" Port) = H1-VDC-2 (Delrin Seat)
 1/2" Size (3/16" Port) = H1-VDC-4Q (Delrin Seat)
 *****
 1/2" Size (1/4" Port) = H1-VDC-4QR (Delrin Seat)

Description of NEEDLE Valve (NE6K-CS-T104)

Type: Needle, Hard Seated, H7 Style (i.e., Metal Seat)
 Valve size: NPS 0.5 -to- NPS 0.75
 Class: N/A (6000 psig)
 Ends: MNPT x FNPT (Male x Female)
 Body: Carbon Steel (ASTM A108)
 Stem: 303 SS (ASTM A581-303)
 Body type: T-body
 Flow Direction: One-Directional
 Orifice Size: 3/16" Dia. Orifice
 Seat: Integral (Metal)
 Disc type: Needle
 Operation: T-Handle
 Stem-Packing: Teflon (O-ring)

The standards are:

DESIGN ASME B16.34
 ENDS MFG STD
 RATING ASME B16.34
 DIMENSIONAL MFG STD

Vendor: Anderson Greenwood
 H7-VIS-44Q (1/2" MNPT x 1/2" FNPT)
 H7-VIS-66Q (3/4" MNPT x 3/4" FNPT)

Description of NEEDLE Valve (NE6K-CS-T105)

Type: Needle (H7), Hard Seated (i.e., Metal Seat)
 Valve size: NPS 0.25 -to- NPS 0.5
 Class: N/A (6000 psig @ 200 F)
 Ends: FNPT x FNPT (Female x Female)
 Body: Carbon Steel (ASTM A108)
 Stem: 303 SS (ASTM A581-303)
 Body type: T-body
 Flow Direction: One-Directional
 Orifice Size: 3/16" Dia. Orifice
 Seat: Integral (Metal)
 Disc type: Needle
 Operation: T-Handle
 Stem-Packing: Teflon (O-ring)

The standards are:

DESIGN ASME B16.34
 ENDS MFG STD
 RATING ASME B16.34
 DIMENSIONAL MFG STD

Vendor: Anderson Greenwood
 H7-VIS-2 (1/4" FNPT x 1/4" FNPT)
 H7-VIS-4Q (1/2" FNPT x 1/2" FNPT)

Description of NEEDLE Valve (NE6K-CS-C801)

Type: Needle, Hard-Seated (i.e., Metal)
 Design Type: Ball Tip Stem
 Valve size: NPS 0.5 -to-NPS 0.5
 Class: N/A (6000 psig @ 200 F)
 Ends: Compression Ferrule Tube, Double Ferrule
 Body: Carbon Steel
 Trim: 303 SS
 Body type:..... T-body
 Disc type:..... Needle
 Flow Direction:..... One-Directional
 Orifice Size: 3/16" Dia. Orifice
 Seat: Integral
 Operation:..... T-Handle
 Stem-Packing: PTFE

The standards are:

DESIGN..... MSS SP-105
 ENDS MFG STD
 RATING MFG STD
 TESTING..... MSS SP-105
 DIMENSIONAL..... MFG STD

Vendor: Anderson Greenwood
 H7-VIS-4ATD (1/2" Tube x 1/2" Tube)

Description of NEEDLE Valve (NE6K-CS-T112)

Type: Needle, Hard Seat (Swagelok, GU Series)
 Valve size: NPS 0.25 -to- NPS 1.0
 Temperature:..... 0 F to +450 F (-17 C to 232 C)
 Class: N/A (6000 psig @ 100 F)
 Ends: MNPT x FNPT (Male x Female)
 Body: Carbon Steel (ASTM A108)
 Trim: 17-4 Stainless Steel
 Body type:..... T-body, Straight Pattern
 Disc type:..... Needle
 Seat: Integral (metal)
 Flow Direction:..... Bi-directional
 Orifice Size: See Below
 Operation:..... T-Handle
 Stem-Packing: PTFE Packing

The standards are:

DESIGN..... ASME B16.34
 ENDS MFG STD
 RATING ASME B16.34
 DIMENSIONAL..... MFG STD

Vendor: Swagelok, GU Series, Straight Pattern, MNPT x FNPT
 1/4" Size (0.20" Orifice) = S-4GUM4-F4
 3/8" Size (0.20" Orifice) = S-4GUM6-F6
 1/2" Size (0.20" Orifice) = S-4GUM8-F8
 *****
 1/2" Size (5/16" Orifice) = S-8GUM8-F8
 3/4" Size (5/16" Orifice) = S-8GUM12-F12
 1" Size (5/16" Orifice) = S-8GUM16-F16
 *****
 3/4" Size (7/16" Orifice) = S-16GUM12-F12
 1" Size (7/16" Orifice) = S-16GUM16-F16



Description of NEEDLE Valve (NE6K-CS-T113)

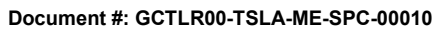
Type: Needle, Hard Seat (Swagelok, GU Series)
Valve size: NPS 0.25 -to- NPS 1.0
Temperature: 0 F to +450 F (-17 C to 232 C)
Class: N/A (6000 psig @ 100 F)
Ends: FNPT x FNPT
Body: Carbon Steel (ASTM A108), Zinc-plated
Trim: 17-4 Stainless Steel
Body type: T-body, Straight Pattern
Disc type: Needle
Seat: Integral (metal)
Orifice Size: See Below
Operation: T-Handle
Stem-Packing: PTFE Packing

The standards are:

DESIGN ASME B16.34
ENDS MFG STD
RATING ASME B16.34
DIMENSIONAL MFG STD

Vendor: Swagelok, GU Series, Straight Pattern, FNPT x FNPT
..... 1/4" Size (0.20" Orifice) = S-4GUF4
..... 3/8" Size (0.20" Orifice) = S-4GUF6
..... 1/2" Size (0.20" Orifice) = S-4GUF8
..... *****
..... 1/2" Size (5/16" Orifice) = S-8GUF8
..... 3/4" Size (5/16" Orifice) = S-8GUF12
..... 1" Size (5/16" Orifice) = S-8GUF16
..... *****
..... 3/4" Size (7/16" Orifice) = S-16GUF12
..... 1" Size (7/16" Orifice) = S-16GUF16

Rev.	Rev. Date	Rev. Description	By:	Check:
0	08 SEPT 2023	Issued for Release	MFS	SBS
1	11 SEPT 2023	Revise Valve Sizes	MFS	SBS



The standards are:

Approved Vendor:.....Red Valve, Series 75, Cast Iron Body, EPDM Sleeve, PI, SS Stem

The standards are:

Approved Vendor:.....Red Valve, **Series 70**, Carbon Steel Body, EPDM Sleeve, PI, SS Stem



Rev.	Rev. Date	Rev. Description	By	Checked
0	18 SEPT 2023	Issued for release	MFS	SBS

End Document #: GCTLR00-TSLA-ME-SPC-00010

Valve Specification for CS Pinch Valves

Description of PLUG Valve (PL01CB500)

PL15-CS-F500

Type: Plug, 2-Port, Sleeved (Non-Lubricated)
 Design Type: Rectangular Port, Top Entry
 Valve Size: NPS 0.5 -to- NPS 3
 Class: ASME 150
 Ends: Flanged RF
 Body: CS (ASTM A105 or ASTM A216-WCB)
 Plug: 316 SS
 Stem: 316 SS
 Sleeve: PTFE
 Design (Other): Fire Safe to API 607 (External Leakage Only)
 Bore-Port: Regular - API 599
 Operation: Lever - High-Hub with Lockout
 Stem Seals: PTFE and 316 SS Diaphragms
 Stem-Packing: PTFE
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: API 599
 ENDS: ASME B16.5
 RATING: MFG STD
 TESTING: API 598
 DIMENSIONAL: API 599

Approved Vendors: Xomox Tuflin, CL150, RF, 2-Port, CS Body, PTFE Sleeved Plug, Lever Op.
 {Size}-067-FT-2-6-P1-W

Description of PLUG Valve (PL01CB501)

PL15-CS-F501

Type: Plug, 2-Port, Sleeved (Non-Lubricated)
 Design Type: Rectangular Port, Top Entry
 Valve Size: NPS 4 -to- NPS 18
 Class: ASME 150
 Ends: Flanged RF
 Body: CS (ASTM A105 or ASTM A216-WCB)
 Plug: 316 SS
 Stem: 316 SS
 Sleeve: PTFE
 Design (Other): Fire Safe to API 607 (External Leakage Only)
 Bore-Port: Regular - API 599
 Operation: Gear
 Stem Seals: PTFE and 316 SS Diaphragms
 Stem-Packing: PTFE
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: API 599
 ENDS: ASME B16.5
 RATING: MFG STD
 TESTING: API 598
 DIMENSIONAL: API 599

Approved Vendors: Xomox Tuflin, CL150, RF, 2-Port, CS Body, PTFE Sleeved Plug, Gear Op.
 {Size}-067EG-FT-2-6-P1-G

Description of PLUG Valve (PL03CB500)

PL30-CS-F500

Type: Plug, 2-Port, Sleeved (Non-lubricated)
 Design Type: Rectangular Port, Top Entry
 Valve Size: NPS 0.5 -to- NPS 3
 Class: ASME 300
 Ends: Flanged RF
 Body: CS (ASTM A105 or ASTM A216-WCB)
 Plug: 316 SS
 Stem: 316 SS
 Sleeve: PTFE
 Design (Other): Fire Safe to API 607 (External Leakage Only)
 Bore-Port: Regular - API 599
 Operation: **Lever** - High-Hub with Lockout
 Stem Seals: PTFE and 316 SS Diaphragms
 Stem-Packing: PTFE
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD

The standards are:

DESIGN: API 599
 ENDS: ASME B16.5
 RATING: MFG STD
 TESTING: API 598
 DIMENSIONAL: API 599

Approved Vendors: Xomox Tufline, CL300, RF, 2-Port, CS Body, PTFE Sleeved Plug, Lever Op.
 {Size}-0367-FT-2-6-P1-W

Description of PLUG Valve (PL03CB501)

PL30-CS-F501

Type: Plug, 2-Port, Sleeved (Non-lubricated)
 Design Type: Rectangular Port, Top Entry
 Valve Size: NPS 4 -to- NPS 18
 Class: ASME 300
 Ends: Flanged RF
 Body: CS (ASTM A105 or ASTM A216-WCB)
 Plug: 316 SS
 Stem: 316 SS
 Sleeve: **PTFE**
 Design (Other): Fire Safe to API 607 (External Leakage Only)
 Bore-Port: Regular - API 599
 Operation: **Gear**
 Stem Seals: PTFE and 316 SS Diaphragms
 Stem-Packing: PTFE
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD

The standards are:

DESIGN: API 599
 ENDS: ASME B16.5
 RATING: MFG STD
 TESTING: API 598
 DIMENSIONAL: API 599

Approved Vendors: Xomox Tufline, CL300, RF, 2-Port, CS Body, PTFE Sleeved Plug, Gear Op.
 {Size}-0367EG-FT-2-6-P1-G

Description of PLUG Valve (PL15-CS-T220)

Type: Plug, 2-Port, Sleeved Plug (Non-Lubricated)
 Sub Type: Rectangular Port, Top Entry
 Ratings: 280 psig @ -20 °F to +100 °F (-29 °C to +37 °C)
 190 psig @ +400 °F (+204 °C)
 Valve Size: NPS 0.5 -to- NPS 2"
 Class: ASME 150
 Ends: **FNPT** (Threaded)
 Body: Carbon Steel (ASTM A216 WCB)
 Plug: 316 SS
 Stem: 316 SS
 Sleeve: **PTFE**
 Design (Other): Fire Safe to API 607 (External Leakage Only)
 Bore-Port: Regular - API 599
 Operation: Lever - High-Hub with Lockout
 Stem Seals: PTFE and 316 SS Diaphragms
 Stem-Packing: PTFE and Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD

The standards are:

DESIGN: API 599
 ENDS: ASME B16.11
 RATING: MFG STD
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: Xomox Tuflin, CL150, 2-Port, CS Body, PTFE Sleeved Plug, FNPT, Lever
 {Size}-066-FT-2-6-P1-W

Description of PLUG Valve (PL30-CS-T230)

Type: Plug, 2-Port, Sleeved Plug (Non-Lubricated)
 Sub Type: Rectangular Port, Top Entry
 Ratings: 720 psig @ -20 °F to +100 °F (-29 °C to +37 °C)
 440 psig @ +400 °F (+204 °C)
 Valve Size: NPS 0.5 -to- NPS 2"
 Class: ASME 300
 Ends: **FNPT** (Threaded)
 Body: Carbon Steel (ASTM A216 WCB)
 Plug: 316 SS
 Stem: 316 SS
 Sleeve: **PTFE**
 Design (Other): Fire Safe to API 607 (External Leakage Only)
 Bore-Port: Regular - API 599
 Operation: Lever - High-Hub with Lockout
 Stem Seals: PTFE and 316 SS Diaphragms
 Stem-Packing: PTFE and Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD

Approved Vendors: Xomox Tuflin, CL300, 2-Port, CS Body, PTFE Sleeved Plug, FNPT, Lever
 {Size}-0366-FT-2-6-P1-W

Rev.	Rev. Date	Rev. Description	By	Checked
0	18 SEPT 2023	Issued for release	MFS	SBS

***** 316 Stainless Steel Valves (316 SS) *****

Description of BALL Valve (BA01ST500): BA15-S16-F500

Type: Ball, Floating, RP
Valve Size: NPS 1" -to- NPS 6"
Service: -29 to 205 °C (-20 to 400 °F)
Class: **ASME 150**
Ends: Flanged RF
Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
Trim: 316 SS
Seat: RPTFE or TFM 1600
Stem 316 SS
Body Type: 2-Piece
Design: Floating Ball, Fire Safe API 607
Bore-Port: **Single Reduced (RP)**
Operation: Lever to 6" NPS
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 – Long Pattern (1" to 4" NPS)
..... ASME B16.10 – Short Pattern (6" only) *** MUST be Short Pattern ***

Approved Vendor:

(1" to 1-1/2" NPS) Ladish (2-Piece, Bar-Stock, RP): P115-L050-GRnn-A-{Size}
(2" to 4" NPS Only) Ladish (2-Piece, Cast, CL150, RP): P715-L050-GRnn-A-{Size} (No 6" NPS Valves allowed)
(1-1/2" to 6" NPS) Apollo (Cast, CL150, RP): Series 87A-100-24, Lever Op.
(1" to 6" NPS) Bray, Flow-Tek, Series RF15: (Cast, CL150, RP): 15-{Size}-11001-SSJG

Description of BALL Valve (BA01ST501): BA15-S16-F501

Type: Ball, Floating, FP
Valve Size: NPS 0.5 -to- NPS 6
Service: -29 to 205 °C (-20 to 400 °F)
Class: **ASME 150**
Ends: Flanged RF
Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
Trim: 316 SS
Seat: RPTFE or TFM 1600
Stem 316 SS
Body Type: 2-Piece
Design: Floating Ball, Fire Safe API 607
Bore-Port: **Full Port (FP)**
Operation: Lever to 6" NPS
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 - Long Pattern (All Sizes)

Approved Vendor:

(1/2" to 1-1/2" NPS) Ladish (2-Piece, CL150, Bar-Stock, FP): P215-L050-GRnn-A-{Size}
(2" to 6" NPS) Ladish (2-Piece, CL150, Cast, FP): P815-L050-GRnn-A-{Size}
(1/2" to 6" NPS) Apollo (2-Piece, CL150, Cast FP): {Size}-87A-200-65
(1/2" to 6" NPS) Bray Flow-Tek, Series F15 (CL150, Cast, FP): 15-{Size}-11000-SSJG

Description of BALL Valve (BA01ST502):..

BAT15-S16-F502

Type: Ball, Trunnion, RP
 Valve Size: NPS 6 -to- NPS 12
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim: 316 SS
 Seat: RPTFE
 Stem 316 SS
 Springs Inconel X-750
 Body Type: 2-Piece
 Design: **Trunnion**, Fire Safe API 607
 Bore-Port: **Single Reduced (RP)**
 Operation: **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - **Long Pattern** (NPS 6 -to- NPS 12)
 DESIGN: *** NO Short Pattern Valves allowed ***

Approved Vendor: WKM (2-Piece, RP): {Size x Size}-370D6-150-RFxRF-23-TGF-23-WG

Description of BALL Valve (BA01ST503):..

BAT15-S16-F503

Type: Ball, Trunnion, FP
 Valve Size: NPS 8 -to- NPS 16
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim: 316 SS
 Seat: RPTFE
 Stem 316 SS
 Body Type: 2-Piece
 Design: **Trunnion**, Fire Safe API 607
 Bore-Port: **Full Port (FP)**
 Operation: **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor: WKM (2-Piece, Cast, CL150, FP): {Size}-370D6-150-RFxRF-23-TGF-23-WG (8" to 16" NPS)

Description of BALL Valve (BA01ST504):..

BA15-S16-F504

Type: Ball, Floating, RP
 Valve Size: NPS 4 -to- NPS 8
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim: 316 SS
 Seat: RPTFE or TFM 1600
 Stem 316 SS
 Body Type: 2-Piece
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Single Reduced (RP)**
 Operation: **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: **ASME B16.10 – Short Pattern (Up to 8" NPS)**

Approved Vendor:

(4" Size only) Ladish (Cast, CL150, RP): P715-G050-GRnn-N-{Size}, Gear Op.
 (4" to 6" NPS sizes only) Apollo (Cast, CL150, RP): Series 87A-100-24, Gear Op. (Valve size limited to 6" max.)
 (4" to 8" NPS Sizes) Bray, Flow-Tek, Series RF15 (Cast, CL150, RP): 15-{Size}-11001-SSJG, Gear Op.

Description of BALL Valve (BA01ST505):..

BA15-S16-F505

Type: Ball, Floating, FP
 Valve Size: NPS 4 -to- NPS 10
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim: 316 SS
 Seat: RPTFE or TFM 1600
 Stem 316 SS
 Body Type: 2-Piece
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Full Port (FP)**
 Operation: **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: **ASME B16.10 - Long Pattern (All Sizes)**

Approved Vendor: Ladish (2-Piece, Cast, CL150, FP): P815-G050-GRnn-N-{Size}, Gear Op.
 Bray, Flow-Tek, F15: (2-Piece, Cast, CL150, FP): 15-{Size}-11000-SSJG, Gear Op.
 WKM (2-Piece, Cast, CL150, FP): {Size}-320F-150-23-TGF-23-WG, Gear Op.



Description of BALL Valve (BA01ST508): BA15-S16-F508
Type: Ball, Floating, RP
Valve Size: NPS 1" -to- NPS 8"
Service: -29 to +205 °C (-20 to +400 °F)
Class: **ASME 150**
Ends: Flanged RF
Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
Trim: 316 SS
Seat: RPTFE or TFM 1600
Stem 316 SS
Body Type: 2-Piece
Design: Floating Ball, Fire Safe API 607
Bore-Port: **Single Reduced (RP)**
Operation: Lever to 4" NPS; Gear Op. for 6" or 8" NPS
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 – Long Pattern (1" to 4")
..... ASME B16.10 – Long Pattern (6" and 8") *** NO Short Pattern Allowed ***

Approved Vendor:

(1" to 1-1/2" NPS).....Ladish (2-Piece, Bar-Stock, RP): P115-L050-GRnn-A-{Size}
(2" to 6" NPS Only)Ladish (2-Piece, Cast, CL150, RP): P715-L050-GRnn-A-{Size} (Gear Op. for 6" NPS)
(1-1/2" to 6" NPS Only)Apollo (2-Piece, Cast, CL150, RP): Series 87A-100-24 (Gear Op. for 6" NPS)
(1" to 6" NPS Only)Bray, Flow-Tek, RF15 (Cast, CL150, RP): 15-{Size}-11001-SSJG (Gear Op. for 6" NPS)
(1" to 8" NPS).....WKM (2-Piece, RP): {Size x Size}-320F-150-23x23-TGF-23 (Gear Op. for 6" or 8" NPS)

Description of BALL Valve (BA15-S16-F511):

Type:.....Ball, Cavity-Filled, Compact, FP
 Valve Size:.....NPS 0.5 -to-NPS 6"
 Service:.....-29 to 205 °C (-20 to 400 °F)
 Class:.....**ASME 150**
 Ends:Wafer Type (for RF flanges)
 Body:316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim:316 SS
 Seat:RPTFE
 Stem:316 SS
 Body Type:.....Split
 Design:.....Floating Ball, Fire Tested API 607
 Cavity Filling:Floating
 Bore-Port:**Full Port (FP)**
 Operation:Lever
 Stem-Packing/Seals:.....Flexible Graphite
 Body Gaskets:Flexible Graphite
 Notes:

- 1) Ball valve is like a double-flanged butterfly, except flange holes are tapped (Not through)
 Use Cap screws rather than stud bolts to mount the valve to the mating RF Flange.
- 2) No substitutions (Sole Source)

The standards are:

DESIGN:.....API 608
 ENDS:.....CL 150,ASME B16.5 (except Tapped Holes)
 RATING:ASME B16.34
 TESTING:API 598
 DIMENSIONAL:Per Manufacturer (Compact Length)

VENDORS: **Flo-Tite Valves: W150-SS-CFF-L** (SS with Cavity Filler)

Description of BALL Valve (BA15-S16-F512):

Type:.....Ball, **Floating, Cavity-Filled, FP**
 Valve Size:.....NPS 0.5 -to-NPS 10
 Temperature:-20 to +450 °F (-29 to 205 °C)
 Rating:275 psig at 100 °F (19.0 bar @ 37.8 °C)
 Class:.....**ASME 150**
 Ends:Flange, RF
 Body:316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim:316 SS
 Seat:RPTFE
 Stem:316 SS, Blow-out Proof Stem
 Body Type:.....2-Piece
 Ball Type:.....Floating Ball
 Bore-Port:**Full Port**
 Other:.....**Cavity Filling**, totally encapsulated ball, PTFE
 Operation:Lever
 Stem-Packing:PTFE
 Body Gaskets:Flexible Graphite
 Notes:This valve is NOT a compact valve

The standards are:

DESIGN:.....API 608
 ENDS:.....CL 150, ASME B16.5
 RATING:ASME B16.34
 TESTING:API 598
 DIMENSIONAL:ASME B16.10 – **Long Pattern (All sizes)**

VENDORS: Morris Valves (CL150, Full Port, Cavity-Filled): CF-FBV-316 SS-{Size}
 **Bray, Flow-Tek, F15 (CL150, Full Port, Cavity-filled): 15-{Size}-11000-SS-JG-NN-0001**

**Description of BALL Valve (BA01ST518):..**

BA15-S16-F518

Type:Ball, Floating, RP
Valve Size:NPS 0.5 -to- NPS 6
Service:-29 to 205 °C (-20 to 400 °F)
Class:**ASME 150**
Ends:Flanged RF
Body:316 SS (ASTM A182-F316 or ASTM A351-CF8M)
Trim:316 SS
Seat:RPTFE
Stem316 SS
Body Type:Split
Design:Floating Ball
Bore-Port:**Single Reduced (RP)**
Operation:Lever
Stem-Packing/Seals:PTFE
Body Gaskets:PTFE

The standards are:

DESIGN:ASME B16.34 or API 608
ENDS:ASME B16.5
RATING:ASME B16.34
TESTING:API 598
DIMENSIONAL:ASME B16.10 - Long Pattern to 6" NPS

Approved Vendor:Ladish (2-Piece, RP): P715-L050-TRnn-A-{Size}

Description of BALL Valve (BA01ST519):..

BA15-S16-F519

Type:Ball, Floating, FP
Valve Size:NPS 0.5 -to- NPS 6
Service:-29 to 205 °C (-20 to 400 °F)
Class:**ASME 150**
Ends:Flanged RF
Body:316 SS (ASTM A182-F316 or ASTM A351-CF8M)
Trim:316 SS
Seat:RPTFE
Stem316 SS
Body Type:Split
Design:Floating Ball
Bore-Port:**Full Port (FP)**
Operation:Lever
Stem-Packing/Seals:PTFE
Body Gaskets:PTFE

The standards are:

DESIGN:ASME B16.34 or API 608
ENDS:ASME B16.5
RATING:ASME B16.34
TESTING:API 598
DIMENSIONAL:ASME B16.10 - Long Pattern

Approved Vendor:Ladish (2-Piece, FP): P815-L050-TRnn-A-{Size}
.....WKM (Style 320F): {Size}-320F-150-23x23-TTF23

Description of BALL Valve (BA01ST522):..

BA15-S16-F522

Type: Ball, Floating, RP
 Valve Size: NPS 4 -to- NPS 10
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim: 316 SS
 Seat: RPTFE
 Stem : 316 SS
 Body Type: 2-Piece
 Design: Floating Ball
 Bore-Port: **Single Reduced (RP)**
 Operation: **Gear**
 Stem-Packing/Seals: PTFE
 Body Gaskets: PTFE

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - **Long Pattern**

Approved Vendor: WKM (2-Piece, RP): {Size x Size}-320F-150-23x23-TTF23

Description of BALL Valve (BA01ST523):..

BA15-S16-F523

Type: Ball, Floating, FP
 Valve Size: NPS 4 -to- NPS 10
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim: 316 SS
 Seat: RPTFE
 Stem : 316 SS
 Body Type: 2-Piece
 Design: Floating Ball
 Bore-Port: **Full Port (FP)**
 Operation: **Gear**
 Stem-Packing/Seals: PTFE
 Body Gaskets: PTFE

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor: Ladish (2-Piece, FP): P815-G050-TRnn-A-{Size}
 WKM (Style 320F): {Size}-320F-150-23x23-TTF23

Description of BALL Valve (BA03ST500): BA30-S16-F500

Type: Ball, Floating, RP

Valve Size: NPS 1" -to- NPS 8"

Service: -29 to 205 °C (-20 to 400 °F)

Class: **ASME 300**

Ends: Flanged RF

Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)

Trim: 316 SS

Seat: RPTFE or TFM 1600

Stem 316 SS

Body Type: 2-Piece

Design: Floating Ball, Fire Safe API 607

Bore-Port: **Single Reduced (RP)**

Operation: Lever to 4" NPS; **Gear Op. 6" to 8" NPS**

Stem-Packing/Seals: Flexible Graphite

Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608

ENDS: ASME B16.5

RATING: ASME B16.34

TESTING: API 598

DIMENSIONAL: ASME B16.10 – Long Pattern (1" to 6" NPS)

..... ASME B16.10 – **Short Pattern** (8" only) *** MUST be Short Pattern ***

Approved Vendor:

(1" to 1-1/2" NPS)..... Ladish (2-Piece, Bar-Stock, RP): P135-L050-GRnn-A-{Size}

(2" to 6" NPS Only) Ladish (2-Piece, Cast, CL300, RP): P735-L050-GRnn-A-{Size} (Gear Op. for 6" NPS)

(1-1/2" to 6" NPS)..... Apollo (2-Piece, Cast, CL300, RP): Series 87A-700-24 (Gear Op. for 6" NPS)

(1" to 8" NPS)..... Bray, Flow-Tek, RF30 (2-Piece, Cast, CL300, RP): 15-{Size}-11021-SSJG (Gear for 6" to 8" NPS)

Description of BALL Valve (BA03ST501): BA30-S16-F501

Type: Ball, Floating, FP

Valve Size: NPS 0.5 -to- NPS 8"

Service: -29 to 205 °C (-20 to 400 °F)

Class: **ASME 300**

Ends: Flanged RF

Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)

Trim: 316 SS

Seat: RPTFE or TFM 1600

Stem 316 SS

Body Type: 2-Piece

Design: Floating Ball, Fire Safe API 607

Bore-Port: **Full Port (FP)**

Operation: Lever to 4" NPS; **Gear Op. 6" to 8" NPS**

Stem-Packing/Seals: Flexible Graphite

Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608

ENDS: ASME B16.5

RATING: ASME B16.34

TESTING: API 598

DIMENSIONAL: ASME B16.10 - **Long Pattern** (All sizes)

Approved Vendor:

(1/2" to 1-1/2" NPS)..... Ladish (2-Piece, CL300, Bar-Stock, FP): P235-L050-GRnn-A-{Size}

(2" to 8" NPS)..... Ladish (2-Piece, Cast, CL300, FP): P835-L050-GRnn-A-{Size} (Gear Op. 6" to 8" NPS)

(1/2" to 8" NPS)..... Apollo (2-Piece, Cast, CL300, FP): {Size}-87A-900-65 (Gear Op. 6" to 8" NPS)

(1/2" to 8" NPS)..... Bray, Flow-Tek, F30 (Cast, CL300, FP): 15-{Size}-11002-SSJG (Gear Op. 6" to 8" NPS)

**Description of BALL Valve (BA03ST502):**

BAT30-S16-F502

Type: Ball, Trunnion, RP
Valve Size: NPS 6" -to- NPS 12"
Service: -29 to 205 °C (-20 to 400 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
Trim: 316 SS
Seat: RPTFE
Stem: 316 SS
Springs: Inconel X-750
Body Type: 2-Piece
Design: **Trunnion**, Fire Safe API 607
Bore-Port: **Single Reduced (RP)**
Operation: **Gear Op.**
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite
O-Rings: Viton

The standards are:

DESIGN: ASME B16.34 or API 608
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 - **Long Pattern** (All sizes – No Short Pattern Allowed)

Approved Vendor:

(6" to 12" NPS)..... [WKM, 370-D6 Series \(2-Piece, Trunnion, CL300, RP\): {Size x Size}-370D6-300-RFxRF-23TVF23-WG](#)**Description of BALL Valve (BA03ST503):**

BAT30-S16-F503

Type: Ball, Trunnion, FP
Valve Size: NPS 6 -to- NPS 16"
Service: -29 to 205 °C (-20 to 400 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
Trim: 316 SS
Seat: RPTFE
Stem: 316 SS
Springs: Inconel X-750
Body Type: 2-Piece
Design: **Trunnion**, Fire Safe API 607
Bore-Port: **Full Port (FP)**
Operation: **Gear**
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite
O-Rings: Viton

The standards are:

DESIGN: ASME B16.34 or API 608
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 - Long Pattern (All Sizes)

Approved Vendors:

(6" NPS to 16" NPS) [WKM \(2-Piece, Trunnion, Cast, CL300, FP\): {Size}-370D6-300-RFxRF-23TVF23-WG](#)
(6" NPS to 16" NPS) [Warren \(2-Piece, Trunnion, Cast, CL300, FP\): {size}-6306-T26-TPR-G](#)

Description of BALL Valve (BA03ST504): BA30-S16-F504

Type: Ball, Floating, RP (Gear Op.)

Valve Size: NPS 3 -to- NPS 8

Service: -29 to 205 °C (-20 to 400 °F)

Class: **ASME 300**

Ends: Flanged RF

Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)

Trim: 316 SS

Seat: RPTFE or TFM 1600

Stem 316 SS

Body Type: 2-Piece

Design: Floating Ball, Fire Safe API 607

Bore-Port: **Red. Port (RP)**

Operation: **Gear Op.**

Stem-Packing/Seals: Flexible Graphite

Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608

ENDS: ASME B16.5

RATING: ASME B16.34

TESTING: API 598

DIMENSIONAL: ASME B16.10 – Long Pattern to 6” NPS

..... ASME B16.10 – **Short Pattern at 8” NPS**

Approved Vendor:

(3” NPS to 6” NPS) Ladish (2-Piece, Cast, CL150 RP): P735-G05D-GRnn-A-{Size}, Gear Op. (No 8” Valves Allowed)

(3” NPS to 6” NPS) APOLLO (2-Piece, Cast, CL150, RP): {Size}-87A-700-24-MG, Gear Op. (No 8” Valves Allowed)

(3” NPS to 6” NPS) Bray, Flow-Tek, RF30 (Cast, CL300, RP): 15-{Size}-11021-SSJG (Gear Op. at 6” NPS)

(3” NPS to 8” NPS) WKM (Style 320F, Cast, CL150, RP): {Size x Size}-320F-300-RFxRF-23-TGF-23-WG, Gear Op.

Description of BALL Valve (BA03ST505): BA30-S16-F505

Type: Ball, Floating, **FP** (Gear Op.)

Valve Size: NPS 4 -to- NPS 10

Service: -29 to 205 °C (-20 to 400 °F)

Class: **ASME 300**

Ends: Flanged RF

Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)

Trim: 316 SS

Seat: RPTFE or TFM 1600

Stem 316 SS

Body Type: 2-Piece

Design: Floating Ball, Fire Safe API 607

Bore-Port: **Full Port (FP)**

Operation: **Gear**

Stem-Packing/Seals: Flexible Graphite

Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608

ENDS: ASME B16.5

RATING: ASME B16.34

TESTING: API 598

DIMENSIONAL: ASME B16.10 - Long Pattern (All Sizes)

Approved Vendor:

(4” NPS to 6” NPS Only) Ladish (2-Piece, Cast, CL300, FP): P735-G053-GRnn-A-{Size}, Gear Op.

(4” NPS to 10” NPS) Apollo (2-Piece, Cast, CL300, FP): {Size}-87A-900-24-MH, Gear Op.

(4” NPS to 6” NPS Only) WKM (Style 320F, CL300, FP): {Size}-320F-300-RFxRF-23-TGF-23-WG, Gear Op.

Description of BALL Valve (BA03ST508): BA30-S16-F508

Type: Ball, Floating, RP

Valve Size: NPS 1" -to- NPS 10"

Service: -29 to +205 °C (-20 to +400 °F)

Class: **ASME 300**

Ends: Flanged RF

Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)

Trim: 316 SS

Seat: RPTFE or TFM 1600

Stem 316 SS

Body Type: 2-Piece

Design: Floating Ball, Fire Safe API 607

Bore-Port: **Single Reduced (RP)**

Operation: Lever to 4" NPS; Gear Op. for 6" to 10" NPS

Stem-Packing/Seals: Flexible Graphite

Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608

ENDS: ASME B16.5

RATING: ASME B16.34

TESTING: API 598

DIMENSIONAL: ASME B16.10 – Long Pattern (1" to 6" NPS)

..... ASME B16.10 – Long Pattern (8" & 10" NPS) *** NO Short Pattern Allowed ***

Approved Vendor:

(1" to 1-1/2" NPS Only) Ladish (2-Piece, Bar-Stock, RP): P135-L050-GRnn-A-{Size}

(2" to 6" NPS Only) Ladish (2-Piece, Cast, CL300, RP): P735-L050-GRnn-A-{Size} (Gear Op. for 6" Only)

(1-1/2" to 10" NPS Only) Apollo (Cast, CL300, RP): Series 87A-700-24, (Gear Op. for 6" to 10" NPS)

(1" to 6" NPS Only) Bray, Flow-Tek, RF15 (Cast, CL300, RP): 15-{Size}-11021-SSJG (Gear Op. for 6" NPS Only)

(1" to 6" NPS Only) WKM (2-Piece, Cast, CL300, RP): {Size x Size}-320F-300-23x23-TGF-23 (Gear Op. for 6" NPS Only)

Description of BALL Valve (BA03ST519):.. BAT30-S16-F519

Type: Ball, Trunnion, RP

Valve Size: NPS 8 -to- NPS 12

Service: -29 to 205 °C (-20 to 400 °F)

Class: **ASME 300**

Ends: Flanged RF

Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)

Trim: 316 SS

Seat: RPTFE

Stem 316 SS

Springs Inconel X-750

Body Type: Split

Design: **Trunnion**, Fire Safe API 607

Bore-Port: **Single Reduced (RP)**

Operation: **Gear**

Stem-Packing/Seals: PTFE

Body Gaskets: PTFE

O-Rings: Viton

The standards are:

DESIGN: ASME B16.34 or API 608

ENDS: ASME B16.5

RATING: ASME B16.34

TESTING: API 598

DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor: WKM (Style 370D6, RP): {Size x Size}-370D6-300-RFxRF-23-TVF-23-WH

Description of BALL Valve (BA03ST520):..

BAT30-S16-F520

Type:Ball, Trunnion, FP
 Valve Size:NPS 6 -to- NPS 16
 Service:-29 to 205 °C (-20 to 400 °F)
 Class:**ASME 300**
 Ends:Flanged RF
 Body:316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim:316 SS
 Seat:RPTFE
 Stem :316 SS
 SpringsInconel X-750
 Body Type:Split
 Design:**Trunnion**
 Bore-Port:**Full Port (FP)**
 Operation:**Gear**
 Stem-Packing/Seals:PTFE
 Body Gaskets:PTFE
 O-Rings:Viton

The standards are:

DESIGN:ASME B16.34 or API 608
 ENDS:ASME B16.5
 RATING:ASME B16.34
 TESTING:API 598
 DIMENSIONAL:ASME B16.10 - Long Pattern

Approved Vendor:WKM (Style 370D6, FP): {Size}-370D6-300-RFxRF-23TVF23-WG

Description of BALL Valve (BA03ST521):..

BA30-S16-F521

Type:Ball, Floating, FP
 Valve Size:NPS 0.5 -to- NPS 4
 Service:-29 to 205 °C (-20 to 400 °F)
 Class:**ASME 300**
 Ends:Flanged RF
 Body:316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim:316 SS
 Seat:RPTFE
 Stem :316 SS
 Body Type:Split
 Design:Floating
 Bore-Port:**Full Port (FP)**
 Operation:Lever
 Stem-Packing/Seals:PTFE
 Body Gaskets:PTFE

The standards are:

DESIGN:ASME B16.34 or API 608
 ENDS:ASME B16.5
 RATING:ASME B16.34
 TESTING:API 598
 DIMENSIONAL:ASME B16.10 - Long Pattern

Approved Vendor:Ladish (2-Piece, FP): P835-L053-TRnn-A-{Size}
APOLLO (2-Piece, FP): {Size}-87A-900 (0.5" NPS to 4" NPS)

Description of BALL Valve (BA03ST522):..

BA30-S16-F522

Type: Ball, Floating, RP
 Valve Size: NPS 0.5 -to- NPS 6
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim: 316 SS
 Seat: RPTFE
 Stem : 316 SS
 Body Type: Split
 Design: Floating Ball
 Bore-Port: **Single Reduced (RP)**
 Operation: Lever
 Stem-Packing/Seals: PTFE
 Body Gaskets: PTFE

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor: Ladish (2-Piece, RP): P735-L053-TRnn-A-{Size}
 APOLLO (2-Piece, RP): {Size}-87A-700 (1.5" NPS to 4" NPS)

Description of BALL Valve (BA03ST523):..

BA30-S16-F523

Type: Ball, Floating, FP
 Valve Size: NPS 3 -to- NPS 8
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim: 316 SS
 Seat: RPTFE
 Stem : 316 SS
 Body Type: Split
 Design: Floating Ball
 Bore-Port: **Full Port (FP)**
 Operation: **Gear**
 Stem-Packing/Seals: PTFE
 Body Gaskets: PTFE

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor: Ladish (2-Piece, FP): P835-G053-TR03-A-{Size}
 APOLLO (2-Piece, FP): {Size}-87A-900-MG (3" NPS to 8" NPS)

Description of BALL Valve (BA03ST524):..

BA30-S16-F524

Type: Ball, Floating, RP
 Valve Size: NPS 4 -to- NPS 8
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim: 316 SS
 Seat: RPTFE
 Stem 316 SS
 Body Type: Split
 Design: Floating Ball
 Bore-Port: **Single Reduced (RP)**
 Operation: **Gear**
 Stem-Packing/Seals: PTFE
 Body Gaskets: PTFE

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor: Ladish (2-Piece, RP): P735-L053-TRnn-A-{Size}
 APOLLO (2-Piece, RP): {Size}-87A-700-MG (4" NPS to 8" NPS)

Description of BALL Valve (BA03SU301): ..

BA30-S16-S301

Type: Ball, Floating, RP
 Valve Size: NPS 0.5 -to- NPS 2
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: SW
 Body: 316L SS (ASTM A182-F316L or ASTM A351-CF3M)
 Trim: 316 SS
 Seat: RPTFE
 Stem 316 SS
 Body Type: 3-Piece Bolted
 Design: Floating Ball
 Bore-Port: **Single Reduced (RP)**
 Operation: Lever
 Stem-Packing/Seals: PTFE
 Body Gaskets: PTFE

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.11
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor:

**Description of BALL Valve (BA06ST500):..**

BA60-S16-F500

Type: Ball, Floating, RP
Valve Size: NPS 1" -to- NPS 4"
Service: -29 to 205 °C (-20 to 400 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
Trim: 316 SS
Seat: RPTFE or TFM 1600
Stem 316 SS
Body Type: 2-Piece
Design: Floating Ball, Fire Safe API 607, AED Seals
Bore-Port: **Single Reduced (RP)**
Operation: Lever
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 - Long Pattern = Short Pattern (for All CL600 Sizes)

Approved Vendor:

(1" to 1-1/2" NPS Only) Ladish (2-Piece, CL600, RP): P165-L053-GRnn-A-{Size}
(2" to 4" NPS Only) Ladish (2-Piece, CL600, RP): P765-L053-GRnn-A-{Size}
(2" to 4" NPS Only) WKM (2-Piece, CL600, RP): {Size x Size}-320F-600-RFxRF-23TGF23-WH

Description of BALL Valve (BA06ST501):..

BA60-S16-F501

Type: Ball, Floating, FP
Valve Size: NPS 0.5 -to- NPS 4"
Service: -29 to 205 °C (-20 to 400 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
Trim: 316 SS
Seat: RPTFE or TFM 1600
Stem 316 SS
Body Type: 2-Piece
Design: Floating Ball, Fire Safe API 607, AED Seals
Bore-Port: **Full Port (FP)**
Operation: Lever
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 - Long Pattern (All Sizes)

Approved Vendor:

(1/2" to 4" NPS Only) Ladish (2-Piece, FP): P865-L053-GRnn-A-{Size}
(1/2" to 4" NPS Only) WKM (2-Piece, FP): {Size}-320F-600-RFxRF-23TGF23-WH

Description of BALL Valve (BA06ST502):..

BAT60-S16-F502

Type: Ball, Trunnion, RP
 Valve Size: NPS 3 -to- NPS 12
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim: 316 SS
 Seat: RPTFE
 Stem : 316 SS
 Springs Inconel X-750
 Body Type: Split
 Design: **Trunnion**, Fire Safe API 607, AED Seals
 Bore-Port: **Single Reduced (RP)**
 Operation: **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite
 O-Rings: Viton AED

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor:

Description of BALL Valve (BA06ST503):..

BAT60-S16-F503

Type: Ball, Trunnion, FP
 Valve Size: NPS 2 -to- NPS 24
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim: 316 SS
 Seat: RPTFE
 Stem : 316 SS
 Springs Inconel X-750
 Body Type: Split
 Design: **Trunnion**, Fire Safe API 607, AED Seals
 Bore-Port: **Full Port (FP)**
 Operation: **Gear**
 Stem-Packing/Seals: Flexible Graphite Body Gaskets: .Flexible Graphite
 O-Rings: Viton AED

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor:

Description of BALL Valve (BA06ST518):..

BA60-S16-F518

Type: Ball, Floating, RP
 Valve Size: NPS 0.5 -to- NPS 4
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim: 316 SS
 Seat: RPTFE
 Stem : 316 SS
 Body Type: Split
 Design: Floating Ball
 Bore-Port: **Single Reduced (RP)**
 Operation: Lever
 Stem-Packing/Seals: PTFE
 Body Gaskets: PTFE

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor: Ladish (2-Piece, RP): P765-L053-TRnn-A-{Size}

Description of BALL Valve (BA06ST519):..

BA60-S16-F519

Type: Ball, Floating, FP
 Valve Size: NPS 0.5 -to- NPS 3
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim: 316 SS
 Seat: RPTFE
 Stem : 316 SS
 Body Type: Split
 Design: Floating Ball
 Bore-Port: **Full Port (FP)**
 Operation: Lever
 Stem-Packing/Seals: PTFE
 Body Gaskets: PTFE

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor: Ladish (2-Piece, FP): P865-L053-TRnn-A-{Size}

**Description of BALL Valve (BA06ST520):..**

BAT60-S16-F520

Type: Ball, Trunnion, RP
Valve Size: NPS 2 -to- NPS 24
Service: -29 to 205 °C (-20 to 400 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
Trim: 316 SS
Seat: RPTFE
Stem : 316 SS
Springs Inconel X-750
Body Type: Split
Design: **Trunnion**
Bore-Port: **Single Reduced (RP)**
Operation: **Gear**
Stem-Packing/Seals: PTFE
Body Gaskets: PTFE

The standards are:

DESIGN: ASME B16.34 or API 608
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor:

Description of BALL Valve (BA06ST521):..

BAT60-S16-F521

Type: Ball, Trunnion, FP
Valve Size: NPS 3 -to- NPS 24
Service: -29 to 205 °C (-20 to 400 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
Trim: 316 SS
Seat: RPTFE
Stem : 316 SS
Springs Inconel X-750
Body Type: Split
Design: **Trunnion**
Bore-Port: **Full Port (FP)**
Operation: **Gear**
Stem-Packing/Seals: PTFE
Body Gaskets: PTFE
O-Rings: Viton

The standards are:

DESIGN: ASME B16.34 or API 608
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor:

Description of BALL Valve (BA06SU200):..

BA60-S16-X210

Type: Ball, Floating, FP
 Valve Size: NPS 0.5 -to- NPS 2
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: SW x FNPT
 Body: 316 SS (ASTM A182-F316/316L or ASTM A351-CF3M)
 Trim: 316 SS
 Seat: RPTFE or TFM 1600
 Stem 316 SS
 Body Type: 3-Piece Bolted
 Design: Floating Ball, Fire Safe API 607, AED Seals
 Bore-Port: **Full Port (FP)**
 Operation: Lever
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: SW: ASME B16.11, FNPT: ASME B1.20.1
 RATING: ASME B16.34
 TESTING: API 598 DIMENSIONAL:
 MFG STD

Approved Vendor: Apollo (3-Piece Bolted, FP): {Size}-86B-100-24-63
 Bray, Triad (3-Piece, CL600, FP): 18-{Size}-11015-SS-JGNN

Description of BALL Valve (BA06SU300):..

BA60-S16-S300

Type: Ball, Floating, FP
 Valve Size: NPS 0.5 -to- NPS 2
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: SW
 Body: 316L SS (ASTM A182-F316L or ASTM A351-CF3M)
 Trim: 316 SS
 Seat: RPTFE or TFM 1600
 Stem 316 SS
 Body Type: 3-Piece Bolted
 Design: Floating Ball, Fire Safe API 607, AED Seals
 Bore-Port: **Full Port (FP)**
 Operation: Lever
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.11
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Apollo (3-Piece Bolted, FP): {Size}-86B-200-24
 Bray, Triad (3-Piece, CL600, FP): 18-{Size}-11011-SS-JGNN



Description of BALL Valve (BA06SU310):..

BA15-S16L-S310

Type: Ball, Floating, FP
Valve Size: NPS 0.5 -to- NPS 2
Service: Low Temperature -73 to 204 °C (-99 to 400 °F)
Class: **ASME 600**
Ends: SW
Body: 316 SS (ASTM A182-F316L or ASTM A351-CF3M)
Trim: 316 SS
Seat: RPTFE
Stem 316 SS
Body Type: 3-Piece Bolted
Design: Floating Ball, Fire Safe API 607, AED Seals
Bore-Port: **Full Port (FP)**
Operation: Lever
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
ENDS: ASME B16.11
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: MFG STD

Approved Vendor: Apollo (3-Piece Bolted):

..... [Bray, Triad \(3-Piece, CL600, FP\): 18-{Size}-11011-SS-JGNN](#)

Description of BALL Valve (BA60-S16-T210) was BA06SU000:

Type: Ball, Floating, FP
 Valve Size: NPS 0.5 -to- NPS 2"
 Temperature: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: FNPT
 Body: 316 SS (ASTM A351, CF8M)
 Trim: 316 SS (ASTM A276, 316 SS)
 Seat: Multi-Seal, RPTFE
 Stem 316 SS, Anti-blowout
 Body Type: 3-Piece
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Full Port (FP)**
 Operation: Lever
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:.....
 DESIGN: ASME B16.34
 ENDS: THRD, ASME B1.20.1
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD.

Approved Vendor: Apollo (3-Piece, CL600, FP): {Size}-86B-100-24
 [Bray, Triad \(3-Piece, CL600, FP\): 18-{Size}-11010-SS-JGNN](#)

Description of BALL Valve (BA60-S16-X210):

Type: Ball, Floating, FP
 Valve Size: NPS 0.5 -to- NPS 2"
 Temperature: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: SW x FNPT
 Body: 316 SS (ASTM A351, CF8M)
 Trim: 316 SS (ASTM A276, 316 SS)
 Seat: Multi-Seal, RPTFE
 Stem 316 SS, Anti-blowout
 Body Type: 3-Piece
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Full Port (FP)**
 Operation: Lever
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite d

The standards are:.....
 DESIGN: ASME B16.34
 ENDS: SW: ASME B16.11, THRD, ASME B1.20.1
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD.

Approved Vendor: Apollo (3-Piece, FP): {Size}-86B-100-24-63
 [Bray, Triad \(3-Piece, CL600, FP\): 18-{Size}-11015-SS-JGNN](#)

 ***** **Start of Duplex Stainless Steel Valves (2205 SS)** *****

Description of BALL Valve (BA01SP500):..... BA15-S22-F500

Type:Ball, Floating, RP
 Valve Size:NPS 1.0 -to- NPS 6"
 Service:-29 to 205 °C (-20 to 400 °F)
 Class:**ASME 150**
 Ends:Flanged RF
 Body:2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim:2205 Duplex SS
 Seat:RPTFE
 Stem2205 Duplex SS
 Body Type:2-Piece
 Design:Floating Ball, Fire Safe API 607
 Bore-Port:**Single Reduced (RP)**
 Operation:Lever to 6" NPS
 Stem-Packing/Seals:Flexible Graphite
 Body Gaskets:Flexible Graphite
 The standards are:
 DESIGN:ASME B16.34 or API 608
 ENDS:ASME B16.5
 RATING:ASME B16.34
 TESTING:API 598
 DIMENSIONAL:ASME B16.10 – Long Pattern (1" to 4" NPS)
ASME B16.10 – Short Pattern (6" NPS) *** MUST be Short Pattern ***
 Approved Vendor:
 (1" to 1-1/2" NPS).....Ladish (2-Piece, Cast, CL150, RP): P115-L550-GRnn-A-{Size}
 (2" to 4" NPS).....Ladish (2-Piece, Cast, CL150, RP): P715-L550-GRnn-A-{Size}
 (1.5" NPS to 6" NPS)Apollo (2-Piece, Cast, CL150, RP): {Size}-87J-100-24
 (1" to 6" NPS).....Bray, Flow-Tek, RF15: (Cast, CL150, RP): 15-{Size}-11001-XXJG

Description of BALL Valve (BA01SP501):..... BA15-S22-F501

Type:Ball, Floating, FP
 Valve Size:NPS 0.5 -to- NPS 8
 Service:-29 to 205 °C (-20 to 400 °F)
 Class:**ASME 150**
 Ends:Flanged RF
 Body:2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim:2205 Duplex SS
 Seat:RPTFE
 Stem2205 Duplex SS
 Body Type:2-Piece
 Design:Floating Ball, Fire Safe API 607
 Bore-Port:**Full Port (FP)**
 Operation:Lever to 4" NPS; Gear Op. on 6" to 8" NPS
 Stem-Packing/Seals:Flexible Graphite
 Body Gaskets:Flexible Graphite
 The standards are:
 DESIGN:ASME B16.34 or API 608
 ENDS:ASME B16.5
 RATING:ASME B16.34
 TESTING:API 598
 DIMENSIONAL:ASME B16.10 - Long Pattern (All Sizes)
 Approved Vendor:
 (1/2" to 1-1/2" NPS).....Ladish (2-Piece, Bar-Stock, FP): P215-L550-GRnn-A-{Size}
 (2" to 6" NPS).....Ladish (2-Piece, Cast, CL150, FP): P815-L550-GRnn-A-{Size} (Gear Op. on 6" NPS)
 (1/2" NPS to 8" NPS)Apollo (2-Piece, Cast, CL150, FP): {Size}-87J-200-24 (Gear Op. 6" or 8" NPS)
 (1" to 8" NPS).....Bray, Flow-Tek, F15: (Cast, CL150, FP): 15-{Size}-11000-XXJG (Gear Op. 6" or 8" NPS)

Description of BALL Valve (BA01SP502):..

BAT15-S22-F502

Type: Ball, Trunnion, RP
 Valve Size: NPS 8 -to- NPS 24
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim: 2205 Duplex SS
 Seat: RPTFE
 Stem 2205 Duplex SS
 Springs Inconel X-750
 Body Type: 2-Piece
 Design: **Trunnion**, Fire Safe API 607
 Bore-Port: **Single Reduced (RP)**
 Operation: **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite
 O-Rings: Viton

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 – Long Pattern (NPS 8 -to- NPS 16)
 ASME B16.10 - Long Pattern (NPS 18 -to- NPS 24)

Approved Vendor:
 (6" to 16" NPS)..... Warren (2-Piece, RP): {size}-6158-T28-TPR-G

Description of BALL Valve (BA01SP503):..

BAT15-S22-F503

Type: Ball, Trunnion, FP
 Valve Size: NPS 6 -to- NPS 16"
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim: 2205 Duplex SS
 Seat: RPTFE
 Stem 2205 Duplex SS
 Springs Inconel X-750
 Body Type: 2-Piece
 Design: **Trunnion**, Fire Safe API 607
 Bore-Port: **Full Port (FP)**
 Operation: **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite
 O-Rings: Viton

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor: Warren (2-Piece, FP): {size}-6158-T28-TPR-G (6" to 16" NPS)

Description of BALL Valve (BA01SP504): . BA15-S22-F504

Type: Ball, Floating, RP
 Valve Size: NPS 4 -to- NPS 8
 Service: - 29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim: 2205 Duplex SS
 Seat: RPTFE
 Stem 2205 Duplex SS
 Body Type: 2-Piece
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Single Reduced (RP)**
 Operation: **Gear Oper.**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: **ASME B16.10 – Short Pattern (4" to 10" NPS)**

Approved Vendor:

(4" NPS to 6" NPS) Apollo (2-Piece, Cast, CL150, RP): {Size}-87J-100-24-MG (Gear Op.)
 (4" NPS to 8" NPS) Bray, Flow-Tek, Series RF15 (Cast, CL150, RP): 15-{Size}-11001-XXJG (Gear Op.)

Description of BALL Valve (BA01SP505):.. BA15-S22-F505

Type: Ball, Floating, FP
 Valve Size: NPS 4 -to- NPS 10
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim: 2205 Duplex SS
 Seat: RPTFE
 Stem 2205 Duplex SS
 Body Type: 2-Piece
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Full Port (FP)**
 Operation: **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: **ASME B16.10 - Long Pattern (All Sizes)**

Approved Vendor:

(4" NPS to 10" NPS) Ladish (2-Piece, FP): P815-G550-GRnn-A-{Size}
 (4" NPS to 6" NPS) Apollo (2-Piece, Cast, CL150, FP): {Size}-87J-200-24-MG (Gear Op.)
 (4" NPS to 8" NPS) Bray, Flow-Tek, Series F15 (Cast, CL150, FP): 15-{Size}-11000-XXJG (Gear Op.)

Description of BALL Valve (BA03SP500):..

BA30-S22-F500

Type: Ball, Floating, RP
 Valve Size: NPS 1.0 -to- NPS 6
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim: 2205 Duplex SS
 Seat: RPTFE
 Stem 2205 Duplex SS
 Body Type: 2-Piece
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Single Reduced (RP)**
 Operation: Lever to 4" NPS; **Gear Op. at 6" NPS**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite
 The standards are:
 DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: **ASME B16.10 – Long Pattern (1" to 6")**
 **ASME B16.10 – Short Pattern (8" only) *** MUST be Short Pattern *****

Approved Vendor:

(1" to 1-1/2" NPS)..... Ladish (2-Piece, Cast, CL300, RP): P135-L550-GRnn-A-{Size}
 (2" to 6" NPS)..... Ladish (2-Piece, Cast, CL300, RP): P735-L550-GRnn-A-{Size}
 (1.5" NPS to 6" NPS) Apollo (2-Piece, Cast, CL300, RP): {Size}-87J-700-24
 (1" to 8" NPS)..... Bray, Flow-Tek, RF30: (Cast, CL300, RP): 15-{Size}-11021-XXJG (Gear Op. 6" NPS)

Description of BALL Valve (BA03SP501):..

BA30-S22-F501

Type: Ball, Floating, FP
 Valve Size: NPS 0.5 -to- NPS 8
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim: 2205 Duplex SS
 Seat: RPTFE
 Stem 2205 Duplex SS
 Body Type: 2-Piece
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Full Port (FP)**
 Operation: Lever to 4" NPS; **Gear Op. for 6" or 8" NPS**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite
 The standards are:
 DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: **ASME B16.10 - Long Pattern (All Sizes)**

Approved Vendor:

(1/2" to 1-1/2" NPS)..... Ladish (2-Piece, Cast, CL300, FP): P235-L550-GRnn-A-{Size}
 (2" to 8" NPS)..... Ladish (2-Piece, Cast, CL300, FP): P835-L550-GRnn-A-{Size} (Gear Op. 6" or 8" NPS)
 (1/2" NPS to 8" NPS) Apollo (2-Piece, Cast, CL300, FP): {Size}-87J-900-24 (Gear Op. 6" or 8" NPS)
 (1/2" to 8" NPS)..... Bray, Flow-Tek, F30: (Cast, CL300, FP): 15-{Size}-11002-XXJG (Gear Op. 6" or 8" NPS)

Description of BALL Valve (BA03SP503):..

BA30-S22-F503

Type: Ball, Floating, FP
 Valve Size: NPS 3 -to- NPS 8
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim: 2205 Duplex SS
 Seat: RPTFE
 Stem 2205 Duplex SS
 Body Type: Split
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Full Port (FP)**
 Operation: **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor:

(3" NPS to 8" NPS) Ladish (2-Piece, FP): P835-L550-GRnn-A-{Size}
 (3" NPS to 6" NPS) Apollo (2-Piece, FP): {Size}-87J-900-24-MG

Description of BALL Valve (BA03SP504):..

BAT30-S22-F504

Type: Ball, Trunnion, RP
 Valve Size: NPS 8 -to- NPS 24
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim: 2205 Duplex SS
 Seat: RPTFE
 Stem 2205 Duplex SS
 Springs Inconel X-750
 Body Type: 2-Piece
 Design: **Trunnion**, Fire Safe API 607
 Bore-Port: **Single Reduced (RP)**
 Operation: **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible
 Graphite
 O-Rings: Viton

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor:

(2" to 16" NPS) Warren (2-Piece, RP): {size}-6308-T28-TPR-G

Description of BALL Valve (BA03SP505):..

BAT30-S22-F505

Type: Ball, Trunnion, FP
 Valve Size: NPS 6 -to- NPS 24
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim: 2205 Duplex SS
 Seat: RPTFE
 Stem 2205 Duplex SS
 Springs Inconel X-750
 Body Type: Split
 Design: **Trunnion**, Fire Safe API 607
 Bore-Port: **Full Port (FP)**
 Operation: **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite
 O-Rings: Viton

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor: Warren (2-Piece, FP): {size}-6308-T28-TPR-G (2" to 16" NPS)

Description of BALL Valve (BA03SP506):..

BA30-S22-F506

Type: Ball, Floating, RP
 Valve Size: NPS 4 -to- NPS 10
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim: 2205 Duplex SS
 Seat: RPTFE
 Stem 2205 Duplex SS
 Body Type: Split
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Single Reduced (RP)**
 Operation: **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor: Ladish (2-Piece, RP): P735-G55D-GRnn-A-{Size}

**Description of BALL Valve (BA06SP500):..**

BA60-S22-F500

Type: Ball, Floating, RP
Valve Size: NPS 1.0 -to- NPS 3
Service: -29 to 205 °C (-20 to 400 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Trim: 2205 Duplex SS
Seat: RPTFE
Stem 2205 Duplex SS
Body Type: Split
Design: Floating Ball, Fire Safe API 607, AED Seals
Bore-Port: **Single Reduced (RP)**
Operation: Lever
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 - Long Pattern (All Sizes)

Approved Vendor:

(1" to 1-1/2" NPS)..... Ladish (2-Piece, Cast, CL600, RP): P165-L550-GRnn-A-{Size}
(2" to 3" NPS)..... Ladish (2-Piece, Cast, CL600, RP): P765-L550-GRnn-A-{Size}

Description of BALL Valve (BA06SP501): .

BA60-S22-F501

Type: Ball, Floating, FP
Valve Size: NPS 0.5 -to- NPS 3
Service: -29 to 205 °C (-20 to 400 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Trim: 2205 Duplex SS
Seat: RPTFE
Stem 2205 Duplex SS
Body Type: Split
Design: Floating Ball, Fire Safe API 607, AED Seals
Bore-Port: **Full Port (FP)**
Operation: Lever
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 - Long Pattern (All Sizes)

Approved Vendor:

(1/2" to 1-1/2" NPS)..... Ladish (2-Piece, Cast, CL600, FP): P265-L550-GRnn-A-{Size}
(2" to 3" NPS)..... Ladish (2-Piece, Cast, CL600, FP): P865-L550-GRnn-A-{Size}

Description of BALL Valve (BA06SP502):..

BAT60-S22-F502

Type: Ball, Trunnion, RP
 Valve Size: NPS 3 -to- NPS 24
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim: 2205 Duplex SS
 Seat: RPTFE
 Stem : 2205 Duplex SS
 Body Type: Split
 Design: **Trunnion**, Fire Safe API 607, AED Seals
 Bore-Port: **Single Reduced (RP)**
 Operation: **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite
 O-Rings: Viton AED

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor:

Description of BALL Valve (BA06SP300):..

BA60-S22-S300

Type: Ball, Floating, FP
 Valve Size: NPS 1 -to- NPS 4
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: SW
 Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim: 2205 Duplex SS
 Seat: RPTFE
 Stem : 2205 Duplex SS
 Body Type: 3-Piece Bolted
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Full Port (FP)**
 Operation: Lever
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.11
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG
 STD

Approved Vendor: APOLLO (3-Piece, FP): {Size}-86J-200-24

Description of BALL Valve (BA60-S22-X210) was BA06SP200:

Type:..... Ball, Floating, FP
 Valve Size: NPS 0.5 -to- NPS 2
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: SW x FNPT
 Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim: 2205 Duplex SS
 Seat: RPTFE
 Stem/Trunnion 2205 Duplex SS
 Body Type: 3-Piece Bolted
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Full Port (FP)**
 Operation: Lever
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: SW: ASME B16.11, FNPT: ASME B1.20.1
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: APOLLO (3-Piece, FP): {Size}-86J-100-24-63

Description of BALL Valve (BA60-S22-T210)

Type:..... Ball, Floating, FP
 Valve Size: NPS 0.5 -to- NPS 2
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: FNPT
 Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim: 2205 Duplex SS
 Seat: RPTFE
 Stem/Trunnion 2205 Duplex SS
 Body Type: 3-Piece Bolted
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Full Port (FP)**
 Operation: Lever
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: SW: ASME B16.11, FNPT: ASME B1.20.1
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: APOLLO (3-Piece, FP): {Size}-86J-100-24

 ***** **Start of Super Duplex Stainless Steel Valves (2507 SS)** *****

Description of BALL Valve (BA01SQ500 ==> BA15-S25-F500)

Type: Ball, Floating, RP
 Valve Size: NPS 1" -to- NPS 6"
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 2507 Super Duplex SS (ASTM A995-5A; ACI CDE3MN)
 Trim: 2507 Super Duplex SS
 Seat: RPTFE
 Stem 2507 Super Duplex SS
 Body Type: 2-Piece
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Standard Port** (i.e., Reduced Port)
 Operation: Lever (up to 6" NPS)
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 – Long Pattern (1" to 4")
 ASME B16.10 – Short Pattern (6" only) *** MUST be Short Pattern ***

Approved Vendor:

(1" to 1-1/2" NPS).....Ladish (2-Piece, Bar-Stock, CL150, RP): P115- L560-GRnn-A-{Size}
 (2" to 4" NPS Only)Ladish (2-Piece, Cast, CL150, RP): P715- L560-GRnn-A-{Size} (No 6" NPS allowed)
 (1-1/2" to 6" NPS).....Apollo (2-Piece, Cast, CL150, RP): Series 87K-100-24
 (1" to 6" NPS).....Bray, Flow-Tek, Series RF15: (Cast, CL150, RP): 15-{Size}-11001-YYJG

Description of BALL Valve (BA01SQ501 ==> BA15-S25-F501)

Type: Ball, Floating, FP
 Valve Size: NPS 1/2" -to- NPS 8"
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 2507 Super Duplex SS (ASTM A995-6A)
 Trim: 2507 Super Duplex SS
 Seat: RPTFE
 Stem 2507 Super Duplex SS
 Body Type: 2-Piece
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Full Port (FP)**
 Operation: Lever (up to 4" NPS); **Gear (6" to 8" NPS)**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern (All Sizes)

Approved Vendor:

(1/2" to 1-1/2" NPS).....Ladish (2-Piece, Bar-Stock, CL150, FP): P215- L560-GRnn-A-{Size}
 (2" to 8" NPS):.....Ladish (2-Piece, Cast, CL150, FP): P815-L560-GRnn-A-{Size}, Lever to 4", Gear 6" to 8"
 (1/2" to 8" NPS).....Apollo (2-Piece, Cast, CL150, FP): 87K-200-24
 (1/2" to 8" NPS).....Bray, Flow-Tek, Series F15 (CL150, Cast, FP): 15-{Size}-11000-YYJG

Description of BALL Valve (BA01SQ508 ==> BA15-S25-F508)

Type: Ball, Floating, RP
 Valve Size: NPS 1" -to- NPS 6"
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 2507 Super Duplex SS (ASTM A995-5A; ACI CDE3MN)
 Trim: 2507 Super Duplex SS
 Seat: RPTFE
 Stem 2507 Super Duplex SS
 Body Type: 2-Piece
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Standard Port** (i.e., Reduced Port)
 Operation: Lever (up to 6" NPS)
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: [ASME B16.10 – Long Pattern Only \(ALL Sizes\)](#)

Approved Vendor:

(1" to 1-1/2" NPS).....Ladish (2-Piece, Bar-Stock, CL150, RP): P115- L560-GRnn-A-{Size}
 (2" to 6" NPS Only)Ladish (2-Piece, Cast, CL150, RP): P715- L560-GRnn-A-{Size} (6" Max Valve Size)
 (1-1/2" to 4" NPS).....Apollo (2-Piece, Cast, CL150, RP): Series 87K-100-24 (4" Max Valve Size)
 (1" to 4" NPS).....Bray, Flow-Tek, Series RF15: (Cast, CL150, RP): 15-{Size}-11001-YYJG (4" Max Valve Size)

Description of BALL Valve (BA03SQ508 ==> BA30-S25-F508)

Type: Ball, Floating, RP
 Valve Size: NPS 1" -to- NPS 10"
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 2507 Super Duplex SS (ASTM A995-5A; ACI CDE3MN)
 Trim: 2507 Super Duplex SS
 Seat: RPTFE
 Stem 2507 Super Duplex SS
 Body Type: 2-Piece
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Standard Port** (i.e., Reduced Port)
 Operation: Lever up to 6" NPS; **Gear Op. for 8" to 10" NPS**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: [ASME B16.10 – Long Pattern \(ALL Sizes\)](#)

Approved Vendor:

(1" to 1-1/2" NPS).....Ladish (2-Piece, Bar-Stock, CL300, RP): P135- L560-GRnn-A-{Size}
 (2" to 6" NPS Only)Ladish (2-Piece, Cast, CL300, RP): P735- L560-GRnn-A-{Size} (6" Max Valve Size)
 (1-1/2" to 10" NPS).....Apollo (2-Piece, Cast, CL300, RP): Series 87K-700-24 (10" Max Valve Size)
 (1" to 6" NPS).....Bray, Flow-Tek, Series RF30: (Cast, CL300, RP): 15-{Size}-11021-YYJG (6" Max Valve Size)

Description of BALL Valve (BA03SQ500 == > BA30-S25-F500)

Type: Ball, Floating, RP
Valve Size: NPS 1" -to- NPS 8"
Service: -29 to 205 °C (-20 to 400 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: 2507 Super Duplex SS (ASTM A995-5A; ACI CDE3MN)
Trim: 2507 Super Duplex SS
Seat: RPTFE
Stem 2507 Super Duplex SS
Body Type: 2-Piece
Design: Floating Ball, Fire Safe API 607
Bore-Port: **Single Reduced (RP)**
Operation: Lever to 4" NPS; Gear 6" to 8" NPS
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 – Long Pattern (1" to 6")
..... ASME B16.10 – Short Pattern (8" only) *** MUST be Short Pattern ***

Approved Vendor:

(1" to 1-1/2" NPS)..... Ladish (2-Piece, Bar-Stock, CL300, RP): P135- L560-GRnn-A-{Size}
(2" to 6" NPS Only) Ladish (2-Piece, Cast, CL300, RP): P735- L560-GRnn-A-{Size} (No 8" NPS allowed)
(1-1/2" to 6" NPS)..... Apollo (2-Piece, Cast, CL300, RP): Series 87K-700-24 (Gear Op. at 6" NPS) **No 8" NPS allowed**
(1" to 8" NPS)..... Bray, Flow-Tek, Series RF30: (Cast, CL300, RP): 15-{Size}-11021-YYJG (Gear Op. 6" to 8" NPS)

Description of BALL Valve (BA03SQ501 == > BA30-S25-F501)

Type: Ball, Floating, FP
Valve Size: NPS 1/2" -to- NPS 8"
Service: -29 to 205 °C (-20 to 400 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: 2507 Super Duplex SS (ASTM A995-5A; ACI CDE3MN)
Trim: 2507 Super Duplex SS
Seat: RPTFE
Stem 2507 Super Duplex SS
Body Type: 2-Piece
Design: Floating Ball, Fire Safe API 607
Bore-Port: **Full Port (FP)**
Operation: Lever to 4" NPS; Gear 6" to 8" NPS
Stem-Packing/Seals: Flexible Graphite
Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 – Long Pattern (All Sizes)

Approved Vendor:

(1/2" to 1-1/2" NPS)..... Ladish (2-Piece, Bar-Stock, CL300, FP): P235- L560-GRnn-A-{Size}
(2" to 6" NPS Only) Ladish (2-Piece, Cast, CL300, FP): P835- L560-GRnn-A-{Size} (No 8" NPS allowed)
(1/2" to 6" NPS)..... Apollo (2-Piece, Cast, CL300, FP): Series 87K-900-24 (Gear Op. for 6" NPS; No 8" NPS allowed)
(1/2" to 8" NPS)..... Bray, Flow-Tek, Series F30: (Cast, CL300, FP): 15-{Size}-11002-YYJG (Gear Op. 6" to 8" NPS)

**Description of BALL Valve (BA06SQ300): .****BA60-S25-S300**

Type:Ball, Floating, FP
Valve Size:NPS 0.5 -to- NPS 2"
Service:-29 to 205 °C (-20 to 400 °F)
Class:**ASME 600**
Ends:SW
Body:2507 Super Duplex SS (ASTM A995-6A; ACI CDE3MN)
Trim:2507 Super Duplex SS (ASTM A995-6A; ACI CDE3MN)
Seat:RPTFE
Stem2507 Super Duplex SS
Body Type:3-Piece, Bolted
Design:Floating Ball, Fire Safe API 607
Bore-Port:**Full Port (FP)**
Operation:Lever
Stem-Packing/Seals:Flexible Graphite
Body Gaskets:Flexible Graphite

The standards are:

DESIGN:ASME B16.34
ENDS:ASME B16.11
RATING:ASME B16.34
TESTING:API 598
DIMENSIONAL:Mfg. Std.

Approved Vendor:APOLLO (3-Piece, FP): {Size}-86K-200-24

Description of BALL Valve (BA60-S25-X210):

Type: Ball, Floating, FP
 Valve Size: NPS 0.5 -to- NPS 2
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: SW x FNPT
 Body: 2507 Super Duplex SS (ASTM A995-6A; ACI CE3MN)
 Trim: 2507 Super Duplex SS (ASTM A995-6A; ACI CE3MN)
 Seat: RPTFE
 Stem 2507 Super Duplex SS
 Body Type: 3-Piece, Bolted
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Full Port (FP)**
 Operation: Lever
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: SW: ASME B16.11, FNPT: ASME B1.20.1
 RATING: ASME B16.34
 TESTING: API 598 DIMENSIONAL:
 MFG STD

Approved Vendor: APOLLO (3-Piece, FP): {Size}-86K-100-24-63

Description of BALL Valve (BA60-S25-T210):

Type: Ball, Floating, FP
 Valve Size: NPS 0.5 -to- NPS 2
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: FNPT
 Body: 2507 Super Duplex SS (ASTM A995-6A; ACI CE3MN)
 Trim: 2507 Super Duplex SS (ASTM A995-6A; ACI CE3MN)
 Seat: RPTFE
 Stem 2507 Super Duplex SS
 Body Type: 3-Piece, Bolted
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Full Port (FP)**
 Operation: Lever
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: SW: ASME B16.11, FNPT: ASME B1.20.1
 RATING: ASME B16.34
 TESTING: API 598 DIMENSIONAL:
 MFG STD

Approved Vendor: APOLLO (3-Piece, FP): {Size}-86K-100-24

 ***** **Start of SMO Duplex Stainless Steel Valves (254 SMO SS)** *****

Description of BALL Valve (BA01SV500):.. BA15-SMO-F500

Type: Ball, Floating, RP
 Valve Size: NPS 2" -to- NPS 6"
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
 Trim: 254 SMO SS
 Seat: RPTFE
 Stem: 254 SMO SS
 Body Type: 2-Piece
 Design: Floating Ball
 Bore-Port: **Single Reduced (RP)**
 Operation: Lever
 Stem-Packing/Seals: PTFE
 Body Gaskets: PTFE

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: **ASME B16.10 – Long Pattern (1" to 4")**
 **ASME B16.10 – Short Pattern (6" only) *** MUST be Short Pattern *****

Approved Vendor:

(2" to 4" NPS) Ladish (2-Piece, RP): P715-L160-TRnn-A-{Size} (No 6" Size allowed)
 (2" to 6" NPS) WKM (2-Piece, RP): {Size x Size}-320F-150-23x23-TTF23

Description of BALL Valve (BA01SV501):.. BA15-SMO-F501

Type: Ball, Floating, FP
 Valve Size: NPS 0.5 -to- NPS 6
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
 Trim: 254 SMO SS
 Seat: RPTFE
 Stem: 254 SMO SS
 Body Type: 2-Piece
 Design: Floating Ball
 Bore-Port: **Full Port (FP)**
 Operation: Lever
 Stem-Packing/Seals: PTFE
 Body Gaskets: PTFE

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: **ASME B16.10 - Long Pattern (All Sizes)**

Approved Vendor: Ladish (2-Piece, FP): P815-L160-TRnn-A-{Size}

Description of BALL Valve (BA01SV502):..

BA15-SMO-F502

Type: Ball, Floating, RP
 Valve Size: NPS 4" -to- NPS 8"
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
 Trim: 254 SMO SS
 Seat: RPTFE
 Stem : 254 SMO SS
 Body Type: 2-Piece
 Design: Floating Ball
 Bore-Port: **Single Reduced (RP)**
 Operation: **Gear**
 Stem-Packing/Seals: PTFE
 Body Gaskets: PTFE

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: **ASME B16.10 – Long Pattern (1" to 4" NPS; also 8" NPS)**
 **ASME B16.10 – Short Pattern (6" NPS only) *** MUST be Short Pattern *****

Approved Vendor: WKM (Style 320F): {Size x Size}-320F-150-23x23-TTF23-WG, Gear Op.

Description of BALL Valve (BA01SV503):..

BA15-SMO-F503

Type: Ball, Floating, FP
 Valve Size: NPS 4 -to- NPS 10
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
 Trim: 254 SMO SS
 Seat: RPTFE
 Stem : 254 SMO SS
 Body Type: 2-Piece
 Design: Floating Ball
 Bore-Port: **Full Port (FP)**
 Operation: **Gear**
 Stem-Packing/Seals: PTFE
 Body Gaskets: PTFE

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: **ASME B16.10 - Long Pattern**

Approved Vendor: Ladish (2-Piece, FP): P815-G160-TRnn-A-{Size}

Description of BALL Valve (BA03SV300):..

BA30-SMO-S300

Type: Ball, Floating, RP
 Valve Size: NPS 0.5 -to- NPS 1.5
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: SW
 Body: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
 Trim: 254 SMO Duplex SS
 Seat: RPTFE
 Stem : 254 SMO Duplex SS
 Body Type: 3-Piece Bolted
 Design: Floating Ball
 Bore-Port: **Single Reduced (RP)**
 Operation: Lever
 Stem-Packing/Seals: PTFE
 Body Gaskets: PTFE

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.11
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor:

Description of BALL Valve (BA03SV500):..

BA30-SMO-F500

Type: Ball, Floating, RP
 Valve Size: NPS 0.5 -to- NPS 6
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
 Trim: 254 SMO Duplex SS
 Seat: RPTFE
 Stem : 254 SMO Duplex SS
 Body Type: 2-Piece
 Design: Floating Ball
 Bore-Port: **Single Reduced (RP)**
 Operation: Lever
 Stem-Packing/Seals: PTFE
 Body Gaskets: PTFE

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor: Ladish (2-Piece, RP): P735-L16D-TRnn-A-{Size}

Description of BALL Valve (BA03SV501):..

BA30-SMO-F501

Type: Ball, Floating, FP
 Valve Size: NPS 0.5 -to- NPS 4
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
 Trim: 254 SMO Duplex SS
 Seat: RPTFE
 Stem : 254 SMO Duplex SS
 Body Type: 2-Piece
 Design: Floating Ball
 Bore-Port: **Full Port (FP)**
 Operation: Lever
 Stem-Packing/Seals: PTFE
 Body Gaskets: PTFE

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor: Ladish (2-Piece, Cast, CL300, FP): P835-L16D-TRnn-A-{Size}

Description of BALL Valve (BA03SV502):..

BA30-SMO-F502

Type: Ball, Floating, RP
 Valve Size: NPS 4 -to- NPS 10
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
 Trim: 254 SMO Duplex SS
 Seat: RPTFE
 Stem : 254 SMO Duplex SS
 Body Type: Split
 Design: Floating Ball
 Bore-Port: **Single Reduced (RP)**
 Operation: **Gear**
 Stem-Packing/Seals: PTFE
 Body Gaskets: PTFE

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor: Ladish (2-Piece, Cast, CL300, RP): P735-G16D-TRnn-A-{Size}

Description of BALL Valve (BA03SV503):..

BA30-SMO-F503

Type: Ball, Floating, FP
 Valve Size: NPS 3 -to- NPS 8
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
 Trim: 254 SMO Duplex SS
 Seat: RPTFE
 Stem : 254 SMO Duplex SS
 Body Type: 2-Piece
 Design: Floating Ball
 Bore-Port: **Full Port (FP)**
 Operation: **Gear**
 Stem-Packing/Seals: PTFE
 Body Gaskets: PTFE

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor: Ladish (2-Piece, Cast, CL300, FP): P835-G16D-TRnn-A-{Size}

Description of BALL Valve (BA03SV504):..

BAT30-SMO-F504

Type: Ball, Trunnion, RP
 Valve Size: NPS 8 -to- NPS 24
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
 Trim: 254 SMO Duplex SS
 Seat: RPTFE
 Stem : 254 SMO Duplex SS
 Springs Inconel X-750
 Body Type: 2-Piece
 Design: **Trunnion**
 Bore-Port: **Single Reduced (RP)**
 Operation: **Gear**
 Stem-Packing/Seals: PTFE
 Body Gaskets: PTFE
 O-Rings: Viton

The standards are:

DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor:



Description of BALL Valve (BA03SV505): ..

BAT30-SMO-F505

Type: Ball, Trunnion, FP
Valve Size: NPS 6 -to- NPS 24
Service: -29 to 205 °C (-20 to 400 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
Trim: 254 SMO Duplex SS
Seat: RPTFE
Stem 254 SMO Duplex SS
Springs Inconel X-750
Body Type: Split
Design: **Trunnion**
Bore-Port: **Full Port (FP)**
Operation: **Gear**
Stem-Packing/Seals: PTFE
Body Gaskets: PTFE

The standards are:

DESIGN: ASME B16.34 or API 608
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor:

***** Start of Press-Fit (Vic-Press) Valves, 316 SS *****

Description of BALL Valve (BA15-S16-P200):

Type: Ball, Press-Fit (Vic-Press for Sch. 10S 316 SS), RP
 Valve Size: NPS 0.5 -to- NPS 2"
 Temperature: -29 to 98 °C (-20 to 210 °F)
 Class: N/A (Pressure Rated, 400 psig)
 Ends: Vic-Press, Press-Fit
 Body: 316 SS (ASTM A351, CF8M)
 Stem: 316 SS (ASTM A351, CF8M)
 Seat: PTFE
 Elastomer: **HNBR**, Default (see below)
 Body Type: 3-Piece
 Design: Floating Ball
 Bore-Port: Conventional (RP)
 Operation: Lever

The standards are:.....
 DESIGN: ASME B16.34
 ENDS: Vic-Press, MFG Std.
 RATING: ASME B16.34
 TESTING: API 598
 CERTIFICATION: NSF 61 (Potable Water)
 DIMENSIONAL: MFG STD.

Approved Vendor: Victaulic, Vic-Press, 400 psig: {Size} - Series P569 – {HNBR}
 HNBR Elastomer (Mfgr default) = -20° to +210°F (-29°C to 98°C), NSF 61,
 Orange Stripes, Hot Water to 180°F (82 °C), hot petroleum & mineral oils.

Description of BALL Valve (BA15-S16-P201):

Type: Ball, Press-Fit (Vic-Press for Sch. 10S 316 SS), RP
 Valve Size: NPS 0.5 -to- NPS 2"
 Temperature: -34 to 121 °C (-30 F to 250 °F)
 Class: N/A (Pressure Rated, 400 psig)
 Ends: Vic-Press, Press-Fit
 Body: 316 SS (ASTM A351, CF8M)
 Stem: 316 SS (ASTM A351, CF8M)
 Seat: PTFE
 Elastomer: **EPDM** (see below)
 Body Type: 3-Piece
 Design: Floating Ball
 Bore-Port: Conventional (RP)
 Operation: Lever

The standards are:.....
 DESIGN: ASME B16.34
 ENDS: Vic-Press, MFG Std.
 RATING: ASME B16.34
 TESTING: API 598
 CERTIFICATION: NSF 61 (Potable Water)
 DIMENSIONAL: MFG STD.

Approved Vendor: Victaulic, Vic-Press, 400 psig: {Size} - Series P569 – {EPDM}
 EPDM Elastomer = -30°F to +250°F (-34°C to +121°C), Green Stripes, NSF 61,
 Hot Water to 180 °F (82 °C). Not compatible with petroleum or steam service.

 ***** **Start of Compression Fitting Ball Valves, 316 SS** *****

Description of BALL Valve (BA00-S16-C800):

Type: Ball, Floating, Compression Fittings, PTFE Seats
 Valve Size: NPS 0.25 -to- NPS 2
 Service Limits: -29 to 205 °C (-20 to 400 °F)
 Pressue Rating: 2200 psig @ 100 °F (151 bar @ 37 °C)
 Ends: Compression Fitting (For use with Tubing)
 Body: 316 SS (ASTM A479 or ASTM A351-CF8M)
 Trim: 316 SS
 Seat: RPTFE
 Stem 316 SS
 Body Type: 3-Piece
 Design: Floating Ball
 Bore-Port: **Single Reduced (RP)**
 Operation: Lever
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34
 ENDS: MFG Std.
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG Std.

Approved Vendor: Swagelok (3-Piece, RP): SS-6nTS-{Size}

Description of BALL Valve (BA00-S16-C802):

Type: Ball, Floating, Compression Fittings, Alloy X-750 Seats
 Valve Size: NPS 0.25 -to- NPS 2
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Compression Fitting (For use with Tubing)
 Body: 316 SS (ASTM A479 or ASTM A351-CF8M)
 Trim: 316 SS
 Seat: RPTFE
 Stem 316 SS
 Body Type: 3-Piece
 Design: Floating Ball
 Bore-Port: **Single Reduced (RP)**
 Operation: Lever
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:

DESIGN: ASME B16.34
 ENDS: MFG. STD.
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG. STD.

Approved Vendor: Swagelok (3-Piece, RP): SS-62TS-{Size}

***** Low Temperature 316 SS Ball Valves *****

Description of BALL Valve (BA01ST540):.. BA15-S16-F540
 Type: Ball, Low Temperature, RP
 Valve Size: NPS 0.5 -to- NPS 6
 Service: Low Temperature -73 to 204 °C (-99 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim: 316 SS
 Seat: RPTFE
 Stem 316 SS
 Body Type: 2-Piece
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Single Reduced (RP)**
 Operation: Lever
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:
 DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor: Ladish (2-Piece, RP): P715-L050-GRnn-N-{Size}

Description of BALL Valve (BA01ST542):.. BA15-S16-F542
 Type: Ball, Low Temperature, RP
 Valve Size: NPS 8 -to- NPS 10
 Service: Low Temperature -73 to 204 °C (-99 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim: 316 SS
 Seat: RPTFE
 Stem 316 SS
 Design: Floating Ball, Fire Safe API 607
 Bore-Port: **Single Reduced (RP)**
 Operation: **Gear**
 Stem-Packing/Seals: Flexible Graphite
 Body Gaskets: Flexible Graphite

The standards are:
 DESIGN: ASME B16.34 or API 608
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10 - Long Pattern

Approved Vendor: Ladish (2-Piece, RP): P715-G050-GRnn-D-N-{Size}

**Description of BALL Valve (BA01ST545):..**

BA15-S16-F545

Type:Ball, Low Temperature, FP
Valve Size:NPS 8 -to- NPS 8
Service:Low Temperature -73 to 204 °C (-99 to 400 °F)
Class:**ASME 150**
Ends:Flanged RF
Body:316 SS (ASTM A182-F316 or ASTM A351-CF8M)
Trim:316 SS
Seat:RPTFE
Stem316 SS
Design:Floating Ball, Fire Safe API 607
Bore-Port:**Full Port (FP)**
Operation:**Gear**
Stem-Packing/Seals:Flexible Graphite
Body Gaskets:Flexible Graphite

The standards are:

DESIGN:ASME B16.34 or API 608
ENDS:ASME B16.5
RATING:ASME B16.34
TESTING:API 598
DIMENSIONAL:ASME B16.10 - Long Pattern

Approved Vendor:Ladish (2-Piece, FP): P815-G050-GRnn-N-{Size}

Note: When Valve Tags are shown as something like "BA03SQ500 == > BA30-S25-F500", it means that the PIP Valve Code (BA03SQ500) should be replaced with the Tesla Valve Code (BA30-S25-F500).

Rev.	Rev. Date	Rev. Description	By	Checked
0	06 SEPT 2023	Issued for release	MFS	SBS
1	15 SEPT 2023	Add Additional Valves	MFS	SBS
2	18 SEPT 2023	Add Duplex Valve	MFS	SBS
3	03 NOV 2023	Add/Revise Valves	MFS	SBS
4	05 FEB 2024	Reorder Valves, Add Compression Valves, Misc. Edits	MFS	SBS
5	26 MAR 2024	Add Valve Tags	MFS	WGS
6	20 MAY 2024	Revise Red. Port Ball Valves & Add BA15-S16-F512	MFS	MFS

***** Start of Resilient Seat Butterfly Valves *****

Description of BUTTERFLY Valve (BUR15-CI-W710)

Type: Butterfly, Resilient Seat, one-piece body, Wafer
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 150**
 Service: General Use, Process Water, Potable Water
 Temperature: -20 °F to +250 °F (-29 °C to +121 °C)
 Rating: [2" to 12" NPS] = 175 psig (12.1 bar) @ 100 °F (38 °C)
 [14" to 24" NPS] = 150 psig (10.3 bar) @ 100 °F (38 °C)
 Ends: **WAFER** (Install between FF Flanges)
 Body: **Cast Iron**, ASTM A126, Class B
 Disc: 316 Stainless Steel, ASTM A351 CF8M
 Stem: 416 Stainless Steel, ASTM A582
 Liner/Seat: EPDM
 Operation: Lever (2" to 4" NPS); Gear (6" to 24" NPS)
 Shut-off: Bidirectional, Bubble-tight
 Stem-Seal: Buna-N p
 Notes: Gaskets NOT required (because seat extends to face of valve)

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 CERTIFICATION: NFS 61, NSF 372
 Face-to-Face: API 609, Category B (Lug Type)

Approved Vendor: Bray, **Series 3W**: 3W-{Size}-11010-{Trim} Trim = 169
 (Wafer, CL150, CI Body, 316 SS Disc, EPDM Seat)

Description of BUTTERFLY Valve (BUR15-DI-L711)

Type: Butterfly, Resilient Seat, One-piece Body, Lug
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 150**
 Service: General Use, Process Water, Potable Water
 Temperature: -20 °F to +250 °F (-29 °C to +121 °C)
 Rating: [2" to 12" NPS] = 175 psig (12.1 bar) @ 100 °F (38 °C)
 [14" to 24" NPS] = 150 psig (10.3 bar) @ 100 °F (38 °C)
 Ends: **LUGGED** (Install between RF Flanges)
 Body: **Ductile Iron**, A536 Gr. 65-45-12
 Disc: 316 Stainless Steel, ASTM A351 CF8M
 Stem: 416 Stainless Steel, A582
 Liner/Seat: EPDM
 Operation: Lever (2" to 4" NPS); Gear (6" to 24" NPS)
 Shut-off: Bidirectional, Bubble-tight
 Stem-Seal: Buna-N
 Notes: Gaskets NOT required (because seat extends to face of valve)

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 CERTIFICATION: NFS 61, NSF 372
 Face-to-Face: API 609, Category B (Lug Type)

Approved Vendor: Bray, **Series 3L**: 3L-{Size}-11010-{Trim} Trim =375
 (Lugged, CL150, Ductile Iron Body, 316 SS Disc, EPDM Seat)

Description of BUTTERFLY Valve (BUR15-DI-L712)

Type: Butterfly, Resilient Seat, One-piece Body, Lug
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 150** (See pressure limits below)
 Service: General Use, Process Water, Potable Water
 Temperature: -20 °F to +250 °F (-29 °C to +121 °C)
 Rating: [2" to 12" NPS] = 175 psig (12 bar) @ 100 °F (38 °C)
 [14" to 24" NPS] = 150 psig (10 bar) @ 100 °F (38 °C)
 Ends: **LUGGED** (Install between RF Flanges)
 Body: **Ductile Iron**, A536 Gr. 65-45-12
 Disc: Ductile Iron with Nylon 11 Coating
 Stem: 416 Stainless Steel, A582
 Liner/Seat: EPDM
 Operation: Lever (2" to 4" NPS); Gear (6" to 24" NPS)
 Shut-off: Bidirectional, Bubble-tight
 Stem-Seal: Buna-N
 Notes: Gaskets NOT required (because seat extends to face of valve)

The standards are:

DESIGN: API 609, Category A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 CERTIFICATION: NFS 61, NSF 372
 Face-to-Face: API 609, Category B (Lug Type)

Approved Vendor: Bray, **Series 3L**: 3L-{Size}-11010-{Trim} Trim =390
 (Lugged, CL150, Ductile Iron Body, DI Disc w/Nylon Coating, EPDM Seat)

Description of BUTTERFLY Valve (BUR15-CI-W701)

Type: Butterfly, Resilient Seat, one-piece body, Wafer
 Valve Size: NPS 3 -to- NPS 20
 Service: Sanitary, Water service
 Class: **ASME 150**
 Temperature: -20 °F to +250 °F (-29 °C to +121 °C)
 Rating: [2" to 12" NPS] = 175 psig (12.1 bar) @ 100 °F (38 °C)
 [14" to 24" NPS] = 150 psig (10.3 bar) @ 100 °F (38 °C)
 Ends: **Wafer** (Install between FF Flanges)
 Body: Cast Iron
 Disc: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Stem: 416 SS
 Seat: **EPDM**
 Operation: Lever (3" to 4" NPS); Gear (6" NPS and larger)
 Shut-off: Bidirectional, Bubble-tight
 Stem Seal: Buna-N
 Notes: Gaskets NOT required (because seat extends to face of valve)

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 CERTIFICATION: NSF 61
 Face-to-Face: API 609, Category B (Wafer Type)

Approved Vendor: Bray, **Series 30:** 30-{Size}-11010-{Trim} Trim = 169
 (Wafer, CL150, Cast Iron Body, 316 SS Disc, **EPDM Seat**)
 Lever (3" to 4" NPS); Gear (6" NPS and larger)

Description of BUTTERFLY Valve (BUR15-DI-L702)

Type: Butterfly, Resilient Seat, one-piece body, Lugged
 Valve Size: NPS 3 -to- NPS 20
 Class: **ASME 150**
 Service: Sanitary, Water service
 Temperature: -20 °F to +250 °F (-29 °C to +121 °C)
 Rating: [2" to 12" NPS] = 175 psig (12.1 bar) @ 100 °F (38 °C)
 [14" to 24" NPS] = 150 psig (10.3 bar) @ 100 °F (38 °C)
 Ends: **Lugged** (Install between RF Flanges)
 Body Mat'l: Ductile Iron
 Body: 2-Piece
 Disc: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Stem: 416 SS
 Seat: **EPDM**
 Operation: Lever (3" to 4" NPS); Gear (6" to 20" NPS)
 Shut-off: Bidirectional, Bubble-tight
 Stem Seal: Buna-N
 Notes: Gaskets NOT required (because seat extends to face of valve)

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 CERTIFICATION: NSF 61
 DIMENSIONS: API 609, Category B (Lug Type)

Approved Vendor: Bray, **Series 31** 31-{Size}-11010-{Trim} Trim = 375
 (Lugged, CL150, Ductile Iron Body, 316 SS Disc, EPDM Seat)
 Lever (3" to 4" NPS); Gear (6" to 20" NPS)

Description of BUTTERFLY Valve (BUR15-S16-W703)

Type: Butterfly, Resilient Seat, 2-piece body, Wafer
 Valve Size: NPS 3 -to- NPS 20
 Class: **ASME 150**
 Temperature: -20 °F to +250 °F (-29 °C to +121 °C)
 Rating: 150 psig (10.3 bar) @ 100 °F (38 °C)
 Ends: **WAFER** (Install between FF Flanges)
 Body: 316 SS (ASTM A351-CF8M)
 Stem: 416 SS
 Seat: PTFE Lined EPDM (with PTFE facing process fluid)
 Stem: Stainless Steel, one-piece
 Operation: **Gear** (Use Lever up to 4" NPS)
 Shut-off: Bidirectional, Bubble-tight
 Stem Seal: Buna-N
 Notes: Gaskets NOT required (because seat extends to face of valve)

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 Face-to-Face: API 609, Category B (Lug Type)

Approved Vendor: Bray, **Series 20** (Wafer, CL150, 316 SS Body, 316 SS Disc/Stem, PTFE Seat) w/Gear Oper (Except as noted above).

Description of BUTTERFLY Valve (BUR15-S16-L704)

Type: Butterfly, Resilient Seat, 2-piece body, Lugged
 Valve Size: NPS 3 -to- NPS 20
 Class: **ASME 150**
 Temperature: -20 °F to +250 °F (-29 °C to +121 °C)
 Rating: 150 psig (10.3 bar) @ 100 °F (38 °C)
 Ends: **Lugged** (Install between RF Flanges)
 Body: 316 SS (ASTM A351-CF8M)
 Stem: 416 SS
 Seat: PTFE Lined EPDM (with PTFE facing process fluid)
 Stem: Stainless Steel, one-piece
 Operation: Lever (3" to 4" NPS); Gear (6" NPS and larger)
 Shut-off: Bidirectional, Bubble-tight
 Stem Seal: Buna-N
 Notes: Gaskets NOT required (because seat extends to face of valve)

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5+
 RATING: ASME B16.34
 TESTING: API 598
 Face-to-Face: API 609, Category B (Lug Type)

Approved Vendor: Bray, **Series 21** (Lugged, CL150, 316 SS Body, 316 SS Disc/Stem, PTFE Seat) w/Gear Oper (Except as noted above).

 ***** End of Resilient Seat Butterfly Valves *****

***** Start of 316 SS, Double Offset Butterfly Valves *****

Description of BUTTERFLY Valve (BF01ST700)

BUD15-S16-W700

Type: Butterfly, Double Offset, High Performance
 Valve Size: NPS 3 -to- NPS 4
 Class: **ASME 150**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 250 psig (17.2 bar) @ 100 °F (37.8 °C)
 Ends: **Wafer** (Install between FF Flanges)
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Disc: 316 SS / RPTFE
 Stem: 17-4 PH SS
 Body Type: **Wafer**
 Design (Other): Fire Safe API 607
 Operation: Lever
 Stem-Packing: Flexible Graphite Rings
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609, Category B

Approved Vendor: Bray (Series 4x): 40-{Size}-11001-068
 w/Lever Oper.

Description of BUTTERFLY Valve (BF01ST701)

BUD15-S16-W701

Type: Butterfly, Double Offset, High Performance
 Valve Size: NPS 6 -to- NPS 24
 Class: **ASME 150**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 250 psig (17.2 bar) @ 100 °F (37.8 °C)
 Ends: **Wafer** (Install between FF Flanges)
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Disc: 316 SS / RPTFE
 Stem: 17-4 PH SS
 Body Type: **Wafer**
 Design (Other): Fire Safe API 607
 Operation: **Gear**
 Stem-Packing: Flexible Graphite Rings
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609, Category B

Approved Vendor: Bray (Series 4x): 40-{Size}-11001-068
 w/Gear Oper.

Description of BUTTERFLY Valve (BF01ST702)

BUD15-S16-L702

Type: Butterfly, Double Offset, High Performance
 Valve Size: NPS 3 -to- NPS 4
 Class: **ASME 150**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 250 psig (17.2 bar) @ 100 °F (37.8 °C)
 Ends: **Lugged** (Install between RF Flanges)
 Body: 316 SS (ASTM A351-CF8M)
 Disc: 316 SS / RPTFE
 Stem: 17-4 PH SS
 Seat: Inconel 718
 Body Type: **Lugged**, Bolt Holes Tapped
 Design (Other): Fire Safe API 607
 Operation: Lever
 Stem-Packing: Flexible Graphite Rings
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609, Category B

Approved Vendor: Bray (Series 4x): 41-{Size}-11001-068
 w/Lever Oper.

Description of BUTTERFLY Valve (BF01ST703)

BUD15-S16-L703

Type: Butterfly, Double Offset, High Performance
 Valve Size: NPS 6 -to- NPS 24
 Class: **ASME 150**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 250 psig (17.2 bar) @ 100 °F (37.8 °C)
 Ends: **Lugged** (Install between RF Flanges)
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Disc: 316 SS / RPTFE
 Stem: 17-4 PH SS
 Body Type: **Lugged**, Bolt Holes Tapped
 Design (Other): Fire Safe API 607
 Operation: **Gear**
 Stem-Packing: Flexible Graphite Rings
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609, Category B

Approved Vendor: Bray (Series 4x): 41-{Size}-11001-068
 w/Gear Oper

Description of BUTTERFLY Valve (BF01ST704)

(BUD15-S16-L724)

Type: Butterfly, Double Offset, High Performance
 Valve size: **NPS 26" -to- NPS 48"**
 Class: **ASME 150**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 250 psig (17.2 bar) @ 100 °F (37.8 °C)
 Ends: **Lugged** (Install between RF Flanges)
 Body: 316 SS (ASTM A351-CF8M)
 Disc: 316 SS / RPTFE
 Stem: 316 SS
 Body Type: **Lugged**, Bolt Holes Tapped
 Design (Other): Fire Safe API 607
 Operation: **Gear**
 Stem-Packing: Flexible Graphite Rings
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN API 609, Category B
 ENDS ASME B16.47, Series B
 RATING MFG STD
 TESTING API 598
 DIMENSIONAL MFG STD

Vendor: Bray (Series 4x): 41-{Size}-11001-068 w/Gear Operator &
 Graphite Packing

Description of BUTTERFLY Valve (BUD15-S16-L725)

Type: Butterfly, Double Offset, High Performance
 Valve size: **NPS 26" -to- NPS 48"**
 Class: **ASME 150**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 250 psig (17.2 bar) @ 100 °F (37.8 °C)
 Ends: **Lugged** (Install between RF Flanges)
 Body: 316 SS (ASTM A351-CF8M)
 Disc: 316 SS / RPTFE
 Stem: 316 SS
 Body Type: **Lugged**, Bolt Holes Tapped
 Design (Other): Fire Safe API 607
 Operation: **Gear**
 Stem-Packing: Flexible Graphite Rings
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN API 609, Category B
 ENDS ASME B16.47, Series A
 RATING MFG STD
 TESTING API 598
 DIMENSIONAL MFG STD

Vendor: Bray (Series 4x): 41-{Size}-11001-068 w/Gear Operator &
 Graphite Packing

Description of BUTTERFLY Valve (BF01ST705)

BUD15-S16-W705

Type: Butterfly, Double Offset, High Performance
 Valve Size: NPS 2.5 -to- NPS 4
 Class: **ASME 150**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 250 psig (17.2 bar) @ 100 °F (37.8 °C)
 Ends: **Wafer** (Install between FF Flanges)
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Disc: 316 SS / RPTFE
 Stem: 17-4 PH SS
 Body Type: **Wafer**
 Design (Other): Not Fire Safe
 Operation: Lever
 Stem-Packing: PTFE
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609, Category B

Approved Vendor: Bray (Series 4x): 40-{Size}-11001-066
 w/Lever Oper & PTFE Packing

Description of BUTTERFLY Valve (BF01ST706)

BUD15-S16-W706

Type: Butterfly, Double Offset, High Performance
 Valve Size: NPS 6 -to- NPS 24
 Class: **ASME 150**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 250 psig (17.2 bar) @ 100 °F (37.8 °C)
 Ends: **Wafer** (Install between FF Flanges)
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Disc: 316 SS / RPTFE
 Stem: 17-4 PH SS
 Body Type: **Wafer**
 Design (Other): Not Fire Safe
 Operation: **Gear**
 Stem-Packing: PTFE
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609, Category B

Approved Vendor: Bray (Series 4x): 40-{Size}-11001-066
 w/Gear Oper & PTFE Packing

Description of BUTTERFLY Valve (BF01ST707)

BUD15-S16-L707

Type: Butterfly, Double Offset, High Performance
 Valve Size: NPS 2.5 -to- NPS 4
 Class: **ASME 150**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 250 psig (17.2 bar) @ 100 °F (37.8 °C)
 Ends: **Lugged** (Install between RF Flanges)
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Disc: 316 SS / RPTFE
 Stem: 17-4 PH SS
 Body Type: **Lugged**, Bolt Holes Tapped
 Design (Other): Not Fire Safe
 Operation: Lever
 Stem-Packing: PTFE
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609, Category B

Approved Vendor: Bray (Series 4x): 41-{Size}-11001-068
 w/Lever Oper & PTFE Packing

Description of BUTTERFLY Valve (BF01ST708)

BUD15-S16-L708

Type: Butterfly, Double Offset, High Performance
 Valve Size: NPS 6 -to- NPS 24
 Class: **ASME 150**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 250 psig (17.2 bar) @ 100 °F (37.8 °C)
 Ends: **Lugged** (Install between RF Flanges)
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Disc: 316 SS / RPTFE
 Stem: 17-4 PH SS
 Body Type: **Lugged**, Bolt Holes Tapped
 Design (Other): Not Fire Safe
 Operation: **Gear**
 Stem-Packing: PTFE
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609, Category B

Approved Vendor: Bray (Series 4x): 41-{Size}-11001-066
 w/Gear Oper & PTFE Packing

Description of BUTTERFLY Valve (BF03ST701)

BUD30-S16-W701

Type: Butterfly, Double Offset, High Performance
 Valve Size: NPS 3 -to- NPS 24
 Class: **ASME 300**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 720 psig (49.6 bar) @ 100 °F (37.8 °C)
 Ends: **Wafer** (Install between FF Flanges)
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Disc: 316 SS / RPTFE
 Stem: 17-4 PH SS
 Body Type: **Wafer**
 Design (Other): Fire Safe API 607
 Operation: **Gear**
 Stem-Packing: Flexible Graphite Rings
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598 DIMENSIONAL:
 API 609, Category B

Approved Vendor: Bray (Series 4x): 42-{Size}-11001-068,
 w/Gear Oper & Graphite Stem Packing

Description of BUTTERFLY Valve (BF03ST703)

BUD30-S16-L703

Type: Butterfly, Double Offset, High Performance
 Valve Size: NPS 3 -to- NPS 24
 Class: **ASME 300**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 720 psig (49.6 bar) @ 100 °F (37.8 °C)
 Ends: **Lugged** (Install between RF Flanges)
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Disc: 316 SS / RPTFE
 Stem: 17-4 PH SS
 Body Type: **Lugged**, Bolt Holes Tapped
 Design (Other): Fire Safe API 607
 Operation: **Gear**
 Stem-Packing: Flexible Graphite Rings
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609, Category B

Approved Vendor: Bray (Series 4x): 43-{Size}-11001-068,
 w/Gear Oper & Graph. Packing

Description of BUTTERFLY Valve (BF03ST706)

BUD30-S16-W706

Type: Butterfly, Double Offset, High Performance
 Valve Size: NPS 3 -to- NPS 24
 Class: **ASME 300**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 720 psig (49.6 bar) @ 100 °F (37.8 °C)
 Ends: **Wafer** (Install between FF Flanges)
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Disc: 316 SS / RPTFE
 Stem: 17-4 PH SS
 Body Type: **Wafer**
 Design (Other): Not Fire Safe
 Operation: **Gear**
 Stem-Packing: PTFE
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609, Category B

Approved Vendor: Bray (Series 4x): 42-{Size}-11001-068
 w/Gear Oper & PTFE Stem Packing

Description of BUTTERFLY Valve (BF03ST708)

BUD30-S16-W708

Type: Butterfly, Double Offset, High Performance
 Valve Size: NPS 3 -to- NPS 24
 Class: **ASME 300**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 720 psig (49.6 bar) @ 100 °F (37.8 °C)
 Ends: **Lugged** (Install between RF Flanges)
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Disc: 316 SS / RPTFE
 Stem: 17-4 PH SS
 Body Type: **Lugged**, Bolt Holes Tapped
 Design (Other): Not Fire Safe
 Operation: **Gear**
 Stem-Packing: PTFE
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609, Category B

Approved Vendor: Bray (Series 4x): 43-{Size}-11001-068,
 w/Gear Oper & PTFE Stem Packing

Description of BUTTERFLY Valve (BUD15-S16-F520)

Type: Butterfly, Double Offset, Double-Flanged
 Valve Size: NPS 6 -to- NPS 20
 Class: **ASME 150**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 250 psig (17.2 bar) @ 100 °F (37.8 °C)
 Ends: **Double-Flanged** (Install between RF Flanges)
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Disc: 316 SS (with RPTFE Seat)
 Stem: 17-4 PH SS
 Body Style: **Double-Flanged**
 Design (Other): Fire Safe API 607
 Operation: **Gear**
 Stem-Packing: Flexible Graphite
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609, Category B

Approved Vendor: Bray (Series 4x): 4A-{Size}-11001-068 ("068" = Fire Safe Trim)
 w/Gear Oper

***** End of 316 SS, Double Offset Butterfly Valves *****

***** Start of DUPLEX SS Double Offset Butterfly Valves *****

Description of BUTTERFLY Valve (BF01SP702)

BUD15-S22-L702

Type: Butterfly, Double Offset, High Performance
 Valve Size: NPS 3 -to- NPS 4
 Class: **ASME 150**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 250 psig (17.2 bar) @ 100 °F (37.8 °C)
 Ends: **Lugged** (Install between RF Flanges)
 Body: 2205 Duplex SS (ASTM A995-4A (UNS J92205))
 Disc: 2205 Duplex SS (ASTM A995-4A (UNS J92205))
 Stem: Duplex SS (ASTM A479 (UNS 32750))
 Seat : RPTFE with Resilient Engergizer
 Body Type: **Lugged**, Bolt Holes Tapped
 Design (Other): Fire Safe API 607
 Operation: Lever
 Stem-Packing: Flexible Graphite Rings
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609, Category B

Approved Vendor: Bray (Series 4x): 41-{Size}-11001-{Trim} Trim = "OMV"
 w/Lever Oper.

Description of BUTTERFLY Valve (BF01SP703)

BUD15-S22-L703

Type: Butterfly, Double Offset, High Performance
 Valve Size: NPS 6 -to- NPS 24
 Class: **ASME 150**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 250 psig (17.2 bar) @ 100 °F (37.8 °C)
 Ends: **Lugged** (Install between RF Flanges)
 Body: 2205 Duplex SS (ASTM A995-4A (UNS J92205))
 Disc: 2205 Duplex SS (ASTM A995-4A (UNS J92205))
 Stem: Duplex SS (ASTM A479 (UNS 32750))
 Seat : RPTFE with Resilient Engergizer
 Body Type: **Lugged**, Bolt Holes Tapped
 Design (Other): Fire Safe API 607
 Operation: **Gear**
 Stem-Packing: Flexible Graphite Rings
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609, Category B

Approved Vendor: Bray (Series 4x): 41-{Size}-11001-{Trim} Trim = "OMV"
 w/Gear Oper

Description of BUTTERFLY Valve (BF03SP702)

BUD30-S22-L702

Type: Butterfly, Double Offset, High Performance
 Valve Size: NPS 3 -to- NPS 4
 Class: **ASME 300**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 720 psig (49.6 bar) @ 100 °F (37.8 °C)
 Ends: **Lugged** (Install between RF Flanges)
 Body: 2205 Duplex SS (ASTM A995-4A (UNS J92205))
 Disc: 2205 Duplex SS (ASTM A995-4A (UNS J92205))
 Stem: Duplex SS (ASTM A479 (UNS 32750))
 Seat : RPTFE with Resilient Engergizer
 Body Type: **Lugged**, Bolt Holes Tapped
 Design (Other): Fire Safe API 607
 Operation: Lever
 Stem-Packing: Flexible Graphite Rings
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609, Category B

Approved Vendor: Bray (Series 4x): 43-{Size}-11001-{Trim} Trim = "OMV"
 w/Lever Oper.

Description of BUTTERFLY Valve (BF03SP703)

BUD30-S22-L703

Type: Butterfly, Double Offset, High Performance
 Valve Size: NPS 6 -to- NPS 24
 Class: **ASME 300**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 720 psig (49.6 bar) @ 100 °F (37.8 °C)
 Ends: **Lugged** (Install between RF Flanges)
 Body: 2205 Duplex SS (ASTM A995-4A (UNS J92205))
 Disc: 2205 Duplex SS (ASTM A995-4A (UNS J92205))
 Stem: Duplex SS (ASTM A479 (UNS 32750))
 Seat : RPTFE with Resilient Engergizer
 Body Type: **Lugged**, Bolt Holes Tapped
 Design (Other): Fire Safe API 607
 Operation: Gear
 Stem-Packing: Flexible Graphite Rings
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609, Category B

Approved Vendor: Bray (Series 4x): 43-{Size}-11001-{Trim} Trim = "OMV"
 w/Gear Oper

Description of BUTTERFLY Valve (BUD15-S22-W710)

Type: Butterfly, Double Offset, High Performance
Valve Size: NPS 3 -to- NPS 4
Class: **ASME 150**
Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
Rating: 250 psig (17.2 bar) @ 100 °F (37.8 °C)
Ends: **Wafer** (Install between FF Flanges)
Body: 2205 Duplex SS (ASTM A995-4A (UNS J92205))
Disc: 2205 Duplex SS (ASTM A995-4A (UNS J92205))
Stem: Duplex SS (ASTM A479 (UNS 32750))
Seat : RPTFE with Resilient Engergizer
Body Type: **Wafer**
Design (Other): Fire Safe API 607
Operation: Lever
Stem-Packing: Flexible Graphite
Notes: 1. Gaskets **are** required on each side of Butterfly Valve
..... 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: API 609, Category B

Approved Vendor: Bray (Series 4x, Wafer): 40-{Size}-11001-{Trim Code} Trim = "OMV"
..... w/Lever Oper.

Description of BUTTERFLY Valve (BUD15-S22-W711)

Type: Butterfly, Double Offset, High Performance
Valve Size: NPS 6 -to- NPS 24
Class: **ASME 150**
Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
Rating: 250 psig (17.2 bar) @ 100 °F (37.8 °C)
Ends: **Wafer** (Install between FF Flanges)
Body: 2205 Duplex SS (ASTM A995-4A (UNS J92205))
Disc: 2205 Duplex SS (ASTM A995-4A (UNS J92205))
Stem: Duplex SS (ASTM A479 (UNS 32750))
Seat : RPTFE with Resilient Engergizer
Body Type: **Wafer**
Design (Other): Fire Safe API 607
Operation: **Gear**
Stem-Packing: Flexible Graphite
Notes: 1. Gaskets **are** required on each side of Butterfly Valve
..... 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: API 609, Category B

Approved Vendor: Bray (Series 4x, Wafer): 40-{Size}-11001-{Trim Code} Trim = "OMV"
..... w/Gear Oper

***** Start of Super Duplex Double-Offset Butterfly Valves *****

Description of BUTTERFLY Valve (BUD15-S25-W700)

Type: Butterfly, Double Offset, High Performance
 Valve Size: NPS 3 -to- NPS 4
 Class: **ASME 150**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 250 psig (17.2 bar) @ 100 °F (37.8 °C)
 Ends: **Wafer** (Install between FF Flanges)
 dy: 2507 SS (ASTM A995-6A))
 Disc: 2507 SS / RPTFE
 Stem: SS
 Body Type: **Wafer**
 Design (Other): Fire Safe API 607
 Operation: Lever
 Stem-Packing: Flexible Graphite Rings
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609, Category B

Approved Vendor: Bray (Series 4x): 40-{Size}-11001-nnn
 w/Lever Oper.

Description of BUTTERFLY Valve (BUD15-S25-W701)

Type: Butterfly, Double Offset, High Performance
 Valve Size: NPS 6 -to- NPS 24
 Class: **ASME 150**
 Temperature: -20 °F to + 400 °F (-29 °C to +204 °C)
 Rating: 250 psig (17.2 bar) @ 100 °F (37.8 °C)
 Ends: **Wafer** (Install between FF Flanges)
 Body: 2507 SS (ASTM A995-6A)
 Disc: 2507 SS / RPTFE
 Stem: 17-4 PH SS
 Body Type: **Wafer**
 Design (Other): Fire Safe API 607
 Operation: **Gear**
 Stem-Packing: Flexible Graphite Rings
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609, Category B

Approved Vendor: Bray (Series 4x): 40-{Size}-11001-nnn
 w/Gear Oper.

***** End of Super Duplex SS, Double Offset Butterfly Valves *****

***** Start of Triple-Offset Butterfly Valves *****

Description of BUTTERFLY Valve (BF01SP712)

BUT15-S22-L712

Type: Butterfly, Triple Offset, High Performance
 Valve Size: NPS 3 -to- NPS 4
 Class: **ASME 150**
 Temperature: -20 °F to + 500 °F (-29 °C to +260 °C)
 Rating: 250 psig (17.2 bar) @ 100 °F (37.8 °C)
 Ends: Lug Style (Install between RF Flanges)
 Body: 2205 Duplex SS (ASTM A995-4A (UNS J92205))
 Body Seat: Duplex (UNS 31803)
 Disc: 2205 Duplex SS (ASTM A995-4A (UNS J92205))
 Seal Ring: Duplex SS, UNS 31803 + Graphite
 Stem: Duplex SS, ASTM A479 (UNS 32760)
 Body Type: Lugged (Bolt holes tapped)
 Design (Other): Fire Safe API 607
 Operation: Lever
 Stem-Packing: Flexible Graphite
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609, Category B

Approved Vendor: Bray, Tri-Lok Triple Offset: L0-{Size}-11075-{Trim} Trim = "RAE"
 w/Lever Oper

Description of BUTTERFLY Valve (BF01SP713)

BUT15-S22-L713

Type: Butterfly, Triple Offset, High Performance
 Valve Size: NPS 6 -to- NPS 24
 Class: **ASME 150**
 Temperature: -20 °F to + 500 °F (-29 °C to +260 °C)
 Rating: 250 psig (17.2 bar) @ 100 °F (37.8 °C)
 Ends: Lug Style (Install between RF Flanges)
 Body: 2205 Duplex SS (ASTM A995-4A (UNS J92205))
 Body Seat: Duplex (UNS 31803)
 Disc: 2205 Duplex SS (ASTM A995-4A (UNS J92205))
 Seal Ring: Duplex SS, UNS 31803 + Graphite
 Stem: Duplex SS, ASTM A479 (UNS 32760)
 Body Type: **Lugged** (Bolt holes tapped)
 Design (Other): Fire Safe API 607
 Operation: **Gear**
 Stem-Packing: Flexible Graphite
 Notes: 1. Gaskets **are** required on each side of Butterfly Valve
 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 609, Category B

Approved Vendor: Bray, Tri-Lok Triple Offset: L0-{Size}-11075-{Trim} Trim = "RAE"
 w/Gear Oper

Description of BUTTERFLY Valve (BF03SP712)

BUT30-S22-L712

Type: Butterfly, Triple Offset, High Performance
Valve Size: NPS 3 -to- NPS 4
Class: **ASME 300**
Temperature: -20 °F to + 500 °F (-29 °C to +260 °C)
Rating: 720 psig (49.6 bar) @ 100 °F (37.8 °C)
Ends: Lug Style (Install between RF Flanges)
Body: 2205 Duplex SS (ASTM A995-4A (UNS J92205))
Body Seat: Duplex (UNS 31803)
Disc: 2205 Duplex SS (ASTM A995-4A (UNS J92205))
Seal Ring: Duplex SS, UNS 31803 + Graphite
Stem: Duplex SS, ASTM A479 (UNS 32760)
Body Type: **Lugged** (Bolt holes tapped)
Design (Other): Fire Safe API 607
Operation: Lever
Stem-Packing: Flexible Graphite
Notes: 1. Gaskets **are** required on each side of Butterfly Valve
..... 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: API 609, Category B

Approved Vendor: Bray, Tri-Lok Triple Offset: L1-{Size}-11075-{Trim} Trim = "RAE"
..... w/Lever Oper

Description of BUTTERFLY Valve (BF03SP713)

BUT30-S22-L713

Type: Butterfly, Triple Offset, High Performance
Valve Size: NPS 6 -to- NPS 24
Class: **ASME 300**
Temperature: -20 °F to + 500 °F (-29 °C to +260 °C)
Rating: 720 psig (49.6 bar) @ 100 °F (37.8 °C)
Ends: Lug Style (Install between RF Flanges)
Body: 2205 Duplex SS (ASTM A995-4A (UNS J92205))
Body Seat: Duplex (UNS 31803)
Disc: 2205 Duplex SS (ASTM A995-4A (UNS J92205))
Seal Ring: Duplex SS, UNS 31803 + Graphite
Stem: Duplex SS, ASTM A479 (UNS 32760)
Body Type: **Lugged** (Bolt holes tapped)
Design (Other): Fire Safe API 607
Operation: **Gear**
Stem-Packing: Flexible Graphite
Notes: 1. Gaskets **are** required on each side of Butterfly Valve
..... 2. Seat Retainer to be located downstream unless otherwise noted

The standards are:

DESIGN: API 609, Category B
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: API 609, Category B

Approved Vendor: Bray, Tri-Lok Triple Offset: L1-{Size}-11075-{Trim} Trim = "RAE"
..... w/Gear Oper

***** End of Triple Offset Butterfly Valves *****



Rev.	RevDate	RevDescription	By	Checked
0	06 SEPT 2023	Issued for release	MFS	SBS
1	08 NOV 2023	Add Resilient & Triple Offset Valves	MFS	SBS
2	26 MAR 2024	Add Valve Tags, Add Section Headers	MFS	WGS

***** 316 SS Dual-Plate Check Valves *****

Description of CHECK Valve (CD01ST510): CHD15-S16-B510

Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 8 -to- NPS 24
 Service: -29 to 316 °C (-20 to 600 °F)
 Class: **ASME 150**
 Ends: **Double-Flanged, RF**
 Body: 316 SS (ASTM A351-CF8M)
 Trim: API 594 **TRIM 12**
 Plate & Hinge Pin: 316 SS
 Seat: Hard Face Seat
 Springs: Inconel X-750
 Stop Pin: 316 SS
 Body Type: **Double Flanged**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H15-CPF3-385E

Description of CHECK Valve (CD01ST512): CHD15-S16-B512

Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 8 -to- NPS 24
 Service: -29 to 316 °C (-20 to 600 °F)
 Class: **ASME 150**
 Ends: Double-Flanged, RF
 Body: 316 SS (ASTM A351-CF8M)
 Trim: API 594 **TRIM 10**
 Plate & Hinge Pin: 316 SS
 Seat: 316 SS Overlay
 Springs: Inconel X-750
 Stop Pin: 316 SS
 Body Type: **Double-Flanged**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek, Dbl-flanged): {Size}-H15-CPF3-233

Description of CHECK Valve (CD03ST510):

CHD30-S16-B510

Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 12 -to- NPS 24
 Service: -29 to 316 °C (-20 to 600 °F)
 Class: **ASME 300**
 Ends: **Double Flanged, RF**
 Body: 316 SS (ASTM A351-CF8M)
 Trim: API 594 **TRIM 12** (316 SS & Hard Faced)
 Plate & Hinge Pin: 316 SS
 Seat: Hard Face Seat
 Springs: Inconel X-750
 Stop Pin: 316 SS
 Body Type: **Double Flanged**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H30-CPF3-385E

Description of CHECK Valve (CD03ST512):

CHD30-S16-B512

Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 10 -to- NPS 12
 Service: -29 to 316 °C (-20 to 600 °F)
 Class: **ASME 300**
 Ends: **Double-Flanged, RF**
 Body: 316 SS (ASTM A351-CF8M)
 Trim: API 594 **TRIM 10** (316 SS Trim)
 Plate & Hinge Pin: 316 SS
 Springs: Inconel X-750
 Stop Pin: 316 SS
 Body Type: **Double Flanged**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H30-CPF3-233

Description of CHECK Valve (CD06ST510):

CHD60-S16-B510

Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 12 -to- NPS 24
 Service: -29 to 316 °C (-20 to 600 °F)
 Class: **ASME 600**
 Ends: Double Flanged, RF
 Body: 316 SS (ASTM A351-CF8M)
 Trim: API 594 **TRIM 12** (316 SS & Hard Faced)
 Plate & Hinge Pin: 316 SS
 Seat: Hard Face Seat
 Springs: Inconel X-750
 Stop Pin: 316 SS
 Body Type: **Double Flanged**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H60-CPF3-385E

Description of CHECK Valve (CD06ST512):

CHD60-S16-B512

Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 12 -to- NPS 24
 Service: -29 to 316 °C (-20 to 600 °F)
 Class: **ASME 600**
 Ends: **Double-Flanged, RF**
 Body: 316 SS (ASTM A351-CF8M)
 Trim: API 594 **TRIM 10** (316 SS Trim)
 Plate & Hinge Pin: 316 SS
 Springs: Inconel X-750
 Stop Pin: 316 SS
 Body Type: **Double Flanged**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H60-CPF3-233

Description of CHECK Valve (CD01ST700): CHD15-S16-W700
 Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Service: -29 to 316 °C (-20 to 600 °F)
 Class: **ASME 150**
 Ends: **Wafer** (Install between RF Flanges)
 Body: 316 SS (ASTM A351-CF8M)
 Trim: API 594 **TRIM 10**
 Plate & Hinge Pin: 316 SS
 Springs: Alloy X-750
 Stop Pin: 316 SS
 Body Type: **Wafer**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H15-CPF-233

Description of CHECK Valve (CD01ST702): CHD15-S16-L702
 Note: For Sizes NPS 8" to 24" on CD01ST702, use Valve tag: CD01ST512 (which is Double-Flanged)
 Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 6
 Service: -29 to 316 °C (-20 to 600 °F)
 Class: **ASME 150**
 Ends: Lugged (Install between RF Flanges)
 Body: 316 SS (ASTM A351-CF8M)
 Trim: API 594 **TRIM 10**
 Plate & Hinge Pin: 316 SS
 Springs: Inconel X-750
 Stop Pin: 316 SS
 Body Type: **Lugged (Thru holes)**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H15-CPF2-233

**Description of CHECK Valve (CD03ST700):**

CHD30-S16-W700

Type: Check, Dual Plate, Retainerless
Valve Size: NPS 2 -to- NPS 24
Service: -29 to 316 °C (-20 to 600 °F)
Class: **ASME 300**
Ends: Wafer (Install between RF Flanges)
Body: 316 SS (ASTM A351-CF8M)
Trim: API 594 **TRIM 10**
Plate & Hinge Pin: 316 SS
Springs: Inconel X-750
Stop Pin: 316 SS
Body Type: **Wafer**
Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H30-CPF-233

Description of CHECK Valve (CD03ST702):

CHD30-S16-L702

Note: For Sizes NPS 12" to 24" on CD03ST702, use Valve tag: CD03ST512 (which is Double-Flanged)

Type: Check, Dual Plate, Retainerless
Valve Size: NPS 2 -to- NPS 10
Service: -29 to 316 °C (-20 to 600 °F)
Class: **ASME 300**
Ends: **Lugged** (Install between RF Flanges)
Body: 316 SS (ASTM A351-CF8M)
Trim: API 594 **TRIM 10**
Plate & Hinge Pin: 316 SS
Springs: Inconel X-750
Stop Pin: 316 SS
Body Type: **Lugged (Thru holes)**
Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H30-CPF2-233

Description of CHECK Valve (CD06ST700):

CHD60-S16-W700

Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Service: -29 to 316 °C (-20 to 600 °F)
 Class: **ASME 600**
 Ends: Wafer (Install between RF Flanges)
 Body: 316 SS (ASTM A351-CF8M)
 Trim: API 594 **TRIM 10**
 Plate & Hinge Pin: 316 SS
 Springs: Inconel X-750
 Stop Pin: 316 SS
 Body Type: **Wafer**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H60-CPF-233

Description of CHECK Valve (CD06ST702):

CHD60-S16-L702

Note: For Sizes NPS 12" to 24" on CD06ST702, use Valve tag: CD06ST512 (which is Double-Flanged)

Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 10
 Service: -29 to 316 °C (-20 to 600 °F)
 Class: **ASME 600**
 Ends: **Lugged** (Install between RF Flanges)
 Body: 316 SS (ASTM A351-CF8M)
 Trim: API 594 **TRIM 10**
 Plate & Hinge Pin: 316 SS
 Springs: Inconel X-750
 Stop Pin: 316 SS
 Body Type: **Lugged (Thru holes)**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H60-CPF2-233

***** Duplex and Super Duplex Dual-Plate Check Valves *****

Description of CHECK Valve (CD01SP500): CHD15-S22-B500

Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 8 -to- NPS 24
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: **Double Flanged, RF**
 Body: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Trim: 2205 Duplex SS
 Hinge Pin: 2205 Duplex SS
 Springs: Inconel X-750
 Stop Pin: 2205 Duplex SS
 Body Type: **Double Flanged**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H15-DZPF3-9

Description of CHECK Valve (CD01SQ500): CHD15-S25-B500

Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: **Double Flanged, RF**
 Body: 2507 Duplex SS (ASTM A995-6A)
 Trim: 2507 Duplex SS
 Springs: Alloy X-750
 Stop Pin: 2507 Duplex SS
 Body Type: **Double Flanged**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H15-EBPF3-9

Description of CHECK Valve (CD03SP500):

CHD30-S22-B500

Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 12 -to- NPS 24
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: **Double Flanged RF**
 Body: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Trim: 2205 Duplex SS
 Hinge Pin: 2205 Duplex SS
 Springs: Inconel X-750
 Stop Pin: 2205 Duplex SS
 Body Type: Double Flanged
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H30-DZPF3-9

Description of CHECK Valve (CD06SP500):

CHD60-S22-B500

Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 12 -to- NPS 24
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: **Double Flanged, RF**
 Body: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Trim: 2205 Duplex SS
 Hinge Pin: 2205 Duplex SS
 Springs: Inconel X-750
 Stop Pin: 2205 Duplex SS
 Body Type: Double Flanged
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H60-DZPF2-9

Description of CHECK Valve (CD01SP700):

CHD15-S22-W700

Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Wafer (Install between RF Flanges)
 Body: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Trim: 2205 Duplex SS
 Hinge Pin: 2205 Duplex SS
 Springs: Inconel X-750
 Stop Pin: 2205 Duplex SS
 Body Type: **Wafer**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H15-DZPF-9

Description of CHECK Valve (CD01SP701):

CHD15-S22-L701

Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 3 -to- NPS 24
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Ends: Lugged (Install between RF Flanges)
 Body: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Trim: 2205 Duplex SS
 Hinge Pin: 2205 Duplex SS
 Springs: Inconel X-750
 Stop Pin: 2205 Duplex SS
 Body Type: **Lugged**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H15-DZPF-9

Description of CHECK Valve (CD01SQ700): CHD15-S25-W700
 Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Temperature: -29 to 205 °C (-20 to 400 °F)
 Ends: **Wafer** (Install between FF Flanges)
 Body: 2507 Duplex SS (ASTM A995-6A)
 Trim: 2507 Duplex SS
 Springs: Alloy X-750
 Stop Pin: 2507 Duplex SS
 Body Type: **Wafer**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H15-EBPF-9

Description of CHECK Valve (CD01SQ701): CHD15-S25-L701
 Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 150**
 Temperature: -29 to 205 °C (-20 to 400 °F)
 Ends: **Lugged**, (Install between RF Flanges)
 Body: 2507 Duplex SS (ASTM A995-6A)
 Trim: 2507 Duplex SS
 Springs: Alloy X-750
 Stop Pin: 2507 Duplex SS
 Body Type: **Lugged**, Tapped holes
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H15-EBPF2-9

Description of CHECK Valve (CD03SP700):

CHD30-S22-W700

Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: Wafer (Install between RF Flanges)
 Body: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Trim: 2205 Duplex SS
 Hinge Pin: 2205 Duplex SS
 Springs: Inconel X-750
 Stop Pin: 2205 Duplex SS
 Body Type: **Wafer**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H30-DZPF-9

Description of CHECK Valve (CD03SP701):

CHD30-S22-L701

Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: Lugged (Install between RF Flanges)
 Body: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Trim: 2205 Duplex SS
 Hinge Pin: 2205 Duplex SS
 Springs: Inconel X-750
 Stop Pin: 2205 Duplex SS
 Body Type: **Lugged**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H30-DZPF2-9

Description of CHECK Valve (CD03SQ700):

CHD30-S25-W700

Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: Wafer (Install between RF Flanges)
 Body: 2507 Super Duplex SS (ASTM A182-F53 or ASTM A995-6A)
 Trim: 2507 Super Duplex SS
 Hinge Pin: 2507 Super Duplex SS
 Springs: Inconel X-750
 Stop Pin: 2507 Super Duplex SS
 Body Type: **Wafer**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek, Wafer): {Size}-H30-EB-PF-9

Description of CHECK Valve (CD03SQ701):

CHD30-S25-L701

Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 300**
 Ends: Lugged (Install between RF Flanges)
 Body: 2507 Super Duplex SS (ASTM A182-F53 or ASTM A995-6A)
 Trim: 2507 Super Duplex SS
 Hinge Pin: 2507 Super Duplex SS
 Springs: Inconel X-750
 Stop Pin: 2507 Super Duplex SS
 Body Type: **Lugged** (with through-bolt holes in body)
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek, Lug): {Size}-H30-EB-PF2-9

Description of CHECK Valve (CD06SP700):

CHD60-S22-W700

Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: Wafer (Install between RF Flanges)
 Body: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Trim: 2205 Duplex SS
 Hinge Pin: 2205 Duplex SS
 Springs: Inconel X-750
 Stop Pin: 2205 Duplex SS
 Body Type: **Wafer**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H60-DZPF-9

Description of CHECK Valve (CD06SP701):

CHD60-S22-L701

Type: Check, Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Service: -29 to 205 °C (-20 to 400 °F)
 Class: **ASME 600**
 Ends: Lugged (Install between RF Flanges)
 Body: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Trim: 2205 Duplex SS
 Hinge Pin: 2205 Duplex SS
 Springs: Inconel X-750
 Stop Pin: 2205 Duplex SS
 Body Type: **Lugged**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H60-DZPF2-9



***** SMO SS Dual-Plate Check Valves *****

Description of CHECK Valve (CD01SV500): CHD15-SMO-B500

Type: Check, Dual Plate, Retainerless
Design Type: Dual Plate, Retainerless
Valve Size: NPS 8 -to- NPS 24
Class: **ASME 150**
Ends: **Double Flanged, RF**
Body: 254 SMO SS (ASTM A351-CK3MCuN)
Trim: 254 SMO SS (ASTM A182-F44 or ASTM A351-CK3MCuN)
Hinge Pin: 254 SMO SS
Springs: Inconel X-750
Stop Pin: 254 SMO SS
Body Type: Double Flanged
Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H15-EAPF3-9

Description of CHECK Valve (CD03SV500): CHD30-SMO-B500

Type: Check, Dual Plate, Retainerless
Design Type: Dual Plate, Retainerless
Valve Size: NPS 12 -to- NPS 24
Class: **ASME 300**
Ends: **Double Flanged, RF**
Body: 254 SMO SS (ASTM A351-CK3MCuN)
Trim: 254 SMO SS (ASTM A182-F44 or ASTM A351-CK3MCuN)
Hinge Pin: 254 SMO SS
Springs: Inconel X-750
Stop Pin: 254 SMO SS
Body Type: Double Flanged
Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H30-EAPF3-9

Description of CHECK Valve (CD01SV700):

CHD15-SMO-W700

Type: Check, Dual Plate, Retainerless
 Design Type: Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 150**
 Ends: Wafer (Install between RF Flanges)
 Body: 254 SMO SS (ASTM A351-CK3MCuN)
 Trim: 254 SMO SS (ASTM A182-F44 or ASTM A351-CK3MCuN)
 Hinge Pin: 254 SMO SS
 Springs: Inconel X-750
 Stop Pin: 254 SMO SS
 Body Type: **Wafer**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H15-EAPF3-9

Description of CHECK Valve (CD01SV701):

CHD15-SMO-L701

Type: Check, Dual Plate, Retainerless
 Design Type: Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 150**
 Ends: Lugged (Install between RF Flanges)
 Body: 254 SMO SS (ASTM A351-CK3MCuN)
 Trim: 254 SMO SS (ASTM A182-F44 or ASTM A351-CK3MCuN)
 Hinge Pin: 254 SMO SS
 Springs: Inconel X-750
 Stop Pin: 254 SMO SS
 Body Type: **Lugged**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H15-EAPF2-9

Description of CHECK Valve (CD03SV700):

CHD30-SMO-W700

Type: Check, Dual Plate, Retainerless
 Design Type: Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 300**
 Ends: Wafer (Install between RF Flanges)
 Body: 254 SMO SS (ASTM A351-CK3MCuN)
 Trim: 254 SMO SS (ASTM A182-F44 or ASTM A351-CK3MCuN)
 Hinge Pin: 254 SMO SS
 Springs: Inconel X-750
 Stop Pin: 254 SMO SS
 Body Type: **Wafer**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H30-EAPF-9

Description of CHECK Valve (CD03SV701):

CHD30-SMO-L701

Type: Check, Dual Plate, Retainerless
 Design Type: Dual Plate, Retainerless
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 300**
 Ends: Lugged (Install between RF Flanges)
 Body: 254 SMO SS (ASTM A351-CK3MCuN)
 Trim: 254 SMO SS (ASTM A182-F44 or ASTM A351-CK3MCuN)
 Hinge Pin: 254 SMO SS
 Springs: Inconel X-750
 Stop Pin: 254 SMO SS
 Body Type: **Lugged**
 Installation: Horizontal - Hinge Pin Vertical or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type A
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: API 594

Approved Vendor: Crane (Duo-Chek): {Size}-H30-EAPF2-9

 ***** End of Dual-Plate Check Valves *****

***** Silent Check Valves *****

Description of CHECK Valve (CHN15-S16-W710):

Type: Check, Silent (Wafer)
 Sub Type: Center Guided, Non-Return
 Valve Size: NPS 1/2" -to- NPS 4" (DN 15 to DN100)
 Class: **ASME 150**
 Temperature: -29 to 205 °C (-20 to 400 °F)
 Rating: 232 psig (16 bar = 16 PN)
 Ends: **Wafer** (Install between RF Flanges)
 Body: 316 SS (ASTM A351, CF8M)
 Disc: 316 SS
 Seat: Metal Seat (Match Disc)
 Springs: Alloy X-750
 Stop Pin: 316 SS
 Body Type: **Wafer**
 Installation: Horizontal Flow or Vertical Upward Flow

The standards are:

DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: Mfg. Std.

Approved Vendor: Gestra (CL150, Silent, Center-Guided, Wafer): {Size}-RK-86

Description of CHECK Valve (CHN15-S16-W712):

Type: Check, Silent, Center-Guided, **Metal Seat**, (Wafer)
 Sub Type: Non-Return
 Valve Size: NPS 2" -to- NPS 12" (DN 50 to DN300)
 Class: **ASME 150**
 Temperature: -29 to 205 °C (-20 to 400 °F)
 Rating: 275 psig @ 100 °F (175 psig @ 400 °F)
 Ends: **Wafer** (Install between RF Flanges)
 Body: 316 SS (ASTM A351, CF8M), Epoxy coated
 Trim: 316 SS
 Seat: Metal Seat, Same as Disc
 Springs: 316 SS
 Body Type: **Wafer**
 Open Area: 110% of Pipe Size area
 Installation: Horizontal Flow or Vertical Upward Flow

The standards are:

DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: Mfg. Std.

Approved Vendor: Keckley (CL150, Non-Return, Wafer): {Size}-CW2R-36-36-3-36

**Description of CHECK Valve (CHN15-S16-W714):**

Type: Check, Silent, Center-Guided, **Soft Seat**, (Wafer)
Sub Type: Non-Return
Valve Size: NPS 2" -to- NPS 12" (DN 50 to DN300)
Class: **ASME 150**
Temperature: -20 to 400 °F (-29 to 204 °C)
Rating: 275 psig @ 100 °F (175 psig @ 400 °F)
Ends: **Wafer** (Install between RF Flanges)
Body: 316 SS (ASTM A351, CF8M), Epoxy coated
Trim: 316 SS
Seat: **Viton** (Default); EPDM limited to 300 °F (Optional)
Springs: 316 SS
Body Type: **Wafer**
Open Area: 110% of Pipe Size area
Installation: Horizontal Flow or Vertical Upward Flow

The standards are:

DESIGN: ASME B16.34
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: Mfg. Std.

Approved Vendor: Keckley (CL150, Non-Return, Wafer): {Size}-CW2R-36-36-4-36 (Viton Seat)
..... Keckley (CL150, Non-Return, Wafer): {Size}-CW2R-36-36-5-36 (EPDM Seat)

***** End of "Silent" Non-Return Check Valves *****

***** Piston Lift Check Valves *****

Description of CHECK Valve (CL03ST500): CHP30-S16-F500

Type: Check, Piston Lift, no Spring
Valve Size: NPS 0.5 -to- NPS 2
Class: **ASME 300**
Ends: Flanged RF
Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
Cover/Bonnet: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
Trim: 316 SS
Bore-Port: **Reduced Port (RP)**
Springs: **None**
Cover/Bonnet Type: Bolted
Body-Bonnet Gaskets: 316 SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
Installation: Horizontal - Cover Up

The standards are:

DESIGN: ASME B16.34
ENDS: ASME B16.11
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Cast, CL300, RP): 5352-3054-AG04-A-{Size}

Description of CHECK Valve (CL03ST501): CHP30-S16-F501

Type: Check, Piston Lift, with Spring
Valve Size: NPS 0.5 -to- NPS 2
Class: **ASME 300**
Ends: Flanged RF
Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
Cover/Bonnet: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
Trim: 316 SS
Bore-Port: **Reduced Port (RP)**
Springs: **Alloy X-750**
Cover/Bonnet Type: Bolted
Body-Bonnet Gaskets: 316 SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
Nuts: MFG STD
Installation: Horizontal - Cover Up

The standards are:

DESIGN: ASME B16.34
ENDS: ASME B16.11
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Cast, CL300, RP): 5352-3054-AG04-S-{Size}

Description of CHECK Valve (CL06ST500):

CHP60-S16-F500

Type: Check, Piston Lift, no Spring
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 600**
 Ends: Flanged RF
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Cover/Bonnet: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim: 316 SS
 Bore-Port: **Reduced Port (RP)**
 Springs: **None**
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Cast, CL600, RP): 5652-3054-AG04-A-{Size}

Description of CHECK Valve (CL06ST501):

CHP60-S16-F501

Type: Check, Piston Lift, with Spring
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 600**
 Ends: Flanged RF
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Cover/Bonnet: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Trim: 316 SS
 Bore-Port: **Reduced Port (RP)**
 Springs: **Alloy X-750**
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Cast, CL600, RP): 5651-3054-AG04-S-{Size}

Description of CHECK Valve (CHP80-S16-T210):

Type:..... Check, Piston Lift, no Spring
 Valve Size:..... NPS 0.5 -to-NPS 2
 Class:..... **ASME 800**
 Ends: FNPT
 Body: 316 SS (ASTM A182, F316)
 Cover/Bonnet:..... 316 SS (ASTM A182, F316)
 Port: **Standard Port** (RP)
 Trim: **TRIM 12**
 Springs: **None**
 Cover/Bonnet Type:..... Bolted
 Body-Bonnet Gaskets:..... SS/Graphite
 Body-Bonnet Bolting:..... Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN:..... API 602
 ENDS:..... ASME B16.5
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... Mfg.Std.

Approved Vendor:..... Bonney Forge: HL-40-8-T
 DSI (Forged, CL800, RP): 4321-E12

Description of CHECK Valve (CHP80-S16-X210):

Type:..... Check, Piston Lift, no Spring
 Valve Size:..... NPS 0.5 -to-NPS 2
 Class:..... **ASME 800**
 Ends: SW x FNPT
 Body: 316 SS (ASTM A182, F316)
 Cover/Bonnet:..... 316 SS (ASTM A182, F316)
 Port: **Standard Port** (RP)
 Trim: **TRIM 12**
 Springs: **None**
 Cover/Bonnet Type:..... Bolted
 Body-Bonnet Gaskets:..... SS/Graphite
 Body-Bonnet Bolting:..... Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN:..... API 602
 ENDS:..... ASME B16.5
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... Mfg.Std.

Approved Vendor:
 Bonney Forge: HL-40-8-SW/T
 DSI (Forged, CL800, RP): 4301-E12

Description of CHECK Valve (CL08SU000): CHP80-S16L-S000

Type: Check, Piston Lift, No Spring (Default)
 Valve Size: NPS 0.5 -to- NPS 2
 Class: API 800
 Ends: SW
 Body: 316L SS (ASTM A182-F316L)
 Cover/Bonnet: 316L SS (ASTM A182-F316L)
 Trim: API 602 **TRIM 12**
 Port: **Standard Port** (RP)
 Springs: **None**
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts API 602
 Nuts API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Forged, CL800, RP): 5851-6A65-AGnn-A-{Size}
 Bonney Forge (Forged, CL800, RP): HL-40-8L-SW, no Spring
 DSI (Forged, CL800, RP): 4311-E12, no Spring

Description of CHECK Valve (CL08SU001): CHP80-S16L-S001

Type: Check, Piston Lift, with Spring
 Valve Size: NPS 0.5 -to- NPS 2
 Class: API 800
 Ends: SW
 Body: 316L SS (ASTM A182-F316L)
 Cover/Bonnet: 316L SS (ASTM A182-F316L)
 Trim: API 602 **TRIM 12**
 Port: **Standard Port** (RP)
 Springs: **Alloy X-750**
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts API 602
 Nuts API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Forged, CL800, RP): 5851-6A65-AGnn-S-{Size}
 Bonney Forge (Forged, CL800, RP): HL-40-8L-SW, with Spring
 DSI (Forged, CL800, RP): 4311-E12, with Spring

Description of CHECK Valve (CL08SP300):

CHP80-S22-S300

Type: Check, Piston Lift, no Spring
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 800**
 Ends: SW
 Body: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Cover/Bonnet: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Trim: 2205 Duplex SS
 Port: **Standard Port** (RP)
 Springs: **None**
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 2205 Duplex SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Forged, RP): 5751-6F54-AGnn-A-{Size}

Description of CHECK Valve (CL08SP301):

CHP80-S22-S301

Type: Check, Piston Lift, with Spring
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 800**
 Ends: SW
 Body: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Cover/Bonnet: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Trim: 2205 Duplex SS
 Port: **Standard Port** (RP)
 Springs: **Alloy X-750**
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 2205 Duplex SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Forged, RP): 5851-6F54-AGnn-S-{Size} with spring

Description of CHECK Valve (CL01SP300):

CHP15-S22-S300

Type: Check, Piston Lift, no Spring
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 800**
 Ends: SW
 Body: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Cover/Bonnet: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Trim: 2205 Duplex SS
 Bore: Reduced Port (RP)
 Springs: **None**
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 2205 Duplex SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: ASME B16.34
 ENDS: ASME B16.11
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Cast, CL800, RP): 5251-3554-AGnn-A-{Size}
 Bonney Forge (Forged, CL800, RP): HL-40-8L-SW, no Spring
 DSI (Forged, CL800, RP): 4311-E12, no Spring

Description of CHECK Valve (CL01SP301):

CHP15-S22-S301

Type: Check, Piston Lift, no Spring
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 800**
 Ends: SW
 Body: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Cover/Bonnet: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Trim: 2205 Duplex SS
 Bore: Reduced Port (RP)
 Springs: **None**
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 2205 Duplex SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD

Installation: Horizontal - Cover Up

The standards are:

DESIGN: ASME B16.34
 ENDS: ASME B16.11
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Cast, CL800, RP): 5251-3554-AGnn-S-{Size}
 Bonney Forge (Forged, CL800, RP): HL-40-8L-SW, with Spring
 DSI (Forged, CL800, RP): 4311-E12, with Spring

Description of CHECK Valve (CL08SC300):

CHP80-S04H-S300

Type: Check, Piston Lift, High-Temp, no Spring
 Valve Size: NPS 0.5 -to- NPS 2
 Service: High Temperature 815 °C (1500 °F)
 Class: **ASME 800**
 Ends: SW
 Body: 304H SS (ASTM A182-F304H or ASTM A351-CF10)
 Cover/Bonnet: 304H SS (ASTM A182-F304H or ASTM A351-CF10)
 Trim: 304H SS
 Port: **Standard Port** (RP)
 Springs: **None**
 Hard Facing: Full
 Cover/Bonnet Type: **Welded**
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Forged, RP): 5751-6A36-AHnn-A-{Size}

Description of CHECK Valve (CL08SC301):

CHP80-S04H-S301

Type: Check, Piston Lift, High-Temp, with Spring
 Valve Size: NPS 0.5 -to- NPS 2
 Service: High Temperature 815 °C (1500 °F)
 Class: **ASME 800**
 Ends: SW
 Body: 304H SS (ASTM A182-F304H or ASTM A351-CF10)
 Cover/Bonnet: 304H SS (ASTM A182-F304H or ASTM A351-CF10)
 Trim: 304H SS
 Port: **Standard Port** (RP)
 Springs: **Alloy X-750**
 Hard Facing: Full
 Cover/Bonnet Type: **Welded**
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Forged, RP): 5751-6A36-AHnn-S-{Size}

Description of CHECK Valve (CL08SQ300): CHP80-S25-S300

Type: Check, Piston Lift, no Spring
 Valve Size: NPS 0.5 -to- NPS 2
 Class: API 800
 Ends: SW
 Body: 2507 Duplex SS (ASTM A182-F53)
 Cover/Bonnet: 2507 Duplex SS (ASTM A182-F53)
 Trim: 2507 Duplex SS
 Port: **Standard Port** (RP)
 Springs: **None**
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 2507 Duplex SS/Graphite
 Body-Bonnet Bolting: Bolts API 602
 Nuts API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Forged, RP): 5851-6F64-AGnn-A-{Size}

Description of CHECK Valve (CL08SQ301): CHP80-S25-S301

Type: Check, Piston Lift, with Spring
 Valve Size: NPS 0.5 -to- NPS 2
 Class: API 800
 Ends: SW
 Body: 2507 Duplex SS (ASTM A182-F53)
 Cover/Bonnet: 2507 Duplex SS (ASTM A182-F53)
 Trim: 2507 Duplex SS
 Port: **Standard Port** (RP)
 Springs: **Alloy X-750**
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 2507 Duplex SS/Graphite
 Body-Bonnet Bolting: Bolts API 602
 Nuts API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Forged, RP): 5851-6F54-AGnn-S-{Size}

Description of CHECK Valve (CL08SU003):

CHP80-S16L-S003

Type: Check, Piston Lift, no Spring
 Valve Size: NPS 0.5 -to- NPS 2
 Class: API 800
 Ends: SW
 Body: 316L SS (ASTM A182-F316L)
 Cover/Bonnet: 316L SS (ASTM A182-F316L)
 Trim: API 602 **TRIM 12**
 Port: Reduced Port (RP)
 Springs: **None**
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts API 602
 Nuts API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Forged, RP): 5851-6A65-AT04-{Size}
 Bonney Forge (Forged, RP): HL-40-8L-SW
 DSI (Forged, RP): 4311-E12

Description of CHECK Valve (CHP80-S16-T101):

Type:..... Check, Piston Lift, no Spring
 Valve Size:..... NPS 0.5 -to-NPS 2
 Class:..... **ASME 800**
 Ends: FNPT
 Body: 316 SS (ASTM A182, F316)
 Cover/Bonnet:..... 316 SS (ASTM A182, F316)
 Port: **Reduced Port (RP)**
 Trim: API 602 **TRIM 12**
 Springs: **None**
 Cover/Bonnet Type:..... Bolted
 Body-Bonnet Gaskets:..... SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN:..... API 602
 ENDS:..... ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: Mfg.Std.

Approved Vendor:..... Bonney Forge (Forged, RP): HL-40-8-SW
 DSI (Forged, RP): 4321-E12

Description of CHECK Valve (CL08SU300):

CHP80-S16L-S300

Type: Check, Piston Lift, no Spring (Default)
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 800**
 Ends: SW
 Body: 316L SS (ASTM A182-F316L)
 Cover/Bonnet: 316L SS (ASTM A182-F316L)
 Trim: API 602 **TRIM 10** (316 SS)
 Port: Red. Port (RP)
 Springs: **None**
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Forged, CL800, RP): 5851-6A64-AGnn-A-{Size}
 Bonney Forge (Forged, CL800, RP): HL-40-8L-SW
 DSI (Forged, CL800, RP): 4311-E10

Description of CHECK Valve (CL08SU301):

CHP80-S16L-S301

Type: Check, Piston Lift, no Spring
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 800**
 Ends: SW
 Body: 316L SS (ASTM A182-F316L)
 Cover/Bonnet: 316L SS (ASTM A182-F316L)
 Trim: API 602 **TRIM 10**
 Port: FULL Port (FP)
 Springs: **None**
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Forged, CL800, FP): 5851-6A61-AGnn-S-{Size}
 Bonney Forged (Forged, CL800, FP): H-40-8L-SW
 DSI (Forged, CL800, FP): 4311FB-E10

Description of CHECK Valve (CL08SU302):

CHP80-S16L-S302

Type: Check, Piston Lift, with Spring
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 800**
 Ends: SW
 Body: 316L SS (ASTM A182-F316L)
 Cover/Bonnet: 316L SS (ASTM A182-F316L)
 Trim: API 602 **TRIM 10**
 Port: Red. Port (RP)
 Springs: **Alloy X-750**
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Forged, CL800, RP): 5851-6A64-AGnn-S-{Size} with Spring
 Bonney Forge (Forged, CL800, RP): HL-40-8L-SW with Spring (Trim #12)
 DSI (Forged, CL800, RP): 4311-E10 with Spring

Description of CHECK Valve (CL08SU303):

CHP80-S16L-S303

Type: Check, Piston Lift, with Spring
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 800**
 Ends: SW
 Body: 316L SS (ASTM A182-F316L)
 Cover/Bonnet: 316L SS (ASTM A182-F316L)
 Trim: API 602 **TRIM 10**
 Port: Full Port (FP)
 Springs: **Alloy X-750**
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Forged, CL800, FP): 5851-6A61-ATnn-S-{Size}
 Bonney Forge (Forged, CL800, FP): H-40-8L-SW with Spring (Trim #12)
 DSI (Forged, CL800, FP): 4311FB-E10 with Spring



***** 316 SS Swing Check Valves *****

Description of CHECK Valve (CS01ST500):

CHS15-S16-F500

Type: Check, Swing
Valve Size: NPS 0.5 -to- NPS 24
Class: **ASME 150**
Ends: Flanged RF
Body: 316 SS (ASTM A351-CF8M)
Cover/Bonnet: 316 SS (ASTMA351-CF8M)
Trim: API 594 **TRIM #12**
Port: Full Port (FP)
Hinge Pin: 316 SS
Disc Washer, Disc Nut, Cotter Pin: 316 SS
Cover/Bonnet Type: Bolted
Body-Bonnet Gaskets: PTFE
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B or ASME B16.34, as applicable
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendor: Ladish (Cast, CL150, FP): 5272-7A52-TTnn-A-{Size}

Description of CHECK Valve (CS01ST501):

CHS15-S16-F501

Type: Check, Swing
Valve Size: NPS 0.5 -to- NPS 24
Class: **ASME 150**
Ends: Flanged RF
Body: 316 SS (ASTM A351-CF8M)
Cover/Bonnet: 316 SS (ASTM A351-CF8M)
Trim: API 594 **TRIM 10**
Port: **FULL Port (FB)**
Hinge Pin: 316 SS
Disc Washer, Disc Nut, Cotter Pin: 316 SS
Cover/Bonnet Type: Bolted
Body-Bonnet Gaskets: 316 SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B or ASME B16.34, as applicable
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendor: Ladish (Cast, CL150, FP): 5272-7571-CGnn-A-{Size}

Description of CHECK Valve (CS01ST502):

CHS15-S16-F502

Type: Check, Swing (Flanged)
 Valve Size: NPS 0.5 -to- NPS 12
 Service: Oxygen
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 316 SS (ASTM A351-CF8M)
 Cover/Bonnet: 316 SS (ASTM A351-CF8M)
 Trim: Alloy 400
 Port: **FULL Port (FB)**
 Hard Facing: Half
 Hinge Pin: 316 SS
 Disc Washer, Disc Nut, Cotter Pin: 316 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: Alloy 400/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B or ASME B16.34, as applicable
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10
 CLEANING: Oxygen Cleaning in accordance with CGA C-4.1 and G-4.4/ASTM G-93

Approved Vendor: Ladish (Cast, CL150, FP): 5272-7051-GGnn-B-{Size}

Description of CHECK Valve (CHS15-S16-F504)

Type: Check, Swing (Flanged)
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 150**
 Temperature: -20 °F to +400 °F (-29 °C to +204 °C)
 Ends: Flanged RF
 Body: 316 SS (ASTMA351-CF8M)
 Cover/Bonnet: 316 SS (ASTMA351-CF8M)
 Trim: **TRIM 12**
 Port: Standard Port (RP)
 Hinge Pin: 316 SS
 Disc Washer, Disc Nut, Cotter Pin: 316 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendor: Bonney Forge (Cast, CL 150, RP): {Size}-1-60-8 RF

Description of CHECK Valve (CHS15-S16-F505)

Type:Check, Swing (Flanged)
 Valve Size:NPS 0.5 -to- NPS 2.0
 Class:**ASME 150**
 Temperature:-20 °F to +400 °F (-29 °C to +204 °C)
 Ends:Flanged RF
 Body:316 SS (ASTMA351-CF8M)
 Cover/Bonnet:316 SS (ASTMA351-CF8M)
 Trim:**TRIM 12**
 Port:**FULL Port (FB)**
 Hinge Pin:316 SS
 Disc Washer, Disc Nut, Cotter Pin:316 SS
 Cover/Bonnet Type:Bolted
 Body-Bonnet Gaskets:316 SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD
 Installation:Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN:..... ASME B16.34
 ENDS:..... ASME B16.5
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:.....ASME B16.10

Approved Vendor:Bonney Forge (Cast, CL 150, FP): {Size}-L1-60-8 RF

Description of CHECK Valve (CS03ST500):

CHS30-S16-F500

Type:Check, Swing
 Valve Size:NPS 0.5 -to- NPS 24
 Class:**ASME 300**
 Ends:Flanged RF
 Body:316 SS (ASTMA351-CF8M)
 Cover/Bonnet:316 SS (ASTMA351-CF8M)
 Trim:API 594 **TRIM 10**
 Port:**FULL Port (FB)**
 Hinge Pin:316 SS
 Disc Washer, Disc Nut, Cotter Pin:316 SS
 Cover/Bonnet Type:Bolted
 Body-Bonnet Gaskets:PTFE
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD
 Installation:Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN:.....API 594 Type B or ASME B16.34, as applicable
 NDS:.....ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10

Approved Vendor:Ladish (Cast, CL300, FP): 5372-7051-ATnn-A-{Size}

Description of CHECK Valve (CS03ST501):

CHS30-S16-F501

Type: Check, Swing
 Valve Size: NPS 0.5 -to- NPS 24
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 316 SS (ASTMA351-CF8M)
 Cover/Bonnet: 316 SS (ASTMA351-CF8M)
 Trim: API 594 **TRIM 10**
 Port: **FULL Port (FB)**
 Hinge Pin: 316 SS
 Disc Washer, Disc Nut, Cotter Pin: 316 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B or ASME B16.34, as applicable
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendor: Ladish (Cast, CL300, FP): 5372-7051-AGnn-A-{Size}

Description of CHECK Valve (CS03ST502):

CHS30-S16-F502

Type: Check, Swing
 Valve Size: NPS 0.5 -to- NPS 12
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Cover/Bonnet: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Port: **FULL Port (FB)**
 Trim: Alloy 400
 Hard Facing: Half
 Hinge Pin: Alloy 400
 Disc Washer, Disc Nut, Cotter Pin: Alloy 400
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: Alloy 400/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B or ASME B16.34, as applicable
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10
 CLEANING: Oxygen Cleaning in accordance with CGA C-4.1 and G-4.4/ASTM G-93

Approved Vendor: Ladish (Cast, CL300, FP): 5372-705K-ATnn-B-{Size}

Description of CHECK Valve (CS06ST500):

CHS60-S16-F500

Type: Check, Swing
 Valve Size: NPS 0.5 -to- NPS 24
 Class: **ASME 600**
 Ends: Flanged RF
 Body: 316 SS (ASTMA351-CF8M)
 Cover/Bonnet: 316 SS (ASTMA351-CF8M)
 Port: **Full Port (FB)**
 Trim: API 594 **TRIM 10**
 Hinge Pin: 316 SS
 Disc Washer, Disc Nut, Cotter Pin: 316 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B or ASME B16.34, as applicable
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendor: Ladish (Cast, CL600, FP): 5672-7051-AGnn-A-{Size}

Description of CHECK Valve (CS06ST501):

CHS60-S16-F501

Type: Check, Swing
 Valve Size: NPS 0.5 -to- NPS 12
 Class: **ASME 600**
 Ends: Flanged RF
 Body: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Cover/Bonnet: 316 SS (ASTM A182-F316 or ASTM A351-CF8M)
 Port: **Full Port (FB)**
 Trim: Alloy 400
 Hard Facing: Half
 Hinge Pin: Alloy 400
 Disc Washer, Disc Nut, Cotter Pin: Alloy 400
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: Alloy 400/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B or ASME B16.34, as applicable ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10
 CLEANING: Oxygen Cleaning in accordance with CGA C-4.1 and G-4.4/ASTM G-93

Approved Vendor: Ladish (Cast, CL600, FP): 5672-705K-AGnn-A-{Size}

Description of CHECK Valve (CS06ST502):

CHS60-S16-F502

Type: Check, Swing
 Valve Size: NPS 0.5 -to- NPS 12
 Class: **ASME 600**
 Ends: Flanged RF
 Body: 316 SS (ASTMA351-CF8M)
 Cover/Bonnet: 316 SS (ASTMA351-CF8M)
 Port: Full Port
 Trim: API 594 **TRIM 10**
 Hinge Pin: 316 SS
 Disc Washer, Disc Nut, Cotter Pin: 316 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316 SS/PTFE
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B or ASME B16.34, as applicable
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendor: Ladish (Cast, CL600, FP): 5672-7051-AT04-A-{Size}

Description of CHECK Valve (CHS80-S16-S300):

Type: Check, Swing, FP
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **API 800**
 Ends: SW
 Body: 316 SS (ASTM A182, F316/316L)
 Cover/Bonnet: 316 SS (ASTM A182, F316/316L)
 Trim: **TRIM #12**
 Bore: Full Port (FP)
 Springs: None
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: Bonney Forge (Forged, CL800, FP): H-60-8-SW
 DSI (Forged, CL800, FP): 4461-FB-E12
 Ladish (Forged, CL800, FP): 5871-6A62-GGnn

Description of CHECK Valve (CHS80-S16-X210):

Type:..... Check, Swing
 Valve Size:..... NPS 0.5 -to-NPS 2
 Class:..... **ASME 800**
 Ends: SW x FNPT
 Body: 316 SS (ASTM A182, F316)
 Cover/Bonnet:..... 316 SS (ASTM A182, F316)
 Port:..... **Standard Port**
 Trim: **TRIM 12**
 Springs: None
 Cover/Bonnet Type:..... Bolted
 Body-Bonnet Gaskets:..... SS/Graphite
 Body-Bonnet Bolting:..... Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN:..... API 602
 ENDS:..... ASME B16.5
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... Mfg.Std.

VENDORS:..... Bonney Forge (Forged, CL800, RP): HL-60-8-SW/T
 DSI (Forged, CL800, RP): 4401-E12

Description of CHECK Valve (CHS80-S16-X211):

Type:..... Check, Swing
 Valve Size:..... NPS 0.5 -to-NPS 2
 Class:..... **ASME 800**
 Ends: SW x FNPT
 Body: SS (ASTM A182, F316)
 Cover/Bonnet:..... SS (ASTM A182, F316)
 Port:..... **Full Port**
 Trim: **TRIM 12**
 Springs: None
 Cover/Bonnet Type:..... Bolted
 Body-Bonnet Gaskets:..... SS/Graphite
 Body-Bonnet Bolting:..... Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN:..... API 602
 ENDS:..... ASME B16.5
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... Mfg.Std.

VENDORS:..... Bonney Forge (Forged, CL800, FP): H-60-8-SW/T
 DSI (Forged, CL800, FP): 4401-FB-E12

Description of CHECK Valve (CHS80-S16-T210):

Type:..... Check, Swing
 Valve Size:..... NPS 0.5 -to-NPS 2
 Class:..... **ASME 800**
 Ends: FNPT
 Body: 316 SS (ASTM A182, F316)
 Cover/Bonnet:..... 316 SS (ASTM A182, F316)
 Port:..... **Standard Port (RP)**
 Trim: **TRIM 12**
 Springs: None
 Cover/Bonnet Type:..... Bolted
 Body-Bonnet Gaskets:..... SS/Graphite
 Body-Bonnet Bolting:..... Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN:..... API 602
 ENDS:..... ASME B16.5
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... Mfg.Std.

VENDORS:..... Bonney Forge (Forged, CL800, RP): HL-60-8-T
 DSI (Forged, CL800, RP): 4421-E12

Description of CHECK Valve (CHS80-S16-T211):

Type:..... Check, Swing
 Valve Size:..... NPS 0.5 -to-NPS 2
 Class:..... **ASME 800**
 Ends: FNPT
 Body: SS (ASTM A182, F316)
 Cover/Bonnet:..... SS (ASTM A182, F316)
 Port:..... **Full Port (FP)**
 Trim: **TRIM 12**
 Springs: None
 Cover/Bonnet Type:..... Bolted
 Body-Bonnet Gaskets:..... SS/Graphite
 Body-Bonnet Bolting:..... Bolts: API 602
 Nuts: API 602
 Installation: Horizontal - Cover Up

The standards are:

DESIGN:..... API 602
 ENDS:..... ASME B16.5
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... Mfg.Std.

VENDORS:..... Bonney Forge (Forged, CL800, FP): H-60-8-T
 DSI (Forged, CL800, FP): 4421-FB-E12 (-D12 also OK)

***** Start of Duplex/Super Duplex Swing-Check Valves *****

Description of CHECK Valve (CS01SP500):

CHS15-S22-F500

Type: Check, Swing
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Cover/Bonnet: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Port: **Full Port (FP)**
 Trim: 2205 Duplex SS
 Hinge Pin: 2205 Duplex SS
 Disc Washer, Disc Nut, Cotter Pin: 2205 Duplex SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 2205 Duplex SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendor: Ladish (Cast, CL150, FP): 5272-7551-CG02-A-{Size}

Description of CHECK Valve (CS01SQ500):

CHS15-S25-F500

Type: Check, Swing
 Valve Size: NPS 0.5 -to- NPS 24
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 2507 Duplex SS (ASTM A995-6A)
 Cover/Bonnet: 2507 Duplex SS (ASTM A995-6A)
 Port: **Full Port (FP)**
 Trim: 2507 Duplex SS
 Disc Washer, Disc Nut, Cotter Pin: 2507 Duplex SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 2507 Duplex SS/Graphite
 Body-Bonnet Bolting: Bolts MFG STD
 Nuts MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B or ASME B16.34, as applicable
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendor: Ladish (Cast, CL150, FP): 5272-7571-CG02-A-{Size}

Description of CHECK Valve (CS03SP500):

CHS30-S22-F500

Type: Check, Swing
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Cover/Bonnet: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Port: **Full Port (FP)**
 Trim: 2205 Duplex SS
 Hinge Pin: 2205 Duplex SS
 Disc Washer, Disc Nut, Cotter Pin: 2205 Duplex SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 2205 Duplex SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendor: Ladish (Cast, CL300, FP): 5372-7551-GG02-A-{Size}

Description of CHECK Valve (CS06SP500):

CHS60-S22-F500

Type: Check, Swing
 Valve Size: NPS 2 -to- NPS 24
 Class: **ASME 600**
 Ends: Flanged RF
 Body: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Cover/Bonnet: 2205 Duplex SS (ASTM A182-F51 or ASTM A995-4A)
 Port: **Full Port (FP)**
 Trim: 2205 Duplex SS
 Hinge Pin: 2205 Duplex SS
 Disc Washer, Disc Nut, Cotter Pin: 2205 Duplex SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 2205 Duplex SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendor: Ladish (Cast, CL600, FP): 5672-7551-AG02-A-{Size}

Description of CHECK Valve (CS01SV300):

CHS15-SMO-S300

Type:Check, Swing
 Valve Size:NPS 0.5 -to- NPS 2
 Class:**ASME 150**
 Ends:SW
 Body:254 SMO SS (ASTMA351-CK3MCuN)
 Cover/Bonnet:254 SMO SS (ASTMA351-CK3MCuN)
 Port:**Reduced Port (RP)**
 Trim:254 SMO SS (ASTM A182-F44 or ASTM A351-CK3MCuN)
 Hinge Pin:254 SMO SS
 Disc Washer, Disc Nut, Cotter Pin:254 SMO SS
 Cover/Bonnet Type:Bolted
 Body-Bonnet Gaskets:PTFE
 Body-Bonnet Bolting:Bolts: MFG STD
 Nuts: MFG STD
 Installation:Horizontal - Cover Up or Vertical - Upward Flow
 The standards are:
 DESIGN:API 602
 ENDS:ASME B16.11
 RATING:ASME B16.34
 TESTING:API 598
 DIMENSIONAL:MFG STD

Approved Vendor:Ladish (Cast, CL150, RP): 5272-3164-CG02-A-{Size}

Description of CHECK Valve (CS01SV500):

CHS15-SMO-F500

Type:Check, Swing
 Valve Size:NPS 2 -to- NPS 24
 Class:**ASME 150**
 Ends:Flanged RF
 Body:254 SMO SS (ASTMA351-CK3MCuN)
 Cover/Bonnet:254 SMO SS (ASTMA351-CK3MCuN)
 Port:**Full Port (FP)**
 Trim:254 SMO SS (ASTM A182-F44 or ASTM A351-CK3MCuN)
 Hinge Pin:254 SMO SS
 Disc Washer, Disc Nut, Cotter Pin:254 SMO SS Cover/Bonnet Type: . Bolted
 Body-Bonnet Gaskets:PTFE
 Body-Bonnet Bolting:Bolts: MFG STD
 Nuts: MFG STD
 Installation:Horizontal - Cover Up or Vertical - Upward Flow
 The standards are:
 DESIGN:API 594 Type B
 ENDS:ASME B16.5
 RATING:ASME B16.34
 TESTING:API 598
 DIMENSIONAL:ASME B16.10

Approved Vendor:Ladish (Cast, CL150, FP): 5272-3161-GT04-A-{Size}

***** Start of Hi-Temp Swing-Check Valves *****

Description of CHECK Valve (CS03SC500): CHS30-S04H-F500

Type: Check, Swing (Hi-Temp)
 Valve Size: NPS 0.5 -to- NPS 2
 Service: High Temperature 815 °C (1500 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 304H SS (ASTM A182-F304H or ASTM A351-CF10)
 Cover/Bonnet: 304H SS (ASTM A182-F304H or ASTM A351-CF10)
 Port: **Reduced Port (RP)**
 Trim: 304H SS
 Hard Facing: Full
 Hinge Pin: 304H SS
 Disc Washer, Disc Nut, Cotter Pin: 304H SS
 Cover/Bonnet Type: Welded
 Installation: Horizontal - Cover Up or Vertical - Upward Flow
 The standards are:
 DESIGN: ASME B16.34 or API 602
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendor: Ladish (Forged, CL300, RP): 5372-3A36-TT02-A-{Size}

Description of CHECK Valve (CS03SC501): CHS30-S04H-F501

Type: Check, Swing (Hi-Temp)
 Valve Size: NPS 3 -to- NPS 24
 Service: High Temperature 815 °C (1500 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 304H SS (ASTM A351-CF10)
 Cover/Bonnet: 304H SS (ASTM A351-CF10)
 Port: **Full Port (FP)**
 Trim: 304H SS
 Hard Facing: Full
 Hinge Pin: 304H SS
 Disc Washer, Disc Nut, Cotter Pin: 304H SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: MFG STD
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:
 DESIGN: API 594 Type B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendor: Ladish (Cast, CL300, FP): 5372-7033-HH02-A-{Size}

Description of CHECK Valve (CS06SC500):

CHS60-S04H-F500

Type: Check, Swing (Hi-temp)
 Valve Size: NPS 0.5 -to- NPS 2
 Service: High Temperature 815 °C (1500 °F)
 Class: **ASME 800**
 Ends: Flanged RF
 Body: 304H SS (ASTM A182-F304H or ASTM A351-CF10)
 Cover/Bonnet: 304H SS (ASTM A182-F304H or ASTM A351-CF10)
 Port: **Reduced Port (RP)**
 Trim: 304H SS
 Hard Facing: Full
 Hinge Pin: 304H SS
 Disc Washer, Disc Nut, Cotter Pin: 304H SS
 Cover/Bonnet Type: **Welded**
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: ASME B16.34 or API 602
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendor: Ladish (Forged, CL800, RP): 5772-6A36-AH04-A-{Size}

Description of CHECK Valve (CS06SC501):

CHS30-S04H-F501

Type: Check, Swing (Hi-Temp)
 Valve Size: NPS 3 -to- NPS 24
 Service: High Temperature 815 °C (1500 °F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: 304H SS (ASTM A351-CF10)
 Cover/Bonnet: 304H SS (ASTM A351-CF10)
 Port: **Reduced Port (RP)**
 Trim: 304H SS
 Hard Facing: Full
 Hinge Pin: 304H SS
 Disc Washer, Disc Nut, Cotter Pin: 304H SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: MFG STD
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendor: Ladish (Forged, CL600, RP): 5672-7A36-AH04-A-{Size}

***** **End of Hi-Temp SS Swing-Check Valves** *****

***** Low-Temp/Cryo Swing-Check Valves *****

Description of CHECK Valve (CS01ST510): CHS15-S16-F510

Type: Check, Swing (Low Temp)
 Valve Size: NPS 3 -to- NPS 24
 Service: Low Temperature -101 °C (-150 °F)
 Class: **ASME 150**
 Ends: Flanged, RF
 Body: 316 SS (ASTMA351-CF8M)
 Cover/Bonnet: 316 SS (ASTMA351-CF8M)
 Port: **Full Port (FP)**
 Trim: API 594 **TRIM 10**
 Hinge Pin: 316 SS
 Disc Washer, Disc Nut, Cotter Pin: 316 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendor: Ladish (Cast, FP): 5272-7051-CG02-A-{Size}

Description of CHECK Valve (CS01ST511): CHS15-S16-F511

Type: Check, Swing (Cryogenic)
 Valve Size: NPS 0.5 -to- NPS 24
 Service: Cryogenic -198 °C (-325 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 316 SS (ASTMA351-CF8M)
 Cover/Bonnet: 316 SS (ASTMA351-CF8M)
 Port: **Full Port (FP)**
 Trim: API 594 **TRIM 12**
 Hinge Pin: 316 SS
 Disc Washer, Disc Nut, Cotter Pin: 316 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B or ASME B16.34, as applicable
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendor: Ladish (Cast, FP): 5292-7051-CG02-A-{Size}

Description of CHECK Valve (CS01SU310):

CHS15-S16L-S310

Type: Check, Swing (Low Temp)
 Valve Size: NPS 0.5 -to- NPS 2
 Service: Low Temperature -101 °C (-150 °F)
 Class: **ASME 150**
 Ends: SW
 Body: 316L SS (ASTMA351-CF3M)
 Cover/Bonnet: 316L SS (ASTMA351-CF3M)
 Port: **Reduced Port (RP)**
 Trim: API 602 **TRIM 10**
 Hinge Pin: 316 SS
 Disc Washer, Disc Nut, Cotter Pin: 316 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: ASME B16.34
 TESTING: MFG STD

Approved Vendor: Ladish (Forged, RP): 5291-6A61-CG04-A-{Size}

Description of CHECK Valve (CS01SU311):

CHS15-S16L-S311

Type: Check, Swing (Cryogenic)
 Valve Size: NPS 0.5 -to- NPS 2
 Service: Cryogenic -198 °C (-325 °F)
 Class: **ASME 150**
 Ends: SW
 Body: 316L SS (ASTMA351-CF3M)
 Cover/Bonnet: 316L SS (ASTMA351-CF3M)
 Port: **Reduced Port (RP)**
 Trim: API 602 **TRIM 12**
 Hinge Pin: 316 SS
 Disc Washer, Disc Nut, Cotter Pin: 316 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Forged, RP): 5291-6A65-CG04-A-{Size}

Description of CHECK Valve (CS03ST510):

CHS30-S16-F510

Type: Check, Swing (Low Temp)
 Valve Size: NPS 3 -to- NPS 24
 Service: Low Temperature -101 °C (-150 °F)
 Class: **ASME 300**
 Ends: Flanged, RF
 Body: 316 SS (ASTMA351-CF8M)
 Cover/Bonnet: 316 SS (ASTMA351-CF8M)
 Port: **Full Port (FP)**
 Trim: API 594 **TRIM 10**
 Hinge Pin: 316 SS
 Disc Washer, Disc Nut, Cotter Pin: 316 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B and MSS SP-134
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendor: Ladish (Cast, FP): 5392-7051-AG04-B-{Size}

Description of CHECK Valve (CS03ST511):

CHS30-S16-F511

Type: Check, Swing (Cryogenic)
 Valve Size: NPS 0.5 -to- NPS 24
 Service: Cryogenic -198 °C (-325 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 316 SS (ASTMA351-CF8M)
 Cover/Bonnet: 316 SS (ASTMA351-CF8M)
 Port: **Full Port (FP)**
 Trim: API 594 **TRIM 12**
 Hard Facing: Half
 Hinge Pin: 316 SS
 Disc Washer, Disc Nut, Cotter Pin: 316 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B or ASME B16.34, as applicable, and MSS SP-134
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendor: Ladish (Cast, FP): 5392-7052-AG04-B-{Size}

Description of CHECK Valve (CS06ST510):

CHS60-S16-F510

Type: Check, Swing (Low Temp)
 Valve Size: NPS 3 -to- NPS 24
 Service: Low Temperature -101 °C (-150 °F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: 316 SS (ASTMA351-CF8M)
 Cover/Bonnet: 316 SS (ASTMA351-CF8M)
 Port: **Full Port (FP)**
 Trim: API 594 **TRIM 10**
 Hinge Pin: 316 SS
 Disc Washer, Disc Nut, Cotter Pin: 316 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 594 Type B
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendor: Ladish (Cast, FP): 5692-7051-AG04-A-{Size}

Description of CHECK Valve (CS03SU310):

CHS30-S16L-S310

Type: Check, Swing (Low Temp)
 Valve Size: NPS 0.5 -to- NPS 2
 Service: Low Temperature -101 °C (-150 °F)
 Class: **ASME 300**
 Ends: SW
 Body: 316L SS (ASTMA351-CF3M)
 Cover/Bonnet: 316L SS (ASTMA351-CF3M)
 Port: **Full Port (FP)**
 Trim: API 602 **TRIM 10**
 Hinge Pin: 316 SS
 Disc Washer, Disc Nut, Cotter Pin: 316 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 602 and MSS SP-134
 ENDS: ASME B16.11
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Cast, FP): 5392-7A61-AG04-A-{Size}

Description of CHECK Valve (CS08SD310):

CHS80-S16-S310

Type: Check, Swing (Low Temp)
 Valve Size: NPS 0.5 -to- NPS 2
 Service: Low Temperature -101 °C (-150 °F)
 Class: **ASME 800**
 Ends: SW
 Body: 316/316L SS (ASTMA182-F316/316L)
 Cover/Bonnet: 316/316L SS (ASTMA182-F316/316L)
 Port: **Reduced Port (RP)**
 Trim: API 602 **TRIM 10**
 Hinge Pin: 316 SS
 Disc Washer, Disc Nut, Cotter Pin: 316 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Forged, RP): 5871-6A55-AG04-A-{Size}

Description of CHECK Valve (CS08SD311):

CHS80-S16-S311

Type: Check, Swing (Cryogenic)
 Valve Size: NPS 0.5 -to- NPS 2
 Service: Cryogenic -198 °C (-325 °F)
 Class: **ASME 800**
 Ends: SW
 Body: 316/316L SS (ASTMA182-F316/316L)
 Cover/Bonnet: 316/316L SS (ASTMA182-F316/316L)
 Port: **Reduced Port (RP)**
 Trim: API 602 **TRIM 12**
 Hinge Pin: 316 SS
 Disc Washer, Disc Nut, Cotter Pin: 316 SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 602 and MSS SP-134
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Forged, RP): 5871-6A55-AG04-A-{Size}

***** Start of SMO SS Swing-Check Valves *****

Description of CHECK Valve (CS03SV300):

CHS30-SMO-S300

Type: Check, Swing
 Valve Size: NPS 0.5 -to- NPS 2
 Class: **ASME 300**
 Ends: SW
 Body: 254 SMO SS (ASTM A351-CK3MCuN)
 Cover/Bonnet: 254 SMO SS (ASTM A351-CK3MCuN)
 Port: **Reduced Port (RP)**
 Trim: 254 SMO SS (ASTM A182-F44 or ASTM A351-CK3MCuN)
 Hinge Pin: 254 SMO SS
 Disc Washer, Disc Nut, Cotter Pin: 254 SMO SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: PTFE
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 602
 ENDS: ASME B16.11
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendor: Ladish (Forged, RP): 5371-6B64-AT04-A-{Size}

Description of CHECK Valve (CS03SV500):

CHS30-SMO-S500

Type: Check, Swing
 Valve Size: NPS 2 -to- NPS 12
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 254 SMO SS (ASTM A351-CK3MCuN)
 Cover/Bonnet: 254 SMO SS (ASTM A351-CK3MCuN)
 Port: **Reduced Port (RP)**
 Trim: 254 SMO SS (ASTM A182-F44 or ASTM A351-CK3MCuN)
 Hinge Pin: 254 SMO SS
 Disc Washer, Disc Nut, Cotter Pin: 254 SMO SS
 Cover/Bonnet Type: Bolted
 Body-Bonnet Gaskets: PTFE
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Installation: Horizontal - Cover Up or Vertical - Upward Flow

The standards are:

DESIGN: API 602
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendor: Ladish (Forged, RP): 5371-7B64-AT04-A-{Size}



Document #: GCTLR00-TSLA-ME-SPC-00014

Valve Specification for SS Check Valves

Rev.	RevDate	RevDescription	By	Checked
0	31 AUG 2023	Issued for release	MFS	SBS
1	18 SEPT 2023	Update Valve Information	MFS	SBS
2	08 NOV 2023	Added Small-bore Check Valves	MFS	SBS
3	26 MAR 2024	Revise Valve Dual Plate Valve Information	MFS	WGS
4	16 AUG 2024	Add Small bore Swing Check Valve	MFS	WGS

End Document #: GCTLR00-TSLA-ME-SPC-00014

Valve Specification for SS Check Valves

Description of DIAPHRAGM Valve (DP01ST500 ==> DM15-S16-F500)

Type: Diaphragm
 Valve size: NPS 0.5 -to-NPS 4
 Service: Slurry, Throttling
 Class: ASME 150
 Rating: 150 psig (10 Bar) to 4" NPS
 Ends: Flanged
 Body: 316 SS (ASTM A351-CF8M/CF3M)
 Diaphragm: PTFE
 Lining: PTFE
 Body type: Straight-through type
 Bonnet type: Bolted
 Disc type: Diaphragm, EPDM
 Design (Other): **Straight-through type**
 Operation: Handwheel
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD

The standards are:

DESIGN ASME B16.34
 ENDS ASME B16.5
 RATING ASME B16.34
 DIMENSIONAL MFG STD

Approved Vendor: Saunders, KB Type (Straight-Thru):
 IB-{Size}-U1-SX-C1

Description of DIAPHRAGM Valve (DM15-S16-F510)

Type: Diaphragm
 Valve size: NPS 0.5 -to-NPS 6
 Service: Slurry, Throttling
 Class: ASME 150
 Rating: 150 psig (10 bar)
 Ends: Flanged
 Body: 316 SS (ASTM A351-CF8M/CF3M)
 Diaphragm: PTFE
 Lining: PFA Lining
 Trim: Stem stainless steel
 Body type: **Weir Type** (Type A)
 Bonnet type: Bolted
 Design (Other): N/A
 Operation: Handwheel, Rising, Plastic
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Note: Gaskets not Required (Liner extends to Face of Flange)

The standards are:

DESIGN ASME B16.34
 ENDS ASME B16.5 (flanged)
 RATING ASME B16.34
 DIMENSIONAL MFG STD

Approved Vendor: Saunders, Type A (**Weir Type**), Rubber Diaphragm, PTFE)
 IA-{Size}-U1-SX-V1-HA-H01

Description of DIAPHRAGM Valve (DM15-S16-T210)

Type: Diaphragm
 Valve size: NPS 0.5 -to-NPS 2
 Service: Slurry, Throttling
 Class: ASME 150
 Rating: 150 psig (10 bar)
 Ends: FNPT
 Body: 316 SS (ASTM A351-CF8M/CF3M)
 Diaphragm: PTFE
 Lining: **Unlined**
 Body type: **Weir Type** (Type A)
 Bonnet type: Bolted
 Design (Other): N/A
 Operation: Handwheel, Rising, Plastic
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD

The standards are:

DESIGN ASME B16.34
 ENDS MFG STD
 RATING ASME B16.34
 DIMENSIONAL MFG STD

Approved Vendor: Saunders, Type A (**Weir Type**), PTFE Diaphragm
 IA-{Size}-53-SW-{PTFE type}

Description of DIAPHRAGM Valve (DM15-S16-T212)

Type: Diaphragm
 Valve size: NPS 0.5 -to-NPS 2
 Service: Slurry, Throttling
 Class: ASME 150
 Rating: 150 psig (10 bar)
 Ends: FNPT
 Body: 316 SS (ASTM A351-CF8M/CF3M)
 Diaphragm: PTFE
 Lining:
 Body type: **Weir Type** (Type A)
 Bonnet type: Bolted
 Design (Other): N/A
 Operation: Handwheel, Rising, Plastic
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD

The standards are:

DESIGN ASME B16.34
 ENDS MFG STD
 RATING ASME B16.34
 DIMENSIONAL MFG STD

Approved Vendor: Saunders, Type A (**Weir Type**), PTFE Diaphragm
 IA-{Size}-53-SW-{PTFE type}

**Description of DIAPHRAGM Valve (DP03SP500)****DM30-S22-F500**

Type: Diaphragm
Valve size: NPS 0.5 -to-NPS 4
Service: Slurry
Class: ASME 300
Ends: Flanged RF
Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Trim: Stem stainless steel
Body type: Straight-through type
Bonnet type: Bolted
Disc type: Diaphragm, EPDM
Design (Other): **Straight-through type**
Operation: Handwheel
Body-Bonnet Gaskets: 316 SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD

The standards are:

DESIGN ASME B16.34
ENDS ASME B16.5
RATING ASME B16.34
DIMENSIONAL MFG STD

Approved Vendor:.....

Description of DIAPHRAGM Valve (DP01SP500 ==> DM15-DI-F500)

Type: Diaphragm, w/Liner
Valve size: NPS 0.5 -to-NPS 4
Service: Slurry
Class: ASME 150
Rating: up to 2" NPS; 240 psig (16 bar),
..... 2.5" to 4" NPS; 150 psig (10 bar)
Range: -23 °F to +185 °F (-10 °C to +85 °C)
Ends: Flanged FF (Liner Extends across Face)
Body: Ductile Iron (SG Iron)
Trim: Stem stainless steel
Body Liner: HRL = Hard Natural Rubber
Body type: Straight-through type
Bonnet type: Bolted
Diaphragm: XA=EPDM
Design (Other): **Straight-through type**
Operation: Handwheel
Body-Bonnet Gaskets: 316 SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD

The standards are:

DESIGN ASME B16.34
ENDS ASME B16.5
RATING ASME B16.34
DIMENSIONAL MFG STD

Approved Vendor: Saunders, KB Type (Straight-Thru):
..... IB-{Size}-U1-ZL-XA

Description of DIAPHRAGM Valve (DP01SQ500 ==> DM15-DI-F501)

Type: Diaphragm, w/Liner
 Valve size: NPS 0.5 -to-NPS 4
 Service: Slurry
 Class: ASME 150
 Rating: up to 2" NPS; 240 psig (16 bar),
 2.5" to 4" NPS; 150 psig (10 bar)
 Range: -23 °F to +185 °F (-10 °C to +85 °C)
 Ends: Flanged FF (Liner Extends across Face)
 Body: Ductile Iron (SG Iron)
 Trim: Stem stainless steel
 Body Liner: HRL = Hard Natural Rubber
 Body type: Straight-through type
 Bonnet type: Bolted
 Diaphragm: XA=EPDM
 Design (Other): **Straight-through type**
 Operation: Handwheel
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD

The standards are:

DESIGN ASME B16.34
 ENDS ASME B16.5
 RATING ASME B16.34
 DIMENSIONAL MFG STD

Approved Vendor: Saunders, KB Type (Straight-Thru):
 IB-{Size}-U1-ZL-XA

Description of DIAPHRAGM Valve (DP03SQ500 ==> DM30-SS-F500)

Type: Diaphragm
 Valve size: NPS 0.5 -to-NPS 4
 Service: Slurry
 Class: ASME 300
 Ends: Flanged RF
 Body: 316 SS
 Trim: Stem stainless steel
 Body Liner: None
 Body type: **Straight-through type**
 Bonnet type: Bolted
 Disc type: Diaphragm, EPDM
 Design (Other): Straight-through type
 Operation: Handwheel
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD

The standards are:

DESIGN ASME B16.34
 ENDS ASME B16.5
 RATING ASME B16.34
 DIMENSIONAL MFG STD

Approved Vendor: Saunders, KB Type (Straight-Thru):

Note: When Valve Tags are shown as something like "DP01ST500 ==> DM15-S16-F500", it means that the PIP Valve Code (DP01ST500) should be replaced with the Tesla Valve Code (DM15-S16-F500).



Document #: GCTLR00-TSLA-ME-SPC-00015

Valve Specification for SS Diaphragm Valves

Rev.	Rev. Date	Rev. Description	By	Checked
0	14 SEPT 2023	Issued for release	MFS	SBS
1	18 SEPT 2023	Correct Valve Tags	MFS	SBS

End Document #: GCTLR00-TSLA-ME-SPC-00015 .

Valve Specification for SS Diaphragm Valves

***** Start of 316 SS Gate Valves *****

Description of GATE Valve (GA01ST500):

GA15-S16-F500

Type: Gate
 Valve Size: NPS 2 -to- NPS 24
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 316 SS (ASTM A351-CF8M)
 Bonnet: 316 SS (ASTM A351-CF8M)
 Trim: API 600 **Trim #12**
 Wedge Type: Solid or Flexible (NPS 0.5 -to- NPS 4), Flexible (NPS 6 and Greater)
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel to 8" NPS; Gear Oper. 10" to 24" NPS
 Stem-Packing: **PTFE**
 Body-Bonnet Gaskets: **PTFE**
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL150, FP): {Size}-37-16-16-HFS-T (10" & larger = Gear Op.)
 Bonney Forge (Cast, CL150, FP): {Size}-1-18S-RF-PTFE, P/G (10" & larger = Gear Op.)
 Warren (Cast, CL150, FP): {Size}-1156-12, w/PTFE (10" & larger = Gear Op.)
 Ladish (Cast, CL150, FP): 8272-2052-{Size} (10" & larger = Gear Op.)

Description of GATE Valve (GA01ST501):

GA15-S16-F501

Type: Gate
 Valve Size: NPS 2 -to- NPS 24
 Service: -29 to 315 °C (-20 to +600 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 316 SS (ASTM A351-CF8M)
 Bonnet: 316 SS (ASTM A351-CF8M)
 Trim: API 600 **Trim #12**
 Wedge Type: Solid or Flexible (NPS 0.5 -to- NPS 4), Flexible (NPS 6 and Greater)
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel to 8" NPS; Gear Oper. From 10" to 24" NPS
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: **316L SS/Graphite**
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL150, FP): {Size}-37-16-16-HFS (10" & larger = Gear Op.)
 Bonney Forge (Cast, CL150, FP): {Size}-1-18S-RF (10" & larger = Gear Op.)
 Warren (Cast, CL150, FP): {Size}-1156-12 (10" & larger = Gear Op.)
 Ladish (Cast, CL150, FP): 8272-2052-CG-nn-A-{Size} (10" & larger = Gear Op.)

Description of GATE Valve (GA01ST502):

GA15-S16-F502

Type: Gate
 Valve Size: NPS 2 -to- NPS 12
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 316 SS (ASTM A351-CF8M)
 Bonnet: 316 SS (ASTM A351-CF8M)
 Trim: API 600 **Trim #11**
 Wedge Type: Solid or Flexible (NPS 0.5 -to- NPS 4), Flexible (NPS 6 and Greater)
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: Alloy 400/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10
 CLEANING: Oxygen Cleaning in accordance with CGA C-4.1 and G-4.4/ASTM G-93
 Approved Vendors: Ladish (Cast, CL150, FP): 8272-205K-{Mfg Std}-B-{Size}

Description of GATE Valve (GA01ST503):

GA15-S16-F503

Type: Gate
 Valve Size: NPS 8 -to- NPS 24
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 316 SS (ASTM A351-CF8M)
 Bonnet: 316 SS (ASTM A351-CF8M)
 Trim: API 600 **Trim #12**
 Wedge Type: Flexible
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: **Gear**
 Stem-Packing: **PTFE**
 Body-Bonnet Gaskets: **PTFE**
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10
 Approved Vendors: DSI (Cast, CL150, FP): {Size}-G-37-16-16-HFS-T (Gear Oper.)
 Bonney Forge (Cast, CL150, FP): {Size}-1-18S-RF-PTFE, P/G, (Gear Op.)
 Warren (Cast, CL150, FP): {Size}-1156-12, w/PTFE (Gear Op.)
 Ladish (Cast, CL150, FP): 8272-2052-TT-nn-{Size} (Gear Oper.)

Description of GATE Valve (GA01ST504):

GA15-S16-F504

Type: Gate
Valve Size: NPS 8 -to- NPS 24
Service: -29 to 315 °C (-20 to +600 °F)
Class: **ASME 150**
Ends: Flanged RF
Body: 316 SS (ASTM A351-CF8M)
Bonnet: 316 SS (ASTM A351-CF8M)
Trim: API 600 **Trim #12**
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: **Gear**
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: 316L SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN: ASME B16.34
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL150, FP): {Size}-G-37-16-16 HFS ([Gear Oper.](#))
..... [Bonney Forge \(Cast, CL150, FP\): {Size}-1-18S-RF-BG \(Gear Oper.\)](#)
..... [Warren \(Cast, CL150, FP\): {Size}-1156-12 \(Gear Op.\)](#)
..... Ladish (Cast, CL150, FP): 8272-2052-{Size} (Gear Oper.)

Description of GATE Valve (GA01ST505):

GA15-S16-F505

Type: Gate
Valve Size: NPS 8 -to- NPS 12
Service: -29 to 204 °C (-20 to +400 °F)
Class: **ASME 150**
Ends: Flanged RF
Body: 316 SS (ASTM A351-CF8M)
Bonnet: 316 SS (ASTM A351-CF8M)
Trim: API 600 **Trim #11**
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: **Gear**
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: Alloy 400/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN: ASME B16.34
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10
CLEANING: Oxygen Cleaning in accordance with CGA C-4.1 and G-4.4/ASTM G-93

Approved Vendors: DSI (Cast, CL150, FP): {Size}-G-37-16-Trim #11
..... Ladish (Cast, CL150, FP): 8272-205K-GG-nn-B-G-{Size}

Description of GATE Valve (GA03ST500):

GA30-S16-F500

Type: Gate
Valve Size: NPS 2 -to- NPS 24
Service: -29 to 204 °C (-20 to +400 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: 316 SS (ASTM A351-CF8M)
Bonnet: 316 SS (ASTM A351-CF8M)
Trim: API 600 **Trim #12**
Wedge Type: Solid or Flexible (NPS 0.5 -to- NPS 4), Flexible (NPS 6 and Greater)
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Handwheel to 8" NPS; Gear Oper. (10" to 24" NPS)
Stem-Packing: **PTFE**
Body-Bonnet Gaskets: **PTFE**
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN: ASME B16.34
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL300, FP): {Size}-23-16-16-HFS-T (>=10" NPS = Gear Oper.)
..... Bonney Forge (Cast, CL300, FP): {Size}-3-18S-RF-PTFE, P/G, (>=10" NPS = Gear Oper.)
..... Warren (Cast, CL300, FP): {Size}-1306-12 w/PTFE (>=10" NPS = Gear Oper.)
..... Ladish (Cast, CL300, FP): 8372-2052-{Size} (>=10" NPS = Gear Oper.)

Description of GATE Valve (GA03ST501):

GA30-S16-F501

Type: Gate
Valve Size: NPS 2 -to- NPS 24
Service: -29 to 315 °C (-20 to +600 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: 316 SS (ASTM A351-CF8M)
Bonnet: 316 SS (ASTM A351-CF8M)
Trim: API 600 **Trim #12**
Wedge Type: Solid or Flexible (NPS 0.5 -to- NPS 4), Flexible (NPS 6 and Greater)
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Handwheel to 8" NPS; Gear 10" to 24"
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: 316L SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN: ASME B16.34
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL300, FP): {Size}-23-16-16-HFS (>=10" NPS = Gear Oper.)
..... Bonney Forge (Cast, CL300, FP): {Size}-3-18S-RF (>=10" NPS = Gear Oper.)
..... Warren (Cast, CL300, FP): {Size}-1306-12 (>=10" NPS = Gear Oper.)
..... Ladish (Cast, CL300, FP): 8372-2052-{Size} (>=10" NPS = Gear Oper.)

Description of GATE Valve (GA03ST502):.

GA30-S16-F502

Type: Gate
Valve Size:..... NPS 2" -to- NPS 12
Service: -29 to 204 °C (-20 to +400 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: 316 SS (ASTM A351-CF8M)
Bonnet: 316 SS (ASTM A351-CF8M)
Trim: API 600 **Trim #11**
Wedge Type: Solid or Flexible (NPS 0.5 -to- NPS 4), Flexible (NPS 6 and Greater)
Bonnet Type: Bolted
Bore-Port: **Standard Port (SP)**
Stem Design: OS&Y
Operation:..... Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:..... Alloy 400/Graphite
Body-Bonnet Bolting:..... Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN:..... ASME B16.34
ENDS: ASME B16.5
RATING:..... ASME B16.34
TESTING:..... API 598
DIMENSIONAL:..... ASME B16.10
CLEANING: Oxygen Cleaning in accordance with CGA C-4.1 and G-4.4/ASTM G-93

Approved Vendors:..... Ladish (Cast, CL300, FP): 8372-205K-{Size} w/Gear Op.

Description of GATE Valve (GA03ST503):.

GA30-S16-F503

Type: Gate
Valve Size:..... NPS 4 -to- NPS 24
Service: -29 to 204 °C (-20 to +400 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: 316 SS (ASTM A351-CF8M)
Bonnet: 316 SS (ASTM A351-CF8M)
Trim: API 600 **Trim #12**
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation:..... **Gear**
Stem-Packing: PTFE
Body-Bonnet Gaskets:..... PTFE
Body-Bonnet Bolting:..... Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN:..... ASME B16.34
ENDS: ASME B16.5
RATING:..... ASME B16.34
TESTING:..... API 598
DIMENSIONAL:..... ASME B16.10

Approved Vendors:..... DSI (Cast, Class 300, FP): {Size}-G-23-16-16-HFS-T ([Gear Oper.](#))
..... Bonney Forge (Cast, CL300, FP): {Size}-3-18S-RF-BG-PTFE ([P/G, Gear Oper.](#))
..... Warren (Cast, CL300, FP): {Size}-1306-12 w/PTFE ([Gear Oper.](#))
..... Ladish (Cast, Class 300, FP): 8372-2052-{Size} ([Gear Oper.](#))

Description of GATE Valve (GA03ST504):

GA30-S16-F504

Type: Gate
 Valve Size: NPS 4 -to- NPS 24
 Service: -29 to 316 °C (-20 to +600 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 316 SS (ASTM A351-CF8M)
 Bonnet: 316 SS (ASTM A351-CF8M)
 Trim: API 600 **Trim #12**
 Wedge Type: Flexible
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: **Gear**
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, Class 300, FP): {Size}-G-23-16-16-HFS ([Gear Oper.](#))
 [Bonney Forge \(Cast, CL300, FP\): {Size}-3-18S-RF-BG \(Gear Oper.\)](#)
 [Warren \(Cast, CL300, FP\): {Size}-1306-12 \(Gear Oper.\)](#)
 Ladish (Cast, CL300, FP): 8372-2052-{Size} ([Gear Oper.](#))

Description of GATE Valve (GA03ST505):

GA30-S16-F505

Type: Gate
 Valve Size: NPS 4 -to- NPS 12
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 316 SS (ASTM A351-CF8M)
 Bonnet: 316 SS (ASTM A351-CF8M)
 Trim: API 600 **Trim #11**
 Wedge Type: Flexible
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: **Gear**
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: Alloy 400/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10
 CLEANING: Oxygen Cleaning in accordance with CGA C-4.1 and G-4.4/ASTM G-93

Approved Vendors: Ladish (Cast, CL300, FP): 8372-205K-GG-nn-B-{Size}

Description of GATE Valve (GA06ST500):.

GA60-S16-F500

Type: Gate
 Valve Size: NPS 0.5 -to- NPS 24 (2" min. For some Valve Mfgr.)
 Service: -29 to 316 °C (-20 to +600 °F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: 316 SS (ASTM A351-CF8M)
 Bonnet: 316 SS (ASTM A351-CF8M)
 Trim: API 600 **Trim #12**
 Wedge Type: Solid or Flexible (NPS 0.5 -to- NPS 4), Flexible (NPS 6 and Greater)
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel **to 6" NPS; Gear Oper. 8" to 24" NPS**
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL 600, FP): {Size}-66-16-16-HFS (8" to 24" NPS Gear Oper.)
 Bonney Forge (Cast, CL600, FP): {Size}-6-18S-RF-BG (8" to 10" NPS Gear Oper.)
 Warren (Cast, CL600, FP): {Size}-1606-12 (8" to 24" NPS Gear Oper.)
 Ladish (Cast, CL600, FP): 8672-2052-CG-nn-{Size} (8" to 24" NPS Gear Oper.)

Description of GATE Valve (GA06ST501):.

GA60-S16-F501

Type: Gate
 Valve Size: NPS 0.5 -to- NPS 12
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: 316 SS (ASTM A351-CF8M)
 Bonnet: 316 SS (ASTM A351-CF8M)
 Trim: API 600 **Trim #11**
 Wedge Type: Solid or Flexible (NPS 0.5 -to- NPS 4), Flexible (NPS 6 and Greater)
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: Alloy 400/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10
 CLEANING: Oxygen Cleaning in accordance with CGA C-4.1 and G-4.4/ASTM G-93

Approved Vendors: Ladish (Cast, CL600, FP): 8672-205K-GG-nn-B-{Size}

**Description of GATE Valve (GA06ST502):.**

GA60-S16-F502

Type: Gate
Valve Size: NPS 0.5 -to- NPS 12
Service: -29 to 204 °C (-20 to +400 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: 316 SS (ASTM A351-CF8M)
Bonnet: 316 SS (ASTM A351-CF8M)
Trim: API 600 **Trim #12**
Wedge Type: Solid or Flexible (NPS 0.5 -to- NPS 4), Flexible (NPS 6 and Greater)
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Handwheel to 6" NPS; Gear Oper. 8" to 12" NPS
Stem-Packing: **PTFE**
Body-Bonnet Gaskets: 316L SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN: ASME B16.34
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL 600, FP): {Size}-66-16-16-HFS-T (8" to 12" NPS Gear Oper.)
..... Bonney Forge (Cast, CL600, FP): {Size}-6-18S-RF-BG-PTFE P/G (8" to 10" Gear Oper.)
..... Warren (Cast, CL600, FP): {Size}-1606-12 w/PTFE (8" to 24" NPS Gear Oper.)
..... Ladish (Cast, CL600, FP): 8672-2052-{Size} w/Gear Op.

Description of GATE Valve (GA06ST503):.

GA60-S16-F503

Type: Gate
Valve Size: NPS 4 -to- NPS 24
Service: -29 to 316 °C (-20 to +600 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: 316 SS (ASTM A351-CF8M)
Bonnet: 316 SS (ASTM A351-CF8M)
Trim: API 600 **Trim #12**
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: **Gear**
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: 316L SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN: ASME B16.34
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Cast, CL600, FP): {Size}-G-66-16-16-HFS (4" to 24" NPS Gear Oper.)
..... Bonney Forge (Cast, CL600, FP): {Size}-6-18S-RF-BG (4" to 10" Gear Oper.)
..... Warren (Cast, CL600, FP): {Size}-1606-12 (4" to 24" NPS Gear Oper.)
..... Ladish (Cast, CL600, FP): 8672-2052-{Size} (4" to 24" Gear Oper.)

Description of GATE Valve (GA06ST504):.

GA60-S16-F504

Type: Gate
Valve Size: NPS 3 -to- NPS 12
Service: -29 to 204 °C (-20 to +400 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: 316 SS (ASTM A351-CF8M)
Bonnet: 316 SS (ASTM A351-CF8M)
Trim: API 600 **Trim #11**
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: **Gear**
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: Alloy 400/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN: ASME B16.34
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10
CLEANING: Oxygen Cleaning in accordance with CGA C-4.1 and G-4.4/ASTM G-93
Approved Vendors: Ladish (Cast, FP): 8672-205K-GG-nn-B-{Size}

Description of GATE Valve (GA06ST505):.

GA60-S16-F505

Type: Gate
Valve Size: NPS 3 -to- NPS 12
Service: -29 to 204 °C (-20 to +400 °F)
Class: **ASME 600**
Ends: Flanged RF
Body: 316 SS (ASTM A351-CF8M)
Bonnet: 316 SS (ASTM A351-CF8M)
Trim: API 600 **Trim #12**
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: **Gear**
Stem-Packing: **PTFE**
Body-Bonnet Gaskets: 316L SS/Graphite
Body-Bonnet Bolting: Bolts MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN: ASME B16.34
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10
Approved Vendors: Ladish (Cast, FP): 8672-2052-{Size} w/Gear Op.

Description of GATE Valve (GA01SU201 ==> GA80-S16-X201):

Type: Gate
Valve Size:..... NPS 0.5 -to- NPS 2
Service: -29 to 316 °C (-20 to +600 °F)
Class: **ASME 800**
Ends: SW x FNPT
Body: 316/316L SS (ASTM, F316/316L)
Bonnet: 316/316L SS
Trim: **API 602 Trim #12**
Wedge Type: Solid
Bonnet Type: Bolted
Bore-Port: **Standard Port (SP)**
Stem Design: OS&Y
Operation:..... Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:..... 316L SS/Graphite
Body-Bonnet Bolting:..... Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN:..... ASME B16.34
ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
RATING:..... ASME B16.34
TESTING:..... API 598
DIMENSIONAL:..... MFG STD

Approved Vendors:..... DSI (Forged, RP): {Size}-4101-E12
..... Bonney Forge (Forged, RP): {Size}-HL-18L-SW/T
..... Ladish (Forged, RP): 8867-3A55-{Size}

Description of GATE Valve (GA01SU300 ==> GA80-S16-S320):

Type: Gate
Valve Size:..... NPS 0.5 -to- NPS 2
Service: -29 to 204 °C (-20 to +400 °F)
Class: **ASME 800**
Ends: SW
Body: 316/316L SS (ASTM A182, F316/F316L)
Bonnet: 316/316L SS
Trim: **API 602 Trim #12**
Wedge Type: Solid
Bonnet Type: Bolted
Bore-Port: **Standard Port (SP)**
Stem Design: OS&Y
Operation:..... Handwheel
Stem-Packing: **PTFE**
Body-Bonnet Gaskets:..... **PTFE**
Body-Bonnet Bolting:..... Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN:..... ASME B16.34
ENDS: ASME B16.11
RATING:..... ASME B16.34
TESTING:..... API 598
DIMENSIONAL:..... MFG STD

Approved Vendors:..... Bonney Forge (Forged, RP): HL-10-8-SW w/PTFE Packing/Gaskets
..... DSI (Forged, RP): {Size}-4111-E12 w/PTFE Packing/Gaskets

**Description of GATE Valve (GA80-S16-S300):**

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Service: -29 to 316 °C (-20 to +600 °F)
Class: **ASME 800**
Ends: SW
Body: 316/316L SS (ASTM A182, F316/316L)
Bonnet: 316/316L SS
Trim: **API 602 Trim #12**
Wedge Type: Solid
Bonnet Type: Bolted
Bore-Port: **Standard (RP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: 316L SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN: ASME B16.34
ENDS: ASME B16.11
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: MFG STD

Approved Vendors: DSI (Forged, RP): {Size}-4111-E12
..... Bonney Forge (Forged, RP): HL-10-8-SW

Description of GATE Valve (GA80-S16-X210):

Type: Gate
 Valve Size: NPS 0.5 -to- NPS 2
 Service: -29 to 316 °C (-20 to +600 °F)
 Class: **API 800**
 Ends: SW x FNPT
 Body: 316/316L SS (ASTM A182-F316/F316L)
 Bonnet: 316/316L SS (ASTM A182-F316/F316L)
 Trim: **API 602 Trim #12**
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API602
 The standards are:
 DESIGN: API 602
 ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD
 Approved Vendors: Bonney Forge (Forged, RP): HL-10-8-SW/T
 DSI (Forged, RP): 4101-E12
 Ladish (Forged, RP): 8867-6A55-{Size}

Description of GATE Valve (GA80-S16-T210):

Type: Gate
 Valve Size: NPS 0.5 -to- NPS 2
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **API 800**
 Ends: FNPT
 Body: 316/316L SS (ASTM A182-F316/F316L)
 Bonnet: 316/316L SS (ASTM A182-F316/F316L)
 Trim: **API 602 Trim #12**
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API602
 The standards are:
 DESIGN: API 602
 ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD
 Approved Vendors: Bonney Forge (Forged, RP): HL-10-8-T
 DSI (Forged, RP): 4121-E12
 Ladish (Forged, RP): 8863-6A55-{Size}

Description of GATE Valve (GA80-S16-X211):

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Service: -29 to 316 °C (-20 to +600 °F)
Class: **API 800**
Ends: SW x FNPT
Body: 316/316L SS (ASTM A182-F316/F316L)
Bonnet: 316/316L SS (ASTM A182-F316/F316L)
Trim: **API 602 Trim #12**
Wedge Type: Solid
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: 316 SS/Graphite
Body-Bonnet Bolting: Bolts: API 602
..... Nuts: API602
The standards are:
DESIGN: API 602
ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
RATING: API 602
TESTING: API 598
DIMENSIONAL: MFG STD
Approved Vendors: Bonney Forge (Forged, FP): H-10-8-SW/T
..... DSI (Forged, FP): 4101-FB-E12
..... Ladish (Forged, FP): 8867-6A52-{Size}

Description of GATE Valve (GA80-S16-T211):

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Service: -29 to 204 °C (-20 to +400 °F)
Class: **API 800**
Ends: FNPT
Body: 316/316L SS (ASTM A182-F316/F316L)
Bonnet: 316/316L SS (ASTM A182-F316/F316L)
Trim: API 602 Trim #12
Wedge Type: Solid
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: 316 SS/Graphite
Body-Bonnet Bolting: Bolts: API 602
..... Nuts: API602
The standards are:
DESIGN: API 602
ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
RATING: API 602
TESTING: API 598
DIMENSIONAL: MFG STD
Approved Vendors: Bonney Forge (Forged, FP): H-10-8-T
..... DSI (Forged, FP): 4121-FB-E12
..... Ladish (Forged, FP): 8863-6A52-{Size}

***** Start of Duplex Stainless Steel Gate Valves *****

Description of GATE Valve (GA01SP500):. GA15-S22-F500

Type: Gate
 Valve Size:..... NPS 0.5 -to- NPS 2
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 2205 Duplex SS (ASTM A182-F51)
 Bonnet: 2205 Duplex SS (ASTM A182-F51)
 Trim: 2205 Duplex SS
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 2205 Duplex
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API602
 The standards are:
 DESIGN: API 602
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: DSI (Forged, RP): 1131-R51
 Ladish (Forged, RP): 8275-3F54-{Size}

Description of GATE Valve (GA01SP501):. (GA15-S22-F502)

Type: Gate
 Valve Size:..... NPS 2 -to- NPS 24
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Bonnet: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim: 2205 Duplex SS
 Wedge Type: Solid or Flexible (NPS 2 -to- NPS 4), Flexible (NPS 6 and Greater)
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 2205 Duplex
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Cast, FP): 8272-3551-{Size} w/Gear Op.



Description of GATE Valve (GA01SP502): .

(GA15-S22-F502)

Type: Gate
Valve Size: NPS 8 -to- NPS 24
Service: -29 to 204 °C (-20 to +400 °F)
Class: **ASME 150**
Ends: Flanged RF
Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Bonnet: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Trim: 2205 Duplex SS
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: **Gear**
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: 2205 Duplex
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN: ASME B16.34
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Cast, FP): 8272-3551-{Size} w/Gear Op.

Description of GATE Valve (GA03SP500):

GA30-S22-F500

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Service: -29 to 204 °C (-20 to +400 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: 2205 Duplex SS (ASTM A182, F51)
Bonnet: 2205 Duplex SS (ASTM A182, F51)
Trim: 2205 Duplex SS
Wedge Type: Solid or Flexible
Bonnet Type: Bolted
Bore-Port: **Standard Port (SP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: 2205 Duplex
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN: ASME B16.34
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors:
..... Ladish (Cast, RP): 8375-3F54-{Size}

Description of GATE Valve (GA03SP501):

GA30-S22-F501

Type: Gate
Valve Size: NPS 2 -to- NPS 24
Service: -29 to 204 °C (-20 to +400 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Bonnet: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Trim: 2205 Duplex SS
Wedge Type: Solid or Flexible (NPS 2 -to- NPS 4), Flexible (NPS 6 and Greater)
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: 2205 Duplex
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN: ASME B16.34
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Cast, FP): 8372-3552-{Size}



Description of GATE Valve (GA03SP502):

GA30-S22-F502

Type: Gate
Valve Size: NPS 4 -to- NPS 24
Service: -29 to 204 °C (-20 to +400 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Bonnet: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Trim: 2205 Duplex SS
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: **Gear**
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: 2205 Duplex
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN: ASME B16.34
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Cast, FP): 8372-3552-{Size} w/Gear Op.

Description of GATE Valve (GA06SP501):

GA60-S22-F501

Type: Gate
 Valve Size: NPS 2 -to- NPS 24
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Bonnet: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim: 2205 Duplex SS
 Hard Facing: Half
 Wedge Type: Solid or Flexible (NPS 2 -to- NPS 4), Flexible (NPS 6 and Greater)
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 2205 Duplex
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Cast, FP): 8672-3552-{Size}

Description of GATE Valve (GA06SP502):

GA60-S22-F502

Type: Gate
 Valve Size: NPS 3 -to- NPS 24
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Bonnet: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim: 2205 Duplex SS
 Hard Facing: Half
 Wedge Type: Flexible
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: **Gear**
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 2205 Duplex
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Cast, FP): 8672-2552-{Size} w/Gear Op.

Description of GATE Valve (GA80-S22-S300), was GA03SP300 or GA01SP300:

Type: Gate
 Valve Size:..... NPS 0.5 -to- NPS 2
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **ASME 800**
 Ends: **SW**
 Body: 2205 Duplex SS (ASTM A182, F51)
 Bonnet: 2205 Duplex SS (ASTM A182, F51)
 Trim: 2205 Duplex SS
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation:..... Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:..... 2205 Duplex
 Body-Bonnet Bolting:..... Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN:..... ASME B16.34
 ENDS: ASME B16.11
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... MFG STD

Approved Vendors:..... Bonney Forge (Forged, RP): HL-10-DD-SW (DD = 2205 Duplex)
 DSI (Forged, RP): 4111-R51
 Ladish (Forged, RP): 8864-3F54-{Size}

Description of GATE Valve (GA80-S22-X210) – was GA03SP200:

Type: Gate
 Valve Size:..... NPS 0.5 -to- NPS 2
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **ASME 800**
 Ends: **SW x FNPT**
 Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Bonnet: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim: 2205 Duplex SS
 Wedge Type: Solid or Flexible
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation:..... Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:..... 2205 Duplex
 Body-Bonnet Bolting:..... Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN:..... ASME B16.34
 ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
 RATING:..... ASME B16.34
 TESTING:..... API 598
 DIMENSIONAL:..... MFG STD

Approved Vendors:..... DSI (CL800, Forged, RP): {Size}-4101-R51
 Ladish (CL800, Forged, RP): 8867-3F54-{Size}

Description of GATE Valve (GA80-S22-X200), was GA01SP200:

Type: Gate
 Valve Size:..... NPS 0.5 -to- NPS 2
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **ASME 800**
 Ends: **SW x FNPT**
 Body: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Bonnet: 2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim: 2205 Duplex SS
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 2205 Duplex
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: DSI: {Size}-4101-R51
 Ladish (CL800, Forged, RP): 8867-3F54-{Size}

Description of GATE Valve (GA80-S22-X210) – was GA08SP200:

Type: Gate
 Valve Size:..... NPS 0.5 -to- NPS 2
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **API 800**
 Ends: **SW x FNPT**
 Body: 2205 Duplex SS (ASTM A182-F51)
 Bonnet: 2205 Duplex SS (ASTMA182-F51)
 Trim: 2205 Duplex SS
 Hard Facing: Half
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 2205 Duplex
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API602
 The standards are:
 DESIGN: API 602
 ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: Bonney Forge (Forged, RP): HL-10-DD-SW/T (DD = F51, Duplex SS)
 Ladish (Forged, RP): 8867-3F55-{Size}
 DSI (Forged, RP): 4101-R51



Description of GATE Valve (GA08SP300):

GA80-S22-S300

Type: Gate
Valve Size: NPS 0.5 -to- NPS 2
Service: -29 to 204 °C (-20 to +400 °F)
Class: **API 800**
Ends: SW
Body: 2205 Duplex SS (ASTM A182-F51)
Bonnet: 2205 Duplex SS (ASTM A182-F51)
Trim: 2205 Duplex SS
Hard Facing: Half
Wedge Type: Solid
Bonnet Type: Bolted
Bore-Port: **Standard Port (SP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: 2205 Duplex
Body-Bonnet Bolting: Bolts: API 602
..... Nuts: API602
The standards are:
DESIGN: API 602
ENDS: ASME B16.11
RATING: API 602
TESTING: API 598
DIMENSIONAL: MFG STD

Approved Vendors: Ladish (Forged, RP): 8864-3F55-{Size}

***** End of Duplex Stainless Steel Gate Valves *****

Description of GATE Valve (GA03SU310):

Type: Gate, Low Temp
 Valve Size: NPS 0.5 -to- NPS 2
 Service: Low Temperature -73 to 204 °C (-99 to 400 °F)
 Class: **ASME 300**
 Ends: SW
 Body: 316L SS (ASTM A351-CF3M)
 Bonnet: 316L SS (ASTM A351-CF3M)
 Trim: API 602 Trim #12
 Wedge Type: Solid or Flexible
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD

The standards are:

DESIGN: ASME B16.34
 ENDS: ASME B16.11
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors:

Description of GATE Valve (GA08SU000):

GA80-S16L-T000

Type: Gate
 Valve Size: NPS 0.25 -to- NPS 0.75
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **API 800**
 Ends: FNPT
 Body: 316L SS (ASTM A182-F316L)
 Bonnet: 316L SS (ASTM A182-F316L)
 Trim: API 602 Trim #12
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API602

The standards are:

DESIGN: API 602
 ENDS: ASME B1.20.1
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: Ladish (Forged, RP): 8863-3A65-{Size}

Description of GATE Valve (GA08SU001):

GA80-S16-T001

Type: Gate
 Valve Size: NPS 0.25 -to- NPS 0.75
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **API 800**
 Ends: FNPT
 Body: 316L SS (ASTM A182-F316L)
 Bonnet: 316L SS (ASTM A182-F316L)
 Trim: API 602 Trim #12
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: PTFE
 Body-Bonnet Gaskets: PTFE
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API602

The standards are:

DESIGN: API 602
 ENDS: ASME B1.20.1
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: Ladish (Forged, RP): 8863-3A65-{Size}

Description of GATE Valve (GA08SU202):

Type: Gate
 Valve Size: NPS 0.5 -to- NPS 2
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **API 800**
 Ends: SW x FNPT
 Body: 316L SS (ASTM A182-F316L)
 Bonnet: 316L SS (ASTM A182-F316L)
 Trim: API 602 Trim #12
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API602

The standards are:

DESIGN: API 602
 ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: Ladish (Forged, RP): 8867-3A65-{Size}

Description of GATE Valve (GA08SU203):

Type: Gate
 Valve Size: NPS 0.5 -to- NPS 2
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **API 800**
 Ends: SW x FNPT
 Body: 316L SS (ASTM A182-F316L)
 Bonnet: 316L SS (ASTM A182-F316L)
 Trim: API 602 Trim #12
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: PTFE
 Body-Bonnet Gaskets: 316 SS/PTFE
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API602
 The standards are:
 DESIGN: API 602
 ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD
 Approved Vendors: Ladish (Forged, RP): 8867-3A65-{Size}

Description of GATE Valve (GA08SU300):

Type: Gate,
 Valve Size: NPS 0.5 -to- NPS 2
 Service: -29 to 316 °C (-20 to +600 °F)
 Class: **API 800**
 Ends: SW
 Body: 316L SS (ASTM A182-F316L)
 Bonnet: 316L SS (ASTM A182-F316L)
 Trim: **API 602 Trim #12**
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API602
 The standards are:
 DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD
 Approved Vendors: Bonney Forge (Forged, RP): HL-10-8L-SW
 Ladish (Forged, RP): 8864-6A65-{Size}

Description of GATE Valve (GA08SU301):

Type: Gate
 Valve Size: NPS 0.5 -to- NPS 2
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **API 800**
 Ends: SW
 Body: 316L SS (ASTM A182-F316L)
 Bonnet: 316L SS (ASTM A182-F316L)
 Trim: **API 602 Trim #12**
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: **PTFE**
 Body-Bonnet Gaskets: PTFE
 Body-Bonnet Bolting: Bolts: API 602
 Nuts: API602
 The standards are:
 DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: Bonney Forge (Forged, RP): HL-10-8L-SW (w/PTFE Packing/Gasketing)
 Ladish (Forged, RP): 8864-6A65-{Size}

Description of GATE Valve (GA09SC501):

Type: Gate, Pressure Seal, Hi-Temp
 Valve Size: NPS 3 -to- NPS 24
 Service: High Temperature -29 to 816 °C (-20 to 1500 °F)
 Class: ASME 900
 Ends: Flanged RF
 Body: 304H SS (ASTM A351-CF10)
 Bonnet: 304H SS (ASTM A351-CF10)
 Trim: 304H SS
 Hard Facing: Full
 Wedge Type: Flexible with Cavity Vent to High-Pressure Side (Indicated) Bonnet Type:
 Pressure Seal
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: MFG STD
 Body-Bonnet Gaskets: MFG STD
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34 and MSS SP-144
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors:

Description of GATE Valve (GA09SC502):

Type: Gate, Pressure Seal, Hi-Temp
 Valve Size: NPS 3 -to- NPS 24
 Service: High Temperature -29 to 816 °C (-20 to 1500 °F)
 Class: ASME 900
 Ends: Flanged RF
 Body: 304H SS (ASTM A351-CF10)
 Bonnet: 304H SS (ASTM A351-CF10)
 Trim: 304H SS
 Hard Facing: Full
 Wedge Type: Flexible with Cavity Vent to High-Pressure Side (Indicated)
 Bonnet Type: Pressure Seal
 Bore-Port: **Full Port (FP)** (FP)
 Stem Design: OS&Y
 Operation: **Gear**
 Stem-Packing: MFG STD
 Body-Bonnet Gaskets: MFG STD
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34 and MSS SP-144
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors:

Description of GATE Valve (GA15-S25-F500):

Type: Gate
 Valve Size: NPS 2" -to- NPS 24
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 2507 Super Duplex SS (ASTM A995, CE3MN)
 Bonnet: 2507 Super Duplex SS (ASTM A995, CE3MN)
 Trim: API 600 Trim #12
 Wedge Type: Solid or Flexible (NPS 0.5 -to- NPS 4), Flexible (NPS 6 and Greater)
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: PTFE
 Body-Bonnet Gaskets: PTFE
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Cast, FP): 8272-3562-{Size}

Description of GATE Valve (GA30-S25-F500):

Type: Gate
 Valve Size: NPS 1/2" -to- NPS 2"
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 2507 Super Duplex SS (ASTM A995, CE3MN)
 Bonnet: 2507 Super Duplex SS (ASTM A995, CE3MN)
 Trim: API 600 Trim #12
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: PTFE
 Body-Bonnet Gaskets: PTFE
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD

The standards are:

DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Cast, FP): 8375-2562-TT-{Size}

Description of GATE Valve (GA30-S25-F501):

Type: Gate
 Valve Size: NPS 3" -to- NPS 24
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 2507 Super Duplex SS (ASTM A995, CE3MN)
 Bonnet: 2507 Super Duplex SS (ASTM A995, CE3MN)
 Trim: API 600 Trim #12
 Wedge Type: Solid or Flexible (NPS 0.5 -to- NPS 4), Flexible (NPS 6 and Greater)
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: PTFE
 Body-Bonnet Gaskets: PTFE
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD

The standards are:

DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Cast, FP): 8372-3562-{Size}

Description of GATE Valve (GA80-S25-X210)

Type: Gate
 Valve Size:..... NPS 0.5 -to- NPS 2
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: ASME 800
 Ends: SW x FNPT
 Body: 2507 Duplex SS (ASTM A182-F53)
 Bonnet: 2507 Duplex SS (ASTM A182-F53)
 Trim: 2507 Duplex SS
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 2507 Duplex
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD

The standards are:

DESIGN: ASME B16.34
 ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: DSI: {Size}-4101-O53
 Ladish (Forged, RP): 8877-3F64-{Size}

Description of GATE Valve (GA80-S25-S300)

Type: Gate
 Valve Size:..... NPS 0.5 -to- NPS 2
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: ASME 800
 Ends: SW
 Body: 2507 Duplex SS (ASTM A182-F53)
 Bonnet: 2507 Duplex SS (ASTM A182-F53)
 Trim: 2507 Duplex SS
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 2507 Duplex
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD

The standards are:

DESIGN: ASME B16.34
 ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: DSI: {Size}-4111-O
 Ladish (Forged-RP): 8874-3F64-{Size}

***** Start of SMO Duplex Stainless Steel *****

Description of GATE Valve (GA01SV300 ==> GA80-SMO-S300):

Type: Gate
Valve Size:..... NPS 0.5 -to- NPS 2
Service: -29 to 204 °C (-20 to +400 °F)
Class: ASME 800
Ends: SW
Body: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
Bonnet: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
Trim: 254 SMO Duplex SS
Wedge Type: Solid or Flexible
Bonnet Type: Bolted
Bore-Port: **Standard Port (SP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: PTFE
Body-Bonnet Gaskets: PTFE
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN: ASME B16.34
ENDS: ASME B16.11
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: MFG STD

Approved Vendors: Ladish (Cast, FP): 8274-3162-{Size}

Description of GATE Valve (GA01SV500):.

GA15-SMO-F500

Type: Gate
Valve Size:..... NPS 0.5 -to- NPS 2
Service: -29 to 204 °C (-20 to +400 °F)
Class: **ASME 150**
Ends: Flanged RF
Body: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
Bonnet: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
Trim: 254 SMO Duplex SS
Wedge Type: Solid or Flexible
Bonnet Type: Bolted
Bore-Port: **Standard Port (SP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: PTFE
Body-Bonnet Gaskets: PTFE
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN: ASME B16.34
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Cast, FP): 8275-3162-{Size}

Description of GATE Valve (GA01SV501):

GA15-SMO-F501

Type: Gate
 Valve Size: NPS 2 -to- NPS 24
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
 Bonnet: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
 Trim: 254 SMO Duplex SS
 Wedge Type: Solid or Flexible (NPS 2 -to- NPS 4), Flexible (NPS 6 and Greater)
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: PTFE
 Body-Bonnet Gaskets: PTFE
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Cast, FP): 8272-3162-{Size}

Description of GATE Valve (GA01SV502):

GA15-SMO-F502

Type: Gate
 Valve Size: NPS 8 -to- NPS 24
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
 Bonnet: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
 Trim: 254 SMO Duplex SS
 Wedge Type: Flexible
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: **Gear**
 Stem-Packing: PTFE
 Body-Bonnet Gaskets: PTFE
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Cast, FP): 8272-3162-{Size} w/Gear Op.

Description of GATE Valve (GA03SV200):

Type: Gate
 Valve Size: NPS 0.5 -to- NPS 2
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **ASME 300**
 Ends: SW x FNPT
 Body: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
 Bonnet: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
 Trim: 254 SMO Duplex SS
 Wedge Type: Solid or Flexible
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: PTFE
 Body-Bonnet Gaskets: PTFE
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors:

Description of GATE Valve (GA03SV501):.

GA30-SMO-F501

Type: Gate
 Valve Size: NPS 2 -to- NPS 12
 Service: -29 to 204 °C (-20 to +400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
 Bonnet: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
 Trim: 254 SMO Duplex SS
 Wedge Type: Solid or Flexible (NPS 2 -to- NPS 4), Flexible (NPS 6 and Greater)
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: PTFE
 Body-Bonnet Gaskets: PTFE
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Cast, FP): 8372-3162-{Size}



Description of GATE Valve (GA03SV502):

GA30-SMO-F502

Type: Gate
Valve Size: NPS 4 -to- NPS 12
Service: -29 to 204 °C (-20 to +400 °F)
Class: **ASME 300**
Ends: Flanged RF
Body: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
Bonnet: 254 SMO Duplex SS (ASTM A351-CK3MCuN)
Trim: 254 SMO Duplex SS
Wedge Type: Flexible
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation: **Gear**
Stem-Packing: PTFE
Body-Bonnet Gaskets: PTFE
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN: ASME B16.34
ENDS: ASME B16.5
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Cast, FP): 8372-3163-{Size} w/Gear Op.

***** Start of Hi-Temp Gate Valves *****

Description of GATE Valve (GA03SC300 ==> GA30-S304H-S300):

Type: Gate, Hi-Temp
 Valve Size: NPS 0.5 -to- NPS 2
 Service: High Temperature -29 to 816 °C (-20 to 1500 °F)
 Class: ASME 800
 Ends: SW
 Body: 304H SS (ASTM A351-CF10)
 Bonnet: 304H SS (ASTM A351-CF10)
 Trim: 304H SS
 Hard Facing: Full
 Wedge Type: Solid or Flexible
 Bonnet Type: Welded
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.11
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: Ladish (Forged, RP): 8774-3A36-{Size}

Description of GATE Valve (GA03SC500):

GA30-S04H-F500

Type: Gate, Hi-Temp
 Valve Size: NPS 0.5 -to- NPS 2
 Service: High Temperature -29 to 816 °C (-20 to 1500 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 304H SS (ASTM A182-F304H)
 Bonnet: 304H SS (ASTM A182-F304H)
 Trim: 304H SS
 Hard Facing: Full
 Wedge Type: Solid or Flexible
 Bonnet Type: Welded
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: MFG STD
 The standards are:
 DESIGN: API 602
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Forged, RP): 8775-3A36-{Size}

Description of GATE Valve (GA03SC501):

GA30-S04H-F501

Type: Gate, Hi-Temp
 Valve Size: NPS 3 -to- NPS 24
 Service: High Temperature -29 to 816 °C (-20 to 1500 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 304H SS (ASTM A351-CF10)
 Bonnet: 304H SS (ASTM A351-CF10)
 Trim: 304H SS
 Hard Facing: Full
 Wedge Type: Flexible
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: MFG STD
 Body-Bonnet Gaskets: MFG STD
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34 or API 600
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors:

Description of GATE Valve (GA03SC502):

GA30-S04H-F502

Type: Gate, Hi-Temp
 Valve Size: NPS 4 -to- NPS 24
 Service: High Temperature -29 to 816 °C (-20 to 1500 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 304H SS (ASTM A351-CF10)
 Bonnet: 304H SS (ASTM A351-CF10)
 Trim: 304H SS
 Hard Facing: Full
 Wedge Type: Flexible
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: **Gear**
 Stem-Packing: MFG STD
 Body-Bonnet Gaskets: MFG STD
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34 or API 600
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors:

Description of GATE Valve (GA09SM500):

GA90-S47H-F500

Type: Gate, Pressure Seal, Hi-Temp
 Valve Size: NPS 3 -to- NPS 24
 Service: High Temperature -29 to 816 °C (-20 to 1500 °F)
 Class: ASME 900
 Ends: Flanged RF
 Body: 347H SS (ASTM A351-CF8C)
 Bonnet: 347H SS (ASTM A351-CF8C)
 Material Remarks: 0.04 Percent Minimum Carbon Content
 Trim: 347H SS
 Hard Facing: Full
 Wedge Type: Flexible with Cavity Vent to High-Pressure Side (Indicated)
 Bonnet Type: Pressure Seal
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: MFG STD
 Body-Bonnet Gaskets: MFG STD
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34 and MSS SP-144
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors:

Description of GATE Valve (GA09SM501):

GA90-S47H-F501

Type: Gate, Pressure Seal, Hi-Temp
 Valve Size: NPS 3 -to- NPS 24
 Service: High Temperature -29 to 816 °C (-20 to 1500 °F)
 Class: ASME 900
 Ends: Flanged RF
 Body: 347H SS (ASTM A351-CF8C)
 Bonnet: 347H SS (ASTM A351-CF8C)
 Material Remarks: 0.04 Percent Minimum Carbon Content Trim: 347H SS
 Hard Facing: Full
 Wedge Type: Flexible with Cavity Vent to High-Pressure Side (Indicated)
 Bonnet Type: Pressure Seal
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: **Gear**
 Stem-Packing: MFG STD
 Body-Bonnet Gaskets: MFG STD
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34 and MSS SP-144
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors:

Description of GATE Valve (GA06SC500):

GA60-S04H-F500

Type: Gate, Hi-Temp
 Valve Size: NPS 0.5 -to- NPS 2
 Service: High Temperature -29 to 816 °C (-20 to 1500 °F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: 304H SS (ASTM A182-F304H)
 Bonnet: 304H SS (ASTM A182-F304H)
 Trim: 304H SS
 Hard Facing: Full
 Wedge Type: Solid
 Bonnet Type: Welded
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: MFG STD
 The standards are:
 DESIGN: API 602
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Forged, RP): 8672-3A36-{Size}

Description of GATE Valve (GA06SC501):

GA60-S04H-F501

Type: Gate, Hi-Temp
 Valve Size: NPS 3 -to- NPS 24
 Service: High Temperature -29 to 816 °C (-20 to 1500 °F)
 Class: **ASME 600**
 Ends: Flanged RF
 Body: 304H SS (ASTM A351-CF10)
 Bonnet: 304H SS (ASTM A351-CF10)
 Trim: 304H SS
 Hard Facing: Full
 Wedge Type: Flexible
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: MFG STD
 Body-Bonnet Gaskets: MFG STD
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Description of GATE Valve (GA08SC200):

GA80-S04H-X200

Type: Gate, Hi-Temp
 Valve Size: NPS 0.5 -to- NPS 2
 Service: High Temperature -29 to 816 °C (-20 to 1500 °F)
 Class: **API 800**
 Ends: SW x FNPT
 Body: 304H SS (ASTM A182-F304H)
 Bonnet: 304H SS (ASTM A182-F304H)
 Trim: 304H SS
 Hard Facing: Full
 Wedge Type: Solid
 Bonnet Type: Welded
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: MFG STD
 The standards are:
 DESIGN: API 602
 ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: Ladish (Forged, RP): 8867-3A36-{Size}

Description of GATE Valve (GA08SC300):

GA80-S04H-S300

Type: Gate, Hi-Temp
 Valve Size: NPS 0.5 -to- NPS 2
 Service: High Temperature -29 to 816 °C (-20 to 1500 °F)
 Class: **API 800**
 Ends: SW
 Body: 304H SS (ASTM A182-F304H)
 Bonnet: 304H SS (ASTM A182-F304H)
 Trim: 304H SS
 Hard Facing: Full
 Wedge Type: Solid
 Bonnet Type: Welded
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: MFG STD
 The standards are:
 DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: Ladish (Forged, RP): 8867-3A36-nn-{Size}

***** Start of Low Temp or Cryo Gate Valves *****

Description of GATE Valve (GA01SD311 ==> GA15-S16-S300):

Type: Gate, Cryogenic
Valve Size:..... NPS 0.5 -to- NPS 2
Service: Cryogenic -196 to 204 °C (-320 to 400 °F)
Class: ASME 800
Ends: SW
Body: 316/316L SS (ASTM A182, F316/F316L)
Bonnet: 316/316L SS
Trim: API 600 Trim #12
Wedge Type: Solid with Cavity Vent to High-Pressure Side (Indicated)
Bonnet Type: Bolted
Design (Other):..... Extended Bonnet for Cryogenic
Bore-Port: **Standard Port (SP)**
Stem Design: OS&Y
Operation:..... Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:..... 316 SS/Graphite
Body-Bonnet Bolting:..... Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN:..... ASME B16.34 and MSS SP-134
ENDS: ASME B16.11
RATING:..... ASME B16.34
TESTING:..... API 598
DIMENSIONAL:..... MFG STD
DIMENSIONAL:..... MFG STD

Approved Vendors:..... DSI: {Size}-4111-EB-E12-22 (with Cavity Vent)
..... Ladish (Cast, FP): 8894-3A55-{MFG STD}-V-{Size} (V = Vented)

Description of GATE Valve (GA01ST510): . GA15-S16-F510

Type: Gate, Low Temp
Valve Size:..... NPS 2" -to- NPS 24
Service: Low Temperature -73 to 204 °C (-99 to 400 °F)
Class: **ASME 150**
Ends: Flanged RF
Body: 316 SS (ASTM A351-CF8M)
Bonnet: 316 SS (ASTM A351-CF8M)
Trim: API 600 **Trim #12**
Wedge Type: Solid or Flexible (NPS 0.5 -to- NPS 4), Flexible (NPS 6 and Greater)
Bonnet Type: Bolted
Bore-Port: **Full Port (FP)**
Stem Design: OS&Y
Operation:..... Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:..... 316L SS/Graphite
Body-Bonnet Bolting:..... Bolts: MFG STD
..... Nuts: MFG STD

The standards are:
DESIGN:..... ASME B16.34
ENDS: ASME B16.5
RATING:..... ASME B16.34
TESTING:..... API 598
DIMENSIONAL:..... ASME B16.10

Approved Vendors:..... DSI: {Size}-G-37-16-trim
..... Ladish (Cast, FP): 8272-205K-{Size}

Description of GATE Valve (GA01ST511):.

GA15-S16-F510

Type: Gate, Cryogenic
 Valve Size: NPS 2" -to- NPS 24
 Service: Cryogenic -196 to 204 °C (-320 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 316 SS (ASTM A351-CF8M)
 Bonnet: 316 SS (ASTM A351-CF8M)
 Trim: API 600 **Trim #12**
 Wedge Type: Solid or Flexible (NPS 0.5 -to- NPS 4), Flexible (NPS 6 and Greater)
 Bonnet Type: Bolted
 Design (Other): Extended Bonnet for Cryogenic
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34 and MSS SP-134
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Cast, FP): 8292-2052-{Size}

Description of GATE Valve (GA01ST512):.

GA15-S16-F512

Type: Gate, Low Temp
 Valve Size: NPS 8 -to- NPS 24
 Service: Low Temperature -73 to 204 °C (-99 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 316 SS (ASTM A351-CF8M)
 Bonnet: 316 SS (ASTM A351-CF8M)
 Trim: API 600 **Trim #12**
 Wedge Type: Flexible
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: **Gear**
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Cast, FP): 8272-2052-{Size} w/Gear Op.

Description of GATE Valve (GA01ST513):

GA15-S16-F513

Type: Gate, Cryogenic
 Valve Size: NPS 8 -to- NPS 24
 Service: Cryogenic -196 to 204 °C (-320 to 400 °F)
 Class: **ASME 150**
 Ends: Flanged RF
 Body: 316 SS (ASTM A351-CF8M)
 Bonnet: 316 SS (ASTM A351-CF8M)
 Trim: API 600 Trim #12
 Wedge Type: Flexible with Cavity Vent to High-Pressure Side (Indicated) Bonnet Type:
 Bolted
 Design (Other): Extended Bonnet for Cryogenic
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: **Gear**
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34 and MSS SP-134
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Cast, FP): 8292-2052-{Mfg Std}-V-{Size}, (V=Vented)

Description of GATE Valve (GA80-S16-S310), was GA01SU310):

Type: Gate, Low Temp
 Valve Size: NPS 0.5 -to- NPS 2
 Service: Low Temperature -73 to 204 °C (-99 to 400 °F)
 Class: ASME 800
 Ends: SW
 Body: 316L SS (ASTM A351-CF3M)
 Bonnet: 316L SS (ASTM A351-CF3M)
 Trim: API 602 **Trim #12**
 Wedge Type: Solid or Flexible
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.11
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors:

Description of GATE Valve (GA03ST510):

GA30-S16-F510

Type: Gate, Low Temp
 Valve Size: NPS 2" -to- NPS 24
 Service: Low Temperature -73 to 204 °C (-99 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 316 SS (ASTM A351-CF8M)
 Bonnet: 316 SS (ASTM A351-CF8M)
 Trim: API 600 Trim #12
 Wedge Type: Solid or Flexible (NPS 0.5 -to- NPS 4), Flexible (NPS 6 and Greater)
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Cast, FP): 8372-2052-{Size}

Description of GATE Valve (GA03ST512):

GA30-S16-F512

Type: Gate, Low Temp
 Valve Size: NPS 4 -to- NPS 24
 Service: Low Temperature -73 to 204 °C (-99 to 400 °F)
 Class: **ASME 300**
 Ends: Flanged RF
 Body: 316 SS (ASTM A351-CF8M)
 Bonnet: 316 SS (ASTM A351-CF8M)
 Trim: API 600 Trim #12
 Wedge Type: Flexible
 Bonnet Type: Bolted
 Bore-Port: **Full Port (FP)**
 Stem Design: OS&Y
 Operation: **Gear**
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34
 ENDS: ASME B16.5
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: ASME B16.10

Approved Vendors: Ladish (Cast, FP): 8372-2052

Description of GATE Valve (GA01SU311):

GA15-S16L-S311

Type: Gate, Cryogenic
 Valve Size: NPS 0.5 -to- NPS 2
 Service: Cryogenic -196 to 204 °C (-320 to 400 °F)
 Class: **ASME 150**
 Ends: SW
 Body: 316L SS (ASTM A351-CF3M)
 Bonnet: 316L SS (ASTM A351-CF3M)
 Trim: API 602 Trim #12
 Wedge Type: Solid or Flexible, with Cavity Vent to High-Pressure Side (Indicated) Bonnet Type:
 Bolted
 Design (Other): Extended Bonnet for Cryogenic
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34 and MSS SP-134
 ENDS: ASME B16.11
 RATING: ASME B16.34
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: Ladish (Cast, FP): 8294-6062-{mfg std}-V-{Size} (V=Vented)

Description of GATE Valve (GA08SD210):

Type: Gate, Low Temp
 Valve Size: NPS 0.5 -to- NPS 2
 Service: Low Temperature -73 to 204 °C (-99 to 400 °F)
 Class: **API 800**
 Ends: SW x FNPT
 Body: 316/316L SS (ASTM A182-F316/316L)
 Bonnet: 316/316L SS (ASTM A182-F316/316L)
 Trim: API 602 Trim #12
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: API 602
 ENDS: SW: ASME B16.11, THRD: ASME B1.20.1
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: Ladish (Forged, RP): 8867-3A55-{Size}

Description of GATE Valve (GA08SD310):

GA80-S16-S310

Type: Gate, Low Temp
 Valve Size: NPS 0.5 -to- NPS 2
 Service: Low Temperature -73 to 204 °C (-99 to 400 °F)
 Class: **API 800**
 Ends: SW
 Body: 316/316L SS (ASTM A182-F316/316L)
 Bonnet: 316/316L SS (ASTM A182-F316/316L)
 Trim: API 602 Trim #12
 Wedge Type: Solid
 Bonnet Type: Bolted
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: API 602
 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: Ladish (Forged, RP): 8864-2A55-HH-{Size}

Description of GATE Valve (GA08SD311):

GA80-S16-S311

Type: Gate, Cryogenic
 Valve Size: NPS 0.5 -to- NPS 2
 Service: Cryogenic -196 to 204 °C (-320 to 400 °F)
 Class: **API 800**
 Ends: SW
 Body: 316/316L SS (ASTM A182-F316/316L)
 Bonnet: 316/316L SS (ASTM A182-F316/316L)
 Trim: API 602 Trim #12
 Wedge Type: Solid with Cavity Vent to High-Pressure Side (Indicated)
 Bonnet Type: Bolted
 Design (Other): Extended Bonnet for Cryogenic
 Bore-Port: **Standard Port (SP)**
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 316L SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD
 The standards are:
 DESIGN: ASME B16.34 and MSS SP-134 ENDS: ASME B16.11
 RATING: API 602
 TESTING: API 598
 DIMENSIONAL: MFG STD

Approved Vendors: Ladish (Forged, RP): 8864-2A55-{Size}



Description of GATE Valve (GA03SD311):

GA30-S16-S311

Type: Gate, Cryogenic
Valve Size: NPS 0.5 -to- NPS 2
Service: Cryogenic -196 to 204 °C (-320 to 400 °F)
Class: **ASME 300**
Ends: SW
Body: 316/316L SS (ASTM A351-CF8M/CF3M)
Bonnet: 316/316L SS (ASTM A351-CF8M/CF3M)
Trim: API 602 Trim #12
Wedge Type: Solid with Cavity Vent to High-Pressure Side (Indicated)
Bonnet Type: Bolted
Design (Other): Extended Bonnet for Cryogenic
Bore-Port: **Standard Port (SP)**
Stem Design: OS&Y
Operation: Handwheel
Stem-Packing: Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets: 316 SS/Graphite
Body-Bonnet Bolting: Bolts: MFG STD
..... Nuts: MFG STD
The standards are:
DESIGN: ASME B16.34 and MSS SP-134 ENDS: ASME B16.11
RATING: ASME B16.34
TESTING: API 598
DIMENSIONAL: MFG STD

Approved Vendors:

***** End of Low Temp or Cryo Gate Valves *****

Rev.	RevDate	RevDescription	By	Checked
0	08 SEPT 2023	Issued for release	MFS	SBS
1	18 SEPT 2023	Revise Valve Information	MFS	SBS
2	26 MAR 2024	Add/Modify Valve Tags	MFS	WGS
3	01 MAY 2024	Add additional Vendors	MFS	MFS

Description of GLOBE Valve (GL01ST500):

GL15-S16-F500

Type:Globe, Mid-Temp
 Valve Size:NPS 2" -to- NPS 12"
 Service:-29 to 316 °C (-20 to +600 °F)
 Class:**ASME 150**
 Ends:Flanged RF
 Body:316 SS (ASTM A351-CF8M)
 Bonnet:316 SS (ASTM A351-CF8M)
 Trim:API 623 **Trim #12**
 Port:Full Port (FP)
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:Handwheel to 6" NPS; Gear to 12" NPS
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10

Approved Vendors:DSI (Cast, CL150, FP): {Size}-133-16-16 HFS (2" to 6" NPS)
DSI (Cast, CL150, FP): {Size}-G-133-16-16 HFS (8" to 12" NPS, Gear Op.)
Bonney Forge (Cast, CL150, FP): {Size}-1-38S-RF (2" to 6" NPS)
Bonney Forge (Cast, CL150, FP): {Size}-1-38S-RF-BG (8" to 12" NPS, Gear Op.)
Warren (Cast, CL150, FP): {Size}- 2156-12 (to 6" Handwheel; 8" to 12" NPS, Gear Op.)
Ladish (Cast, CL150, FP): 7272-3052-PGnn-A-{Size}

Description of GLOBE Valve (GL01ST501):

GL15-S16-F501

Type:Globe, Normal Temp.
 Valve Size:NPS 2" -to- NPS 16"
 Service:-29 to 204 °C (-20 to +400 °F)
 Class:**ASME 150**
 Ends:Flanged RF
 Body:316 SS (ASTM A351-CF8M)
 Bonnet:316 SS (ASTM A351-CF8M)
 Trim:API 623 **Trim #12**
 Port:Full Port (FP)
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:PTFE
 Body-Bonnet Gaskets:PTFE
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10

Approved Vendors:Ladish (Cast, CL150, FP): 7272-3052-TTnn-A-{Size}
DSI (Cast, CL150, FP): {Size}-133-16-16 HFS-T (2" to 6" NPS)
DSI (Cast, CL150, FP): {Size}-G-133-16-16 HFS-T (8" to 12" NPS)

**Description of GLOBE Valve (GL01ST502):**

GL15-S16-F502

Type:Globe
Valve Size:NPS 2 -to- NPS 16
Service:-29 to 316 °C (-20 to +600 °F)
Class:**ASME 150**
Ends:Flanged RF
Body:316 SS (ASTM A351-CF8M)
Bonnet:316 SS (ASTM A351-CF8M)
Trim:API 623 **Trim #11**
Port:Full Port (FP)
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:Handwheel
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:316L SS/Graphite
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:ASME B16.34
ENDS:ASME B16.5
RATING:ASME B16.34
TESTING:API 598
DIMENSIONAL:ASME B16.10
CLEANING:Oxygen Cleaning in accordance with CGA C-4.1 and G-4.4/ASTM G-93

Approved Vendors:Ladish (Cast, CL150, FP): 7272-305K-GGnn-B-{Size}

Description of GLOBE Valve (GL01ST503):

GL15-S16-F503

Type:Globe
Valve Size:NPS 6 -to- NPS 16
Service:-29 to 316 °C (-20 to +600 °F)
Class:**ASME 150**
Ends:Flanged RF
Body:316 SS (ASTM A351-CF8M)
Bonnet:316 SS (ASTM A351-CF8M)
Trim:API 623 **Trim #12**
Port:Full Port (FP)
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:**Gear**
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:316L SS/Graphite
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:ASME B16.34
ENDS:ASME B16.5
RATING:ASME B16.34
TESTING:API 598
DIMENSIONAL:ASME B16.10

Approved Vendors:DSI (Cast, CL150, FP): {Size}-G-133-16-16 HFS (6" to 12" NPS)
.....Ladish (Cast, CL150, FP): 7272-3052-GGnn-G-{Size}

**Description of GLOBE Valve (GL01ST504):**

GL15-S16-F504

Type:Globe
Valve Size:NPS 6 -to- NPS 16
Service:-29 to 204 °C (-20 to +400 °F)
Class:**ASME 150**
Ends:Flanged RF
Body:316 SS (ASTM A351-CF8M)
Bonnet:316 SS (ASTM A351-CF8M)
Trim:API 623 **Trim #12**
Port:Full Port (FP)
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:**Gear**
Stem-Packing:PTFE
Body-Bonnet Gaskets:PTFE
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
ENDS:ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10

Approved Vendors:DSI (Cast, CL150, FP): {Size}-G-133-16-16 HFS-T (6" to 12" NPS)
.....Ladish (Cast, CL150, FP): 7272-8052-TTnn-G-{Size}

Description of GLOBE Valve (GL01ST505):

GL15-S16-F505

Type:Globe
Valve Size:NPS 6 -to- NPS 16
Service:-29 to 316 °C (-20 to +600 °F)
Class:**ASME 150**
Ends:Flanged RF
Body:316 SS (ASTM A351-CF8M)
Bonnet:316 SS (ASTM A351-CF8M)
Trim:API 623 **Trim #11**
Port:Full Port (FP)
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:**Gear**
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:316L SS/Graphite
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
ENDS:ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10
CLEANING:.....Oxygen Cleaning in accordance with CGA C-4.1 and G-4.4/ASTM G-93

Approved Vendors: Ladish (Cast, CL150, FP): 7272-805K-GGnn-B-G-{Size}

Description of GLOBE Valve (GL15-S16-F520):

Type:Globe, Integral Flange
 Valve Size:NPS 0.5 -to- NPS 2
 Service:-29 to 316 °C (-20 to +600 °F)
 Class:**ASME 150**
 Ends:Flanged, RF
 Body:316 SS (ASTM A182-F316)
 Bonnet:316 SS (ASTM A182-F316)
 Trim:API **Trim #12**
 Body Type:T-body
 Port:**Standard Port (Conventional Port)**
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:ASME B16.10

Approved Vendors:Bonney Forge (Forged, CL150, RP): L1-30-8-RF
DSI (Forged, CL150, RP): {Size}-1231-D12

Description of GLOBE Valve (GL15-S16-F521):

Type:Globe, Integral Flange
 Valve Size:NPS 0.5 -to- NPS 2
 Service:-29 to 316 °C (-20 to +600 °F)
 Class:**ASME 150**
 Ends:Flanged, RF
 Body:316 SS (ASTM A182-F316)
 Bonnet:316 SS (ASTM A182-F316)
 Trim:API **Trim #12**
 Body Type:T-body
 Port:**Full Port**
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:ASME B16.10

Approved Vendors:Bonney Forge (Forged, CL150, FP): 1-30-8-RF
DSI (Forged, CL150, FP): {Size}-1281-FB-D12

**Description of GLOBE Valve (GL03ST500):**

GL30-S16-F500

Type:Globe
Valve Size:NPS 2" -to- NPS 12
Service:-29 to 316 °C (-20 to +600 °F)
Class:**ASME 300**
Ends:Flanged RF
Body:316 SS (ASTM A351-CF8M)
Bonnet:316 SS (ASTM A351-CF8M)
Trim:API 623 **Trim #12**
Port:Full Port (FP)
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:Handwheel to 6" NPS; Gear to 12" NPS
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:316L SS/Graphite
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
ENDS:ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10 (NPS 0.5 -to- NPS 12)

Approved Vendors:DSI (Cast, CL300, FP): {Size}-141-16-16 HFS (2" to 6" NPS, Gear Op to 12")
.....Bonney Forge (Cast, CL300, FP): {Size}-3-38S-RF (2" to 6" NPS)
.....Bonney Forge (Cast, CL300, FP): {Size}-3-38S-RF-BG (8" to 12" NPS, Gear Op.)
.....Warren (Cast, CL300, FP): {Size}- 2306-12 (to 6" Handwheel; 8" to 12" NPS, Gear)
.....Ladish (Cast, CL300, FP): 7372-8052-PGnn-A-{Size}

Description of GLOBE Valve (GL03ST501):

GL30-S16-F501

Type:Globe
Valve Size:NPS 2" -to- NPS 16
Service:-29 to 204 °C (-20 to +400 °F)
Class:**ASME 300**
Ends:Flanged RF
Body:316 SS (ASTM A351-CF8M)
Bonnet:316 SS (ASTM A351-CF8M)
Trim:API 623 **Trim #12**
Port:Full Port (FP)
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:Handwheel
Stem-Packing:PTFE
Body-Bonnet Gaskets:PTFE
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
ENDS:ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10 (NPS 0.5 -to- NPS 12)
.....MFG STD (NPS 14 -to- NPS 16)

Approved Vendors:DSI (Cast, CL300, FP): {Size}-141-16-16 HFS-T (2" to 12" NPS)
.....Ladish (Cast, CL300, FP): 7372-8052-TTnn-A-{Size}

**Description of GLOBE Valve (GL03ST502):**

GL30-S16-F502

Type:Globe
Valve Size:NPS 2 -to- NPS 16
Service:-29 to 316 °C (-20 to +600 °F)
Class:**ASME 300**
Ends:Flanged RF
Body:316 SS (ASTM A351-CF8M)
Bonnet:316 SS (ASTM A351-CF8M)
Trim:API 623 **Trim #11**
Port:Full Port (FP)
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:Handwheel
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:Alloy 400/Graphite
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:ASME B16.34
ENDS:ASME B16.5
RATING:ASME B16.34
TESTING:API 598
DIMENSIONAL:ASME B16.10 (NPS 0.5 -to- NPS 12)
.....MFG STD (NPS 14 -to- NPS 16)
CLEANING:Oxygen Cleaning in accordance with CGA C-4.1 and G-4.4/ASTM G-93

Approved Vendors: Ladish (Cast, CL300, FP): 7372-805K-GPnn-G-{Size}

Description of GLOBE Valve (GL03ST503):

GL30-S16-F503

Type:Globe
Valve Size:NPS 4 -to- NPS 16
Service:-29 to 316 °C (-20 to +600 °F)
Class:**ASME 300**
Ends:Flanged RF
Body:316 SS (ASTM A351-CF8M)
Bonnet:316 SS (ASTM A351-CF8M)
Trim:API 623 **Trim #12**
Port:Full Port (FP)
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:**Gear**
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:316L SS/Graphite
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:ASME B16.34
ENDS:ASME B16.5
RATING:ASME B16.34
TESTING:API 598
DIMENSIONAL:ASME B16.10 (NPS 4 -to- NPS 12)
.....MFG STD (NPS 14 -to- NPS 16)

Approved Vendors:DSI (Cast, CL300, FP): {Size}-G-141-16-16 HFS (4" to 16" NPS)
.....Ladish (Cast, CL300, FP): 7372-8052-PGnn-G-{Size}



Description of GLOBE Valve (GL03ST504):

GL30-S16-F504

Type:Globe
Valve Size:NPS 4 -to- NPS 16
Service:-29 to 204 °C (-20 to +400 °F)
Class:**ASME 300**
Ends:Flanged RF
Body:316 SS (ASTM A351-CF8M)
Bonnet:316 SS (ASTM A351-CF8M)
Trim:API 623 **Trim #12**
Port:Full Port (FP)
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:**Gear**
Stem-Packing:PTFE
Body-Bonnet Gaskets:PTFE
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
ENDS:ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10 (NPS 4 -to- NPS 12)
.....MFG STD (NPS 14 -to- NPS 16)

Approved Vendors:DSI (Cast, CL300, FP): {Size}-G-141-16-16 HFS-T (4" to 12" NPS)
.....Ladish (Cast, CL300, FP): 7372-8052-TTnn-G-{Size}

Description of GLOBE Valve (GL03ST505):

GL30-S16-F505

Type:Globe
Valve Size:NPS 4 -to- NPS 16
Service:-29 to 316 °C (-20 to +600 °F)
Class:**ASME 300**
Ends:Flanged RF
Body:316 SS (ASTM A351-CF8M)
Bonnet:316 SS (ASTM A351-CF8M)
Trim:API 623 **Trim #11**
Port:Full Port (FP)
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:**Gear**
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:Alloy 400/Graphite
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
ENDS:ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10 (NPS 4 -to- NPS 12)
.....MFG STD (NPS 14 -to- NPS 16)
CLEANING:.....Oxygen Cleaning in accordance with CGA C-4.1 and G-4.4/ASTM G-93

Approved Vendors:Ladish (Cast, CL300, FP): 7372-805K-PGnn-G-{Size}

Description of GLOBE Valve (GL03SU300):

GL30-S16L-S300

Type:Globe
 Valve Size:NPS 0.5 -to- NPS 2
 Service:-29 to 204 °C (-20 to +400 °F)
 Class:**ASME 300**
 Ends:SW
 Body:316L SS (ASTM A351-CF3M)
 Bonnet:316L SS (ASTM A351-CF3M)
 Trim:API 602 **Trim #12**
 Port:Full Port (FP)
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:PTFE
 Body-Bonnet Gaskets:PTFE
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
 ENDS:ASME B16.11
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Ladish (Cast, CL300, FP): 7361-6062-TTnn-A-{Size}

Description of GLOBE Valve (GL03SU301):

GL30-S16L-S301

Type:Globe
 Valve Size:NPS 0.5 -to- NPS 2
 Service:-29 to 316 °C (-20 to +600 °F)
 Class:**ASME 300**
 Ends:SW
 Body:316L SS (ASTM A351-CF3M)
 Bonnet:316L SS (ASTM A351-CF3M)
 Trim:API 602 **Trim #12**
 Port:Full Port (FP)
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
 ENDS:ASME B16.11
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Ladish (Cast, CL300, FP): 7361-6062-PGnn-A-{Size}

Description of GLOBE Valve (GL06ST500):... GL60-S16-F500

Type:Globe
Valve Size:NPS 2" -to- NPS 12
Service:-29 to 316 °C (-20 to +600 °F)
Class:**ASME 600**
Ends:Flanged RF
Body:316 SS (ASTM A351-CF8M)
Bonnet:316 SS (ASTM A351-CF8M)
Trim:API 623 **Trim #12**
Port:Full Port (FP)
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:Handwheel to 6" NPS; Gear to 12" NPS
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:316 SS/Graphite
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:ASME B16.34
ENDS:ASME B16.5
RATING:ASME B16.34
TESTING:API 598
DIMENSIONAL:ASME B16.10 - Long Pattern (NPS 0.5 -to- NPS 10)
.....MFG STD (NPS 12 -to- NPS 14)

Approved Vendors:DSI (Cast, CL600, FP): {Size}-161-16-16 HFS (2" to 8" NPS, Gear at 8" NPS)
.....Bonney Forge (Cast, CL600): {Size}-6-38S-RF (2" to 6" NPS, Handwheel)
.....Bonney Forge (Cast, CL600): {Size}-6-38S-RF-BG (8" NPS, Gear Op.)
.....Warren (Cast, CL600): {Size}- 2606-12 (2" to 6" Handwheel; 8" to 12" NPS, Gear Op.)
.....Ladish (Cast, CL600, FP): 7672-3052-PGnn-A-{Size}

Description of GLOBE Valve (GL06ST501):... GL60-S16-F501

Type:Globe
Valve Size:NPS 2" -to- NPS 14
Service:-29 to 316 °C (-20 to +600 °F)
Class:**ASME 600**
Ends:Flanged RF
Body:316 SS (ASTM A351-CF8M)
Bonnet:316 SS (ASTM A351-CF8M)
Trim:API 623 **Trim #12**
Port:Full Port (FP)
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:Handwheel to 6" NPS; Gear to 12" NPS
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:316 SS /Graphite
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:ASME B16.34
ENDS:ASME B16.5
RATING:ASME B16.34
TESTING:API 598
DIMENSIONAL:ASME B16.10 - Long Pattern (NPS 0.5 -to- NPS 10)
.....MFG STD (NPS 12 -to- NPS 14)

Approved Vendors:DSI (Cast, CL600, FP): {Size}-161-16-16 HFS (2" to 8" NPS, Gear at 8" NPS)
.....Warren (Cast, CL600): {Size}- 2606-12 (2" to 6" Handwheel; 8" to 12" NPS, Gear Op.)
.....Bonney Forge (Cast, CL600): {Size}-6-38S-RF (2" to 6" NPS, Handwheel)
.....Bonney Forge (Cast, CL600): {Size}-6-38S-RF-BG (8" NPS, Gear Op.)
.....Ladish (Cast, CL600, FP): 7672-3052-PGnn-A-{Size}

Description of GLOBE Valve (GL06ST502):

GL60-S16-F502

Type:Globe
 Valve Size:NPS 2" -to- NPS 14
 Service:-29 to 204 °C (-20 to +400 °F)
 Class:**ASME 600**
 Ends:Flanged RF
 Body:316 SS (ASTM A351-CF8M)
 Bonnet:316 SS (ASTM A351-CF8M)
 Trim:API 623 **Trim #12**
 Port:Full Port (FP)
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:PTFE
 Body-Bonnet Gaskets:PTFE
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:ASME B16.10 - Long Pattern (NPS 0.5 -to- NPS 10)
MFG STD (NPS 12 -to- NPS 14)

Approved Vendors:DSI (Cast, CL600, FP): {Size}-161-16-16 HFS-T (2" to 8" NPS)
Ladish (Cast, CL600, FP): 7672-8052-TT01-A-{Size}

Description of GLOBE Valve (GL06ST503):

GL60-S16-F503

Type:Globe
 Valve Size:NPS 3 -to- NPS 14
 Service:-29 to 316 °C (-20 to +600 °F)
 Class:**ASME 600**
 Ends:Flanged RF
 Body:316 SS (ASTM A351-CF8M)
 Bonnet:316 SS (ASTM A351-CF8M)
 Trim:API 623 **Trim #12**
 Port:Full Port (FP)
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:**Gear**
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:ASME B16.10 - Long Pattern (NPS 3 -to- NPS 12)
MFG STD (NPS 14 -to- NPS 14)

Approved Vendors:DSI (Cast, CL600, FP): {Size}-G-161-16-16 HFS (3" to 8" NPS)
Ladish (Cast, CL600, FP): 7672-8052-PGnn-G-{Size}

Description of GLOBE Valve (GL06ST504):

GL60-S16-F504

Type:Globe
 Valve Size:NPS 3 -to- NPS 14
 Service:-29 to 316 °C (-20 to +600 °F)
 Class:**ASME 600**
 Ends:Flanged RF
 Body:316 SS (ASTM A351-CF8M)
 Bonnet:316 SS (ASTM A351-CF8M)
 Trim:API 623 **Trim #11**
 Port:Full Port (FP)
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:**Gear**
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:Alloy 400/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:ASME B16.10 - Long Pattern (NPS 3 -to- NPS 10)
MFG STD (NPS 12 -to- NPS 14)
 CLEANING:Oxygen Cleaning in accordance with CGA C-4.1 and G-4.4/ASTM G-93

Approved Vendors:Ladish (Cast, FP): 7672-305K-PGnn-G-{Size}

Description of GLOBE Valve (GL06ST505):

GL60-S16-F505

Type:Globe
 Valve Size:NPS 3 -to- NPS 14
 Service:-29 to 204 °C (-20 to +400 °F)
 Class:**ASME 600**
 Ends:Flanged RF
 Body:316 SS (ASTM A351-CF8M)
 Bonnet:316 SS (ASTM A351-CF8M)
 Trim:API 623 **Trim #12**
 Port:Full Port (FP)
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:**Gear**
 Stem-Packing:PTFE
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:ASME B16.10 - Long Pattern (NPS 3 -to- NPS 10)
MFG STD (NPS 12 -to- NPS 14)

Approved Vendors:Ladish (Cast, FP): 7672-8052-TTnn-G-{Size}

Description of GLOBE Valve (GL80-S16-X210)

Type:Globe
 Valve Size:NPS 0.5 -to- NPS 2
 Service:-29 to 316 °C (-20 to +600 °F)
 Class:**API 800**
 Ends:SW x FNPT
 Body:316/316L SS (ASTM A182-F316/316L)
 Bonnet:316/316L SS (ASTM A182-F316/316L)
 Trim:API 602 **Trim #12**
 Body Type:T-body
 Port:.....**Standard Port (RP)**
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....API 602
 ENDS:ASME B1.20.1
 RATING:.....API 602
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Bonney Forge (Forged, RP): HL-30-8-SW/T
DSI (Forged, RP): 4201-E12

Description of GLOBE Valve (GL80-S16-T210)

Type:Globe
 Valve Size:NPS 0.5 -to- NPS 2
 Service:-29 to 204 °C (-20 to +400 °F)
 Class:**API 800**
 Ends:FNPT (Threaded)
 Body:316/316L SS (ASTM A182-F316/316L)
 Bonnet:316/316L SS (ASTM A182-F316/316L)
 Trim:API 602 **Trim #12**
 Body Type:T-body
 Port:.....**Standard Port (RP)**
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....API 602
 ENDS:ASME B1.20.1
 RATING:.....API 602
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Bonney Forge (Forged, RP): HL-30-8-T
DSI (Forged, RP): 4221-E12

**Description of GLOBE Valve (GL08SU000):**

GL80-S16L-T100

Type:Globe
Valve Size:NPS 0.25 -to- NPS 0.75
Service:-29 to 316 °C (-20 to +600 °F)
Class:**API 800**
Ends:FNPT
Body:316L SS (ASTM A182-F316L)
Bonnet:316L SS (ASTM A182-F316L)
Trim:API 602 **Trim #12**
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:Handwheel
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:316 SS/Graphite
Body-Bonnet Bolting:Bolts: API 602
.....Nuts: API602

The standards are:

DESIGN:.....API 602
ENDS:ASME B1.20.1
RATING:.....API 602
TESTING:.....API 598
DIMENSIONAL:MFG STD

Approved Vendors:Ladish (Forged): 7860-6A65-CGnn-A-{Size}

Description of GLOBE Valve (GL08SU001):

GL80-S16L-T101

Type:Globe
Valve Size:NPS 0.25 -to- NPS 0.75
Service:-29 to 204 °C (-20 to +400 °F)
Class:**API 800**
Ends:FNPT
Body:316L SS (ASTM A182-F316L)
Bonnet:316L SS (ASTM A182-F316L)
Trim:API 602 **Trim #12**
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:Handwheel
Stem-Packing:PTFE
Body-Bonnet Gaskets:PTFE
Body-Bonnet Bolting:Bolts: API 602
.....Nuts: API602

The standards are:

DESIGN:.....API 602
ENDS:ASME B1.20.1
RATING:.....API 602
TESTING:.....API 598
DIMENSIONAL:MFG STD

Approved Vendors:Ladish (Forged): 7860-6A65-TTnn-A-{Size}

**Description of GLOBE Valve (GL08SU300):**

GL80-S16L-S300

Type:Globe
Valve Size:NPS 0.5 -to- NPS 2
Service:-29 to 316 °C (-20 to +600 °F)
Class:**API 800**
Ends:SW
Body:316L SS (ASTM A182-F316L)
Bonnet:316L SS (ASTM A182-F316L)
Trim:API 602 **Trim #12**
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Port:Standard
Stem Design:OS&Y
Operation:Handwheel
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:316 SS/Graphite
Body-Bonnet Bolting:Bolts: API 602
.....Nuts: API602

The standards are:

DESIGN:API 602
ENDS:ASME B16.11
RATING:API 602
TESTING:API 598
DIMENSIONAL:MFG STD

Approved Vendors:Ladish (Forged): 7861-6A65-CGnn-A-{Size}

Description of GLOBE Valve (GL80-S16-S300):

Type:Globe
Valve Size:NPS 0.5 -to- NPS 2
Service:-29 to 316 °C (-20 to +600 °F)
Class:**API 800**
Ends:SW
Body:316L SS (ASTM A182-F316)
Bonnet:316L SS (ASTM A182-F316)
Trim:API 602 **Trim #12**
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Port:Standard (RP)
Stem Design:OS&Y
Operation:Handwheel
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:316 SS/Graphite
Body-Bonnet Bolting:Bolts: API 602
.....Nuts: API602

The standards are:

DESIGN:API 602
ENDS:ASME B16.11
RATING:API 602
TESTING:API 598
DIMENSIONAL:MFG STD

Approved Vendors:Bonney Forge (Forged, RP): HL-30-8-SW
.....DSI (Forged, RP); {size}-4211-D12 (-E12 OK also)

Description of GLOBE Valve (GL09ST500):

GL90-S16-F500

Type:Globe
 Valve Size:NPS 3 -to- NPS 12
 Service:-29 to 316 °C (-20 to +600 °F)
 Class:**ASME 900**
 Ends:Flanged RF
 Body:316 SS (ASTM A351-CF8M)
 Bonnet:316 SS (ASTM A351-CF8M)
 Trim:API 623 **Trim #12**
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10 - Long Pattern

Approved Vendors:Ladish (Cast): 7972-8052-GGnn-A-{Size}

Description of GLOBE Valve (GL09ST501):

GL90-S16-F501

Type:Globe
 Valve Size:NPS 3 -to- NPS 12
 Service:-29 to 316 °C (-20 to +600 °F)
 Class:**ASME 900**
 Ends:Flanged RF
 Body:316 SS (ASTM A351-CF8M)
 Bonnet:316 SS (ASTM A351-CF8M)
 Trim:API 623 **Trim #12**
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:**Gear**
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10 - Long Pattern

Approved Vendors:Ladish (Cast): 7972-8052-GGnn-G-{Size}



Description of GLOBE Valve (GL15SD300):

GL1.5K-S16-S300

Type:Globe
Valve Size:NPS 0.5 -to- NPS 2
Service:-29 to 204 °C (-20 to +400 °F)
Class:**ASME 1500**
Ends:SW
Body:316/316L SS (ASTM A182-F316/316L)
Bonnet:316/316L SS (ASTM A182-F316/316L)
Trim:API 602 **Trim #12**
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:Handwheel
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:316 SS/Graphite
Body-Bonnet Bolting:Bolts: API 602
.....Nuts: API602

The standards are:

DESIGN:.....API 602
ENDS:ASME B16.11
RATING:.....API 602
TESTING:.....API 598
DIMENSIONAL:MFG STD

Approved Vendors:Ladish (Forged, RP): 7561-6A55-CGnn-A-{Size}

Description of GLOBE Valve (GL15ST500):

GL1.5K-S16-F500

Type:Globe
 Valve Size:NPS 0.5 -to- NPS 2
 Service:-29 to 204 °C (-20 to +400 °F)
 Class:**ASME 1500**
 Ends:Flanged RF
 Body:316 SS (ASTM A351-CF8M)
 Bonnet:316 SS (ASTM A351-CF8M)
 Trim:API 602 **Trim #12**
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10 - Long Pattern

Approved Vendors:Ladish (Cast, FP): 7572-6055-GGnn-A-{Size}

Description of GLOBE Valve (GL15ST501):

GL1.5K-S16-F501

Type:Globe
 Valve Size:NPS 2 -to- NPS 12
 Service:-29 to 204 °C (-20 to +400 °F)
 Class:**ASME 1500**
 Ends:Flanged RF
 Body:316 SS (ASTM A351-CF8M)
 Bonnet:316 SS (ASTM A351-CF8M)
 Trim:API 623 **Trim #12**
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10 - Long Pattern

Approved Vendors:Ladish (Cast, FP): 7572-8055-GGnn-A-{Size}



Description of GLOBE Valve (GL15ST502):

GL1.5K-S16-F502

Type:Globe
Valve Size:NPS 3 -to- NPS 12
Service:-29 to 204 °C (-20 to +400 °F)
Class:**ASME 1500**
Ends:Flanged RF
Body:316 SS (ASTM A351-CF8M)
Bonnet:316 SS (ASTM A351-CF8M)
Trim:API 623 **Trim #12**
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:**Gear**
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:316L SS/Graphite
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
ENDS:ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:ASME B16.10 - Long Pattern

Approved Vendors:Ladish (Cast, FP): 7572-8055-GGnn-G-{Size}



*****S22 = 2205 Duplex SS Valves*****

Description of GLOBE Valve (GL01SP500):

GL15-S22-F500

Type:Globe
Valve Size:NPS 2" -to- NPS 16"
Service:-29 to 204 °C (-20 to +400 °F)
Class:**ASME 150**
Ends:Flanged RF
Body:2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Bonnet:2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Trim:2205 Duplex SS
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:Handwheel
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:2205 Duplex
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
ENDS:ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10

Approved Vendors:Ladish (Cast): 7272-3553-PGnn-A-{Size}

Description of GLOBE Valve (GL01SP501):

GL15-S22-F501

Type:Globe
Valve Size:NPS 6 -to- NPS 16
Service:-29 to 204 °C (-20 to +400 °F)
Class:**ASME 150**
Ends:Flanged RF
Body:2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Bonnet:2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Trim:2205 Duplex SS
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:**Gear**
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:2205 Duplex
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
ENDS:ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10

Approved Vendors:Ladish (Cast): 7272-3553-PGnn-G-{Size}

Description of GLOBE Valve (GL03SP500):

GL30-S22-F500

Type:Globe
 Valve Size:NPS 2 -to- NPS 16
 Service:-29 to 204 °C (-20 to +400 °F)
 Class:**ASME 300**
 Ends:Flanged RF
 Body:2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Bonnet:2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim:2205 Duplex SS
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:2205 Duplex
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10 (NPS 0.5 -to- NPS 12)
MFG STD (NPS 14 -to- NPS 16)

Approved Vendors:Ladish (Cast, FP): 7372-3551-PG04-A-{Size}

Description of GLOBE Valve (GL03SP501):

GL30-S22-F501

Type:Globe
 Valve Size:NPS 4 -to- NPS 16
 Service:-29 to 204 °C (-20 to +400 °F)
 Class:**ASME 300**
 Ends:Flanged RF
 Body:2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Bonnet:2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
 Trim:2205 Duplex SS
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:**Gear**
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:2205 Duplex
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10 (NPS 4 -to- NPS 12)
MFG STD (NPS 14 -to- NPS 16)

Approved Vendors:Ladish (Cast, FP): 7372-3551-PGnn-G-{Size}

**Description of GLOBE Valve (GL03SP300):**

GL30-S22-S300

Type:Globe
Valve Size:NPS 0.5 -to- NPS 2
Service:-29 to 204 °C (-20 to +400 °F)
Class:**ASME 300**
Ends:SW
Body:2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Bonnet:2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Trim:2205 Duplex SS
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:Handwheel
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:2205 Duplex
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
ENDS:.....ASME B16.11
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....MFG STD

Approved Vendors:Ladish (Cast, CL 300, FP): 7361-3551-PGnn-A-{Size}

Description of GLOBE Valve (GL06SP501):

GL60-S22-S300

Type:Globe
Valve Size:NPS 3 -to- NPS 14
Service:-29 to 204 °C (-20 to +400 °F)
Class:**ASME 600**
Ends:Flanged RF
Body:2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Bonnet:2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Trim:2205 Duplex SS
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:**Gear**
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:2205 Duplex
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
ENDS:.....ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10 - Long Pattern (NPS 3 -to- NPS 10)
.....MFG STD (NPS 12 -to- NPS 14)

Approved Vendors:Ladish (Cast, FP): 7672-3551-PGnn-G-{Size}

**Description of GLOBE Valve (GL08SP300):**

GL80-S22-S300

Type:Globe
Valve Size:NPS 0.5 -to- NPS 2
Service:-29 to 204 °C (-20 to +400 °F)
Class:**API 800**
Ends:SW
Body:2205 Duplex SS (ASTM A182-F51)
Bonnet:2205 Duplex SS (ASTM A182-F51)
Trim:2205 Duplex SS
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:Handwheel
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:2205 Duplex
Body-Bonnet Bolting:Bolts: API 602
.....Nuts: API 602

The standards are:

DESIGN:.....API 602
ENDS:ASME B16.11
RATING:.....API 602
TESTING:.....API 598
DIMENSIONAL:MFG STD

Approved Vendors:Ladish (Forged): 7861-6F54-CGnn-A-{Size}

Description of GLOBE Valve (GL09SP500):

GL90-S22-F500

Type:Globe
Valve Size:NPS 3 -to- NPS 12
Service:-29 to 204 °C (-20 to +400 °F)
Class:**ASME 900**
Ends:Flanged RF
Body:2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Bonnet:2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Trim:2205 Duplex SS
Hard Facing:Half
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:Handwheel
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:2205 Duplex
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
ENDS:ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:ASME B16.10 - Long Pattern

Approved Vendors:Ladish (Cast): 7972-3552-GGnn-A-{Size}

**Description of GLOBE Valve (GL15SP500):**

GL1.5K-S22-F500

Type:Globe
Valve Size:NPS 2 -to- NPS 12
Service:-29 to 204 °C (-20 to +400 °F)
Class:**ASME 1500**
Ends:Flanged RF
Body:2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Bonnet:2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Trim:2205 Duplex SS
Hard Facing:Half
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:Handwheel
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:2205 Duplex
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
ENDS:ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10 - Long Pattern

Approved Vendors:Ladish (Cast, FP): 7572-3555-GGnn-A-{Size}

Description of GLOBE Valve (GL15SP501):

GL1.5K-S22-F501

Type:Globe
Valve Size:NPS 3 -to- NPS 12
Service:-29 to 204 °C (-20 to +400 °F)
Class:**ASME 1500**
Ends:Flanged RF
Body:2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Bonnet:2205 Duplex SS (ASTM A995-4A (ACI CD3MN))
Trim:2205 Duplex SS
Hard Facing:Half
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:**Gear**
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:2205 Duplex
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
ENDS:ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10 - Long Pattern

Approved Vendors:Ladish (Cast, FP): 7572-3555-GGnn-G-{Size}

***** 2507 Super Duplex SS*****

Description of GLOBE Valve (GL01SQ500 ==> GL15-S25-F500)

Type:	Globe
Valve Size:	NPS 2" -to- NPS 16"
Service:	-29 to 204 °C (-20 to +400 °F)
Class:	ASME 150
Ends:	Flanged RF
Body:	2507 Super Duplex SS (ASTM A995-5A (CE3MN)
Bonnet:	2507 Super Duplex SS (ASTM A995-5A (CE3MN)
Trim:	2205 Super Duplex SS
Body Type:	T-body
Port:	Reduced Port
Bonnet Type:	Bolted
Disc Type:	Conical
Stem Design:	OS&Y
Operation:	Handwheel
Stem-Packing:	Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:	2507 Super Duplex
Body-Bonnet Bolting:	Bolts: MFG STD
	Nuts: MFG STD

The standards are:

DESIGN:.....	ASME B16.34
ENDS:.....	ASME B16.5
RATING:.....	ASME B16.34
TESTING:.....	API 598
DIMENSIONAL:.....	ASME B16.10

Approved Vendors: Ladish (CL150, RP, Cast): 7275-3564-PGnn-A-{Size}

Description of GLOBE Valve (GL03SQ500 ==> GL30-S25-F500):

Type:	Globe
Valve Size:	NPS 2" -to- NPS 16"
Service:	-29 to 204 °C (-20 to +400 °F)
Class:	ASME 300
Ends:	Flanged RF
Body:	2507 Duplex SS (ASTM A995-5A (CE3MN)
Bonnet:	2507 Duplex SS (ASTM A995-5A (CE3MN)
Trim:	2205 Duplex SS
Port:	Reduced Port
Body Type:	T-body
Bonnet Type:	Bolted
Disc Type:	Conical
Stem Design:	OS&Y
Operation:	Handwheel
Stem-Packing:	Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:	2205 Duplex
Body-Bonnet Bolting:	Bolts: MFG STD
	Nuts: MFG STD

The standards are:

DESIGN:.....	ASME B16.34
ENDS:	ASME B16.5
RATING:	ASME B16.34
TESTING:.....	API 598
DIMENSIONAL:.....	ASME B16.10

Approved Vendors: Ladish (CL300, RP, Cast): 7375-3564-PGnn-A-{Size}

**Description of GLOBE Valve (GL08SQ300 ==> GL80-S25-S300)**

Type:Globe
Valve Size:NPS 0.5 -to- NPS 2
Service:-29 to 204 °C (-20 to +400 °F)
Class:**API 800**
Ends:SW
Body:2507 Super Duplex SS (ASTM A182-F53)
Bonnet:2507 Super Duplex SS (ASTM A182-F53)
Trim:2507 Super Duplex SS
Body Type:T-body
Port:Standard Port
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:Handwheel
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:2507 Super Duplex
Body-Bonnet Bolting:Bolts: API 602
.....Nuts: API 602

The standards are:

DESIGN:.....API 602
ENDS:ASME B16.11
RATING:.....API 602
TESTING:.....API 598
DIMENSIONAL:MFG STD

Approved Vendors:Ladish (Forged): 7861-6F64-CGnn-A-{Size}

***** Low Temp. & Cryo Globe Valves *****

Description of GLOBE Valve (GL01ST510):

GL15-S16-F510

Type:Globe, Low-Temp
Valve Size:NPS 3 -to- NPS 16
Service:Low Temperature -73 to 204 °C (-99 to 400 °F)
Class:**ASME 150**
Ends:Flanged RF
Body:316 SS (ASTM A351-CF8M)
Bonnet:316 SS (ASTM A351-CF8M)
Trim:API 623 **Trim #12**

Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:Handwheel
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:316L SS/Graphite
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
ENDS:.....ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10

Approved Vendors:DSI: {Size}-133-16-16 HFS (3" to 16" NPS)
.....Ladish (Cast): 7272-8052-PGnn-A

Description of GLOBE Valve (GL01ST511):

GL15-S16-F511

Type:Globe, Cryogenic
Valve Size:NPS 0.5 -to- NPS 16
Service:Cryogenic -196 to 204 °C (-320 to 400 °F)
Class:**ASME 150**
Ends:Flanged RF
Body:316 SS (ASTM A351-CF8M)
Bonnet:316 SS (ASTM A351-CF8M)
Trim:API 623 **Trim #12**

Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Design (Other):Extended Bonnet for Cryogenic
Stem Design:OS&Y
Operation:Handwheel
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:316L SS/Graphite
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34 and MSS SP-134
ENDS:.....ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10

Approved Vendors:DSI: {Size}-G-133-16-16 HFS-EB (6" to 16" NPS)
.....NEWCO: S152-1/2U-G-EB-CR
.....Ladish (Cast): 7292-8052-PGnn-A

**Description of GLOBE Valve (GL01ST512):**

GL15-S16-F512

Type:Globe, Low-Temp
Valve Size:NPS 6 -to- NPS 16
Service:Low Temperature -73 to 204 °C (-99 to 400 °F)
Class:**ASME 150**
Ends:Flanged RF
Body:316 SS (ASTM A351-CF8M)
Bonnet:316 SS (ASTM A351-CF8M)
Trim:API 623 **Trim #12**
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:**Gear**
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:316L SS/Graphite
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
ENDS:.....ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10

Approved Vendors:DSI: {Size}-G-133-16-16 HFS (6" to 16" NPS)
.....Ladish (Cast): 7272-8052-PGnn-G
.....

Description of GLOBE Valve (GL01ST513):

GL15-S16-F513

Type:Globe, Cryogenic
Valve Size:NPS 6 -to- NPS 16
Service:Cryogenic -196 to 204 °C (-320 to 400 °F)
Class:**ASME 150**
Ends:Flanged RF
Body:316 SS (ASTM A351-CF8M)
Bonnet:316 SS (ASTM A351-CF8M)
Trim:API 623 **Trim #12**
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Design (Other):Extended Bonnet for Cryogenic
Stem Design:OS&Y
Operation:**Gear**
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:316L SS/Graphite
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34 and MSS SP-134
ENDS:.....ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10

Approved Vendors:DSI: {Size}-G-133-16-16 HFS-EB (6" to 16" NPS)
.....NEWCO: S152-1/2U-G-EB-CR-GO
.....Ladish (Cast): 7292-8052-PGnn-G

Description of GLOBE Valve (GL03ST510):

GL30-S16-F510

Type:Globe, Low-Temp
 Valve Size:NPS 3 -to- NPS 16
 Service:Low Temperature -73 to 204 °C (-99 to 400 °F)
 Class:**ASME 300**
 Ends:Flanged RF
 Body:316 SS (ASTM A351-CF8M)
 Bonnet:316 SS (ASTM A351-CF8M)
 Trim:API 623 **Trim #12**
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10 (NPS 3 -to- NPS 12)
MFG STD (NPS 14 -to- NPS 16)

Approved Vendors:Ladish (Cast, FP): 7372-8052-PGnnAG-{Size}

Description of GLOBE Valve (GL03ST511):

GL30-S16-F511

Type:Globe, Cryogenic
 Valve Size:NPS 2" -to- NPS 16
 Service:Cryogenic -196 to 204 °C (-320 to 400 °F)
 Class:**ASME 300**
 Ends:Flanged RF
 Body:316 SS (ASTM A351-CF8M)
 Bonnet:316 SS (ASTM A351-CF8M)
 Trim:API 623 **Trim #12**
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Design (Other):Extended Bonnet for Cryogenic
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34 and MSS SP-134
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10 (NPS 0.5 -to- NPS 12)
MFG STD (NPS 14 -to- NPS 16)

Approved Vendors:DSI: {Size}-141-16-16 HFS-EB (2" to 16" NPS)
Ladish (Cast, FP): 7392-8052-PGnn-A-{Size}

Description of GLOBE Valve (GL03ST512):

GL30-S16-F512

Type:Globe, Low-Temp
Valve Size:NPS 4 -to- NPS 16
Service:Low Temperature -73 to 204 °C (-99 to 400 °F)
Class:**ASME 300**
Ends:Flanged RF
Body:316 SS (ASTM A351-CF8M)
Bonnet:316 SS (ASTM A351-CF8M)
Trim:API 623 **Trim #12**
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:**Gear**
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:316L SS/Graphite
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
ENDS:ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10 (NPS 4 -to- NPS 12)
.....MFG STD (NPS 14 -to- NPS 16)

Approved Vendors:Ladish (Cast, FP): 7372-8052-PGnn-G-{Size}

Description of GLOBE Valve (GL03ST513):

GL30-S16-F513

Type:Globe, Cryogenic
Valve Size:NPS 4 -to- NPS 16
Service:Cryogenic -196 to 204 °C (-320 to 400 °F)
Class:**ASME 300**
Ends:Flanged RF
Body:316 SS (ASTM A351-CF8M)
Bonnet:316 SS (ASTM A351-CF8M)
Trim:API 623 **Trim #12**
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Design (Other):Extended Bonnet for Cryogenic
Stem Design:OS&Y
Operation:**Gear**
Stem-Packing:Flexible Graphite, Corrosion Inhibited
Body-Bonnet Gaskets:316L SS/Graphite
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34 and MSS SP-134
ENDS:ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10 (NPS 4 -to- NPS 12)
.....

Approved Vendors:Ladish (Cast, FP): 7392-8052-PGnn-G-{Size}

Description of GLOBE Valve (GL01SD311):

GL15-S16-S311

Type:Globe, Cryogenic
 Valve Size:NPS 0.5 -to- NPS 2
 Service:Cryogenic -196 to 204 °C (-320 to 400 °F)
 Class:**ASME 150**
 Ends:SW
 Body:316/316L SS (ASTMA351-CF8M/CF3M)
 Bonnet:316/316L SS (ASTMA351-CF8M/CF3M)
 Trim:API 602 **Trim #12**
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Design (Other):Extended Bonnet for Cryogenic
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316 SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34 and MSS SP-134
 ENDS:ASME B16.11
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:DSI: 1211-EB-E12-22
Ladish (Cast, CL150): 7291-3052-GGnn-A-{Size}
Ladish (Forged, CL 800): 7891-6A55-CGnn-A-{Size}

Description of GLOBE Valve (GL03SD311):

GL30-S16-S311

Type:Globe, Cryogenic
 Valve Size:NPS 0.5 -to- NPS 2
 Service:Cryogenic -196 to 204 °C (-320 to 400 °F)
 Class:**ASME 300**
 Ends:SW
 Body:316/316L SS (ASTMA351-CF8M/CF3M)
 Bonnet:316/316L SS (ASTMA351-CF8M/CF3M)
 Trim:API 602 **Trim #12**
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Design (Other):Extended Bonnet for Cryogenic
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316 SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
MFG STD

The standards are:

DESIGN:.....ASME B16.34 and MSS SP-134
 ENDS:ASME B16.11
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:NEWCO OIC: S152-1/2U-G-EB-CR
Ladish (Cast, CL 300, FP): 7392-6052-PGnn-A-{Size}

Description of GLOBE Valve (GL06ST510):

GL60-S16-F510

Type:Globe, Low-Temp
 Valve Size:NPS 3 -to- NPS 14
 Service:Low Temperature -73 to 204 °C (-99 to 400 °F)
 Class:**ASME 600**
 Ends:Flanged RF
 Body:316 SS (ASTM A351-CF8M)
 Bonnet:316 SS (ASTM A351-CF8M)
 Trim:API 623 **Trim #12**
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:ASME B16.10 - Long Pattern (NPS 3 -to- NPS 10)
MFG STD (NPS 12 -to- NPS 14)

Approved Vendors:Ladish (Cast, FP): 7672-8052-PGnn-A-{Size}

Description of GLOBE Valve (GL06ST511):

GL60-S16-F511

Type:Globe, Cryogenic
 Valve Size:NPS 2" -to- NPS 14
 Service:Cryogenic -196 to 204 °C (-320 to 400 °F)
 Class:**ASME 600**
 Ends:Flanged RF
 Body:316 SS (ASTM A351-CF8M)
 Bonnet:316 SS (ASTM A351-CF8M)
 Trim:API 623 **Trim #12**
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Design (Other):Extended Bonnet for Cryogenic
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34 and MSS SP-134
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:ASME B16.10 - Long Pattern (NPS 0.5 -to- NPS 10)
MFG STD (NPS 12 -to- NPS 14)

Approved Vendors:Ladish (Cast, FP): 7692-8052-GGnn-A

Description of GLOBE Valve (GL06ST512):

GL60-S16-F512

Type:Globe, Low-Temp
 Valve Size:NPS 3 -to- NPS 14
 Service:Low Temperature -73 to 204 °C (-99 to 400 °F)
 Class:**ASME 600**
 Ends:Flanged RF
 Body:316 SS (ASTM A351-CF8M)
 Bonnet:316 SS (ASTM A351-CF8M)
 Trim:API 623 **Trim #12**
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:**Gear**
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:ASME B16.10 - Long Pattern (NPS 3 -to- NPS 10)
MFG STD (NPS 12 -to- NPS 14)

Approved Vendors:Ladish (Cast, FP): 7672-8052-GGnn-G

Description of GLOBE Valve (GL06ST513):

GL60-S16-F513

Type:Globe, Cryogenic
 Valve Size:NPS 3 -to- NPS 14
 Service:Cryogenic -196 to 204 °C (-320 to 400 °F)
 Class:**ASME 600**
 Ends:Flanged RF
 Body:316 SS (ASTM A351-CF8M)
 Bonnet:316 SS (ASTM A351-CF8M)
 Trim:API 623 **Trim #12**
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Design (Other):Extended Bonnet for Cryogenic
 Stem Design:OS&Y
 Operation:**Gear**
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34 and MSS SP-134
 ENDS:ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:ASME B16.10 - Long Pattern (NPS 3 -to- NPS 10)
MFG STD (NPS 12 -to- NPS 14)

Approved Vendors:Ladish (Cast, FP): 7692-8052-GGnn-G

Description of GLOBE Valve (GL03SU310):

GL30-S16L-S310

Type:Globe, Low-Temp
 Valve Size:NPS 0.5 -to- NPS 2
 Service:Low Temperature -73 to 204 °C (-99 to 400 °F)
 Class:**ASME 300**
 Ends:SW
 Body:316L SS (ASTM A351-CF3M)
 Bonnet:316L SS (ASTM A351-CF3M)
 Trim:API 602 **Trim #12**
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
 ENDS:ASME B16.11
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Ladish (Cast, FP): 7361-6062-PGnn-A-{Size}

Description of GLOBE Valve (GL03SU311):

GL30-S16L-S311

Type:Globe, Cryogenic
 Valve Size:NPS 0.5 -to- NPS 2
 Service:Cryogenic -196 to 204 °C (-320 to 400 °F)
 Class:**ASME 300**
 Ends:SW
 Body:316L SS (ASTM A351-CF3M)
 Bonnet:316L SS (ASTM A351-CF3M)
 Trim:API 602 **Trim #12**
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Design (Other):Extended Bonnet for Cryogenic
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34 and MSS SP-134
 ENDS:ASME B16.11
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Ladish (Cast, FP): 7391-6062-PGnn-A-{Size}

Description of GLOBE Valve (GL08SD310):

GL80-S16-S310

Type:Globe, Low-Temp
 Valve Size:NPS 0.5 -to- NPS 2
 Service:Low Temperature -73 to 204 °C (-99 to 400 °F)
 Class:**API 800**
 Ends:SW
 Body:316/316L SS (ASTM A182-F316/316L)
 Bonnet:316/316L SS (ASTM A182-F316/316L)
 Trim:API 602 **Trim #12**
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....API 602
 ENDS:ASME B16.11
 RATING:.....API 602
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Bonney Forge (Forged, RP): HL-30-8-SW
Ladish (Forged): 7861-6A55-CGnn-A-{Size}

Description of GLOBE Valve (GL08SD311):

GL80-S16-S311

Type:Globe, Cryogenic
 Valve Size:NPS 0.5 -to- NPS 2
 Service:Cryogenic -196 to 204 °C (-320 to 400 °F)
 Class:**API 800**
 Ends:SW
 Body:316/316L SS (ASTM A182-F316/316L)
 Bonnet:316/316L SS (ASTM A182-F316/316L)
 Trim:API 602 **Trim #12**
 Body Type:T-body
 Bonnet Type:Bolted
 Disc Type:Conical
 Design (Other):Extended Bonnet for Cryogenic
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets:316L SS/Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:.....API 602 and MSS SP-134
 ENDS:ASME B16.11
 RATING:.....API 602
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Ladish (Forged): 7891-6A55-CGnn-A-{Size} Cryo Valve

***** 254 SMO Duplex SS Valves *****

Description of GLOBE Valve (GL01SV500):

GL15-SMO-F500

Type:	Globe
Valve Size:	NPS 2 -to- NPS 16
Service:	-29 to 204 °C (-20 to +400 °F)
Class:	ASME 150
Ends:	Flanged RF
Body:	254 SMO Duplex SS (ASTM A351-CK3MCuN)
Bonnet:	254 SMO Duplex SS (ASTM A351-CK3MCuN)
Trim:	254 SMO Duplex SS
Body Type:	T-body
Bonnet Type:	Bolted
Disc Type:	Conical
Stem Design:	OS&Y
Operation:	Handwheel
Stem-Packing:	PTFE
Body-Bonnet Gaskets:	PTFE
Body-Bonnet Bolting:	Bolts: MFG STD
	Nuts: MFG STD

The standards are:

DESIGN:.....	ASME B16.34
ENDS:.....	ASME B16.5
RATING:.....	ASME B16.34
TESTING:.....	API 598
DIMENSIONAL:.....	ASME B16.10

Approved Vendors: Ladish (Cast): 7272-3161-TTnn-A

Description of GLOBE Valve (GL01SV501):

GL15-SMO-F501

Type:	Globe
Valve Size:	NPS 6 -to- NPS 16
Service:	-29 to 204 °C (-20 to +400 °F)
Class:	ASME 150
Ends:	Flanged RF
Body:	254 SMO Duplex SS (ASTM A351-CK3MCuN)
Bonnet:	254 SMO Duplex SS (ASTM A351-CK3MCuN)
Trim:	254 SMO Duplex SS
Body Type:	T-body
Bonnet Type:	Bolted
Disc Type:	Conical
Stem Design:	OS&Y
Operation:	Gear
Stem-Packing:	PTFE
Body-Bonnet Gaskets:	PTFE
Body-Bonnet Bolting:	Bolts: MFG STD
	Nuts: MFG STD

The standards are:

DESIGN:.....	ASME B16.34
ENDS:.....	ASME B16.5
RATING:.....	ASME B16.34
TESTING:.....	API 598
DIMENSIONAL:.....	ASME B16.10

Approved Vendors:Ladish (Cast): 7272-3161-TTnn-G



Document #: GCTLR00-TSLA-ME-SPC-00018

Valve Specification for SS Globe Valves

Description of GLOBE Valve (GL03SV300):

GL30-SMO-S300

Type:Globe
Valve Size:NPS 0.5 -to- NPS 2
Service:-29 to 204 °C (-20 to +400 °F)
Class:**ASME 300**
Ends:SW
Body:254 SMO Duplex SS (ASTM A351-CK3MCuN)
Bonnet:254 SMO Duplex SS (ASTM A351-CK3MCuN)
Trim:254 SMO Duplex SS
Body Type:T-body
Bonnet Type:Bolted
Disc Type:Conical
Stem Design:OS&Y
Operation:Handwheel
Stem-Packing:PTFE
Body-Bonnet Gaskets:PTFE
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34
ENDS:ASME B16.11
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10

Approved Vendors:Ladish (Cast, FP): 7361-3161-TTnn-A-{Size}

 ***** Hi Temp Globe Valves *****

Description of GLOBE Valve (GL08SM300): GL80-S47H-S300

Type:Globe, Hi-Temp
 Valve Size:NPS 0.5 -to- NPS 2
 Service:High Temperature -29 to 816 °C (-20 to 1500 °F)
 Class:**API 800**
 Ends:SW
 Body:347H SS (ASTM A182-F347H)
 Bonnet:347H SS (ASTM A182-F347H)
 Trim:347H SS
 Hard Facing:Full
 Body Type:T-body
 Bonnet Type:Welded
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:MFG STD

The standards are:
 DESIGN:.....API 602
 ENDS:ASME B16.11
 RATING:.....API 602
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Ladish (Forged): 7761-6B26-PGnn-A-{Size}

Description of GLOBE Valve (GL09SM500): GL90-S47H-F500

Type:Globe, Hi-Temp
 Valve Size:NPS 3 -to- NPS 12
 Service:High Temperature -29 to 816 °C (-20 to 1500 °F)
 Class:**ASME 900**
 Ends:Flanged RF
 Body:347H SS (ASTM A351-CF8C)
 Bonnet:347H SS (ASTM A351-CF8C)
 Material Remarks:0.04 percent minimum carbon content Trim: 347H SS
 Hard Facing:Full
 Body Type:T-body
 Bonnet Type:Pressure Seal
 Disc Type:Conical
 Stem Design:OS&Y
 Operation:Handwheel
 Stem-Packing:MFG STD
 Body-Bonnet Gaskets:MFG STD
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:
 DESIGN:.....ASME B16.34 and MSS SP-144 ENDS: ASME B16.5
 RATING:.....ASME B16.34
 TESTING:.....API 598
 DIMENSIONAL:.....ASME B16.10 - Short Pattern

Approved Vendors:Ladish (Cast): 7922-3123-HHnn-A-{Size}

**Description of GLOBE Valve (GL09SM501):**

GL90-S47H-F501

Type:Globe, Hi-Temp
Valve Size:NPS 3 -to- NPS 12
Service:High Temperature -29 to 816 °C (-20 to 1500 °F)
Class:**ASME 900**
Ends:Flanged RF
Body:347H SS (ASTM A351-CF8C)
Bonnet:347H SS (ASTM A351-CF8C)
Material Remarks:0.04 percent minimum carbon content Trim: 347H SS
Hard Facing:Full
Body Type:T-body
Bonnet Type:Pressure Seal
Disc Type:Conical
Stem Design:OS&Y
Operation:**Gear**
Stem-Packing:MFG STD
Body-Bonnet Gaskets:MFG STD
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34 and MSS SP-144
ENDS:ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10 - Short Pattern

Approved Vendors:Ladish (Cast): 7922-3123-HHnn-G-{Size}

Description of GLOBE Valve (GL15SK300):

GL1.5K-S21H-S300

Type:Globe
Valve Size:NPS 0.5 -to- NPS 2
Service:High Temperature -29 to 816 °C (-20 to 1500 °F)
Class:**ASME 1500**
Ends:SW
Body:321H SS (ASTM A182-F321H)
Bonnet:321H SS (ASTM A182-F321H)
Trim:321H SS
Hard Facing:Full
Body Type:T-body
Bonnet Type:Welded
Disc Type:Conical
Stem Design:OS&Y
Operation:Handwheel
Stem-Packing:MFG STD

The standards are:

DESIGN:.....API 602
ENDS:ASME B16.11
RATING:.....API 602
TESTING:.....API 598
DIMENSIONAL:.....MFG STD

Approved Vendors:Ladish (Forged): 7161-6B86-HHnn-A-{Size}

**Description of GLOBE Valve (GL15SM500):**

GL1.5K-S47H-F500

Type:Globe, Hi-Temp
Valve Size:NPS 0.5 -to- NPS 2
Service:High Temperature -29 to 816 °C (-20 to 1500 °F)
Class:**ASME 1500**
Ends:Flanged RF
Body:347H SS (ASTM A182-F347H)
Bonnet:347H SS (ASTM A182-F347H)
Trim:347H SS
Hard Facing:Full
Body Type:T-body
Bonnet Type:Welded
Disc Type:Conical
Stem Design:OS&Y
Operation:Handwheel
Stem-Packing:MFG STD

The standards are:

DESIGN:.....API 602
ENDS:.....ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10 - Long Pattern

Approved Vendors:Ladish (Forged): 7122-3B26-HHnn-A-{Size}

Description of GLOBE Valve (GL15SM501):

GL1.5K-S47H-F501

Type:Globe, Hi-Temp
Valve Size:NPS 3 -to- NPS 12
Service:High Temperature -29 to 816 °C (-20 to 1500 °F)
Class:**ASME 1500**
Ends:Flanged RF
Body:347H SS (ASTM A351-CF8C)
Bonnet:347H SS (ASTM A351-CF8C)
Material Remarks:0.04 percent minimum carbon content
Trim:347H SS
Hard Facing:Full
Body Type:T-body
Bonnet Type:Pressure Seal
Disc Type:Conical
Stem Design:OS&Y
Operation:Handwheel
Stem-Packing:MFG STD
Body-Bonnet Gaskets:MFG STD
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34 and MSS SP-144
ENDS:.....ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10 - Short Pattern

Approved Vendors:Ladish (Cast, FP): 7522-3033-HHnn-A-{Size}



Document #: GCTLR00-TSLA-ME-SPC-00018

Valve Specification for SS Globe Valves

Description of GLOBE Valve (GL15SM502):

GL1.5K-S47H-F502

Type:Globe, Hi-Temp
Valve Size:NPS 3 -to- NPS 12
Service:High Temperature -29 to 816 °C (-20 to 1500 °F)
Class:**ASME 1500**
Ends:Flanged RF
Body:347H SS (ASTM A351-CF8C)
Bonnet:347H SS (ASTM A351-CF8C)
Material Remarks:0.04 percent minimum carbon content Trim: 347H SS
Hard Facing:Full
Body Type:T-body
Bonnet Type:Pressure Seal
Disc Type:Conical
Stem Design:OS&Y
Operation:**Gear**
Stem-Packing:MFG STD
Body-Bonnet Gaskets:MFG STD
Body-Bonnet Bolting:Bolts: MFG STD
.....Nuts: MFG STD

The standards are:

DESIGN:.....ASME B16.34 and MSS SP-144 ENDS: ASME B16.5
RATING:.....ASME B16.34
TESTING:.....API 598
DIMENSIONAL:.....ASME B16.10 - Short Pattern

Approved Vendors:Ladish (Cast, FP): 7522-3033-HHnn-G-{Size}

Rev.	Rev. Date	Rev. Description	By	Checked
0	01 SEPT 2023	Issued for release	MFS	SBS
1	18 SEPT 2023	Updated Valve Information	MFS	SBS
2	08 NOV 2023	Reorder Valve information, Add Misc. Valves	MFS	SBS
3	26 MAR 2024	Add Service Temp for Valves, Minor Updates	MFS	WGS
4	25 APR 2024	Add additional valve options	MFS	MFS
5	10 JUL 2024	Revise Trim on Tag GL06ST501, Add port sizes on valves	MFS	WGS

End Document #: GCTLR00-TSLA-ME-SPC-00018

Valve Specification for SS Globe Valves

Description of BALL Valve (BA00-CPV-S300):

Type: Ball, True Union 2000, CPVC
 Valve Size: NPS 0.5 -to- **NPS 8"**
 Rating: 235 psig (16.2 bar) @ 73 °F (23 °C) up to 2" NPS
 Rating: 150 psig (10.3 bar) @ 73 °F (23 °C) 2-1/2" to **8" NPS**
 Service: -29 to 82.2 °C (-20 to 180 °F)
 Class: Pressure Rated
 Ends: SW Ends
 Body: CPVC
 Trim: CPVC
 Seat: EPDM
 Stem CPVC
 Body Type: 2-Piece
 Bore-Port: Single Reduced
 Operation: Lever, Plastic Handle
 Stem-Packing/Seals: O-ring, EPDM

The standards are:

DESIGN: Mfg. Std.
 ENDS: ASTM F439 (Socket)
 RATING: Mfg. Std.
 TESTING: API 598
 CERTIFICATION: NSF-14 & NSF-61
 DIMENSIONAL: Mfg. Std.

Approved Vendor: 1/2" to 2" NPS (CPVC, Socket Ends) = Spears, True Union 2000, Series 1829-{Size}-C
 2-1/2" to 8" NPS (CPVC, Socket Ends) = Spears, True Union 2000, Series 1822-{Size}-C

Description of BALL Valve (BA00-CPV-F500):

Type: Ball, True Union 2000, CPVC
 Valve Size: NPS 0.5 -to- NPS 6"
 Rating: 150 psig (10.3 bar) @ 73 °F (23 °C) up to 6" NPS
 Service: -29 to 82.2 °C (-20 to 180 °F)
 Class: ASME 150
 Ends: FF Ends
 Body: CPVC
 Trim: CPVC
 Seat: EPDM
 Stem CPVC
 Bore-Port: Full Port
 Operation: Lever, Plastic Handle
 Stem-Packing/Seals: O-ring, EPDM

The standards are:

DESIGN: Mfg. Std.
 ENDS: ASME B16.5 Pattern
 RATING: Mfg. Std.
 TESTING: API 598
 CERTIFICATION: NSF-14 & NSF-61
 DIMENSIONAL: Mfg. Std.

Approved Vendor: 1/2" to 6" NPS (CPVC, CL 150, FF Ends) = Spears, True Union 2000, Series 1823-{Size}-C

Description of GATE Valve (GA00-CPV-S300):

Type: Gate, CPVC
 Valve Size: NPS 1/2" -to- NPS 4"
 Rating: 235 psig (16.2 bar) @ 73 °F (23 °C)
 Service: -29 to +82.2 °C (-20 to +180 °F)
 Class: Pressure Rated
 Ends: Socket Ends
 Body: CPVC
 Gate: CPVC
 Seat: EPDM Seat
 Stem CPVC
 Bore-Port: Full Port
 Operation: PP Lever, Plastic Handle
 Stem-Packing/Seals: O-ring, EPDM

The standards are:

DESIGN: Mfg. Std.
 ENDS: ASTM F439 (Socket)
 RATING: Mfg. Std.
 TESTING: API 598
 CERTIFICATION: NSF-14 & NSF-61
 DIMENSIONAL: Mfg. Std.

Approved Vendor:

..... (Gate Valve, CPVC, Socket Ends) Spears, Series 2022-{Size}-C

Description of GATE Valve (GA00-CPV-F500):

Type: Gate, CPVC
 Valve Size: NPS 1/2" -to- NPS 4"
 Rating: 150 psig (10.3 bar) @ 73 °F (23 °C)
 Service: -29 to +82.2 °C (-20 to +180 °F)
 Class: ASME 150
 Ends: **FF Flange**
 Body: CPVC
 Gate: CPVC
 Seat: EPDM Seat
 Stem CPVC
 Bore-Port: Full Port
 Operation: PP Lever, Plastic Handle
 Stem-Packing/Seals: O-ring, EPDM

The standards are:.....

DESIGN: Mfg. Std.
 ENDS: ASME B16.5 Pattern
 RATING: Mfg. Std.
 TESTING: API 598
 CERTIFICATION: NSF-14 & NSF-61
 DIMENSIONAL: Mfg. Std.

Approved Vendor:

..... (Gate Valve, CPVC, EPDM Seat, FF Ends) Spears, Series 2023-{Size}-C

Description of GLOBE Valve (GL00-CPV-F500):

Type:Globe, CPVC, (Flanged Ends)
 Valve Size:NPS 2-1/2" -to- NPS 6"
 Rating:150 psig (10.3 bar) @ 73 °F (23 °C)
 Service:-29 to +82.2 °C (-20 to +180 °F)
 Class:ASME 150
 Ends:**FF Flange**
 Body:CPVC
 Gate:CPVC
 Seat:EPDM Seat
 StemCPVC
 Bore-Port:Standard Port
 Operation:Handwheel, Plastic Handle (all sizes)
 Stem-Packing/Seals:O-ring, EPDM

The standards are:.....
 DESIGN:Mfg. Std.
 ENDS:ASME B16.5 Pattern
 RATING:Mfg. Std.
 TESTING:API 598
 CERTIFICATION:NSF-14 & NSF-61
 DIMENSIONAL:Mfg. Std.

Approved Vendor:.....(Globe Valve, CPVC, EPDM Seat, FF Ends) Spears, Series 6023-{Size}-C

Description of BASKET STRAINER Valve (STR00-CPV-F500):

Type:Basket Strainer, CPVC, (Flanged Ends)
 Valve Size:NPS 1/2" -to- NPS 12"
 Rating:up to 8" NPS: 150 psig (10.3 bar) @ 73 °F (23 °C)
 Rating:10" to 12" NPS: 50 psig (3.4 bar) @ 73 °F (23 °C)
 Service:-29 to +82.2 °C (-20 to +180 °F)
 Class:ASME 150
 Ends:**FF Flange**
 Body:CPVC
 Screen:.....1/8" perforated Strainer Basket (#6 Mesh Code)
 Seal:EPDM Seat

The standards are:
 DESIGN:Mfg. Std.
 ENDS:ASME B16.5 Pattern
 RATING:Mfg. Std.
 TESTING:API 598
 CERTIFICATION:NSF-14 & NSF-61
 DIMENSIONAL:Mfg. Std.

Approved Vendor:
Basket Strainer, CPVC, EPDM Seal, SS Screen, FF Ends)
Spears, Series BS-23S6-{Size}-C

Description of BALL-CHECK Valve (CHB00-CPV-S300):

Type: Ball-Check Valve, True Union, CPVC
 Valve Size: NPS 1/2" -to- NPS 4"
 Rating: 235 psig (16.2 bar) @ 73 °F (23 °C) up to 2" NPS
 Rating: 150 psig (10.3 bar) @ 73 °F (23 °C) 2-1/2" to 6" NPS
 Service: -29 to +82.2 °C (-20 to +180 °F)
 Class: Pressure Rated
 Ends: Socket Ends
 Body: CPVC
 Gate: CPVC
 Seat: EPDM Seat
 Stem CPVC
 Bore-Port: Full Port
 Operation: No operator
 Stem-Packing/Seals: O-ring, EPDM

The standards are:.....
 DESIGN: Mfg. Std.
 ENDS: ASTM F439 (Socket)
 RATING: Mfg. Std.
 TESTING: API 598
 CERTIFICATION: NSF-14 & NSF-61
 DIMENSIONAL: Mfg. Std.

Approved Vendor: (Ball-Check Valve, Sch. 80 CPVC, Socket Ends): Spears, True Union, Series 2229-{Size}-C

Description of BALL-CHECK Valve (CHB00-CPV-F500):

Type: Ball-Check Valve, True Union, CPVC
 Valve Size: NPS 1/2" -to- NPS 4"
 Rating: 150 psig (10.3 bar) @ 73 °F (23 °C)
 Service: -29 to +82.2 °C (-20 to +180 °F)
 Class: ASME 150
 Ends: FF Ends
 Body: CPVC
 Gate: CPVC
 Seat: EPDM Seat
 Stem CPVC
 Bore-Port: Full Port
 Operation: No operator
 Stem-Packing/Seals: O-ring, EPDM

The standards are:.....
 DESIGN: Mfg. Std.
 ENDS: ASME B16.5 Pattern
 RATING: Mfg. Std.
 TESTING: API 598
 CERTIFICATION: NSF-14 & NSF-61
 DIMENSIONAL: Mfg. Std.

Approved Vendor: (Ball-Check Valve, Sch. 80 CPVC, FF Ends): Spears, True Union, Series 2223-{Size}-C

Description of SWING-CHECK Valve (CHB00-CPV-F500):

Type: Check Valve, Socketing Type, CPVC
 Valve Size: NPS 1/2" -to- NPS 6"
 Rating: 150 psig (10.3 bar) @ 73 °F (23 °C)
 Service: -29 to +82.2 °C (-20 to +180 °F)
 Class: ASME 150
 Ends: **FF Ends**
 Body: CPVC
 Disc: CPVC
 Seat: EPDM Seat
 Stem CPVC
 Bore-Port: Full Port
 Operation: No operator
 Stem-Packing/Seals: O-ring, EPDM

The standards are:.....
 DESIGN: Mfg. Std.
 ENDS:..... ASME B16.5 Pattern
 RATING: Mfg. Std.
 TESTING: API 598
 CERTIFICATION: NSF-14 & NSF-61
 DIMENSIONAL: Mfg. Std.

Approved Vendor:
 (Check Valve, Swing, CPVC, FF Ends): Spears, Series 4423-{Size}-C

Description of Y-CHECK Valve (CHY00-CPV-S300):

Type: Y-Check Valve, True Union, CPVC
 Valve Size: NPS 1/2" -to- NPS 4"
 Rating: 235 psig (16.2 bar) @ 73 °F (23 °C) up to 2" NPS
 Rating: 150 psig (10.3 bar) @ 73 °F (23 °C) 2-1/2" to 6" NPS
 Service: -29 to +82.2 °C (-20 to +180 °F)
 Class: ASME 150
 Ends: Socket Ends
 Body: CPVC
 Gate: CPVC
 Seat: EPDM Seat
 Stem CPVC
 Bore-Port: Full Port
 Operation: No operator
 Stem-Packing/Seals: O-ring, EPDM

The standards are:.....
 DESIGN: Mfg. Std.
 ENDS:..... ASTM F439 (Socket)
 RATING: Mfg. Std.
 TESTING: API 598
 CERTIFICATION: NSF-14 & NSF-61
 DIMENSIONAL: Mfg. Std.

Approved Vendor: (Y-Check Valve, Sch. 80 CPVC, Socket Ends): Spears, True Union, Series 1622-{Size}-C

Description of BUTTERFLY Valve (BU00-CPV-L700):

Type: Butterfly, CPVC
 Valve Size: NPS 0.5 -to- NPS 8"
 Rating: 150 psig (10.3 bar) @ 73 °F (23 °C)
 Service: -29 to 82.2 °C (-20 to 180 °F)
 Class: ASME 150
 Ends: LUGGED Ends (Mates to FF Flanges)
 Body: CPVC
 Trim: CPVC
 Seat: EPDM
 Stem CPVC
 Bore-Port: Full Port
 Operation: Lever (up to 3" NPS); Gear (3-1/2" to 8" NPS)
 Stem-Packing/Seals: O-ring
 Gasket NOT required (built into Valve)

The standards are:.....
 DESIGN: Mfg. Std.
 ENDS: ASME B16.5
 RATING: Mfg. Std.
 TESTING: API 598
 CERTIFICATION: NSF-14 & NSF-61
 DIMENSIONAL: Mfg. Std.

Approved Vendor: 1/2" to 3" NPS (CPVC, Lugged Ends) = Spears, Series 721311, Lever Op.
 3-1/2" to 8" NPS (CPVC, Lugged Ends) = Spears, Series 721312, Gear Oper.

Description of NEEDLE Valve (NE00-CPV-S300):

Type: Needle, CPVC
 Valve Size: NPS 1/2" -to- NPS 4"
 Rating: 235 psig (16.2 bar) @ 73 °F (23 °C)
 Service: -29 to +82.2 °C (-20 to +180 °F)
 Class: Pressure Rated
 Ends: Socket Ends
 Body: CPVC
 Gate: CPVC
 Seat: EPDM Seat
 Stem CPVC
 Bore-Port: Red. Port
 Operation: Plastic Handle
 Stem-Packing/Seals: O-ring, EPDM

The standards are:.....
 DESIGN: Mfg. Std.
 ENDS: ASTM F439 (Socket)
 RATING: Mfg. Std.
 TESTING: API 598
 CERTIFICATION: NSF-14 & NSF-61
 DIMENSIONAL: Mfg. Std.

Approved Vendor:
 (Needle Valve, CPVC, SW Ends) Spears, Series 5591-{Size}-C



Note: When Valve Tags are shown as something like "BA03SQ500 == > BA30-S25-F500", it means that the PIP Valve Code (BA03SQ500) should be replaced with the Tesla Valve Code (BA30-S25-F500).

Rev.	Rev. Date	Rev. Description	By	Checked
0	10 OCT 2023	Issued for release	MFS	SBS
1	10 May 2024	Add additional Valves, minor edits	MFS	MFS

Description of KNIFE GATE Valve (GK01ST500 ==> KF15-S16-W700) – Bray, Series 752

Type:.....Knife Gate, Two-Piece Bolted (Wafer)
 Valve size:NPS 2 -to- NPS 24
 Service:.....Slurry
 Class:.....ASME 150
 Range:-20 °F to +248 °F (-29 °C to +120 °C)
 Rating:150 psig (10 bar)
 Ends:**Wafer** (Install be RF Flanges)
 Body:316 SS (ASTM A351-CF8M)
 Gate:316 SS
 Perimeter Seat:**EPDM, U-Shaped**
 Stem:Stainless steel
 Packing Seal:PTFE impregnated fiber
 Flow:Bi-directional
 Design (Other):Closed Body, zero leakage
 Operation:2" NPS to 8" NPS = Handwheel, Direct Mounted
> 8" NPS = Bevel Gear
 Body Bolting:Bolts: MFG STD
Nuts: MFG STD

Notes:

- 1)Gaskets ARE required for this knife gate valve
- 2)Wafer Body uses through-holes and threaded holes for bolting (see IOM)

The standards are:

DESIGNMSS SP-81
 Face-to-FaceMSS SP-81
 ENDSASME B16.5, CL150
 RATINGASME B16.34
 TESTING:MSS SP-81

Approved Vendor:Bray Knife Gate, CL150, Series 752, Specify pressure rating
Handwheel Oper (to 8" NPS); Gear (to 24" NPS)

Description of KNIFE Valve (GK01SP500 ==> KF15-S22-W700) – Bray, Series 752

Type:.....Knife Gate, Two-Piece Bolted (Wafer)
 Valve size:NPS 2 -to- NPS 24
 Service:.....Slurry
 Class:.....ASME 150
 Range:-20 °F to +248 °F (-29 °C to +120 °C)
 Rating:150 psig (10 bar) = Default; 240 psig (16 bar) = Optional
 Ends:**Wafer** (Install be RF Flanges)
 Body:2205 Duplex SS (ASTM A995-4A, ACI CD3MN)
 Gate:2205 Duplex SS
 Perimeter Seat:**EPDM, U-Shaped**
 Stem:Stainless steel
 Packing Seal:PTFE impregnated fiber
 Flow:Bi-directional
 Design (Other):Closed Body, zero leakage
 Operation:2" NPS to 8" NPS = Handwheel, Direct Mounted
> 8" NPS = Bevel Gear
 Body Bolting:Bolts: MFG STD
Nuts: MFG STD

Notes:

- 1)Gaskets ARE required for this knife gate valve
- 2)Wafer Body uses through-holes and threaded holes for flange bolting (see IOM)

The standards are:

DESIGNMSS SP-81
 Face-to-FaceMSS SP-81
 ENDSASME B16.5, CL150
 RATINGASME B16.34
 TESTING:MSS SP-81

Approved Vendor:Bray Knife Gate, CL150, Series 752, Specify pressure rating (150 psig = default)
Handwheel Oper (to 8" NPS); Gear (to 24" NPS)



Description of KNIFE Valve (GK01SQ500 ==> KF15-S25-W700) – Bray, Series 752

Type:	Knife Gate, Two-Piece Bolted (Wafer)
Valve size:	NPS 2 -to- NPS 16
Service:	Slurry
Class:	ASME 150
Range:	-20 °F to +248 °F (-29 °C to +120 °C)
Rating:	150 psig (10 bar) = Default; 240 psig (16 bar) = Optional
Ends:	Wafer (Install be RF Flanges)
Body:	2507 Super Duplex SS (ASTM A995-5A, ACI CE3MN)
Gate:	2507 Super Duplex SS
Perimeter Seat:	EPDM Seat , U-shaped
Stem:	Stainless steel
Packing Seal:	PTFE impregnated fiber
Flow:	Bi-directional
Design (Other):	Closed Body, zero leakage
Operation:	2" NPS to 8" NPS = Handwheel, Direct Mounted
.....	> 8" NPS = Bevel Gear
Body Bolting:	Bolts: MFG STD
.....	Nuts: MFG STD

- 1) Gaskets ARE required for this knife gate valve
- 2) Wafer Body uses through-holes and threaded holes for flange bolting (see IOM)

The standards are:

DESIGN.....	MSS SP-81
Face-to-Face.....	MSS SP-81
ENDS.....	ASME B16.5, CL150
RATING.....	ASME B16.34
TESTING:.....	MSS SP-81

Approved Vendor:Bray Knife Gate, CL150, Series 752, Specify pressure rating (150 psig = default)
Handwheel Oper (to 8" NPS); Gear (to 24" NPS)

Description of KNIFE Valve (GK01DF510 ==> KF15-DI-F510) – Bray, Series 762

Type: Knife Gate, Two-Piece Bolted Flange
 Valve size: NPS 3 -to- NPS 24
 Service: Slurry, Heavy Duty
 Class: ASME 150
 Range: -20 °F to +302 °F (-29 °C to +150 °C)
 Rating: 100 psig (7 bar)
 Ends: **Flanged, FF** (with tapped holes in each valve flange)
 Body: **Ductile Iron** (ASTM A536, Gr. 65-45-12)
 Gate: 316 Stainless Steel (ASTM A351 CF8M)
 Stem: Stainless steel
 Sleeve: **EPDM** (Elastomer)
 Flow: Bi-directional
 Design (Other): **Push-through** (with bottom flush plate & tapped ports)
 Operation: 2" NPS to 8" NPS = Handwheel, Direct Mounted
 > 8" NPS = Bevel Gear
 Body Bolting: Bolts: MFG STD / Nuts: MFG STD

Notes: Flange bolting uses threaded bolts which screw into the valve body flange
 Gaskets are NOT required (because elastomer extends across face of flange)
 Install valve between **FF** flanges

The standards are:

DESIGN MFG STD
 Face-to-Face MFG STD
 ENDS ASME B16.5, CL 150
 RATING ASME B16.34
 TESTING MFG STD

Approved Vendor: Bray Slurryshield® Knife Gate, **Model 762**, CL150, DI Body, 316 SS Gate,
 EPDM Sleeve, Handwheel Oper (to 8" NPS); Gear (to 24" NPS)

Description of KNIFE Valve (GK01ST510 ==> KF15-S16-F510) – Bray, Series 762

Type: Knife Gate, Two-Piece Bolted Flange
 Valve size: NPS 3 -to- NPS 24
 Service: Slurry, Heavy Duty
 Class: ASME 150
 Range: -20 °F to +302 °F (-29 °C to +150 °C)
 Rating: 100 psig (7 bar)
 Ends: **Flanged, FF** (Elastomer extends across face of flange)
 Body: **316 SS** (ASTM A351-CF8M)
 Gate: 316 Stainless Steel
 Stem: Stainless steel
 Sleeve: **EPDM** (Elastomer)
 Flow: Bi-directional
 Design (Other): **Push-through** (with bottom flush plate & tapped ports)
 Operation: 2" NPS to 8" NPS = Handwheel, Direct Mounted
 > 8" NPS = Bevel Gear
 Body Bolting: Bolts: MFG STD / Nuts: MFG STD

Notes: Flange Bolting uses threaded bolts which screw into the valve body flange
 Gaskets are NOT required (because elastomer extends across face of flange)
 Install valve between **FF** flanges

The standards are:

DESIGN MFG STD
 Face-to-Face MFG STD
 ENDS ASME B16.5, CL 150
 RATING ASME B16.34
 TESTING MFG STD

Approved Vendor: Bray Slurryshield® Knife Gate, **Model 762**, CL150, 316 SS Body & Gate,
 EPDM Sleeve, Handwheel Oper (to 8" NPS); Gear (to 24" NPS)

Description of KNIFE Valve (GK01SQ510 ==> KF15-S22-F510) – Bray, Series 762

Type:	Knife Gate, Two-Piece Bolted Flange
Valve size:	NPS 3 -to- NPS 24
Service:	Slurry, Heavy Duty
Class:	ASME 150
Range:	-20 °F to +302 °F (-29 °C to +150 °C)
Ratings:	232 psig (16 bar) for 3" to 18" NPS; 150 psig (10 bar) for 20" to 24" NPS
Ends:	Flanged, FF (Elastomer extends across face of flange)
Body:	2205 Duplex SS (ASTM A995-4A, ACI CD3MN)
Gate:	2205 Duplex SS
Stem:	Stainless steel
Sleeve:	EPDM (Elastomer)
Flow:	Bi-directional
Design (Other):	Push-through (with bottom flush plate & tapped ports)
Operation:	2" NPS to 8" NPS = Handwheel, Direct Mounted > 8" NPS = Bevel Gear
Body Bolting:	Bolts: MFG STD / Nuts: MFG STD
Notes:	Install valve between FF companion flanges. Gaskets are not needed (i.e. Sleeve extends to Face of Flange)
The standards are:	
DESIGN	MFG STD
Face-to-Face	MFG STD
ENDS	ASME B16.5, CL 150
RATING	ASME B16.34
TESTING	MFG STD
Approved Vendor:	Bray Slurryshield® Knife Gate, Model 762 , CL150, 2205 Duplex SS Body Duplex SS Gate, EPDM, Handwheel Oper (to 8" NPS); Gear (to 24" NPS)

- End of Bray Slurryshield, Series 762, Knife Gate Valves

Description of KNIFE Valve (for 316 SS pipe class) (GK01DF500 ==> KF15-DI-F520) – Clarkson KGA+

Type:	Knife Gate
Valve size:	NPS 3 -to- NPS 24
Class:	ASME 150
Service:	Slurry
Temperature:	Max 180 °F (82 °C)
Ratings:	3" to 24" NPS = 100 psig (6.9 bar)
Ends:	Flange (Install between FF Flanges)
Body:	Ductile iron A536, Epoxy coated
Gate:	316 Stainless Steel
Sleeve:	EPDM Sleeve
Stem:	316 Stainless Steel
Design (Other):	Open bottom end, push-thru style, w/splash drain plate
Flow:	Bi-directional
Operation:	Handwheel Oper. (up to 6" NPS) Gear Op. (8" to 24" NPS)
Lockout:	Open & closed lockout/tagout positions
Body-Bolting:	Bolts: MFG STD Nuts: MFG STD
Notes:	Flange gaskets are not required (because sleeve extends to face of flange) but they may be used (to prevent EPDM from tearing off of face)
The standards are:	
DESIGN	MFG STD
ENDS	ASME B16.5
RATING	ASME B16.34
DIMENSIONAL	MEG STD

Approved Vendor:.....Clarkson, Model KGA, CL150: DI Body, 316 SS Gate, EPDM Sleeve



Approved Vendor:.....Clarkson, Model KGA+, CL150: DI Body, Duplex Gate, EPDM Sleeve

Description of KNIFE Valve (for duplex pipe class) (GK01DF502 ==> KF15-DI-522) – Clarkson KGA+

Approved Vendor:.....Clarkson, Model KGA+, CL150: DI Body, Super Duplex Gate, EPDM Sleeve

----- End of Clarkson, Model KGA+, Knife Gate Valves -----

Description of KNIFE Valve (GK01ST501 = to ==> KF15-S16-W721) – DSS, Model KGV

Type: Knife Gate, Bolted
Valve size: NPS 2 -to- NPS 24
Service: Slurry
Class: ASME 150
Range: -20 °F to +250 °F (-29 °C to +121 °C)
Ends: **Wafer (Install between RF Flanges)**
Body: **316 SS** (ASTM A351-CF8M)
Gate: 316 Stainless Steel
Seals: **EPDM**
Scrapers: Phenolic
Flow: Bi-directional
Design (Other): **Closed Body**, Guided Shear Gate
Operation: 2" NPS to 8" NPS = Handwheel
..... > 8" NPS = Bevel Gear
Body Bolting: Bolts: MFG STD Nuts: MFG STD
Notes:
1) Gaskets ARE required for this knife gate valve
2) Wafer Body uses threaded holes for flange bolting

The standards are:

DESIGN MFG STD
Face-to-Face MSS SP-135, Long Pattern
ENDS ASME B16.5
RATING ASME B16.34

Approved Vendor: DSS Severe Service Knife Gate, Model SSKGV
..... SV1L-{Size}-SS-SS-EP-1-{HW or BG}

Description of KNIFE Valve (GK01SP501 ==> KF15-S22-W722) – DSS, Model KGV

Type: Knife Gate, Bolted
Valve size: NPS 2 -to- NPS 24
Service: Slurry
Class: ASME 150
Range: -20 °F to +250 °F (-29 °C to +121 °C)
Ends: **Wafer (Install between RF Flanges)**
Body: **2205 Duplex SS** (ASTM A995-4A, ACI CD3MN)
Gate: 2205 Duplex SS
Seals: **EPDM**
Scrapers: Phenolic
Flow: Bi-directional
Design (Other): **Closed Body**, Guided Shear Gate
Operation: 2" NPS to 8" NPS = Handwheel
..... > 8" NPS = Bevel Gear
Body Bolting: Bolts: MFG STD Nuts: MFG STD
Notes:
1) Gaskets ARE required for this knife gate valve
2) Wafer Body uses threaded holes for flange bolting

The standards are:

DESIGN MFG STD
ENDS ASME B16.5
Face-to-Face MSS SP-135, Long Pattern
RATING ASME B16.34

Approved Vendor: DSS Severe Service Knife Gate, Model SSKGV
..... SV1L-{Size}-S22-22-EP-1-{HW or BG}

Description of KNIFE Valve (GK01SQ501 ==> KF15-S25-W723) – DSS, Model KGV

Type: Knife Gate, Bolted
 Valve size: NPS 2 -to-NPS 24
 Service: Slurry
 Class: ASME 150
 Range: -20°F to +250 °F (-29°C to +121 °C)
 Ends: **Wafer** (Install between RF Flanges)
 Body: **2507 Super Duplex SS** (ASTM A995-6A, ACI CE3MN)
 Gate: 2507 Super Duplex SS
 Seals: EPDM
 Scrapers: Phenolic
 Flow: Bi-directional
 Design (Other): **Closed Body**, Guided Shear Gate
 Operation: 2" NPS to 8" NPS = Handwheel
 > 8" NPS = Bevel Gear
 Body Bolting: Bolts: MFG STD Nuts: MFG STD
 Notes:
 1) Gaskets ARE required for this knife gate valve
 2) Wafer Body uses threaded holes for flange bolting

The standards are:

DESIGN MFG STD
 ENDS ASME B16.5
 Face-to-Face MSS SP-135, Long Pattern
 RATING ASME B16.34

Approved Vendor: DSS Severe Service Knife Gate Valve, Model SSKGV
 SV1L-{Size}-25-25-EP-1-{HW or BG}

----- End of DSS Severe Service Knife Gate Valves -----

Rev.	Rev. Date	Rev. Description	By	Checked
0	14 SEPT 2023	Issued for release	MFS	SBS
1	08 NOV 2023	General Revision	MFS	SBS
2	04 DEC 2023	Revise Ductile Iron Knife Gate Data	MFS	SBS

End Document #: GCTLR00-TSLA-ME-SPC-00019 .. Valve Specification for SS Knife Gate Valves

**Description of NEEDLE Valve (NE6K-S16-T100)**

Type: Needle, General Service, Soft Seat, H1 Style
Valve size: NPS 0.5 -to- NPS 1.0
Service: Nitrogen, water, etc.
Class: N/A (6000 psig @ 200 F)

Ends: MNPT x FNPT (Male x Female)
Body: 316 SS (ASTM A479-316)
Trim: 316 SS (ASTM A276-316)
Body type: T-body
Disc type: Needle
Seat: Delrin (standard)
Flow Direction: Bi-directional
Orifice Size: 3/8" Dia. Orifice (see P/N's below)
Operation: T-Handle
Stem-Packing: Teflon Packing

The standards are:

DESIGN ASME B16.34
ENDS MFG STD
RATING ASME B16.34
DIMENSIONAL MFG STD

Vendor: Anderson Greenwood, H1 Style, MNPT x FNPT
..... 1/2" Size (3/8" Port) = H1-VDS-44 (Delrin Seat)
..... 3/4" Size (3/8" Port) = H1-VDS-66Q (Delrin Seat)
..... 1" Size (3/8" Port) = H1-VDS-88Q (Delrin Seat)

Description of NEEDLE Valve (NE6K-S16-T101)

Type: Needle, General Service, Soft Seat, H1 Style
Valve size: NPS 0.5 -to- NPS 1.0
Service: Nitrogen, water, etc.
Class: N/A (6000 psig @ 200 F)
Ends: FNPT x FNPT (Female x Female)
Body: 316 SS (ASTM A479-316)
Trim: 316 SS (ASTM A276-316)
Body type: T-body
Disc type: Needle
Seat: Delrin (standard)
Flow Direction: Bi-directional
Orifice Size: 3/8" Dia. Orifice (see P/N's below)
Operation: T-Handle
Stem-Packing: Teflon Packing

The standards are:

DESIGN ASME B16.34
ENDS MFG STD
RATING ASME B16.34
DIMENSIONAL MFG STD

Vendor: Anderson Greenwood, H1 Style, FNPT x FNPT
..... 1/2" Size (3/8" Port) = H1-VDS-4 (Delrin Seat)
..... 3/4" Size (3/8" Port) = H1-VDS-6Q (Delrin Seat)
..... 1" Size (3/8" Port) = H1-VDS-8Q (Delrin Seat)

Description of NEEDLE Valve (NE6K-S16-T102)

Type: Needle, Soft Seat, H1 Style
 Valve size: NPS 0.25 -to- NPS 0.5
 Service: Nitrogen, water, etc.
 Class: N/A (6000 psig @ 200 F)
 Ends: MNPT x FNPT
 Body: 316 SS (ASTM A479-316)
 Trim: 316 SS (ASTM A276-316)
 Body type:..... T-body
 Disc type:..... Needle
 Seat: Delrin (standard)
 Flow Direction:..... Bi-directional
 Orifice Size: 3/16 or 1/4" Dia. Orifice (see P/N's below)
 Operation:..... T-Handle
 Stem-Packing: Teflon Packing

The standards are:

DESIGN ASME B16.34
 ENDS MFG STD
 RATING ASME B16.34
 DIMENSIONAL MFG STD

Vendor: Anderson Greenwood, H1 Style, MNPT x FNPT
 1/4" Size (3/16" Port) = H1-VDS-22 (Delrin Seat)
 1/2" Size (3/16" Port) = H1-VDS-44Q (Delrin Seat)
 *****
 1/2" Size (1/4" Port) = H1-VDS-44QR (Delrin Seat)

Description of NEEDLE Valve (NE6K-S16-T103)

Type: Needle, Soft Seat, H1 Style
 Valve size: NPS 0.25 -to- NPS 0.5
 Service: Nitrogen, water, etc.
 Class: N/A (6000 psig @ 200 F)
 Ends: FNPT x FNPT
 Body: 316 SS (ASTM A479-316)
 Trim: 316 SS (ASTM A276-316)
 Body type:..... T-body
 Disc type:..... Needle
 Seat: Delrin (standard)
 Flow Direction:..... Bi-Directional
 Orifice Size: 3/16" (std.) or 1/4" Dia. (opt.) Orifice
 Operation:..... T-Handle
 Stem-Packing: Teflon Packing

The standards are:

DESIGN ASME B16.34
 ENDS MFG STD
 RATING ASME B16.34
 DIMENSIONAL MFG STD

Vendor: Anderson Greenwood, H1 Style, FNPT x FNPT
 1/4" Size (3/16" Port) = H1-VDS-2 (Delrin Seat)
 1/2" Size (3/16" Port) = H1-VDS-4Q (Delrin Seat)
 *****
 1/2" Size (1/4" Port) = H1-VDS-4QR (Delrin Seat)

Description of NEEDLE Valve (NE6K-S16-T104)

Type: Needle, Hard Seated, H7 Style (i.e., Metal Seat)
 Valve size: NPS 0.5 -to- NPS 0.75
 Class: N/A (6000 psig)
 Ends: MNPT x FNPT (Male x Female)
 Body: 316 SS (ASTM A479-316)
 Stem: 316 SS (ASTM A276-316)
 Body type:..... T-body
 Flow Direction:..... One-Directional
 Orifice Size: 3/16" Dia. Orifice
 Seat: Integral (Metal)
 Disc type:..... Needle
 Operation:..... T-Handle
 Stem-Packing: Teflon (O-ring)

The standards are:

DESIGN ASME B16.34
 ENDS MFG STD
 RATING ASME B16.34
 DIMENSIONAL MFG STD

Vendor: Anderson Greenwood
 H7-VIS-44Q (1/2" MNPT x 1/2" FNPT)
 H7-VIS-66Q (3/4" MNPT x 3/4" FNPT)

Description of NEEDLE Valve (NE6K-S16-T105)

Type: Needle (H7), Hard Seated (i.e., Metal Seat)
 Valve size: NPS 0.25 -to- NPS 0.5
 Class: N/A (6000 psig @ 200 F)
 Ends: FNPT x FNPT (Female x Female)
 Body: 316 SS (ASTM A479-316)
 Stem: 316 SS (ASTM A276-316)
 Body type:..... T-body
 Flow Direction:..... One-Directional
 Orifice Size: 3/16" Dia. Orifice
 Seat: Integral (Metal)
 Disc type:..... Needle
 Operation:..... T-Handle
 Stem-Packing: Teflon (O-ring)

The standards are:

DESIGN ASME B16.34
 ENDS MFG STD
 RATING ASME B16.34
 DIMENSIONAL MFG STD

Vendor: Anderson Greenwood
 H7-VIS-2 (1/4" FNPT x 1/4" FNPT)
 H7-VIS-4Q (1/2" FNPT x 1/2" FNPT)

Description of NEEDLE Valve (ND6K-S16-C801)

Type: Needle, Hard-Seated (i.e., Metal)
 Design Type: Ball Tip Stem
 Valve size: NPS 0.5 -to-NPS 0.5
 Class: N/A (6000 psig @ 200 F)
 Ends: Compression Ferrule Tube, Double Ferrule
 Body: 316 SS
 Trim: 316 SS
 Body type: T-body
 Disc type: Needle
 Flow Direction: One-Directional
 Orifice Size: 3/16" Dia. Orifice
 Seat: Integral
 Operation: T-Handle
 Stem-Packing: PTFE

The standards are:

DESIGN MSS SP-105
 ENDS MFG STD
 RATING MFG STD
 TESTING MSS SP-105
 DIMENSIONAL MFG STD

Vendor: Anderson Greenwood
 H7-VIS-4ATD (1/2" Tube x 1/2" Tube)

Description of NEEDLE Valve (NE6K-S16-T112)

Type: Needle, Hard Seat (Swagelok, GU Series)
 Valve size: NPS 0.25 -to- NPS 1.0
 Temperature: 0 F to +450 F (-17 C to 232 C)
 Class: N/A (6000 psig @ 100 F)
 Ends: FNPT x FNPT
 Body: 316 SS (ASTM A479-316)
 Trim: 316 SS (ASTM A276-316)
 Body type: T-body, Straight Pattern
 Disc type: Needle
 Seat: Integral (metal)
 Orifice Size: See P/N's below
 Operation: T-Handle
 Stem-Packing: PTFE Packing

The standards are:

DESIGN ASME B16.34
 ENDS MFG STD
 RATING ASME B16.34
 DIMENSIONAL MFG STD

Vendor: Swagelok, GU Series, Straight Pattern, FNPT x FNPT
 1/4" Size (0.20" Orifice) = SS-4GUF4
 3/8" Size (0.20" Orifice) = SS-4GUF6
 1/2" Size (0.20" Orifice) = SS-4GUF8
 *****
 1/2" Size (5/16" Orifice) = SS-8GUF8
 3/4" Size (5/16" Orifice) = SS-8GUF12
 1" Size (5/16" Orifice) = SS-8GUF16
 *****
 3/4" Size (7/16" Orifice) = SS-16GUF12
 1" Size (7/16" Orifice) = SS-16GUF16

Description of NEEDLE Valve (NE6K-S16-T113)

Type: Needle, Hard Seat (Swagelok, GU Series)
 Valve size: NPS 0.25 -to- NPS 1.0
 Temperature: 0 F to +450 F (-17 C to 232 C)
 Service: Nitrogen, water, etc.
 Class: N/A (6000 psig @ 100 F)
 Ends: MNPT x FNPT (Male x Female)
 Body: 316 SS (ASTM A479-316)
 Trim: 316 SS (ASTM A276-316)
 Body type: T-body, Straight Pattern
 Disc type: Needle
 Seat: Integral (metal)
 Flow Direction: Bi-directional
 Orifice Size: See below
 Operation: T-Handle
 Stem-Packing: PTFE Packing

The standards are:

DESIGN ASME B16.34
 ENDS MFG STD
 RATING ASME B16.34
 DIMENSIONAL MFG STD

Vendor: Swagelok, GU Series, Straight Pattern, MNPT x FNPT
 1/4" Size (0.20" Orifice) = SS-4GUM4-F4
 3/8" Size (0.20" Orifice) = SS-4GUM6-F6
 1/2" Size (0.20" Orifice) = SS-4GUM8-F8
 *****
 1/2" Size (5/16" Orifice) = SS-8GUM8-F8
 3/4" Size (5/16" Orifice) = SS-8GUM12-F12
 1" Size (5/16" Orifice) = SS-8GUM16-F16
 *****
 3/4" Size (7/16" Orifice) = SS-16GUM12-F12
 1" Size (7/16" Orifice) = SS-16GUM16-F16

Description of NEEDLE Valve (ND01ST500) (NE15-S16-F500)

Type: Needle, Metal Seat, Flanged
 Valve size: NPS 0.5 -to-NPS 2
 Class: ASME 150 (PN 20)
 Service: Nitrogen, water
 Temperature: -20 F to + 482 °F (-29 °C to + 250 °C)
 Rating: 275 psig (19 bar)
 Ends: Flanged RF, CL150 Pattern
 Body: 316/316L SS (ASTM A182, Grade F316)
 Seat: 316 SS
 Trim: API TRIM 10
 Body type: T-body
 Bonnet type: Bolted
 Disc type: Needle
 Stem Design: OS&Y
 Operation: Handwheel, Plastic
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD

The standards are:

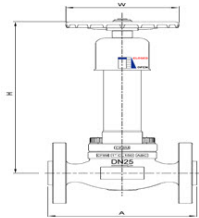
DESIGN ASME B16.34
 ENDS ASME B16.5
 RATING ASME B16.34
 DIMENSIONAL MFG STD

Approved Vendor: HY-LOK USA in Houston, TX. (Country-of-Origin: Korea)

..... DN-{Size}-PN20, CL150: Sizes 1/2" to 1" NPS

Added Vendor: Zavero (Country-of-Origin: Italy)

..... DN-{Size}-PN20, CL150: Sizes 1/2" to 2" NPS



1-3 of 3 Products



Sort by: Default

Compare

Product	Body Material	Class	DN (mm)	A (mm)	H (mm)	Ø W (mm)	Weight RF (kg)
☐ DN-1/2"-PN 20	ASTM A351 CF8M, CF3M	PN 20 - Class 150 lbs	1/2"	114	198	80	2.93
☐ DN-3/4"-PN 20	ASTM A351 CF8M, CF3M	PN 20 - Class 150 lbs	3/4"	140	260	100	5.00
☐ DN-1"-PN 20	ASTM A351 CF8M, CF3M	PN 20 - Class 150 lbs	1"	140	260	100	5.44

Description of NEEDLE Valve (ND01ST502)

(NE15-S16-F502)

Type: Needle, Metal Seat, Flanged
 Valve size: NPS 0.5 -to-NPS 2
 Service: Nitrogen, water
 Class: ASME 150 (PN 20)
 Ends: Flanged RF, CL150 Pattern
 Body: 316/316L SS (ASTM A182, F316 or ASTM A351-CF8M)
 Trim: API TRIM 12
 Body type: T-body
 Bonnet type: Bolted
 Disc type: Needle
 Stem Design: OS&Y
 Operation: Handwheel, Plastic
 Stem-Packing: PTFE/ Flexible Graphite
 Body-Bonnet: MFG. STD.

The standards are:

DESIGN ASME B16.34
 ENDS ASME B16.5
 RATING ASME B16.34
 DIMENSIONAL MFG STD

Approved Vendor: ZA.VE.RO Industrial Valves (Country-of-Origin: Italy)
 {NPS Size}-PN20, CL150, RF

Description of NEEDLE Valve (ND03ST500)

(NE30-S16-F500)

Type: Needle, Metal Seat, Flanged
 Valve size: NPS 0.5 -to- NPS 2
 Class: ASME 300 (PN 50)
 Service: Nitrogen, water
 Temperature: -20 F to + 482 °F (-29 °C to + 250 °C)
 Rating: 720 psig (50 bar)
 Ends: Flanged RF, CL150 Pattern
 Body: 316/316L SS (ASTM A182, Gr. F316)
 Seat: 316 SS
 Trim: API TRIM 10
 Body type: T-body
 Bonnet type: Bolted
 Disc type: Needle
 Stem Design: OS&Y
 Operation: Handwheel
 Stem-Packing: Flexible Graphite, Corrosion Inhibited
 Body-Bonnet Gaskets: 316 SS/Graphite
 Body-Bonnet Bolting: Bolts: MFG STD
 Nuts: MFG STD

The standards are:

DESIGN ASME B16.34
 ENDS ASME B16.5
 RATING ASME B16.34
 DIMENSIONAL MFG STD

Approved Vendor: HY-LOK USA in Houston, TX. (Country-of-Origin: Korea)
 DN-{Size}-PN50, CL300: Size Range: 1/2" to 1" NPS
 Added Vendor: Za.ve.ro (Country-of-Origin: Italy)
 DN-{Size}-PN50, CL300: Size Range: 1/2" to 2" NPS



Rev.	Rev. Date	Rev. Description	By	Checked
0	31 AUG 2023	Issued for release	MFS	SBS
1	19 MAY 2023	Minor Updates	MFS	SBS
2	11 SEPT 2023	Add Anderson Greenwood, General Needle Valves	MFS	SBS
3	08 NOV 2023	Add Flanged Needle Valves	MFS	SBS
4	11 MAR 2024	Add CL300 Flanged Needle Valves	MFS	SBS

End Document #: GCTLR00-TSLA-ME-SPC-00020**Valve Specification for SS Needle Valves**

Description of PINCH Valve (PC01DF500 ==> PN15-CI-F500):

Type:..... Pinch Valve
 Valve size: NPS 2 -to-NPS 12
 Service:..... Slurry, Throttling
 Class:..... ASME 150
 Rating: 150 psig to 6" NPS; 125 psig at 8" NPS, 100 psig at 10" to 12" NPS
 Ends:..... FF Flanged, Drilled & Tapped
 Body:..... Cast iron, Epoxy Coated
 Sleeves:..... G=Gum Rubber (Standard)
 Stem: Stainless Steel
 Indicator:..... Position Indicator (PI)
 Port: Full Port
 Flow: Bi-directional, bubble-tight shutoff
 Design (Other): N/A
 Operation: Handwheel
 Body Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Optional sleeve mat'ls; E=EPDM, N=Neoprene, B=Buna-N, or F=Fluoroelastomer

The standards are:

DESIGN MFG STD
 ENDS ASME B16.5
 FACE-TO-FACE ASME B16.10
 RATING ASME B16.34

Approved Vendor:..... Red Valve, Series 75, Cast Iron Body, {Elastomer Sleeve}, PI, SS Stem

Description of PINCH Valve (PN15-CS-F500)

Type:..... Pinch Valve
 Valve size: NPS 2 -to-NPS 12
 Service:..... Slurry, Throttling
 Class:..... ASME 150
 Rating: 150 psig to 6" NPS; 125 psig at 8" NPS, 100 psig at 10" to 12" NPS
 Ends:..... FF Flanged, Drilled & Tapped
 Body:..... Carbon Steel, Epoxy Coated
 Sleeves:..... G=Gum Rubber (Default Sleeve)
 Stem: Stainless Steel
 Indicator:..... Position Indicator (PI)
 Port: Full Port
 Flow: Bi-directional, bubble-tight shutoff
 Design (Other): N/A
 Operation: Handwheel
 Body Bolting: Bolts: MFG STD
 Nuts: MFG STD
 Optional sleeve mat'ls; E=EPDM, N=Neoprene, B=Buna-N, or F=Fluoroelastomer

The standards are:

DESIGN MFG STD
 ENDS ASME B16.5
 FACE-TO-FACE ASME B16.10
 RATING ASME B16.34

Approved Vendor: Red Valve, **Series 70**, Carbon Steel Body, {Elastomer Sleeve}, PI, SS Stem



Note: When Valve Tags are shown as something like " PC01DF500 ==> PN15-CI-F500", it means that the PIP Valve Code (PC01DF500) should be replaced with the Tesla Valve Code (PN15-CI-F500).

Rev.	Rev. Date	Rev. Description	By	Checked
0	14 SEPT 2023	Issued for release	MFS	SBS
1	18 SEPT 2023	Revise Valve Information	MFS	SBS

End Document #: GCTLR00-TSLA-ME-SPC-00021Valve Specification for SS Pinch Valves

Description of PLUG Valve (PL01ST500 ==> PL15-S16-F500)

Type:.....Plug, 2-Port, Sleeved (Non-Lubricated)
 Sub Type:.....Rectangular Port, Top Entry
 Rating:280 psig @ -20 °F to +100 °F (-29 °C to +37 °C); 190 psig @ +400 °F (+204 °C)
 Valve Size:NPS 0.5 -to- NPS 3
 Class:ASME 150
 Ends:.....Flanged RF
 Body:316 SS (ASTM A351-CF8M)
 Plug:316 SS
 Stem:316 SS
 Sleeve:PTFE
 Design (Other):Fire Safe to API 607 (External Leakage Only)
 Bore-Port:Regular Pattern - API 599
 Operation:Lever - High-Hub with Lockout
 Stem Seals:PTFE and 316 SS Diaphragms
 Stem-Packing:PTFE and Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:API 599
 ENDS:.....ASME B16.5
 RATING:MFG STD
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Xomox Tuflin, CL150, 2-Port, Sleeved Plug Valve
{Size}-067-FT-6-6-P1-W

Description of PLUG Valve (PL01ST501 ==> PL15-S16-F501)

Type:.....Plug, 2-Port, Sleeved (Non-Lubricated)
 Sub Type:.....Rectangular Port, Top Entry
 Rating:280 psig @ -20 °F to +100 °F (-29 °C to +37 °C); 190 psig @ +400 °F (+204 °C)
 Valve Size:NPS 4 -to- NPS 20
 Class:ASME 150
 Ends:.....Flanged RF
 Body:316 SS (ASTM A351-CF8M)
 Plug:316 SS
 Stem:316 SS
 Sleeve:PTFE
 Design (Other):Fire Safe to API 607 (External Leakage Only)
 Bore-Port:Regular Pattern - API 599
 Operation:Gear
 Stem Seals:PTFE and 316 SS Diaphragms
 Stem-Packing:PTFE and Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:API 599
 ENDS:.....ASME B16.5
 RATING:MFG STD
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Xomox Tuflin, CL150, 2-Port, Sleeved Plug Valve
{Size}-067-FT-6-6-P1-EG

Description of PLUG Valve (PL01ST550 => PL15-S16-F550)

Type:.....Plug, 2-Port, Sleeved Plug (Non-Lubricated), Fully Jacketed
 Sub Type:.....Rectangular Port, Top Entry
 Rating:280 psig @ -20 °F to +100 °F (-29 °C to +37 °C); 190 psig @ +400 °F (+204 °C)
 Valve Size:NPS 1 -to- NPS 4
 Class:ASME 150
 Ends:.....Flanged RF & Oversized Reducing Flanges
 Body:316 SS (ASTM A351-CF8M)
 Plug:316 SS
 Stem:316 SS
 Sleeve:PTFE
 Jacket:Fabricated or Integrally Cast and 2 NPTF Connections
 Bore-Port:.....Regular Pattern
 Operation:Lever - High-Hub with Lockout
 Stem Seals:PTFE
 Stem-Packing:.....PTFE
 Body-Bonnet Gaskets:PTFE
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:MFG STD
 ENDS:ASME B16.5
 RATING:MFG STD
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Xomox Tuflin Sleeved Plug Valve, Jacketed
{Size+1}-{Size}-{Size+1}-067-FJ-6-6-P1-W
3" Jacketed Plug Example (w/4" flanges): 4-3-4-067-FJ-6-6-P1-W

Description of PLUG Valve (PL01ST551 => PL15-S16-F551)

Type:.....Plug, 2-Port, Sleeved Plug (Non-Lubricated), Fully Jacketed
 Sub Type:.....Rectangular Port, Top Entry
 Rating:280 psig @ -20 °F to +100 °F (-29 °C to +37 °C); 190 psig @ +400 °F (+204 °C)
 Valve Size:NPS 6 -to- NPS 12
 Class:ASME 150
 Ends:.....Flanged RF & Oversized Reducing Flanges
 Body:316 SS (ASTM A351-CF8M)
 Plug:316 SS
 Stem:316 SS
 Sleeve:PTFE
 Jacket:Fabricated or Integrally Cast and 2 NPTF Connections
 Bore-Port:.....Regular Pattern
 Operation:Gear
 Stem Seals:PTFE
 Stem-Packing:.....PTFE
 Body-Bonnet Gaskets:PTFE
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:MFG STD
 ENDS:ASME B16.5
 RATING:MFG STD
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Xomox Tuflin® Sleeved Plug Valve, 2-Port, Type A, Jacketed
{Size+1}-{Size}-{Size+1}-067-FJ-6-6-P1-EG
8" Jacketed Plug Example (w/10" flanges): 10-8-10-067-FJ-6-6-P1-EG

Description of PLUG Valve (PL03ST500 => PL30-S16-F500)

Type:.....Plug, 2-Port, Sleeved (Non-Lubricated)
 Sub Type:.....Rectangular Port, Top Entry
 Rating:720 psig @ -20 °F to +100 °F (-29 °C to +37 °C); 440 psig @ +400 °F (+204 °C)
 Valve Size:NPS 0.5 -to- NPS 3
 Class:ASME 300
 Ends:.....Flanged RF
 Body:316 SS (ASTM A351-CF8M)
 Plug:316 SS
 Stem:316 SS
 Sleeve:PTFE
 Design (Other):Fire Safe to API 607 (External Leakage Only)
 Bore-Port:Regular Pattern - API 599
 Operation:Lever - High-Hub with Lockout
 Stem Seals:PTFE and 316 SS Diaphragms
 Stem-Packing:PTFE and Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:API 599
 ENDS:ASME B16.5
 RATING:MFG STD
 TESTING:API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Xomox Tuflin, CL300, 2-Port, Sleeved Plug Valve, Lever Op.
{Size}-0367-FT-6-6-P1-W

Description of PLUG Valve (PL03ST501 => PL30-S16-F501)

Type:.....Plug, 2-Port, Sleeved (Non-Lubricated)
 Sub Type:.....Rectangular Port, Top Entry
 Rating:720 psig @ -20 °F to +100 °F (-29 °C to +37 °C); 440 psig @ +400 °F (+204 °C)
 Valve Size:NPS 4 -to- NPS 20
 Class:ASME 300
 Ends:.....Flanged RF
 Body:316 SS (ASTM A351-CF8M)
 Plug:316 SS
 Stem:316 SS
 Sleeve:PTFE
 Design (Other):Fire Safe to API 607 (External Leakage Only)
 Bore-Port:Regular Pattern - API 599
 Operation:Gear
 Stem Seals:PTFE and 316 SS Diaphragms
 Stem-Packing:PTFE and Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:API 599
 ENDS:ASME B16.5
 RATING:MFG STD
 TESTING:API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Xomox Tuflin, CL300, 2-Port, Sleeved Plug Valve, Gear Op.
{Size}-0367-FT-6-6-P1-EG

Description of PLUG Valve (PL15-S16-T220)

Type:.....Plug, 2-Port, Sleeved Plug (Non-Lubricated)
 Sub Type:.....Rectangular Port, Top Entry
 Rating:280 psig @ -20 °F to +100 °F (-29 °C to +37 °C); 190 psig @ +400 °F (+204 °C)
 Valve Size:NPS 0.5 -to- NPS 2"
 Class:ASME 150
 Ends:.....FNPT (Threaded)
 Body:316 SS (ASTM A351-CF8M)
 Plug:316 SS
 Stem:316 SS
 Sleeve:.....PTFE
 Design (Other):Fire Safe to API 607 (External Leakage Only)
 Bore-Port:.....Regular - API 599
 Operation:Lever - High-Hub with Lockout
 Stem Seals:PTFE and 316 SS Diaphragms
 Stem-Packing:.....PTFE and Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:API 599
 ENDS:.....ASME B16.11
 RATING:MFG STD
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Xomox Tuflin, CL150, 2-Port, Sleeved Plug Valve, FNPT Ends
{Size}-066-FT-6-6-P1-W

Description of PLUG Valve (PL30-S16-T230)

Type:.....Plug, 2-Port, Sleeved Plug (Non-Lubricated)
 Sub Type:.....Rectangular Port, Top Entry
 Rating:720 psig @ -20 °F to +100 °F (-29 °C to +37 °C); 440 psig @ +400 °F (+204 °C)
 Valve Size:NPS 0.5 -to- NPS 2"
 Class:ASME 300
 Ends:.....FNPT (Threaded)
 Body:316 SS (ASTM A351-CF8M)
 Plug:316 SS
 Stem:316 SS
 Sleeve:.....PTFE
 Design (Other):Fire Safe to API 607 (External Leakage Only)
 Bore-Port:.....Regular - API 599
 Operation:Lever - High-Hub with Lockout
 Stem Seals:PTFE and 316 SS Diaphragms
 Stem-Packing:.....PTFE and Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:API 599
 ENDS:.....ASME B16.11
 RATING:MFG STD
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Xomox Tuflin, CL300, 2-Port, Sleeved Plug Valve, FNPT Ends
{Size}-0366-FT-6-6-P1-W

Description of PLUG Valve (PL15-S16-F530)

Type:.....Plug, 2-Port, **Fully-Lined** Plug Valve (Non-Lubricated)
 Sub Type:.....Rectangular Port, Top Entry
 Rating:270 psig @ -20 °F to +100 °F (-29 °C to +37 °C); 190 psig @ +400 °F (+204 °C)
 Valve Size:NPS 1 -to- NPS 3
 Class:ASME 150
 Ends:.....FF flange with PFA Liner on End of FF
 Body:316 SS (ASTM A351-CF8M)
 Plug:316 SS
 Stem:316 SS
 Plug Liner:.....PFA (Perfluoroalkoxy)
 Body Liner:PFA (Perfluoroalkoxy)
 Design (Other):.....N/A
 Bore-Port:.....Regular Pattern - API 599
 Operation:Lever - High-Hub with Lockout
 Stem Seals:PTFE and 316 SS Diaphragms
 Stem-Packing:.....PTFE and Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:API 599
 ENDS:.....ASME B16.11
 RATING:MFG STD
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Xomox Tuflined Lined Plug Valves, CL150, 2-Port, Flanged Ends, Lever Op.
{Size}-061-TFM-6-P6-6-P6-WY

Description of PLUG Valve (PL15-S16-F531)

Type:.....Plug, 2-Port, **Fully-Lined** Plug Valve (Non-Lubricated)
 Sub Type:.....Rectangular Port, Top Entry
 Rating:270 psig @ -20 °F to +100 °F (-29 °C to +37 °C); 190 psig @ +400 °F (+204 °C)
 Valve Size:NPS 4" -to- NPS 12"
 Class:ASME 150
 Ends:.....FF flange with PFA Liner on End of FF
 Body:316 SS (ASTM A351-CF8M)
 Plug:316 SS
 Stem:316 SS
 Plug Liner:.....PFA (Perfluoroalkoxy)
 Body Liner:PFA (Perfluoroalkoxy)
 Design (Other):.....N/A
 Bore-Port:.....Regular Pattern - API 599
 Operation:Lever - High-Hub with Lockout
 Stem Seals:PTFE and 316 SS Diaphragms
 Stem-Packing:.....PTFE and Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:API 599
 ENDS:.....ASME B16.11
 RATING:MFG STD
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Xomox Tuflined Lined Plug Valves, CL150, 2-Port, Flanged Ends, Gear Op.
{Size}-061-TFM-6-P6-6-P6-GZ

Description of PLUG Valve (PL30-S16-F540)

Type:.....Plug, 2-Port, **Fully-Lined** Plug Valve (Non-Lubricated)
 Sub Type:.....**Rectangular Port**, Top Entry
 Rating:720 psig @ -20 °F to +100 °F (-29 °C to +37 °C); 190 psig @ +400 °F (+204 °C)
 Valve Size:NPS 1 -to- NPS 3
 Class:ASME 300
 Ends:.....FF flange (PFA Liner extends beyond face of Flange)
 Body:316 SS (ASTM A351-CF8M)
 Plug:316 SS
 Stem:316 SS
 Plug Liner:.....PFA (Perfluoroalkoxy)
 Body Liner:PFA (Perfluoroalkoxy)
 Design (Other):.....N/A
 Bore-Port:.....Regular Pattern - API 599
 Operation:Lever - High-Hub with Lockout
 Stem Seals:PTFE and 316 SS Diaphragms
 Stem-Packing:.....PTFE and Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:API 599
 ENDS:ASME B16.11
 RATING:MFG STD
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Xomox Tuflined Lined Plug Valves, CL300, 2-Port, Flanged Ends, Lever Op.
{Size}-0361-TFM-6-P6-6-P6-WY

Description of PLUG Valve (PL30-S16-F541)

Type:.....Plug, 2-Port, **Fully-Lined** Plug Valve (Non-Lubricated)
 Sub Type:.....**Rectangular Port**, Top Entry
 Rating:720 psig @ -20 °F to +100 °F (-29 °C to +37 °C); 190 psig @ +400 °F (+204 °C)
 Valve Size:NPS 4" -to- NPS 6"
 Class:ASME 300
 Ends:.....FF flange (PFA Liner extends beyond face of Flange)
 Body:316 SS (ASTM A351-CF8M)
 Plug:316 SS
 Stem:316 SS
 Plug Liner:.....PFA (Perfluoroalkoxy)
 Body Liner:PFA (Perfluoroalkoxy)
 Design (Other):.....N/A
 Bore-Port:.....Regular - API 599
 Operation:Lever - High-Hub with Lockout
 Stem Seals:PTFE and 316 SS Diaphragms
 Stem-Packing:.....PTFE and Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:API 599
 ENDS:ASME B16.11
 RATING:MFG STD
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Xomox Tuflined Lined Plug Valves, CL300, 2-Port, Flanged Ends, Gear Op.
{Size}-0361-TFM-6-P6-6-P6-GZ

Description of PLUG Valve - for Duplex SS Pipe - (PL01SP500 ==> PL15-S22-F500)

Type:.....Plug, 2-Way, Sleeved (Non-Lubricated)
 Sub Type:.....**Rectangular Port, Top Entry**
 Rating:265 psig @ -20 °F to +100 °F (-29 °C to +37 °C)
 Valve Size:NPS 1 -to- NPS 12
 Class:ASME 150
 Ends:Flanged RF
 Body:2205 Duplex SS (ASTM A995-4A, ACI CD3MN)
 Plug:2205 Duplex SS (ASTM A995-4A, ACI CD3MN)
 Cover:A351C
 Top Seal:.....T600 & 304 SS
 Stem:316 SS
 Sleeve:**T600 – Carbon Filled TFM**
 Bore-Port:.....Rectangular Pattern
 Design (Other):.....Fire Safe to API 607 (External Leakage Only)
 Operation:Lever – Lever Op to 3” NPS; Gear Op. for 4” to 12” NPS
 Stem Seals:PTFE and 316 SS Diaphragms
 Stem-Packing:.....PTFE and Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:API 599
 ENDS:ASME B16.5
 RATING:MFG STD
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Xomox Tuflin, CL150, 2-Port, Sleeved Plug Valve
{Size}-067-FT-27-27-P20-W (1” to 3” NPS)
{Size}-067-FT-27-27-P20-EG (4” to 12” NPS)

Description of PLUG Valve – for Duplex SS Pipe (PL01SP501 ==> PL15-S22-F501)

Type:.....Plug, 2-Way, Full-Port, Sleeved (Non-Lubricated)
 Sub Type:.....**Round Bore, Top Entry**
 Rating:265 psig @ -20 °F to +100 °F (-29 °C to +37 °C)
 Valve Size:NPS 1 -to- NPS 12
 Class:ASME 150
 Ends:Flanged RF
 Body:2205 Duplex SS (ASTM A995-4A, ACI CD3MN)
 Plug:2205 Duplex SS (ASTM A995-4A, ACI CD3MN)
 Cover:A351C
 Top Seal:.....T600 & 304 SS
 Stem:316 SS
 Sleeve:**T600 – Carbon Filled TFM**
 Bore-Port:.....**Full Port, Round Bore**
 Design (Other):.....Fire Safe to API 607 (External Leakage Only)
 Operation:Lever – Lever Op to 3” NPS; Gear Op. for 4” to 12” NPS
 Stem Seals:PTFE and 316 SS Diaphragms
 Stem-Packing:.....PTFE and Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:API 599
 ENDS:ASME B16.5
 RATING:MFG STD
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Xomox Tuflin, CL150, 2-Port, Full-Port, Round Bore, Sleeved Plug Valve
{Size}-2067-FT-27-27-P20-W (1” to 3” NPS)
{Size}-2067-FT-27-27-P20-EG (4” to 12” NPS)

Description of PLUG Valve – for Duplex SS Pipe (PL03SP500 ==> PL30-S22-F500)

Type:.....Plug, 2-Way, Sleeved (Non-Lubricated)
 Sub Type:.....**Rectangular Port, Top Entry**
 Rating:.....720 psig @ -20 °F to +100 °F (-29 °C to +37 °C). Reduced pressure at 400 F
 Valve Size:NPS 1 -to- NPS 8
 Class:**ASME 300**
 Ends:.....Flanged RF
 Body:2205 Duplex SS (ASTM A995-4A, ACI CD3MN)
 Plug:2205 Duplex SS (ASTM A995-4A, ACI CD3MN)
 Cover:A351C
 Top Seal:.....T600 & 304 SS
 Stem:316 SS
 Sleeve:T600 – Carbon Filled TFM
 Bore-Port:.....**Rectangular Pattern**
 Design (Other):.....Fire Safe to API 607 (External Leakage Only)
 Operation:Lever – Lever Op to 3” NPS; Gear Op. for 4” to 8” NPS
 Stem Seals:PTFE and 316 SS Diaphragms
 Stem-Packing:.....PTFE and Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:API 599
 ENDS:.....ASME B16.5
 RATING:MFG STD
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Xomox Tuflin, CL150, 2-Way, Sleeved Plug Valve
{Size}-0367-FT-27-27-P20-W (1” to 3” NPS)
{Size}-0367-FT-27-27-P20-EG (4” to 8” NPS)

Description of PLUG Valve – for Duplex SS Pipe (PL03SP501 ==> PL30-S22-F501)

Type:.....Plug, 2-Way, **Full-Port**, Sleeved (Non-Lubricated)
 Sub Type:.....**Round Bore, Top Entry**
 Rating:.....720 psig @ -20 °F to +100 °F (-29 °C to +37 °C). Reduced pressure at 400 F
 Valve Size:NPS 1 -to- NPS 8
 Class:**ASME 300**
 Ends:.....Flanged RF
 Body:2205 Duplex SS (ASTM A995-4A, ACI CD3MN)
 Plug:2205 Duplex SS (ASTM A995-4A, ACI CD3MN)
 Cover:A351C
 Top Seal:.....T600 & 304 SS
 Stem:316 SS
 Sleeve:T600 – Carbon Filled TFM
 Bore-Port:.....**Full Port, Round Bore**
 Design (Other):.....Fire Safe to API 607 (External Leakage Only)
 Operation:Lever – Lever Op to 3” NPS; Gear Op. for 4” to 8” NPS
 Stem Seals:PTFE and 316 SS Diaphragms
 Stem-Packing:.....PTFE and Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:API 599
 ENDS:.....ASME B16.5
 RATING:MFG STD
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Xomox Tuflin, CL150, 2-Way, Full-Port, Sleeved Plug Valve, w/T600 Sleeve
{Size}-20367-FT-27-27-P20-W (1” to 3” NPS)
{Size}-20367-FT-27-27-P20-EG (4” to 8” NPS)

Description of PLUG Valve - for Super Duplex SS (PL01SQ500 ==> PL15-S25-F500)

Type:.....Plug, 2-Port, Rectangular Port, Sleeved (Non-Lubricated)
 Sub Type:.....**Rectangular Port, Top Entry**
 Rating:265 psig @ -20 °F to +100 °F (-29 °C to +37 °C)
 Valve Size:NPS 1 -to- NPS 12
 Class:ASME 150
 Ends:Flanged RF
 Body:2507 Super Duplex SS (ASTM A995-5A, ACI CE3MN)
 Plug:2507 Super Duplex SS (ASTM A995-5A, ACI CE3MN)
 Cover:Super Duplex SS
 Top Seal:.....T600 & 304 SS
 Stem:316 SS
 Sleeve:**T600 – Carbon Filled TFM**
 Bore-Port:.....Rectangular Pattern, Full Port
 Design (Other):.....Fire Safe to API 607 (External Leakage Only)
 Operation:Lever – Lever Op to 3” NPS; Gear Op. for 4” to 12” NPS
 Stem Seals:PTFE and 316 SS Diaphragms
 Stem-Packing:.....PTFE and Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:API 599
 ENDS:ASME B16.5
 RATING:MFG STD
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Flowserve Durco “G4Z” Sleeved Plug Valve, CL150, RF Flange, 2-Port
{Size}-G4Z11C-ZZ0-{TFE}-0-{Wrench} (1” to 3” NPS)
{Size}- G4Z11C-ZZ0-{TFE}-0-{Gear} (4” to 12” NPS)

Description of PLUG Valve - for Super Duplex SS (PL03SQ500 ==> PL30-S25-F500)

Type:.....Plug, 2-Way, Rectangular Port, Sleeved (Non-Lubricated)
 Sub Type:.....**Rectangular Port, Top Entry**
 Rating:720 psig @ -20 °F to +100 °F (-29 °C to +37 °C)
 Valve Size:NPS 1 -to- NPS 8
 Class:**ASME 300**
 Ends:Flanged RF
 Body:2507 Super Duplex SS (ASTM A995-5A, ACI CE3MN)
 Plug:2507 Super Duplex SS (ASTM A995-5A, ACI CE3MN)
 Cover:Super Duplex SS
 Top Seal:.....T600 & 304 SS
 Stem:316 SS
 Sleeve:**T600 – Carbon Filled TFM**
 Bore-Port:.....Rectangular Pattern, Full Port
 Design (Other):.....Fire Safe to API 607 (External Leakage Only)
 Operation:Lever – Lever Op to 3” NPS; Gear Op. for 4” to 8” NPS
 Stem Seals:PTFE and 316 SS Diaphragms
 Stem-Packing:.....PTFE and Graphite
 Body-Bonnet Bolting:Bolts: MFG STD
Nuts: MFG STD

The standards are:

DESIGN:API 599
 ENDS:ASME B16.5
 RATING:MFG STD
 TESTING:.....API 598
 DIMENSIONAL:MFG STD

Approved Vendors:Flowserve Durco “G4Z” Sleeved Plug Valve, CL300, RF Flange, 2-Port
{Size}-G4Z31C-ZZ0-{TFE}-0-{Wrench} (1” to 3” NPS)
{Size}- G4Z31C-ZZ0-{TFE}-0-{Gear} (4” to 8” NPS)



Note: When Valve Tags are shown as something like " PL01ST500 ==> PL15-S16-F500", it means that the PIP Valve Code (PL01ST500) should be replaced with the Tesla Valve Code (PL15-S16-F500).

Rev.	Rev. Date	Rev. Description	By	Checked
0	07 SEPT 2023	Issued for release	MFS	SBS
1	18 SEPT 2023	Revised Valve Tag	MFS	SBS
2	08 NOV 2023	Add Valves Tags	MFS	SBS
3	10 NOV 2023	Revise Round-Bore Sleeved Plug Valves	MFS	SBS